

# HOLOGRAPHIC CONSTRUCTION OF THE UNIVERSE FROM ITS PLANE BOUNDARY

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TO MY WIFE EFTYXIA

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# Chapter 1

## PREFACE

The Universe observed today does not seem to be finite, unchanging, and eternal; that is, it has not remained the same from the moment of its creation until today. The observation of the Universe raises a multitude of questions that were raised centuries ago without having been answered to this day, some of which are the following:

1. **Where did all that we observe around us come from?**
2. **Was there a Creator who made all of this, or was it created from nothing, and if so, how?**
3. **What are the rules for structuring space and the objects within it that determine everything that we perceive with our senses?**
4. **Is everything we see objective or is it the result of unknown brain processes?**

We will try to give answers to all questions of this type, using another method (the full scientific method) which is a combination of the method proposed by Plato (through the allegory of the cave) and the method proposed by Galileo (scientific method), so that the conclusions from this new method are consistent with experimental and observational data.

Observational data show that in the Universe the production of each form of matter at present is so small, compared to previous eras. The density of matter tends to zero due to the expansion of empty space. This conclusion was derived from observations of both the distant and recent past of the Universe. Studying the distant past of the Universe, it is found that the density of matter-energy was many orders of magnitude greater than the energy density of all other forms. As observations approach the present day, the density of matter and the density of a type of energy that observers have called dark energy, were found to be equal.

The term "dark energy" stems from the fact that it is not observed directly—unlike the energy released by solar flares—but indirectly, through the accelerating expansion of the Universe. The conclusion that emerges from the process of comparing the two densities is that in the future the density of dark energy will become many orders of magnitude greater than the density of matter and will be the main component of the Universe. The above conclusions are based on the methodology that has been developed for the observation of the past of the Universe, which we will examine immediately. How an observer can see today what was happening in the recent or distant past in a region of the Universe is nothing more than studying the light from nearby or distant bright celestial bodies.

For example, a phenomenon such as a solar flare that occurs on the Sun at some point in time is observed on Earth after about 8 minutes and 30 seconds, that is, the time it takes light to travel the distance from the Sun to the Earth. Through the information carried by the light, we see a snapshot of the Sun's recent past. With the same reasoning, the study of the light coming from distant stars or Galaxies or other celestial bodies shows what they were like in their distant past.

At the beginning of the last century, the American astronomer Edwin Hubble analyzed the light spectrum of the chemical elements that compose distant galaxies and determined their velocities. The dominant idea at that time was that the Universe is static and eternal, so Hubble expected to find an equal number of galaxies moving towards the Earth as those moving away from it. However, what he found was that almost all galaxies are moving away from the Earth and the further they are from it, the greater their speed. This observation was formulated through a relationship that expresses the ratio of the speed of a galaxy and the distance of the observer from it and was called Hubble law. The removal of a galaxy is determined by the fact that the color of the light we receive from it shifts towards the red (redshift). This means that the distance ( $\lambda$ ) between two consecutive wave peaks (the wavelength) increases ( $\lambda > \lambda'$ ), as shown in Figure 1.1. This phenomenon was interpreted as an expansion of the space in which the wave moves.

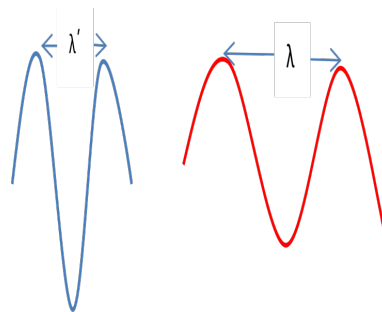


Figure 1.1: Redshift

Based on this observational data, Hubble proposed that the Universe is expanding, so the model of a static Universe that had dominated until then began to shake. After observations that lasted decades, it was found that the expansion of the Universe is not at a constant but at a constantly increasing speed, that is, with an acceleration whose rate, however, is constantly decreasing. From this observational data, namely the change in the density of matter due to the expansion of the Universe, it is immediately understandable that in the Universe the proportion of its components and their density change over time, so it is neither unchanging nor eternal. These new data lead to a fundamental revision of the foundation of Physical theories on the one hand and of philosophical concepts on the other. Although one may claim that philosophy has today been replaced by science and especially by Physics, the philosophical questions that were raised centuries ago can provide the impetus for a different approach to natural phenomena through the prism of modern discoveries.

Let us begin with a brief review of the perceptions formed to date on the above issues and how man, through the systematic observation and classification of celestial phenomena, came to demolish one by one the strange and often absurd ideals of centuries.

To explain the creation of the Universe, all ancient mythologies assumed the existence of anthropomorphic, supernatural beings who constructed the world. This view led, through images and representations of everyday human activities, on the one hand to the interpretation of the creation of the Universe, through some supernatural "Creator" and on the other hand to the attribution of conflicting characteristics and properties to the Creator "as they please". This is one version of the creation of the Universe. In Greek Mythology there is a differentiation regarding the beginning of everything, where according to Hesiod's Theogony it is defined that in the beginning there was Chaos. Chaos was the "Space that contained in seed" everything that would later constitute the Universe, according to the work of Jean Richepin "Greek Mythology".

The attribution of human form to this Being and to his descendants was a law so that people could interpret natural phenomena through processes and representations understandable to them. Thus, people of all religions created their gods "in their image and likeness". Today, the concept-foundation for these supernatural beings is outdated because the questions that arise from such a foundation cannot be answered. The only thing one can hold on to is the assumption that in the beginning there was only Space, according to Greek Mythology. Minimizing this assumption we will consider that this space is the simplest possible, so we will see how the complexity of the three-dimensional World that we observe today at a microscopic and macroscopic level can arise. The other version that remains is that what we call the Universe today was created from "Nothing" or "Vacuum".

The word "vacuum" in physics today has a different meaning from the

word "nothing" in colloquial speech, where it denotes a completely empty whole. In physics, the vacuum is understood as a state characterized by the presence of "virtual particles" which, under appropriate conditions, can manifest in our "real world". This again seems equally absurd as the interpretation of the creation of the Universe by supernatural beings made in the image and likeness of man or vice versa, that is, that men are made in the image and likeness of supernatural beings. This is the first big question. Let us bypass for the moment the question of the beginning of the creation of the Universe and let us ask the next question:

**Since the Universe is evolving, what will be the end of the Universe that we observe today?**

The most "logical" answer to this question has two parts, which are the following: In the first case of the creation of the Universe by a Creator who exists outside and beyond the Universe, the end of the Universe will depend on the moods of the Creator. Thus the Creator will decide or has decided whether the Universe will continue to expand to infinity, or will begin to contract to an infinitesimal area at which point the Big Crunch will occur, or finally before the Big Crunch a new Big Bang will occur and the Universe will begin to expand again.

This means that the Universe will oscillate between these two states, which is called the Big Bounce.

In the second case, the Universe will start from Nothing and end up where it started, that is, in Nothing. The question that arises is how we will determine the characteristics and properties of Nothing. Here we will adopt and concretize Hesiod's hypothesis that Nothing or Chaos is the simplest possible space from which the creation of Everything will begin, as we will see in detail below. If we accept the first interpretation of the creation of the Universe, then the contradiction arises that the supernatural Being has freedom of will to decide on the fate of humans while humans do not because they are bound by the Creator's dispositions for the fate of the Universe and therefore their own. This view, on the one hand, contradicts the freedom of will that every human being considers to have in his daily activities and, on the other hand, destroys the two-way correspondence of the construction of human beings in the image and likeness of the Creator. If we consider that the Universe that we "observe" began from Nothing, then there should be a plausible answer as to how all that we observe around us was created and how all of this will eventually cease to exist, returning to Nothing. When we talk about the extinction of everything, we do not mean the extinction of Earth's flora and fauna from the excessive heat of the Sun when it "bloats" due to a lack of nuclear fuel and becomes a "red giant," the boundaries of which will reach the Earth. Red giants represent an intermediate stage in the evolution of stars similar to our Sun. Having completely converted the hydrogen in their core into helium and other chemical elements, they expand up to 100 times their original size. Obviously, under these extreme conditions for life



on Earth, there will be no human observers to ask such questions. This situation will resemble Hell as described by the first “Christian philosophers” who, when asked questions about creation, would scare the questioners by saying that “God is preparing Hell for those who ask such questions”.

But even if people manage to avoid their local fate, that is, the destruction of our Solar System as we know it today, by moving to a planet in another solar system with similar conditions to ours, the problem is not solved but simply postponed. The disappearance of everything will occur due to the almost zero density of visible matter (Stars, Galaxies, etc.) and dark matter due to the eternal expansion of the Universe, with the consequence that the finite amount of matter constitutes the grains of an invisible cosmic dust.

This picture is similar to the picture of someone dropping a small spoonful of salt into a large lake of fresh water and then tasting the lake water. Obviously, the lake water will not taste salty because the density of the salt (mass of salt per volume of lake water) tends to zero. The reason for the initial and subsequent expansion of the Universe is due to the fact that space should not display any feature when observed, thus maintaining the “structure of Nothingness”. The eternal expansion of the Universe also has as a consequence the fall of its temperature. As we will see, the temperature of the flat boundary of the Universe is proportional to the curvature of the circle that surrounds it, which continuously decreases with the expansion of this circle, as we will see in Appendix 3. Thus, the Universe will be led, among other things, to thermal death due to cold.

Already, as we know, the temperature of the Cosmic Microwave Background Radiation (CMBR) has dropped to 2.72 degrees Kelvin above absolute zero on the Kelvin scale today, from about  $10^{32}$  degrees Kelvin, i.e. 100,000,000,000,000,000,000,000,000,000,000 degrees Kelvin that it had at the beginning of the creation of the Universe. As is obvious, following the second alternative path, we must start from Nothing to create a theory from which Everything is created and interpreted, and from Everything the theory must evolve so as to end up in the initial state, i.e. Nothing. According to the above, this theory should be based on the minimum number of properties that arise from Nothing that will be described by an immaterial idea, such as a mathematical space without any special features.

This second alternative path implies that the Universe may be an illusion created in the human brain reminiscent of the Hindu philosophy. The word Maya, in Sanskrit means magic or illusion and is a fundamental concept in Hindu philosophy. Maya originally expressed the “magic power” through which a god can make people believe as real what is an illusion. The “power” is just a word where the cause, the purpose of its existence and the process of its action are not clarified. This concept later came to mean the powerful force that creates in people the cosmic illusion that the apparent world is real. Through this worldview, the question is asked:

**“Is there objective reality or is what we perceive as objective reality some unknown brain functions?”**

An analytical formulation of this question is Plato’s allegory of the cave where there are two types of observers, the bound and the free observers, as shown in the following Figure.



Figure 1.2: Plato’s Allegory of the Cave

The bound observers are inside the cave and are tied in such a way that they “see” only the shadows of the various objects or entities that are depicted on the two-dimensional walls of the cave, due to a fire that burns constantly behind their backs. Thus, the bound observers believe that these shadows represent reality.

If some of them are freed from their bonds and go outside the cave (free observers) then they “see” the various objects or entities as three-dimensional objects or three-dimensional entities.

Thus, the free observers believe, from their own visual point of view, that these represent reality.



Figure 1.3: Meadow sown with the plant kanola

The meadow in Figure 1.3 is sown with canola. The bound observers, seeing all the plants macroscopically, consider that they form a two-dimensional homogeneous space.

**Macroscopically, the space perceived by bound observers has**

**fewer dimensions but provides more knowledge..**

The free observers, seeing only the neighborhood of each plant microscopically, consider it as a three-dimensional object, so by extending this image everywhere they consider that the space as a whole is also three-dimensional (3D).

**Microscopically, the space perceived by free observers has more dimensions but provides less knowledge..**

The question is whether these two views can be unified into one satisfying the observations of both groups of observers and how this can be done given that human observers are single entities. An answer to this question can be given by investigating the human brain by separating it into the part where it receives information and the part that processes it by creating images of the environment in which it exists. Information about the environment is initially stored on the two-dimensional surface of the cerebral cortex. The cerebral cortex, that is, the outer layer of the brain, plays a key role in memory, attention, perceptual awareness, thought, language and consciousness. Through this process, the original two-dimensional image of the environment is transformed into another three-dimensional image. How this is done is, at present, an unknown brain function.

Plato's two groups of observers can be mapped to the two processes of the cerebral cortex and coexist in the same individual. The unification of the views of the two groups of observers can be done by making a one-to-one correspondence between the geometric characteristics of the flat geometric shapes, through which shadows are simulated, and the characteristics of the three-dimensional objects that the human brain "realizes" as "real". If free observers cannot give an explanation for some phenomena they observe, then they reduce them to problems of two-dimensional space for the bound observers to solve. If we accept the illusion as a brain function, such as a mental conception (idea or construction of a theory) that exists even though it can be considered as non-real, then from Nothing, considered as a mathematical space, a theory can be constructed that contains and explains Everything as we will see later. The explanation of Everything means that the conclusions of the geometric theories constructed by the bound observers should be checked by corresponding theories of the free observers. They should also be checked with the control instruments of the free observers, that is, experimentally or observationally. If, on the one hand, the conclusions of the two theories coincide with each other and, on the other hand, the conclusions of the theories of free observers agree with the results of experimental or observational tests, then and only then should the theory be accepted. Plato suggested to the students of his Academy that they depict the luminous objects of the night sky and their formations on a flat sheet of paper and correspond to geometric shapes. Finally, they should solve the geometric problems that arise from the geometric shapes. The solutions to these problems should be used to improve people's everyday lives.



Figure 1.4: Observation of the night sky (Novikov,1992)

A similar method is used to this day by the Aborigines, the indigenous inhabitants of Australia, to find Emu eggs for their food (Emu is a species of Australian bird) and also to find the shortest paths on the Earth's surface, for transporting goods from one place to another. This method, which is based on the depiction of the "star path formations" of the night sky on a sheet of paper, strangely enough and without obvious scientific justification, works very well in serving the needs of the Aborigines. The question that arises is the following:

**What must the structure of the "walls" of Plato's cave be like for the human mind to perceive the space within which it lives and observes?**

Today we have reached such a point of knowledge where we know from observations that the Universe as a whole must be a two-dimensional flat space that surrounds the three-dimensional space that we perceive with our senses (as is the case with the surface of the Earth and its interior).

The claim that many authors invoke that the Ancient Greeks had not discovered the scientific method, that is, a method of verifying theories, is wrong. The process that Plato proposed is not the precursor of the scientific method but a different methodology for exploring the Universe. Also, this methodology is a specialization of the allegory of the cave where the bound observers determine the appropriate geometric shapes similar to the shadows that they consider as objective entities and through two-way correspondences the free observers find the three-dimensional entities that they also consider as objective entities. Galileo based himself on the general principles of this methodology, that is, on the two-way correspondences between the elements of two sets, to establish, in three-dimensional space only, another principle called the scientific method. According to Galileo, for a theory to be valid in three-dimensional space, the conclusions of the theory of the free observers must coincide with the results of their experimental or observational tests. Our view is that for a theory to be acceptable, two steps must be taken. The first step is that there must be a two-way correspondence between the theory of bound and the theory of free observers (Plato's Method). The second step

is that the scientific method must be valid in three-dimensional space. We will call the methodology that unifies Plato's and Galileo's approaches the "complete scientific method".

In this work, we will initially follow the methodology developed by Plato, that is, we will apply the first step and then we will invoke the scientific method in three-dimensional space, that is, the second step.

Through this process, we will apply the complete scientific method, in order to understand, from the answers we will get, the greatest mystery we call the Universe. The purpose of this book is to formulate a self-consistent theory that can interpret observational and experimental data and provide plausible answers about the creation, evolution, and ultimate fate of the Universe through a combination of philosophical questions and modern discoveries. Man's attempt to understand Nature led him to observe natural phenomena, record the conclusions of their study and finally construct models through which he simulated natural phenomena. Early models of interpretation of natural phenomena include Mythologies of all ancient peoples where for almost every natural phenomenon there was an anthropomorphic being with supernatural powers that caused or stopped a natural phenomenon. For example, Indian and Greek Mythology, which are related in many ways, are structured on these views, where the "human passions" of supernatural beings and the interpretation of natural phenomena are mixed. In Greek Mythology, the battle between the Gods of Olympus and the Giants demonstrates the intense geological phenomena that had occurred in a specific geological period, creating the current geological landscape. The relationship between the undulations of the Earth's crust (mountains-valleys) and earthquakes has been captured in Greek Mythology by the fact that the Goddess Athena "buried" Enceladus, who was the leader of the Giants, under Sicily. The volcanoes of Sicily are mythologically linked to the "wrath of Enceladus". In these models, over the centuries, all kinds of "human activities" of supernatural beings have been separated from the natural phenomena caused by these entities. Thus, the concept of supernatural beings has been replaced by the impersonal concept of "Nature" and their activities by the concept of "natural laws" which were considered the cause of the appearance of natural phenomena.

This resulted in the emergence of a new type of model where the reason for the existence of physical laws was sought to understand the structure and functioning of Nature. These models "stored" the observations and knowledge that their builders had acquired from the study of natural phenomena, encoded by simple shapes, for future use. The initial models of the new type were simply geometric models, simulating the movements of celestial bodies. The processing of geometric shapes was done through Mathematics, opening up new ways of understanding Nature. Despite their simplicity, these geometric models provided predictions that were quite satisfactory for the requirements of their time. Over the centuries, the use of models has

expanded to all areas of human activities and it has become understood that only through them can one approach and explore natural phenomena. Through the interaction of models and new observations of natural phenomena, the models were enriched with new data and methods that helped to more fully understand and describe Nature. The criterion for the validity of a model and the optimal simulation of nature by these models was and is the best correspondence between the numerical values of physical constants through theoretical predictions and their experimental verifications. Since models simulate natural phenomena, then all the processes of natural phenomena should be depicted in the models. But this means that if all these processes were known or a specific law by which these processes occur, it would not be necessary to construct the model. Nature does not easily reveal its “laws”, therefore the construction of a model requires the establishment of axioms or principles that imply the hidden laws of Nature. The difference between these two visions lies in the positioning of the cause-effect relationship related to the real world or to objective reality. This dilemma was posed by Plato through the allegory of the cave, defining two groups of observers, one consisting of bound observers inside a cave and the other of free observers outside it.

Bound observers believe that reality is determined by the two-dimensional shadows of entities and their changes on the walls of the cave, and there is a cause-effect relationship that determines their existence. These observers believe that objective reality is related to a two-dimensional generator space from which their three-dimensional representations are created.

The group of free observers considers reality to be intertwined with their existence, since their presence determines the measurement of the physical quantities required to verify the theories they construct to interpret natural phenomena and, on the other hand, they themselves can participate in the cause-effect relationship that causes natural phenomena. In other words, free observers consider the observed world to be *autonomous*. A third answer to this question is that spaces of all dimensions are similar to each other, and from the space with the smallest dimensions, all other spaces are created sequentially through simple mappings. These spaces are similar to each other because of this construction, so it makes no sense to ask which one is related to objective reality.

- In this paper we will show, among other things, how one can derive all the physical theories of a three-dimensional space from a two-dimensional flat space through simple bidirectional mappings of simple geometric shapes. This process paves the way for extending three-dimensional space into a four-dimensional space (with or without curvature). The initial construction of a three-dimensional space from a two-dimensional space is based on the mapping of specific geometric objects of the flat two-dimensional space to “holograms” of a three-dimensional space, which are identified with experimentally confirmed physical entities.

This paper attempts to integrate all the so-called Physical theories, including Gravity, through the correlation of the geometric objects of a two-dimensional flat space ( $\mathbf{P}^2$ ), and the physical entities of a three-dimensional space ( $3D$ ). Evidence for certain propositions or relationships is not given in many cases in detail for the economy of space, but is given in the appendices or references.

In this work we will initially follow the methodology developed by Plato, that is, we will apply the first step and then we will invoke the scientific method in three-dimensional space, that is, the second step.

Through this process we will apply the full scientific method, in order to understand, from the answers we will get, the greatest mystery we call the Universe. The object of this book is to formulate a self-consistent theory that can interpret the observational and experimental data and give plausible answers for the creation, evolution and ultimate fate of the Universe through a combination of philosophical questions and modern discoveries.

## **THE ROLE OF THE BORDER OF THE UNIVERSE**

According to what we have said about the role of the 2D-boundary of the Universe in the creation of the holographic Universe, its 3D-interior arises from the structure of the 2D-boundary. Thus, answers are given to questions that cannot be answered by examining them only through the properties of the 3D-space. The big issue is that 3D-observers, having no idea how to solve problems that arise during the study of 3D-space, begin to make definitions, axioms, laws and theories to solve them. This process leads to dead ends because some definition or axiom or law or theory contradicts some previous definition or axiom or law or theory, so something new is sought to replace the old one. Thus, Newton's theory refers to the processes and causes of motion of bodies, when they move towards or from the Earth's surface or in a random direction, defining the anthropocentric quantities:

1. Force
2. Mass (Inertial and Gravitational)
3. Bodies, i.e. entities that are considered to consist of mass
4. Fields that replace forces, etc.

Newton's theory has limits both downwards and upwards, i.e. it cannot be applied to the microcosm and the macrocosm respectively. To fill this gap, Electromagnetism was developed by Maxwell, Quantum Mechanics by Schrödinger, Dirac, Heisenberg, Pauli, etc., and Relativity by Einstein. The big question is how all these theories for the entire spectrum of 3D-space are unified into one theory without each of them constructing definitions, axioms and laws for its proper functioning. What solves the problems presented

in 3D-space is the participation of the boundary of 3D-space in the whole process. So the first thing that needs to be done is to find, on the one hand, the most suitable 2D-unit for building the 2D-boundary and, on the other hand, the process of building it from this unit. In this second process, the observations of free observers should be used to construct the boundary of the Universe. The main observations of free observers are:

1. The definition of a non-point building unit of the 2D-boundary where the Planck length will be used for its construction which defines the concept of distance and
2. The self-similar building of the 2D-boundary that appears in all structures of 3D-space.

The whole process is reminiscent of Plato's allegory of the cave where the bound observers study the shadows of the 3D-objects on the 2D-boundary of the cave and when they are freed from their bonds and become free observers then they study the 3D representations of the 2D-geometric objects. The free observers can correlate the observations they had made as bound observers with their current observations. From these observations they find correlations between the two spaces and therefore answers to the problems that arise in the 3D-space. Let's look at some examples. The free observers cannot give answers without the help of the bound observers. Let's look at some of them:

1. How do we find the best geometric unit to create the flat 2D-boundary of the Universe?
2. What is space and time and how are they created?
3. How do the geometric objects of the 2D flat space correspond to entities of the 3D flat space?
4. Why do the 3 generations of particles exist and why do their masses have the values that have been found experimentally?
5. Why do the ionization energies of the  $H_2$  isotopes theoretically have the same numerical values that we measure experimentally?
6. Why do we continue to use the concept of mass even though we know that macroscopic bodies are 99% empty? What can the concept of mass correspond to in the 2D-boundary of the Universe?
7. What are the values of the masses  $m_{(\nu_e)}, m_{(\nu_\mu)}, m_{(\nu_\tau)}$  of the three generations of neutrinos  $\nu_e, \nu_\mu, \nu_\tau$ ?
8. What is the reason for the "attraction" of two electrically neutral bodies? Why do two hydrogen atoms, neutral in terms of charge and spin, combine to form the hydrogen molecule  $H_2$ ?



9. Why do we not encounter in all cases a "union of bodies" but a rotation of a particle around another body as happens for example with the electron that rotates around the nucleus of the hydrogen atom microscopically or of the planets with the Sun macroscopically?
10. What modifications must bound observers make to a circle or an ellipse so that on the one hand they get only one value for each of the 3 geometric quantities in each of its quadrants and on the other hand this value coincides numerically with the value of the corresponding quantity that free observers experimentally measure in an elementary particle?
11. If from one quadrant of a circle or ellipse the total charge  $q_c$ , the total mass  $m_c$  and the total orientation of  $\hat{O}_c$  arise, then how do the other three quadrants of the circle or ellipse behave?
12. What is the geometric reason for the expansion of  $P^2$ -space and consequently the expansion of 3D-space? Does the so-called Dark Energy play a role in the expansion of the Universe or is there another cause?
13. What is the geometric reason for the composition of simple geometric objects into increasingly complex geometric objects?
14. How can neutral overall geometric objects, such as circles and ellipses, whose total charge and total orientation are zero, combine with each other to create even more complex geometric objects?
15. Why do quarks carry fractional charges?
16. What is the combination of charges and orientations of the quadrants of the circles of radii  $r = 1$  and  $r = 2$  such that z.c. $P^2$  is satisfied for the charge and orientation?
17. Does the electron always behave like a wave with charge  $q = -1$  and orientation  $\hat{O} = \hat{e}_3$ , or can it also behave like a particle or a wave-particle?
18. What is the contribution of antimatter to the structure of the Universe?
19. How do we find the value of the fine texture constant theoretically?

Despite the extraordinary successes of modern physics, many of the deepest questions remain unanswered. In this book we will examine in detail the correspondences between the simple and complex geometric shapes of the 2D-boundary of the Universe with the simple and complex entities of 3D-space, obtaining answers to unsolved problems of 3D-space.



**Part I**

**GEOMETRIC OBJECTS**



## Chapter 2

# SPACES

### 2.1 Nothingness or

Every Euclidean space  $\mathbf{R}_0^n$ ,  $n = 1, 2, 3 \dots$  with  $n$  dimensions is defined as the space consisting of geometric points, each of which by definition occupies zero space. These geometric points are the “building units” of  $\mathbf{R}_0^n$  and are located next to each other, “covering” the entire  $\mathbf{R}_0^n$ . From this construction,  $\mathbf{R}_0^n$  is defined as “Nothingness” or as “Empty Space”, which has the following properties:

1. “Compact space”, i.e., without gaps between its building units.
2. Each of its directions is the same as any other (homogeneity or translational symmetry).
3. Each rotation of it around a structural unit leaves it unchanged (isotropy-rotational symmetry).
4. Each geometric point or region of it is characterized by zero quantities (Zero criterion of  $\mathbf{R}_0^n$  (z.c. $\mathbf{R}_0^n$ )).
5. The shape of each part of  $\mathbf{R}_0^n$  resembles the shape of the whole  $\mathbf{R}_0^n$  and vice versa.

The spaces  $\mathbf{R}_0^n$  are static spaces, that is, they cannot evolve into another space, so they are considered as reference spaces.

### 2.2 Pseudo-Euclidean spaces

The need to search for some space different from the Euclidean spaces  $\mathbf{R}_0^n$  arose, on the one hand, from the need to introduce the concept of the distance between two macroscopic objects (as free observers realize it today) and, on the other hand, from Zeno’s paradoxes, whose results contradict common

sense. We will first examine the two most important of Zeno's paradoxes, which are the Achilles-tortoise paradox and The Arrow paradox.

1. **Achilles-tortoise paradox** In a straight-line race, we have two runners: Achilles (A), who runs at a constant speed, and the tortoise (T), who runs at a much lower constant speed than Achilles. Since the tortoise is much slower than Achilles, it takes an initial lead over him before the race begins. At a given time, the race begins.

In order for Achilles to reach the tortoise, he must first reach the point from where the tortoise started, or else travel the distance defined by the geometric points that determine their initial positions. When Achilles has traveled this distance, the tortoise will have moved a little further. Achilles is therefore obliged to travel the new distance that separates him from the tortoise. When Achilles has covered this new distance, the tortoise will have covered a new distance in the time it took Achilles to cover the previous distance.

The distance between them becomes smaller and smaller, but the fast Achilles can never catch up with the slow tortoise because the tortoise will always have gone a little further. This paradox contradicts common sense, which says that the fast Achilles will catch up with and overtake the slow tortoise in a short time. This paradox is based on the following two assumptions:

- a) The two runners move at a constant speed, which is almost impossible to happen in the real world, and
- b) In the space in which the two runners move, distances are defined that have one dimension (length), consisting of dimensionless geometric points that each have zero length or zero area.

Let us examine these two assumptions. Initially, we assume that the speeds of the two runners can be made, to a very good approximation, macroscopically constant, so there is no problem with the first assumption.

The second assumption, however, cannot be considered true because it is structurally inconsistent. The inconsistency lies in the fact that distances are constructed from dimensionless geometric points, that is, distances with one dimension or two dimensions.

To overcome this impasse, we must define a geometric entity with dimensions as a "point". If we see the race from above, then the height of the two runners does not matter because the dimension of their height is lost; that is, we see two two-dimensional dots moving. Thus, we can consider two runners moving on a "line" of 2D-space consisting of "points" that are regions of infinitesimal area. Therefore, if an

infinitesimal region is defined as a point in  $R^2$  (we will define this infinitesimal region as a Planck point in the following), then both runners after starting will be at some point in the same infinitesimal region. Thus, the faster Achilles will reach and overtake the slower tortoise when both runners are at the same Planck point, in agreement with the reality that free observers realize.

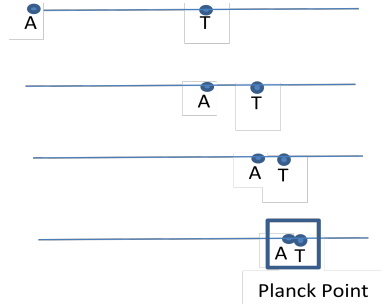


Figure 2.1: The Achilles-tortoise paradox

2. **The Arrow Paradox** The arrow paradox expresses the fact that the movement of an arrow is impossible if we make the assumption that a time interval consists of time points or moments that are considered to be of zero duration. The time intervals ( $\Delta t$ ) are the sum of moments so that their total duration is zero. Thus, at any time, an arrow that appears to free observers to be moving with a constant speed  $u$  occupies zero space  $\Delta s$ . Thus, we get zero distance  $\Delta s = u\Delta t = 0$ .

The arrow, to actually move, requires a period of time greater than a “point in time” that has zero duration. Thus, at any moment, the arrow must be at rest and not moving, contrary to what free observers see in their “reality”. With the same arguments, as mentioned in the previous paradox, an infinitesimal but non-zero time interval can be defined. Therefore, the “moment of time” has duration and the arrow moves, so there is no paradox.

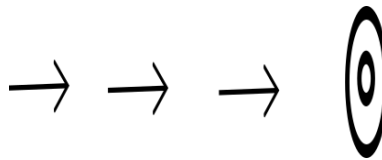


Figure 2.2: The Arrow Paradox

3. **Quantum Zeno Effect** In 1977, Indian physicists Sudarshan and

Misra, studying quantum mechanics, discovered that the dynamical evolution of a quantum system could be stopped by continuously observing the system. This phenomenon is usually referred to as the “Quantum Zeno Effect”, as it strongly resembles the paradox of Zeno’s arrow. Sometimes this phenomenon is formulated as follows: “A system cannot change while it is being observed”. One can “freeze” the time evolution of the system by observing it often enough in its known initial state.

The question that arises is: Does the observed system actually freeze, or can its evolution not be observed on an infinitesimal scale? According to the above, it is impossible to observe it on such a small scale, giving the impression that the system is slowing down its evolution. According to Zeno’s previous paradoxes, three-dimensional free observers “see macroscopically” that the various objects are spatially and temporally distant from each other. Thus, the **concept of distance and time is created for them, and their units are also determined.**

## 2.3 Structure of space

To explain the creation of the Universe, free observers, through the concept of distance, consider a space whose points are not close to each other but are separated from each other by the same non-zero distance  $l$ . Thus, as the farthest past of the Universe, they consider not the space  $\mathbf{R}_0^n$  (Euclidean space) but the space  $\mathbf{R}_l^n$  (Pseudo-Euclidean space), which consists of an infinite set of geometric points that are spaced apart by the same distance  $l$ . The space  $\mathbf{R}_l^n$  is a subset of  $\mathbf{R}_0^n$ , so it should satisfy similar properties to the space  $\mathbf{R}_0^n$  as its subspace. To define the space  $\mathbf{R}_l^n$ , the following should be done:

1. Define the dimensions of  $\mathbf{R}_l^n$ , i.e., the value of  $n$ .
2. Define the properties of  $\mathbf{R}_l^n$  through the properties of  $\mathbf{R}_0^n$ .
3. Define the distance unit  $l_u$  and the time unit  $t_u$  in  $\mathbf{R}_l^n$ .
4. Define its structural unit, different from the geometric point.
5. Define its structural process.

The 5 previous conditions are described in detail as follows:

1. **Definition of the dimensions of  $\mathbf{R}_l^n$ , i.e., finding the value of  $n$ :** We will start from the first non-trivial flat two-dimensional space  $\mathbf{R}_l^2$ , which the bound observers call **Planck space** ( $\mathbf{P}^2$ ).

The reasons we choose the two-dimensional space  $\mathbf{P}^2$  are:



- a) Earth observers perceive in their everyday life a three-dimensional space (**holographic space H**).
- b) The space they are in and observe is a two-dimensional flat part of the Earth's surface (**space c**).

Thus, in the first case they function as **free observers**, while in the second case they function as **bound observers**, according to .

Earthly observers observe phenomena in three-dimensional space and “transfer” them to bound observers who explain them in two-dimensional space. The same applies to the Universe. The Universe as a whole is a flat two-dimensional space as perceived by bound observers, and its interior is a three-dimensional space as perceived by free observers. We will mention in Chapter 2 and the corresponding Appendix how the flatness of the entire Universe arises.

2. **Definition of the properties of  $\mathbf{P}^2$  through the properties of  $\mathbf{R}^n$ :** The properties of  $\mathbf{P}^2$  are the following:

- “**Compact space**”, i.e., without gaps between its structural units. This means that the perimeter of the structural unit consists of straight lines.
- Each of its directions is the same as any other (**homogeneity-translational symmetry**). This means that we should get as much **translational symmetry** as possible.
- Every rotation of  $\mathbf{P}^2$  around one of its structural units leaves it unchanged (**isotropy-rotational symmetry**). This means as **much rotational symmetry as possible**.
- Every geometric point or region of it is characterized by zero quantities (**Zero criterion of  $\mathbf{R}_l^n$**  (z.c. $\mathbf{R}_l^n$ )).
- The shape of each part of  $\mathbf{R}_l^n$  resembles the shape of the whole  $\mathbf{R}_l^n$  and vice versa (self-similar construction process).

3. **Definition of the distance unit  $l_u$  and the time unit  $t_u$  in  $\mathbf{P}^2$ :**

The length and time units of the space  $\mathbf{P}^2$  are currently found by free observers through the universal constants  $(G, c, \hbar)$ . Thus, free observers find the value of the minimum distance from the relation:

$$l_p = \sqrt{\frac{G\hbar}{c^3}} = 1.62 \cdot 10^{-35} m$$

and call it the **Planck length**. They also find the value of the minimum time duration to create the initial structural unit of  $\mathbf{P}^2$  with the value:

$$t_p = \sqrt{\frac{G\hbar}{c^5}} = 5.4 \cdot 10^{-44} \text{sec}$$

This structural unit is called the **Planck time**.

4. **Define the structural unit of  $\mathbf{P}^2$  different from the geometric point:** The structural unit of  $\mathbf{P}^2$  should have the following characteristics:
  - A. Be a finite geometric object that has equal side lengths and equal angles, and occupies non-zero space.
  - B. Incorporate the concept of distance and time.

The equal lengths of sides and the number of equal angles of rotation of this structural unit lead us to the choice of the geometric shape that the initial structural unit should have. The geometric shapes with these characteristics are the regular polygons with sides of length  $l_p$  and their creation time the time interval  $t_p$ .

The simplest regular polygons of  $\mathbf{P}^2$  are the equilateral triangle and the square. We will examine these two shapes and choose from these two that shape that satisfies most of the five previous criteria. The first, second, and fifth criteria are satisfied by both shapes in the same way. The third criterion is more satisfied by the square because the rotations we can make to the square within its circumscribed circle so that it remains unchanged are 4, while for the equilateral triangle, following the same procedure, it is 3. The fourth criterion has to do with the circumscribed circle in both shapes, through which the curvature of the space of each shape is determined.

The circumscribed circle has a radius which is proportional to the square root of the area of the two shapes. In general, the reciprocal of the measure of the radius  $r_c$  of the circumscribed circle in each Planck patch is called the curvature. Curvature always has a positive value and is defined by the relation  $k_c = \frac{1}{r_c}$  only if the radius measure  $r_c$  of the circle has a non-zero value, where the units of  $k_c$  are  $[k_c] = [L^{-1}]$ . In the case of spaces  $\mathbf{R}_0^n$ , curvature cannot be defined because the geometric points have  $r_c = 0$ . The larger the radius  $r_c$  of the circle that encloses a geometric shape, the smaller the value of its curvature will be. Based on the dimensional unit  $[L^{-1}]$  that curvature has, there are different expressions that it can have. Free observers define the curvature of a circle (holographic circle  $H$ ) of radius  $r_c$  as  $k_H = \frac{1}{r_c}$ , while bound observers for the same circle define its curvature as  $k_c = \lambda_c \frac{r_c}{l_p^2}$ , where  $\lambda_c$  is a dimensionless parameter that will be defined below. For free observers, the definition of  $k_H$  also determines the mechanism for zeroing the value of the curvature, which has to do with:

- I. The largest possible initial radius of the circumscribed circle, and
- II. The fastest expansion of the initial structural unit, resulting in the creation of the space  $\mathbf{P}^2$ .

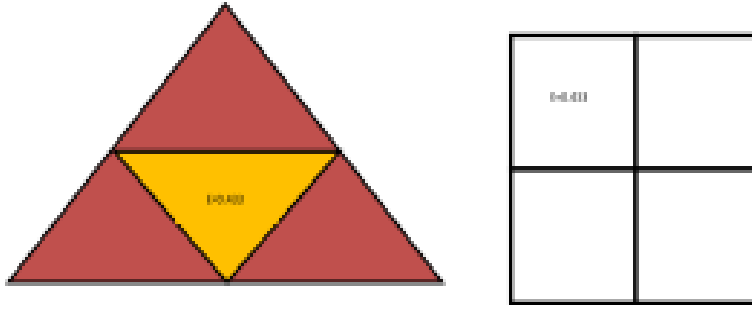


Figure 2.3: Equilateral Triangle-Square

Thus, we choose the square as the building block of  $\mathbf{P}^2$  because it has a larger area than the equilateral triangle when these two shapes are constructed from equal-length sides  $l_P$ , as shown in the previous figure.

5. **Define the building process of  $\mathbf{P}^2$ :** The process of building  $\mathbf{P}^2$  requires self-similarity, that is, the shape of a building block at each stage of expansion of the space is similar to the shape of its initial building block and vice versa. The building block at each stage of creation of the space  $\mathbf{P}^2$  is called a **Planck patch**. From the rate of expansion of the chosen building block, which is the square, the concept of time is introduced. The space  $\mathbf{P}^2$  expands through a self-similar process. Free observers, investigating the structure of all living organisms today, have found that the majority of structures follow a self-similar process of their construction (self-similar shapes or **fractals**), identical to the process of building space[6]. Each self-similar process of construction is described by the proposition:

**The whole resembles the part and the part resembles the whole.**

Such structures consist of almost all the systems of the body of free observers, such as, for example, their circulatory system, their lungs, their cerebral cortex, etc., as is also the case in all animals and plants of the Earth's ecosystem[6].

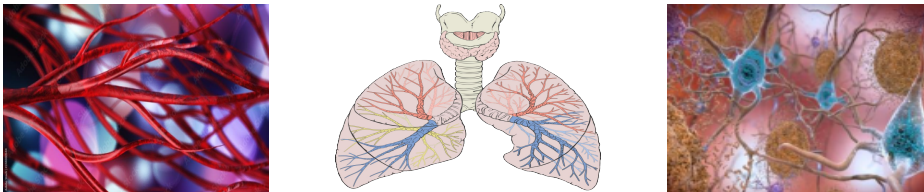


Figure 2.4: Tree shape for blood–Tree shape for lungs–Tree shape for Neurons

Let us consider the example of the geometric construction of a fern leaf. The procedure we follow is as follows: We take a straight line segment with a length of, e.g., 7cm, which we call the generator. We take identical copies of the generator but with a shorter length, that is, two copies of 6cm, two copies of 5cm, two copies of 4cm, two copies of 3cm, etc., and place them symmetrically at equidistant points of the generator, as shown in the top row of the next left Figure 2.5.

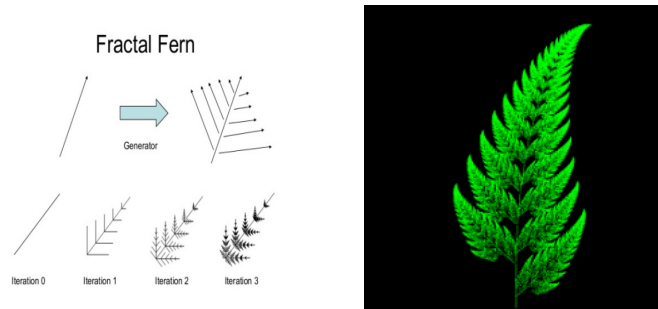


Figure 2.5: Geometric construction of a fern - Final fern shape

We consider each of these new straight segments as a new generator and apply the same procedure as the initial one, taking the 2 previous figures. The figure on the right is a purely geometric figure that looks like a “photograph of a real” fern branch in our 3D world. Another example is the identical Russian dolls ( or Matrioskas) that are each inside the other, expressing self-similarity at different scales.



Figure 2.6: Babouskas

The Fibonacci sequence[15] and self-similarity appear almost everywhere in the Universe, both in the microcosm and in the macrocosm, as shown in the following figures. The reason we will use the Fibonacci sequence is that it is also constructed through a self-similar process, that is, each term of it is the sum of its two previous terms. Thus, this sequence is used in the process of finding the numerical value of the fine structure constant and the masses of the three generations of elementary particles, as we will see later.

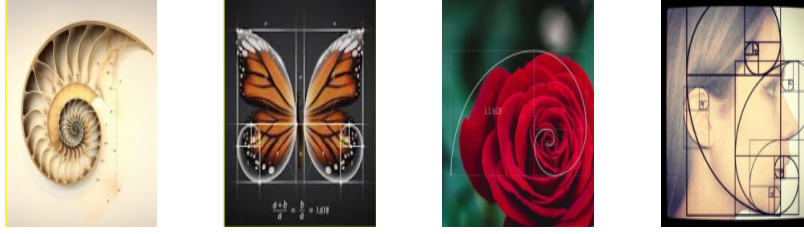


Figure 2.7: Shapes resulting from the Fibonacci sequence

Another example is the painting by the Japanese painter Katsushika Hokusai, which depicts a wave in the next Figure 2.8.



Figure 2.8: Great Wave of Kanagawa

Free observers studying all these structures make the assumption that these structures follow a self-similar building process because, like the Universe as a whole, its parts are also structured through a self-similar process, as shown in the following Figure 2.9.

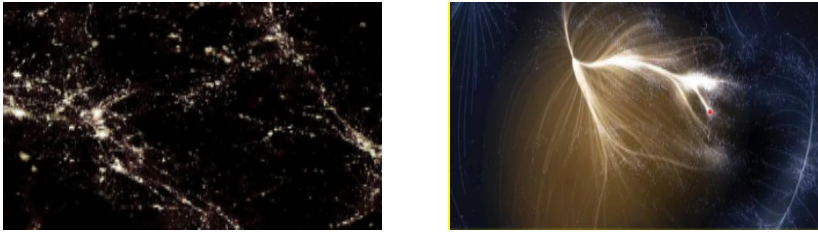


Figure 2.9: Tree-structured Galaxy - Tree-structured Galaxy Clusters

The red dot in the right figure represents our Galaxy. We will examine the space  $\mathbf{P}^2$  thoroughly, determining with the current data the value of the minimum distance  $l_p$  and the minimum time  $t_p$ [5]. Since the circumscribed circle of each structural unit exhibits curvature, that is, a quantity with a non-zero value, then we will examine whether other quantities with non-zero values appear in this circle. The question is:

**To what do free observers correspond the concept of curvature in three-dimensional space?**



## Chapter 3

# GEOMETRICAL OBJECTS-HOLOGRAMS

### 3.1 Geometric quantities in circles and ellipses

In 3D-space, free observers experimentally find for each elementary particle or hologram three quantities, which are its charge ( $q_H$ ), its spin ( $\mathbf{S}_H$ ), and its holographic mass ( $M_H$ ).

In the space  $\mathbf{P}^2$ , bound observers define in each quadrant of a circle the quantities we call charge ( $q_c$ ), orientation ( $\hat{O}_c$ ), and curvature ( $\kappa_c$ ).

Free observers experimentally find for each elementary particle only one value for these three quantities. Bound observers find four values for each of these three quantities; that is, as many as there are quadrants of a circle. The question that arises is:

**What modifications should the bound observers make to a circle or an ellipse so that, on the one hand, they get only one value for each of the three geometric quantities and, on the other hand, this value coincides numerically with the value of the corresponding quantity experimentally measured by the free observers?**

If in each circle of radius  $a$  or curvature  $\kappa_c = \frac{1}{a}$  a Cartesian coordinate system  $(x, y)$  is defined at its center, then the circle is divided into four equal quadrants ( $1^\circ, 2^\circ, 3^\circ, 4^\circ$ ), each characterized by the corresponding vector as follows:

$$\begin{array}{ll} \mathbf{X}_q = \begin{array}{l} 1^\circ \rightarrow (\alpha, \alpha) \\ 2^\circ \rightarrow (-\alpha, \alpha) \\ 3^\circ \rightarrow (-\alpha, -\alpha) \\ 4^\circ \rightarrow (\alpha, -\alpha) \end{array} & \mathbf{X}_\mu = \begin{array}{l} 1^\circ \rightarrow (\kappa_c, \kappa_c) \\ 2^\circ \rightarrow (-\kappa_c, \kappa_c) \\ 3^\circ \rightarrow (-\kappa_c, -\kappa_c) \\ 4^\circ \rightarrow (\kappa_c, -\kappa_c) \end{array} \\ \text{(a)} & \text{(b)} \end{array}$$

Figure 3.1: Charge vectors - Mass vectors

These vectors are called **characteristic vectors** of the quadrants of the circles. Also, in each circle, according to the moving plane methodology, as we will see in the Appendices, the bound observers define the constant vector  $\mathbf{I} = (\frac{1}{3}, \frac{2}{3})$ . From  $\mathbf{I}$  and the characteristic vectors  $\mathbf{X}_q$  and  $\mathbf{X}_\mu$  of the circles of radius  $a$  and  $\kappa_c = \frac{1}{a}$ , the bound observers obtain the scalar and vector quantities as follows:

- The scalar quantity  $q_c$  from the inner product of  $\mathbf{I}$  and  $\mathbf{X}_q$  for the quadrants of the circles of radius  $a$ , which is called **charge** and is given by the relation:

$$q_c = \mathbf{I} \cdot \mathbf{X}_q = \frac{1}{3}(a + 2a)$$

- The scalar quantity  $\mu_c$  from the inner product of  $\mathbf{I}$  and  $\mathbf{X}_\mu$  for the quadrants of the circles of radius  $\kappa_c$ , which is called **mass** and is given by the relation:

$$\mu_c = \lambda_c \cdot \mathbf{I} \cdot \mathbf{X}_\mu = \frac{\lambda_c}{3}(\kappa_c + 2\kappa_c)$$

where  $\lambda_c$  is a constant.

- The unit vector quantity from the outer product of  $\mathbf{I}$  and  $\mathbf{X}_q$  for the quadrants of the circles, which is called **charge orientation**. The charge orientation is given by the relations:

$$\hat{O}_q = \frac{\mathbf{I} \times \mathbf{X}_q}{|\mathbf{I} \times \mathbf{X}_q|} = \frac{a - 2a}{|a - 2a|} \cdot \hat{\mathbf{e}}_3$$

- The unit vector quantity from the outer product of  $\mathbf{I}$  and  $\mathbf{X}_\mu$  for the quadrants of the circles, which is called **mass orientation**. The mass orientation is given by the relations:

$$\hat{O}_\mu = \frac{\mathbf{I} \times \mathbf{X}_\mu}{|\mathbf{I} \times \mathbf{X}_\mu|} = \frac{\kappa_c - 2\kappa_c}{|\kappa_c - 2\kappa_c|} \cdot \hat{\mathbf{e}}_3$$

From the charge and mass orientations, we define the total orientation:

$$\hat{O}_{t_i} = \hat{O}_{q_i} + \hat{O}_{\mu_i}$$

and the average orientation:

$$\hat{O}_{m_i} = \frac{\hat{O}_{t_i}}{2}$$

for each of its quadrants.

From the above relations, it follows that in each circle and ellipse opposite charges, opposite masses, and opposite unit orientations appear for the opposite characteristic vectors of their quadrants, as shown in the following Tables 3.2 and 3.3.



|    | χ.δ.<br>φορτίου | χ.δ.<br>μάζας            | Τιμές<br>φορτίων     | Τιμές μαζών                             | Προσανατολισμός<br>φορτίου        | Προσανατολισμός<br>μάζας            | Μέσος<br>Προσανατολισμός          | Ολικός<br>Προσανατολισμός          |
|----|-----------------|--------------------------|----------------------|-----------------------------------------|-----------------------------------|-------------------------------------|-----------------------------------|------------------------------------|
| 1° | $(a, a)$        | $(\kappa_c, \kappa_c)$   | $q_1 = a$            | $\mu_1 = \lambda_c \kappa_c$            | $\hat{\theta}_{q_1} = -\hat{e}_3$ | $\hat{\theta}_{\mu_1} = -\hat{e}_3$ | $\hat{\theta}_{m_1} = -\hat{e}_3$ | $\hat{\theta}_{t_1} = -2\hat{e}_3$ |
| 2° | $(-a, a)$       | $(-\kappa_c, \kappa_c)$  | $q_2 = \frac{a}{3}$  | $\mu_2 = \lambda_c \frac{\kappa_c}{3}$  | $\hat{\theta}_{q_2} = +\hat{e}_3$ | $\hat{\theta}_{\mu_2} = +\hat{e}_3$ | $\hat{\theta}_{m_2} = +\hat{e}_3$ | $\hat{\theta}_{t_2} = +2\hat{e}_3$ |
| 3° | $(-a, -a)$      | $(-\kappa_c, -\kappa_c)$ | $q_3 = -a$           | $\mu_3 = -\lambda_c \kappa_c$           | $\hat{\theta}_{q_3} = +\hat{e}_3$ | $\hat{\theta}_{\mu_3} = +\hat{e}_3$ | $\hat{\theta}_{m_3} = +\hat{e}_3$ | $\hat{\theta}_{t_3} = +2\hat{e}_3$ |
| 4° | $(a, -a)$       | $(\kappa_c, -\kappa_c)$  | $q_4 = -\frac{a}{3}$ | $\mu_4 = -\lambda_c \frac{\kappa_c}{3}$ | $\hat{\theta}_{q_4} = -\hat{e}_3$ | $\hat{\theta}_{\mu_4} = -\hat{e}_3$ | $\hat{\theta}_{t_4} = -\hat{e}_3$ | $\hat{\theta}_{t_4} = -2\hat{e}_3$ |

Figure 3.2: Geometric quantities of quadrants of circles

| τεταρτημόρια | χ.δ.<br>φορτίου | Τιμές φορτίων             | Προσανατολισμός φορτίου                                    |
|--------------|-----------------|---------------------------|------------------------------------------------------------|
| 1°           | $(a, b)$        | $q_1 = \frac{a + 2b}{3}$  | $\hat{\theta}_{q_1} = \frac{b - 2a}{ b - 2a } \hat{e}_3$   |
| 2°           | $(-a, b)$       | $q_2 = \frac{-a + 2b}{3}$ | $\hat{\theta}_{q_2} = \frac{b + 2a}{ b - 2a } \hat{e}_3$   |
| 3°           | $(-a, -b)$      | $q_3 = -\frac{a + 2b}{3}$ | $\hat{\theta}_{q_3} = \frac{-b + 2a}{ -b + 2a } \hat{e}_3$ |
| 4°           | $(a, -b)$       | $q_4 = \frac{a - 2b}{3}$  | $\hat{\theta}_{q_4} = \frac{-b - 2a}{ -b - 2a } \hat{e}_3$ |

Figure 3.3: Characteristic vectors of quadrants of ellipses

### 3.2 Creation of geometric objects

Free observers, according to their experimental data, find that there are only three matter particles, namely the electron  $e$ , the quark  $d$ , and the quark  $u$ , which have corresponding charges and spins with values:

$$q_e = -1, \quad \hat{\mathbf{S}}_e = -\hat{\mathbf{e}}_3$$

$$q_d = -\frac{1}{3}, \quad \hat{\mathbf{S}}_d = -\hat{\mathbf{e}}_3$$

$$q_u = \frac{2}{3}, \quad \hat{\mathbf{S}}_u = +\hat{\mathbf{e}}_3$$

From these three elementary particles of matter, all simple solid particles are formed, and from them all complex particles or bodies that have zero charge and spin.

Thus, for example, free observers form the structure  $uud$ , which has charge  $q_{uud} = 1$  and spin  $\hat{\mathbf{S}}_{uud} = +\hat{\mathbf{e}}_3$ , which they call the **nucleus**. If we add to this structure the electron with charge  $q_e = -1$  and spin  $\hat{\mathbf{S}}_e = -\hat{\mathbf{e}}_3$ , then the overall structure has zero charge  $q_{uud-e} = 0$  and zero spin  $\hat{\mathbf{S}}_{uud-e} = 0$ .

The electron is not of the same nature as the quarks because it displays an integer value for its charge, so it does not merge with the nucleus but

rotates around it, fulfilling, on the one hand, the criterion of isotropy of 3D-space and, on the other hand, the zero criterion of  $\mathbf{P}^2$  (z.c. $\mathbf{P}^2$ ).

In order to have the same image of the complex structure  $uud$  as free observers, bound observers correspond these matter particles to the quadrants of the circles with radii 1 and 2 that have characteristic vectors  $(-1, -1)$ ,  $(1, -1)$ , and  $(-2, 2)$ , which belong to the 3rd, 4th, and 2nd quadrants of the corresponding circles. From these characteristic vectors, the geometric objects that have the following charges and orientations are obtained, respectively:

$$\begin{aligned} q_e &= -1, & \hat{\mathbf{O}}_e &= +\hat{\mathbf{e}}_3 \\ q_d &= -\frac{1}{3}, & \hat{\mathbf{O}}_d &= -\hat{\mathbf{e}}_3 \\ q_u &= \frac{2}{3}, & \hat{\mathbf{O}}_u &= +\hat{\mathbf{e}}_3 \end{aligned}$$

The total charge is zero, but the orientation has the value:

$$\hat{\mathbf{O}}_{uud-e} = +2\hat{\mathbf{e}}_3$$

Bound observers search for the shape of the geometric object corresponding to the electron so that it is compatible with the charge and spin of the electron found experimentally. The only case that is compatible with these data is that the shape of the geometric object corresponding to the electron is the ellipse (1,2), whose 4th quadrant with characteristic vector  $(1, -2)$  gives  $q_e = -1$  and  $\hat{\mathbf{O}}_e = +\hat{\mathbf{e}}_3$ . Thus, this aggregate has  $q_{uud-e} = 0$  and  $\hat{\mathbf{O}}_{uud-e} = +\hat{\mathbf{e}}_3$ , fulfilling the zero criterion of  $\mathbf{P}^2$  in the geometric model. The question that arises from such a view is:

**How do the other three quadrants of a circle or an ellipse affect the total charge  $q_c$ , the total mass  $\mu_c$ , and the total orientation  $\hat{\mathbf{O}}_c$  of each circle or ellipse to which they belong?**

First of all, two of the three quadrants of the circle corresponding to each quark and of the ellipse corresponding to the electron have opposite characteristic vectors, so they give zero charge and zero orientation, as shown in the following figures. In order to zero the charge and orientation of the third quadrant, a gluon or photon quadrant that has an opposite characteristic vector from it must approach it, thus zeroing its charge and orientation.

Overall, the circle or ellipse corresponding to a gluon or photon has zero charge and zero orientation, but one quadrant of it can “merge” with the quadrant of another geometric object with an opposite characteristic vector from it, creating a new structure. We will see the same structure in the case of the formation of a covalent bond between two neutral and identical hydrogen atoms to create the molecule  $H_2$ .

The structure of the geometric objects corresponding to electrons and quarks is in agreement with the experimental data of free observers, where

each electron is combined with only one photon, as well as each quark is combined with only one gluon. According to the above process, the way is opened for the creation of increasingly complex structures.

In general, the free quadrant of a circle surrounding a cluster of geometric objects can be joined to the free quadrant of another circle surrounding another cluster of geometric objects whose characteristic vectors are opposite. This creates an even more complex structure. As long as each of the two clusters is free, free observers measure the values of  $q_H, M_H, S_H$  of their holograms.

The clusters mentioned above are presented in the following figures for the case of a  $u$ -quark and a gluon (Figure 3.4), a  $d$ -quark and a gluon (Figure 3.5), and finally an electron ( $e$ ) and a photon (Figure 3.6). In Figure 3.6, the ellipses corresponding to the electron and the photon are due to the fact that there is no quadrant of the circle of radius 1 that gives a charge and orientation with a value of  $-1$ .

**How do the other three quadrants of a circle or an ellipse affect the total charge  $q_c$ , the total mass  $\mu_c$ , and the total orientation  $\hat{O}_c$  of each circle or ellipse they belong to?**

From this figure (ellipse) corresponding to the electron and the photon, their wave nature emerges because the linear ellipse becomes part of a linear hyperbola as the bases of each cone shrink. The hyperbola is expressed through an equation formally similar to the wave equation. Thus, we obtain the correspondence:

**Linear ellipse  $\rightarrow$  segment of linear hyperbola  $\rightarrow$  wave**

The following figures give the active quadrants of the circles of radii 1 and 2 that are joined together to form complex structures.

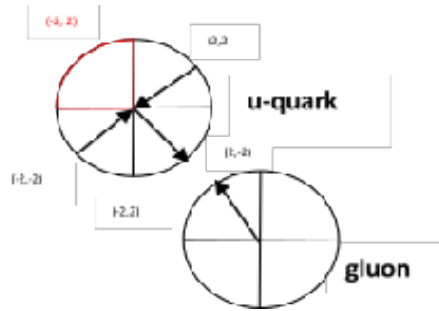


Figure 3.4: Union of quadrants of a  $u$ -quark and a gluon

In figures, the opposite arrows on the same line show the opposite characteristic vectors of two quadrants. Gluons correspond to circles of radii  $1l_p$  and  $2l_p$ , and photons to ellipses with characteristic vectors  $(1l_p, 2l_p)$  or  $(2l_p, 1l_p)$ .

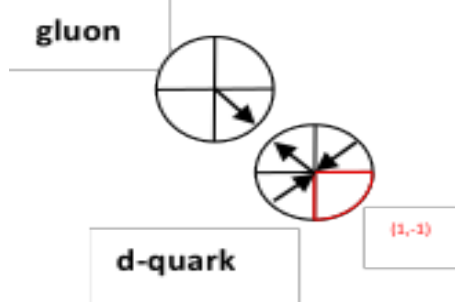
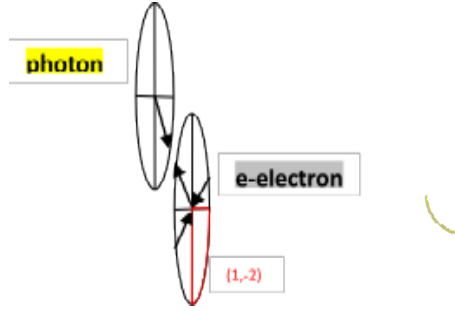
Figure 3.5: Union of quadrants of a  $d$ -quark and a gluon

Figure 3.6: Union of quadrants of an ellipse electron and a photon

Another case is that we have two concentric circles, where the outer circle has twice the radius of the inner circle.

In a stage of contraction, the outer circle becomes the ellipse  $(1l_p, 2l_p)$  or  $(2l_p, 1l_p)$  along the axes  $(x, y)$  that surrounds the inner circle of radius  $1l_p$ . This creates the composite geometric object consisting of a circle and an almost linear ellipse. Since the circle generally corresponds to a particle and the ellipse to a wave because it can be considered as a segment of a linear hyperbola (according to the above correspondence), then this composite geometric object of circle-ellipse is called a **wave-particle** by bound observers. Free observers use the same name for its hologram.

The charge  $(q)$  and the orientation  $\hat{\mathbf{O}}$  of the wave-particle come only from one quadrant of the circle of radius  $a = 1$ , as we will see later. The characteristic vectors of the ellipse  $(2,1)$  that give zero charge belong to its 2nd and 4th quadrants. Thus, for the formation of wave-particles, we take only those as shown in the following Table 3.1 for the ellipses.

| Quadrants | Characteristic vector | $q, \hat{\mathbf{O}}$                                 |
|-----------|-----------------------|-------------------------------------------------------|
| $2^o$     | $(-2, 1)$             | $q_2^E = 0, \quad \hat{\mathbf{O}}_2 = +\mathbf{e}_3$ |
| $4^o$     | $(2, -1)$             | $q_4^E = 0, \quad \hat{\mathbf{O}}_4 = -\mathbf{e}_3$ |

Table 3.1: Characteristic vectors of quadrants of the ellipse  $(\pm 2, \mp 1)$

A first issue that arises with the elementary particles found by free observers is that each of them is accompanied by two other particles that have the same charge and spin but different mass from it. Why this happens is an unsolved mystery for free observers. But for bound observers, there is an answer for the existence of these three particles. Each circle of radius 1 or 2 satisfies a 3rd-degree equation in terms of curvature, which is called the **physical equation of conic sections**. The solution of this equation gives three roots, so we get three curvatures corresponding to three masses.

However, there is another procedure for finding the curvatures and therefore the masses of the generations of elementary particles, which is based on the self-similar process of expansion-contraction of Planck patches, which we will mention in the next section. A second question is why the  $u, d$  quarks and the  $e$  electron combine together. The answer to this question is given by the bound observers and is: **to satisfy z.c. $\mathbf{P}^2$** .

The correspondences of the above quantities between bound and free observers are defined as follows:

$$\begin{aligned} q_c &\leftrightarrow q_H \\ \hat{\mathbf{O}}(c, t) &\leftrightarrow \hat{\mathbf{S}}_H \\ \kappa_c &\leftrightarrow M_H \end{aligned}$$

where by  $H$  we denote the hologram of the geometric object in 3D-space and by  $c$  the geometric object in  $\mathbf{P}^2$ .

### 3.3 Expansion-Contraction of 2D space

The question that arises from the above is the following:

**What is the geometric reason for the expansion of  $\mathbf{P}^2$  and the synthesis of simple geometric objects into increasingly complex geometric objects?**

The expansion of space has to do with the continuously decreasing value of the initial enormous curvature  $\kappa_{c,0}$  of the circumscribed circle of the initial Planck point

For free observers is:

$$\kappa_{H,0} = \frac{1}{r_{c,0}} = \frac{1}{10^{-35}m} = 10^{35}m^{-1}$$

For bound observers with  $r_{c,0} = l_p$  and  $\lambda_{c,0} = 1$  is also:

$$\kappa_{c,0} = \lambda_{c,0} \cdot \frac{r_{c,0}}{l_p^2} = \frac{l_p}{l_p^2} = 10^{35}m^{-1}$$

The reduction of the initial enormous curvature of the Planck point is a consequence of satisfying the z.c. $\mathbf{P}^2$  for curvature so that the  $\mathbf{P}^2$  space in the final stage of its creation becomes the space  $R_0^2$ .

The satisfaction of the z.c. $\mathbf{P}^2$  for curvature explains the cause of the continuous expansion of  $\mathbf{P}^2$  and consequently (due to correspondence) of 3D-space, and not by introducing a mysterious cause that free observers called “**Dark Energy**”.

The expansion of space is determined by the criterion of self-similarity, as shown in the following Figure 3.7. Free observers call holograms of the geometric shapes of  $\mathbf{P}^2$  space the “physical objects” that are perceived in 3D-space. The expansion process of  $\mathbf{P}^2$  will be modified at its beginning due to the phenomenon of **inflation**; that is, a sudden expansion of space (as we will examine in the corresponding Appendix) and will continue its maximization process through a self-similar process.



Figure 3.7: Self-similar space creation process

The creation of each new square is done by shifting the initial square to the right (or left) and rotating or transposing these two squares to create a new square, as shown in the following Figure 3.8.

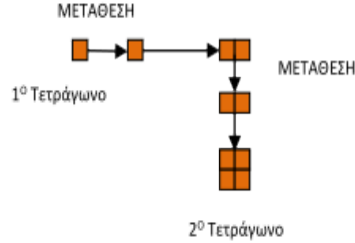


Figure 3.8: Creating a new square from the previous square

In general, each new square creates four times the space occupied by the previous square, as shown in Figure 3.7, so due to the self-similar structure, the total space expands. Each complete square that is created is called a **generator square** for the creation of the next square.

The circumscribed circle that encloses each new generator square has a larger radius and therefore a smaller curvature than the circumscribed circle that encloses the previous generator square. Thus, the expansion of space is due to the tendency for continuous reduction of the curvature of the circle surrounding each generator square, with the final result of its curvature being zeroed to satisfy z.c. $\mathbf{P}^2$ .

Let there be a generator square in which there are the circles surrounding all the previous generator squares. These circles are randomly scattered in the new space created by the new generator square, along with the new circles created in the three new squares.

In order for the curvature of the generator square to appear in the various regions of the new space where these circles are scattered, then the circles are gathered together so that the curvature of the circle surrounding them becomes almost equal to the curvature of the circle of their generator square. This means that locally the space shrinks, creating new, more complex structures. This process is called the **space shrinkage stage** microscopically. These two stages are shown in the next two Figures 3.9 (a) and (b).

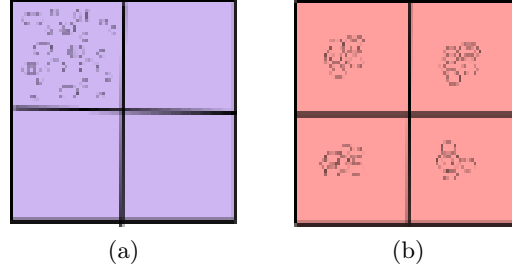


Figure 3.9: Creation of clusters of geometric objects

### 3.4 Creation of Visible Matter

1. According to the bound observers who see only circles of radii  $r_1 = 1l_p$  and  $r_2 = 2l_p$ , in the new space created in an expansion stage, new and old circles of radii  $r_1, r_2$  are scattered. If only one quadrant of these circles is free, as mentioned earlier, then it carries a charge and orientation. These circles are defined as geometric objects corresponding to particles.
2. At the beginning of one of the initial stages of contraction, there are, in addition to individual circles, concentric circles of radii  $r_1, r_2$ . The circles of radius  $r_2$  are contracted along one of their diameters, thus creating the first composite particles consisting of a circle of radius  $r_1$  and an ellipse  $(r_1, r_2)$  surrounding the circle of radius  $r_1$ .

To create the simplest stable composite geometric object from a quadrant of the circle and a quadrant of the ellipse, it should exhibit only the charge from the circle, and two opposite orientations from the corresponding quadrants of the circle and ellipse.

If the single quadrant of a circle exhibiting charge  $q_c$  and orientation  $\hat{\mathbf{O}}_c$  is joined to a quadrant of the ellipse having charge  $q_E = 0$  and

orientation  $\hat{\mathbf{O}}_E = -\hat{\mathbf{O}}_c$ , then the total charge is  $q_c$  and the total orientation of the geometric object is zero. This creates a stable complex geometric object with  $q_t = q_c$  and  $\hat{\mathbf{O}}_t = 0\hat{\mathbf{e}}_3$ . These complex geometric objects correspond to wave-particles in 3D-space.

3. During the shrinkage stage, individual ellipses are also created from compressed circles along one of their diameters, which have characteristic vectors  $(r_1, r_2)$ . These linear ellipses are considered as segments of linear hyperbolas during the linearization of a cone. Since each linear hyperbola is described by an equation similar to the wave equation, then these geometric objects correspond to waves.

We will start from the geometric objects that are the circles of radius  $1l_p$  and  $2l_p$  and the ellipse with semiaxes  $(1l_p, 2l_p)$ , in the quadrants of which the quantities we called charge, orientation, and the quantity we called mass are defined, for each circle as a whole, as shown in the following tables.

According to the theoretical and experimental findings of free observers, it has been established that there are elementary particles that exhibit charges  $q = \pm 1$ ,  $q = \pm \frac{1}{3}$ ,  $q = \pm \frac{2}{3}$  and orientations  $\hat{\mathbf{O}}_c = \pm \hat{\mathbf{e}}_3$ . The question that arises is the following:

**How can neutral overall geometric objects, such as circles and ellipses, whose total charge and total orientation are zero, be combined with each other to create even more complex geometric objects?**

The answer to this question was given earlier, where only one quadrant is “active”. Two quadrants out of the remaining three quadrants have opposite characteristic vectors, so they give zero values for charge, mass, and orientation. The third quadrant is combined with the quadrant of a photon or a gluon, whose characteristic vectors are opposite.

In the following tables, we consider only the quadrants of the circles that give only the known elementary particles, i.e.,  $e, d, u$  and their antiparticles  $\bar{e}, \bar{d}, \bar{u}$ .

| Quadrants      | Characteristic vectors | Charge, Orientation                                           |
|----------------|------------------------|---------------------------------------------------------------|
| 1 <sup>o</sup> | (1, 1)                 | $q_1^c = 1, \quad \hat{O}_1 = -\hat{\mathbf{e}}_3$            |
| 2 <sup>o</sup> | (-1, 1)                | $q_2^c = \frac{1}{3}, \quad \hat{O}_2 = +\hat{\mathbf{e}}_3$  |
| 3 <sup>o</sup> | (-1, -1)               | $q_3^c = -1, \quad \hat{O}_3 = +\hat{\mathbf{e}}_3$           |
| 4 <sup>o</sup> | (1, -1)                | $q_4^c = -\frac{1}{3}, \quad \hat{O}_4 = -\hat{\mathbf{e}}_3$ |

Table 3.2: Characteristic vectors of quadrants of a circle of radius  $r = 1$



| Quadrants | Characteristic vectors | Charge, Orientation                                  |
|-----------|------------------------|------------------------------------------------------|
| $2^o$     | $(-2, 2)$              | $q_2^c = \frac{2}{3}, \quad \hat{O}_2 = +\hat{e}_3$  |
| $4^o$     | $(2, -2)$              | $q_4^c = -\frac{2}{3}, \quad \hat{O}_4 = -\hat{e}_3$ |

Table 3.3: Characteristic vectors of quadrants of a circle of radius  $r = 2$ 

From the above tables, the answer to the question of free observers arises: **“Why do quarks carry fractional charges?”**

### Correlation of quadrants with particles

Preons were invented by Jogesh Pati and Abdus Salam in 1974. In particle physics, preons are hypothetical point particles, considered as sub-components of quarks and leptons. In the geometric model, preons correspond to the quadrants of a circle or ellipse, which compensate the charge and orientation of one quadrant of another circle or ellipse. Each quadrant of a circle is characterized by a characteristic vector, which gives positive or negative charge and positive or negative orientation. Preons exhibit four kinds of quantities: plus, anti-plus, zero, and anti-zero, which correspond to the geometric quantities positive charge, negative charge, positive orientation, and negative orientation in the four quadrants of a circle or ellipse. Quarks and leptons have three generations each, as do preons and their three generations. Preons correspond to the three quadrants of the circles or ellipses that are joined together or to quarks or leptons to give their charge and orientation values.

## 3.5 Creation of the Hydrogen Atom

To combine the characteristic vector  $(-2, 1)$  in the 2nd quadrant of the ellipse that gives zero charge and positive orientation with a characteristic vector of the circle of radius  $r = 1$ , then this must have a negative orientation. Thus, we get the two possible cases: Through this process and for  $r = 1$ , the

$$\begin{aligned}
 (-2, 1) + \left\{ \begin{array}{l} (1, 1) \\ (1, -1) \end{array} \right\} &\rightarrow \left\{ \begin{array}{l} q_t = 1, \hat{\mathbf{O}}_t = 0\hat{\mathbf{e}}_3 \\ q_t = -\frac{1}{3}, \hat{\mathbf{O}}_t = 0\hat{\mathbf{e}}_3 \end{array} \right\} \\
 (2, -1) + \left\{ \begin{array}{l} (-1, -1) \\ (-1, 1) \end{array} \right\} &\rightarrow \left\{ \begin{array}{l} q_t = -1, \hat{\mathbf{O}}_t = 0\hat{\mathbf{e}}_3 \\ q_t = \frac{1}{3}, \hat{\mathbf{O}}_t = 0\hat{\mathbf{e}}_3 \end{array} \right\}
 \end{aligned}$$

Figure 3.10: Relations for charge and orientation

geometric objects corresponding to the wave-particles  $e, d$  are formed, which have charges  $q_e = -1, q_d = -\frac{1}{3}$  and orientations  $\hat{\mathbf{O}}_t = 0\hat{\mathbf{e}}_3$ . Their anti-wave-particles are  $\bar{e}, \bar{d}$  and have charges  $q_{\bar{e}} = 1, q_{\bar{d}} = \frac{1}{3}$  and orientation  $\hat{\mathbf{O}}_t = \hat{\mathbf{e}}_3$ .

For the circle of radius  $r = 2$ , we take the 2nd quadrant with characteristic vector  $(-2, 2)$ , which has charge  $q_u = \frac{2}{3}$  and orientation  $\hat{\mathbf{O}}_2 = +\hat{\mathbf{e}}_3$ . The particle that we consider from the 3rd quadrant of the circle of radius  $r = 1$  has charge  $q_3^c = -1$ . The question that arises is:

**What is the combination of charges and orientations of the quadrants of the circles of radii  $r = 1$  and  $r = 2$  such that z.c. $\mathbf{P}^2$  for charge and orientation is satisfied?**

The procedure is to merge specific quadrants of the circles of radii  $r = 1$  and  $r = 2$  together so as to give a total of  $q = +1$ . The other three quadrants do not affect the charge and orientation because, on the one hand, two quadrants of them merge with each other and, on the other hand, the third quadrant merges with the appropriate quadrant of a gluon.

Let's see how the quadrants of the circles of radii  $r = 1$  and  $r = 2$  can create complex geometric objects that will display a total charge of  $+1$  and a positive orientation of  $+\mathbf{e}_3$ . The  $u$ -quark arises from the 2nd quadrant of the circle of radius  $r = 2$ , which gives  $q_u = \frac{2}{3}$ ,  $\hat{\mathbf{O}}_u = +\hat{\mathbf{e}}_3$ , and the  $d$ -quark arises from the 4th quadrant of the circle of radius  $r = 1$ , which gives  $q_d = -\frac{1}{3}$ ,  $\hat{\mathbf{O}}_d = -\hat{\mathbf{e}}_3$ .

When the two circles of radius  $r = 2$  are rotated so that their 2nd quadrants approach the 4th quadrant of the circle of radius  $r = 1$ , then an aggregate of circles is created that free observers call a **proton** ( $p$ ). The proton has a charge and orientation given by the relations:

$$q_p = q_u + q_u + q_d = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = 1$$

$$\hat{\mathbf{O}}_p = \hat{\mathbf{O}}_u + \hat{\mathbf{O}}_u + \hat{\mathbf{O}}_d = +\hat{\mathbf{e}}_3$$

In order to satisfy z.c. $\mathbf{P}^2$ , this aggregate must approach the quadrant of a geometric object (circle, circle-ellipse, ellipse) which gives  $q = -1$  and  $\hat{\mathbf{O}} = -\hat{\mathbf{e}}_3$ . As can be seen from the tables, for the value of the charge  $q = -1$ , there is no quadrant of a circle of radius 1 that gives  $\hat{\mathbf{O}} = -\hat{\mathbf{e}}_3$ . Also, no wave-particle can be combined with this aggregate because they all have  $\hat{\mathbf{O}} = 0\hat{\mathbf{e}}_3$ . Thus, we conclude that the only geometric shape that remains is a quadrant of an ellipse.

The only case of satisfying the above requirement is to join them with the 4th quadrant of the ellipse (1,2) with characteristic vector  $(1, -2)$  that has charge  $q = -1$  and orientation  $\hat{\mathbf{O}} = -\hat{\mathbf{e}}_3$ . We call this geometric object an electron ( $e$ ), and in this case, it behaves like a wave. The complex geometric object  $uud$  consisting of specific quadrants of the circles of radii 1 and 2, we call a proton ( $p$ ) or hydrogen nucleus.

The total geometric aggregate of the quadrants of the three circles and the ellipse satisfying z.c. $\mathbf{P}^2$  for charge and orientation is called a Hydrogen atom ( $H$ ).

The Hydrogen atom is, on the one hand, the simplest complex structure and, on the other hand, the most abundant chemical element that we have observed in the Universe at the present time. Based on the previous Figures and Tables, the simplest complex geometric object corresponding to the Hydrogen atom is created as shown in the following Figure 3.11.

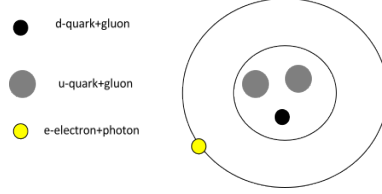


Figure 3.11: The Hydrogen atom

### 3.6 Hydrogen Isotopes

From the process of formation of the Hydrogen atom ( $H$ ), the question arises:

**Does the electron always behave like a wave with charge  $q = -1$  and orientation  $\hat{\mathbf{O}} = -\hat{\mathbf{e}}_3$ ?**

As we can see from the above tables, the electron could behave like a particle with charge  $q = -1$  and orientation  $\hat{\mathbf{O}} = +\hat{\mathbf{e}}_3$  or like a wave-particle with charge  $q = -1$  and orientation  $\hat{\mathbf{O}} = 0\hat{\mathbf{e}}_3$ , respectively.

However, in both of these cases, z.c. $\mathbf{P}^2$  is not satisfied overall for the orientation. This observation leads us to search for other geometric structures that will have zero overall charge and zero orientation. From the above tables, it follows that taking the structure  $ddu$  for the 4th quadrant of the two circles of radius  $r = 1$  and the 2nd quadrant of the circle of radius  $r = 2$ , we get:

$$q_n = q_u + q_d + q_d = \frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$$

$$\hat{\mathbf{O}}_n = \hat{\mathbf{O}}_u + \hat{\mathbf{O}}_d + \hat{\mathbf{O}}_d = +\hat{\mathbf{e}}_3 - \hat{\mathbf{e}}_3 - \hat{\mathbf{e}}_3 = -\hat{\mathbf{e}}_3$$

The complex geometric object  $udd$  consisting of specific quadrants of circles of radii 1 and 2 is called a **neutron** ( $n$ ). Combining the proton with a neutron, we get the structure  $pn$  with charge  $q_{pn} = 1$  and orientation:

$$\hat{\mathbf{O}}_{pn} = \hat{\mathbf{O}}_p + \hat{\mathbf{O}}_n = +\hat{\mathbf{e}}_3 - \hat{\mathbf{e}}_3 = 0\hat{\mathbf{e}}_3$$

Combining the proton with two neutrons, we get the structure  $pnn$  with charge  $q_{pnn} = 1$  and orientation:

$$\hat{\mathbf{O}}_{pnn} = \hat{\mathbf{O}}_p + \hat{\mathbf{O}}_n + \hat{\mathbf{O}}_n = +\hat{\mathbf{e}}_3 - \hat{\mathbf{e}}_3 - \hat{\mathbf{e}}_3 = -\hat{\mathbf{e}}_3$$

The  $pn$  structure is combined with an electron that is a wave-particle with  $q_e = -1$ ,  $\hat{\mathbf{O}} = 0\hat{\mathbf{e}}_3$  and is called Deuterium ( $D$ ). The  $pnn$  structure is combined with an electron that is a particle with  $q_e = -1$  and  $\hat{\mathbf{O}} = +\hat{\mathbf{e}}_3$  and is called Tritium ( $T$ ). Hydrogen has three isotopes, which are shown in the following Figure 3.12.

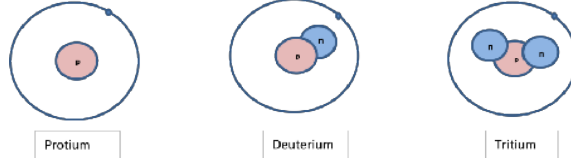


Figure 3.12: Isotopes of Hydrogen

Free observers of today have experimentally established, and bound observers theoretically from ionization energies, as we will see in paragraph 4.4, that the electron of Hydrogen is removed from its nucleus relatively easily, the electron of Deuterium a little more difficult, and the electron of Tritium very difficult. This experimental data is in agreement with the nature of the electron in all three aforementioned cases, which are wave, wave-particle, and particle respectively. We will see in paragraph 4.4 how the values of the ionization energies found experimentally are theoretically derived through the geometric model.

### 3.7 The Helium Atom

In the case where the nucleus has a charge of  $q = 2$ , then we get the helium atom ( $He$ ) with two protons. The question that arises is how many neutrons the nucleus of this structure should have.

1. If the nucleus of this structure has no neutrons, then its orientation will be  $\hat{\mathbf{O}} = 2\hat{\mathbf{e}}_3$ . Each of the two electrons that must be added to this nucleus in order for the overall structure to have  $q = 0$  must also have an orientation  $\hat{\mathbf{O}} = -\hat{\mathbf{e}}_3$ . Thus, the overall structure will also have an orientation  $\hat{\mathbf{O}} = 0\hat{\mathbf{e}}_3$ . But in this case, both electrons behave like waves with the same orientations. This has the consequence that both electrons separate from each other, moving away from the nucleus to satisfy z.c. $\mathbf{P}^2$ . Therefore, this structure is not stable.
2. If the nucleus of this structure has a neutron ( $ppn$ ), then its orientation will be  $\hat{\mathbf{O}} = +\hat{\mathbf{e}}_3$ . The two electrons that are bound in this structure will have orientations  $\hat{\mathbf{O}} = -\hat{\mathbf{e}}_3$  and  $\hat{\mathbf{O}} = 0\hat{\mathbf{e}}_3$ ; that is, one behaves like a wave and the other like a wave-particle. This means that this structure will be more stable than the structure in the previous case due to the wave-particle nature of the single electron.

3. If the nucleus of the  $2p$  structure has two neutrons ( $ppnn$ ), then its orientation is  $\hat{\mathbf{O}} = 0\hat{\mathbf{e}}_3$ . The two electrons associated with this structure must have a zero total orientation  $\hat{\mathbf{O}} = 0\hat{\mathbf{e}}_3$ . Thus, one of them behaves as a wave and the other as a particle. This structure is more stable than the structures in the two previous cases. The last two structures are the isotopes of helium.

The abundance of an isotope as measured today is greater the more stable the structure of that isotope. According to what we mentioned at the beginning of this section, it is possible to create the complex geometric object with three protons ( $ppp$ ) with charge  $q = 3$ , which we call Lithium ( $Li$ ). The creation of these geometric objects and their holograms was called in 3D-space **primordial nucleosynthesis**.

### 3.8 Percentages of Hydrogen and Helium Atoms in the Universe

Having the structure of these two complex geometric objects, corresponding to the hydrogen atom ( $H$ ) and the helium atom ( $He$ ), bound observers can find their percentages in a Planck  $P_9$  patch of the matter-dominated epoch. We choose the Planck patch  $P_9$  because it resulted from an isothermal and adiabatic change in the origin of  $\mathbf{P}^2$  space, which free observers called **Inflation**. The number of circles  $c_1, c_2$  in the Planck patch  $P_9$  is given by the general relation:

$$c_j = (10 - 2 \cdot j)^2$$

where the number 10 denotes the geometric points per side of  $P_9$ . Thus, for  $j = 1, 2$ , we get:

$$c_1 = (10 - 2 \cdot 1)^2 = 64$$

$$c_2 = (10 - 2 \cdot 2)^2 = 36$$

The number of  $u$ -quarks is equal to the number of circles with radius  $r = 2$ ; that is, it is 36. The number of circles with radius  $r = 1$  is 64 and corresponds to the number of electrons and  $d$ -quarks together. We will find the number of hydrogen atoms and helium atoms formed in  $P_9$ . The hydrogen atom has a proton consisting of two  $u$ -quarks and one  $d$ -quark. The helium atom has two protons and two neutrons consisting of a total of six  $u$ -quarks and six  $d$ -quarks, i.e.:

$$H: 2u + 1d$$

$$He: 6u + 6d$$

Since each of these structures must have zero charge, then the geometric objects corresponding to  $1e$  and  $2e$  electrons respectively will be added to

them. To have both structures have the same number of  $u$ -quarks in  $H$  and  $He$ , the number of  $H$  atoms is multiplied by 3, so we get:

$$3H: 6u + 3d$$

$$He: 6u + 6d$$

To exhaust all the  $u$ -quarks of  $P_9$  while maintaining the ratio 3 to 1, we multiply both terms of the above relations by 3, obtaining a quantity of 12 complex geometric objects, namely:

$$9H: 18u + 9d$$

$$3He: 18u + 18d$$

The remaining geometric objects are 22  $c_1$  circles, of which 11 correspond to the  $e$  electrons and 11 to the  $d$ -quarks.  $H$  and  $He$  have the following percentages:

$$H: \frac{9}{12} = 0.75 \quad \text{or} \quad 75\%$$

$$He: \frac{3}{12} = 0.25 \quad \text{or} \quad 25\%$$

The same percentages as those of the geometric objects  $H$  and  $He$  have been measured today for their corresponding physical entities (or holograms) in the Universe.

### 3.9 Creation of molecules

From the above, it follows that the reversal or zeroing of the orientation of the electrons can be done relatively easily to satisfy z.c. $\mathbf{P}^2$ . This process, combined with the interaction of electron orientations, can explain the attraction or repulsion of macroscopic bodies differently. The question that arises is the following:

**Why do two hydrogen atoms that are neutral in charge and spin combine to form the hydrogen molecule  $H_2$ ?**

The answer to this question will be given by bound observers and is the following:

Let there be a Planck patch in a state of contraction, where there are only two  $H$  atoms. The two  $H$  atoms approach each other. In this case, a quadrant of the ellipse corresponding to the electron of each hydrogen atom interacts with the appropriate quadrant of the nucleus of the other hydrogen atom so that the two  $H$  atoms come closer together. From this approach of the two atoms, we obtain the couplings  $p_1^\uparrow + e_2^\downarrow$  and  $p_2^\uparrow + e_1^\downarrow$ , in order to satisfy, on the one hand, the z.c. $\mathbf{P}^2$  and, on the other hand, the two atoms  $H$  occupy a smaller space during the stage of the shrinkage process.

Thus, a bond develops between these two atoms which is called a **covalent bond** because both the nuclei (poles) of these two atoms  $H$  are similar. Therefore, we get:

$$\begin{aligned} H^{(1)}: p_1^\uparrow + e_2^\downarrow \\ H^{(2)}: p_2^\uparrow + e_1^\downarrow \end{aligned}$$

Through this process, the hydrogen molecule  $H^{(1)} - H^{(2)}$  or  $H_2$  is obtained.

Let us consider the case where the nuclei of two atoms are different, as for example in the case where we have a sodium atom ( $Na$ ) and a chlorine atom ( $Cl$ ) that combine to form a sodium chloride molecule. The atoms of  $Na$  and  $Cl$  have the following structures:

$$\begin{aligned} Na: (11p^\uparrow + 12n^\downarrow) + 1e^\uparrow + 10e^{\uparrow\downarrow} \\ Cl: (17p^\uparrow + 18n^\downarrow) + 1e^\uparrow + 16e^{\uparrow\downarrow} \end{aligned}$$

The single electron in both structures has the particle nature with an upward orientation to nullify the overall orientation of both nuclei. The remaining electrons have the wave-particle nature, therefore they have zero orientation.

During the shrinkage process, the atoms of  $Na$  sodium and  $Cl$  chlorine come together to reduce the space they occupy, when initially far apart. This is expressed through the change in the nature of an equal number of electrons in the atoms  $Na$  and  $Cl$ .

Thus, if the  $n_i$  electrons of  $Na$  become ellipses (take on their wave nature) with an orientation of  $O^\downarrow$ , then also the  $n_i$  electrons of  $Cl$  become circles (take on their particle nature) with an orientation of  $O^\uparrow$ . Thus, they satisfy, on the one hand, the z.c. $\mathbf{P}^2$  and, on the other hand, the stage of the contraction process. This bond is called **Heteropolar or Ionic** because the nuclei (poles) of the two atoms are different.

Through the interactions of the electron orientations (changing their nature), increasingly complex structures are formed to satisfy the z.c. $\mathbf{P}^2$  in each Planck patch during the shrinkage process.

In the case of a metallic bond, the nuclei have a positive charge and the electrons have a negative charge and revolve around the nuclei to satisfy the isotropy criterion. In the case of sodium ( $Na$ ), for example, the orientation of the nucleus of a sodium atom is downward ( $O^\downarrow$ ) and the orientation of the single electron is upward ( $O^\uparrow$ ). There are therefore two causes (charge and orientation) for the movement of electrons between the nuclei, giving sodium its metallic properties.

The above processes (expansion-contraction) and the satisfaction of z.c. $\mathbf{P}^2$  explain the formation of increasingly complex geometric structures.





## Chapter 4

# QUESTIONS-ANSWERS

### 4.1 Structure of the Universe

In the present era, after many years of theories and experiments, free observers have concluded that matter consists of microscopic particles that they initially called **atoms**, that is, particles that are not divided into smaller parts. Thus, the periodic table of about 100 elements was constructed, each of which consists of a central region called the nucleus and a multitude of light particles that move around it, which were called **electrons** ( $e$ ).

Through the technology developed by the free observers, they were able to reveal that the nucleus is not solid but consists of a multitude of particles called **protons** ( $p$ ) and **neutrons** ( $n$ ). By further investigating the structure of protons  $p$  and neutrons  $n$ , they found that they are also composed of two even smaller particles, **quarks**, which were called the  $u$ -quark and  $d$ -quark, as mentioned in the previous chapter. To make visible matter, only three particles are enough, namely the electron  $e$  and the  $u, d$  quarks.

However, in addition to these three particles that all matter is made of, another particle was discovered, the **electron neutrino** ( $\nu_e$ ), which does not interact with matter. Free observers on Earth receive neutrinos that come from the nuclear reactions taking place in the Sun's core and then escape from it. It was experimentally established by free observers that each of these four particles belongs to a family that has two other members sharing all their characteristics except for their "mass". The other two members of the same family have a greater mass than the first elementary particle from which visible matter is made.

It was also experimentally established that for each of the members of the three generations, there are also their antiparticles; that is, those that have all the characteristics of the particle but with opposite values (antimatter). Free observers on Earth cannot answer the following questions:

1. **Why do these three generations of particles and antimatter exist?**

2. **Why do their masses have these values?**
3. **What is the contribution of antimatter to the structure of the Universe?**

From the observations of free observers, it was also found that the Universe they can observe is only about 5% of the total Universe. The remaining 95% consists of something they called dark matter (about 27%) and something they called dark energy (about 68%), to which the accelerating expansion of the Universe is “due”, according to the observations of free observers. In the following paragraphs, we will examine all these issues through the reasoning of bound observers who study the World as a two-dimensional flat space, such as the Universe as a whole.

## 4.2 Visible Matter

After centuries of efforts, people today know many things about everyday (visible) matter. But it is true that there is still much that they do not know about it. The most important of these issues are:

1. **Why, out of the 12 elementary particles that have been discovered, only three of them ( $d$ -quark,  $u$ -quark,  $e$ -electron) make up all the ordinary matter that we observe around us?**
2. **Where are the remaining nine particles needed and why do they exist?**

With the methods that scientists of three-dimensional space use today and the theories they have invented, there is no answer to this question. Scientists know that appropriate combinations of quarks  $d$  and  $u$  create the nuclei of atoms, and when the electrons  $e$  combine with them, then atoms are formed. From atoms, molecules are formed, and from molecules, free observers and all the complex physical objects that they observe microscopically and macroscopically are formed. These 12 elementary particles of matter can be arranged in a table of the form:

| Particle Type    | 1st     | 2nd       | 3rd        | Charge |
|------------------|---------|-----------|------------|--------|
| Leptons (matter) | $e$     | $\mu$     | $\tau$     | $-1$   |
| Quarks (matter)  | $u$     | $c$       | $t$        | $2/3$  |
| Quarks (matter)  | $d$     | $s$       | $b$        | $-1/3$ |
| Leptons (energy) | $\nu_e$ | $\nu_\mu$ | $\nu_\tau$ | $0^+$  |

Table 4.1: Elementary particles, generations and their properties

Each column of the three generations consists of three matter particles and one with almost no mass and charge that can be considered as a “particle” of energy. This idea was conceived by the ancient Greeks who believed that:

Macroscopically, everything consists of three basic elements of matter with increasing mass value for the same volume and one element of energy, as shown in the following Table 4.2.

| Nature of building elements | Elements |
|-----------------------------|----------|
| Matter                      | Air      |
| Matter                      | Water    |
| Matter                      | Earth    |
| Energy                      | Fire     |

Table 4.2: Ancient table of building elements of matter

The precious three questions can be answered through the geometric model as follows: The circles that play the main role in the geometric model are the solution of a 3rd-degree equation (**physical equation of conic sections**) which has three solutions, with the curvature ( $\kappa$ ) of the circles being unknown, as shown in Appendix 17.2. Each of the three curvatures corresponds to a mass, according to the ratio of mass and curvature. Thus, the geometric model gives us an answer as to why there are three curvatures (masses) for each of the three generations.

From the moving plane method, an answer is given to the question of why quarks have fractional charges with these values. Also, the existence of the three natures of geometric objects reflects the general process of expansion-contraction of the Universe through the self-similarity of the part to the whole as follows:

$$\left\{ \begin{array}{l} A. \text{Circle} \rightarrow \text{particle (shrinkage)} \\ B. \text{Ellipse} \rightarrow \text{wave (expansion)} \\ C. \text{Circle} + \text{Ellipse} \rightarrow \text{waveparticle (shrinkage + expansion)} \end{array} \right\}$$

Figure 4.1: Properties of geometric objects due to the Expansion-Contraction of  $\mathbf{P}^2$

### 4.3 Mass values of the 3 generations of elementary particles

An unanswered question of free observers is:

**Why do the masses of the 3 generations of elementary particles have the values found experimentally?**

Each wave-particle in 3D-space corresponds in the space  $\mathbf{P}^2$  to a circle enclosed by an ellipse. First, we consider the unit circle with radius  $a_1 = 1$ .

For two perpendicular radii, the circle is defined by the pair  $(a_1 = 1, b_1 = 1) = (1, 1)$ .

The simplest ellipse that encloses it has a minor semi-axis length  $a_2 = b_1 = 1$  and a major semi-axis length, perpendicular to the minor semi-axis,  $b_2 = a_1 + b_1 = 2$ . For two perpendicular semi-axes  $a_1, b_1$ , the ellipse is defined by the pair  $(a_2, b_2) = (1, 2)$ .

The next ellipse will have a minor semi-axis length  $a_3 = b_2 = 2$  and a major semi-axis length perpendicular to the minor semi-axis,  $b_3 = a_2 + b_2 = 3$ . For two perpendicular semi-axes  $a_3, b_3$ , the ellipse is defined by the pair  $(a_3, b_3) = (2, 3)$ .

The next ellipse will have a minor semi-axis length  $a_4 = b_3 = 3$  and a major semi-axis length perpendicular to the minor semi-axis,  $b_4 = a_3 + b_3 = 5$ . For two perpendicular semi-axes  $a_4, b_4$ , the ellipse is defined by the pair  $(a_4, b_4) = (3, 5)$ .

This construction of circles and the ellipses surrounding them is based on the Fibonacci sequence because we start with the circle  $(1, 1)$ , then with the ellipse  $(1, 2)$ , then with the ellipse  $(2, 3)$ , and finally with the ellipse  $(3, 5)$ . Deleting the 1st term in each parenthesis except the first, we get the Fibonacci sequence:

$$(1, 1)(, 2)(, 3)(, 5), \dots \rightarrow 1, 1, 2, 3, 5, \dots$$

Each term of the Fibonacci sequence, from the 3rd onward, is the sum of its two previous terms. This way of creating each member of the Fibonacci sequence involves the self-similar process, as happens with the construction of the  $\mathbf{P}^2$  space where from a Planck patch we get its next one by merging four identical Planck patches. This is the reason why we choose the Fibonacci sequence.

In the geometric model, the  $d, u$ -quarks are depicted with circles of radii  $r_d = 1l_p, r_u = 2l_p$  respectively, and the  $e$  leptons with circles of radius  $r_d = 1l_p$  in the space  $\mathbf{P}^2$ . Each of these generator-circles is characterized by a quadrant that gives a specific charge and orientation, as we saw in Chapter 2. Also, each of the three original circles expands and contracts, resulting in three new circles that maintain the charge and orientation of the generator-circle. The only thing that changes in each member is the radius of the circle resulting from the expansion-contraction processes.

The fact that the masses of the three members of each generation differ has to do with the wave nature of the ellipse that surrounds each circle. When the ellipse in a stage of expansion or contraction acquires a greater or lesser length respectively, then the length of the radius of the circle that surrounds the circle-ellipse aggregate increases or decreases, resulting in a decrease or increase in its curvature. Thus, in 3D-space, the masses of the three members of each generation change.

We will first do our calculations only with the radius of the circle surrounding the circle-ellipse cluster, and then we will find the change in the

length of this radius. Thus, we will examine the three members of the quarks  $d, u$  and then the three members of the lepton  $e$ . The characteristic vector of the quark  $d$  is (1,-1) (4th quadrant of the circle of radius 1) and the characteristic vector of the quark  $u$  is (-2,2) (2nd quadrant of the circle of radius 2) respectively. From the characteristic vectors of the quadrants of the two circles, we obtain their charge and orientation, which are:

For the  $d$ -quark:

$$q_d = -\frac{1}{3}, \quad \hat{\mathbf{O}}_d = -\hat{\mathbf{e}}_3$$

For the  $u$ -quark:

$$q_u = \frac{2}{3}, \quad \hat{\mathbf{O}}_u = +\hat{\mathbf{e}}_3$$

The circles with radii  $r_d = 1$  and  $r_u = 2$  are created in some Planck patch which expands and contracts, according to the self-similar construction of  $\mathbf{P}^2$  space. According to this procedure, for each Planck patch, the geometric objects within it will expand and contract sequentially. For geometric objects, the expansion and contraction process is determined by the following two coefficients:

- **The circle expansion coefficient  $O^i$ , where  $i$  is the radius of the circle surrounding the cluster of geometric objects. The circle expansion coefficient is defined as:  $O^i = 10^i$ .**
- **The correlation coefficient  $c_{(i,j)}$  between the concentric circles with radii  $r_i$  and  $r_j$ . This coefficient is equivalently expressed by the ellipse with semiaxes  $(r_i, r_j)$ , which relates these two circles. The correlation coefficient is defined as the product of the radii of these circles, i.e.,  $c_{(i,j)} = r_i \cdot r_j$ .**

We combine these two factors as a product or quotient depending on whether we have expansion (product), which means an increase in space, or contraction (quotient), which means a decrease in the space occupied by geometric objects in both cases. The quarks  $d, u$  are considered as the first two members of concentric circles, with radii equal to the second and third members of the Fibonacci sequence, and their combination (1,2) with the ellipse between them with semiaxes 1 and 2, as in the following figure.

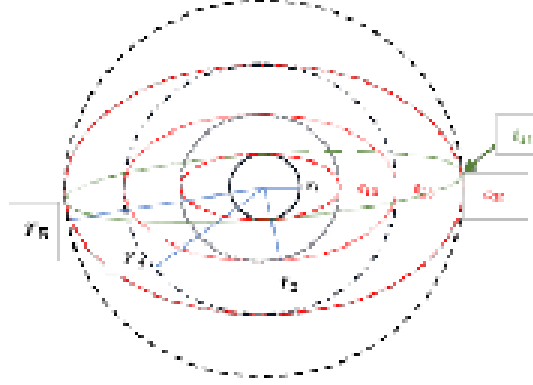


Figure 4.2: Concentric circles-ellipses

The procedure we follow has the following two stages:

### 1. EXPANSION PROCESS

The expansion coefficient  $O^i$  has as exponents the terms of the Fibonacci sequence. During the expansion process, there is only one result, which is the product of the radius of the circle being expanded times the expansion coefficient. The  $d$ -quark will expand by the expansion factor  $O^1$  two consecutive times, according to the first two terms of the Fibonacci sequence (1,1). The circle of radius  $r_1 = 1$  corresponding to the  $d$ -quark undergoes an expansion of its radius by:

$$r_d \rightarrow (r_1 \cdot O^1) \cdot O^1 = 1 \cdot 100 = 100$$

The  $u$ -quark will be expanded by the expansion factor  $O^2$  according to the third term of the Fibonacci sequence, namely 2. Thus, the circle of radius 2 corresponding to the  $u$ -quark undergoes an expansion of its radius by:

$$r_u \rightarrow (r_2 \cdot O^2) = 2 \cdot 100 = 200$$

Finally, the circle of radius 2 surrounding the complex geometric object consisting of the ellipse with semiaxes (1,2), which surrounds the circle with radius 1, will be expanded by the expansion factor  $O^3$  according to the fourth term of the Fibonacci sequence, which is 3. Thus, the circle of radius 2 surrounding the ellipse corresponding to the electron  $e$  undergoes an expansion of its radius by:

$$e_{(1,2)} \rightarrow 2 \cdot 10^3$$

The reciprocals of the radii of these circles are the curvatures of the circles surrounding these individual geometric objects or their aggregates, according

to the bound observers. These curvatures correspond to the quantity that the free observers call mass.

## 2. SHRINKING PROCESS

During the shrinking process of a Planck patch, there are two stages because  $O^i, c_{(i,j)}$  form two quotients ( $Q$ ), which are  $Q_b = \frac{O^i}{c_{(i,j)}}$  (1st stage) and  $Q_s = \frac{c_{(i,j)}}{O^i}$  (2nd stage). The quotient  $Q_b$  is larger than the quotient  $Q_s$  ( $Q_b > Q_s$ ), so the circle of radius  $Q_b$  has a smaller curvature than the circle of radius  $Q_s$ . Thus, the shrinkage process consists of two stages:

### 1st shrinkage stage (Weak shrinkage or smaller curvature)

The area of geometric objects extends from the first ellipse (1,2) to the last ellipse (3,5). The unit of the expansion coefficient  $O^1$  will be reduced by the ellipse correlation factors of the first and last ellipse, giving the radii of two new circles. Thus, in the 1st stage of the contraction process for quarks, we first obtain the quotient of the expansion coefficient  $O^1$  with the correlation coefficient of the first ellipse  $c_{1,2}$ , and then the quotient of  $O^1$  with the correlation coefficient of the last ellipse  $c_{3,5}$ , according to the bound observers. From this process, we obtain the radii of two circles surrounding these geometric objects, as well as their curvatures, which are proportional to the masses of two physical entities.

- **First**, we take the quotient of the expansion coefficient of the first circle  $O^1$  and the correlation coefficient  $c_{1,2}$  of the first ellipse:

$$Q_{[1,(1,2)]} = \frac{O^1}{c_{1,2}} = 5$$

- **Second**, we take the quotient of the expansion coefficient of the last circle  $O^3$  and the correlation coefficient  $c_{3,5}$  of the last ellipse:

$$Q_{[1,(3,5)]} = \frac{O^1}{c_{3,5}} = \frac{2}{3}$$

- **Third**, for a lepton with dominant wave nature from the contraction process, we take the smallest possible value determined by the quotient of the expansion coefficient  $O^1$  to the correlation coefficient  $c_{1,1}$  of the circle of radius 1:

$$Q_{[1,(1,1)]} = \frac{O^1}{c_{1,1}} = 10$$

### 2nd shrinkage stage (Strong shrinkage or greater curvature)

During the 2nd stage of the contraction process for quarks, we take the inverse of the relation used during the 1st stage of the contraction process, decreasing the radius of the circles. Thus, we obtain relations that apply to the first ellipse (1,2) and the last ellipse (2,3). In this case, the last circle is the one with radius 3 and not the circle with radius 5, due to the previous weak shrinkage process of the 1st stage.

- **First**, we obtain the quotient of the correlation coefficient  $c_{1,2}$  of the first ellipse with the denominator of the expansion coefficient of the first circle  $O^1$ :

$$Q_{[(1,2),1]} = \frac{c_{1,2}}{O^1} = 0.2$$

- **Second**, we obtain the quotient of the correlation coefficient  $c_{2,3}$  with the denominator of the expansion coefficient of the last circle  $O^3$ . Thus, we get:

$$Q_{[(2,3),3]} = \frac{c_{2,3}}{O^3} = 6 \cdot 10^{-3}$$

- **Third**, for a lepton, we take the quotient of the correlation coefficient  $c_{1,5}$  for the ellipse between the first and the last circle with radius  $r_5$  because, for an electron as a wave-particle, this is the last circle. Thus, we get:

$$Q_{[(1,5),1]} = \frac{c_{1,5}}{O^1} = 0.5$$

Through these geometric processes, the three generations of elementary particles are determined, as shown in the tables below.

| <b><i>d</i>-generation</b> | <b><i>u</i>-generation</b> | <b><i>e</i>-generation</b> |
|----------------------------|----------------------------|----------------------------|
| $c_{1,1} \cdot O^2$        | $c_{1,2} \cdot O^2$        | $c_{1,2} \cdot O^3$        |
| $\frac{O^1}{c_{1,2}}$      | $\frac{O^1}{c_{3,5}}$      | $\frac{O^1}{c_{1,1}}$      |
| $\frac{c_{1,2}}{O^1}$      | $\frac{c_{2,3}}{O^3}$      | $\frac{c_{1,5}}{O^1}$      |

Table 4.3: Radii of circles corresponding to the 3 generations of elementary particles  $d, u, e$

Substituting the values of  $O^1, O^2, O^3, c_{1,1}, c_{1,2}, c_{2,3}, c_{3,5}$ , we obtain the values in the following table:



### 4.3. MASS VALUES OF THE 3 GENERATIONS OF ELEMENTARY PARTICLES 53

| <b><i>d</i>-generation</b> | <b><i>u</i>-generation</b>                                   | <b><i>e</i>-generation</b>          |
|----------------------------|--------------------------------------------------------------|-------------------------------------|
| 100                        | 200                                                          | 2000                                |
| $\left(\frac{5}{1}\right)$ | $\left(\frac{2}{3}\right)$                                   | $\left(\frac{10}{1}\right)$         |
| $\left(\frac{1}{5}\right)$ | $\left(\frac{3}{2}\right) \cdot \left(\frac{4}{10^3}\right)$ | $\left(\frac{1}{10}\right) \cdot 5$ |

Table 4.4: Table of numerical values of circle radii corresponding to the 3 generations of elementary particles  $d, u, e$

To compare the experimental findings of free observers for the masses of elementary particles (in units of  $\text{GeV}/c^2$ ) with the theoretical findings of bound observers for the curvatures of the corresponding geometric objects, we put the experimental values of the masses next to the theoretical values of the curvatures in the following tables. Thus, we see that the values of the mass and the curvature are the same or almost the same numbers.

| <b>Curvature</b> | <b>Mass</b>  |
|------------------|--------------|
| $k_d = 0.01$     | $m_d = 0.01$ |
| $k_s = 0.2$      | $m_s = 0.2$  |
| $k_b = 5$        | $m_b = 4.7$  |

Table 4.5: Values of  $d, s, b$

| <b>Curvature</b> | <b>Mass</b>   |
|------------------|---------------|
| $k_u = 0.005$    | $m_u = 0.005$ |
| $k_c = 1.5$      | $m_c = 1.5$   |
| $k_t = 166.6$    | $m_t = 170$   |

Table 4.6: Values of  $u, c, t$

| <b>Curvature</b> | <b>Mass</b>      |
|------------------|------------------|
| $k_e = 0.0005$   | $m_e = 0.0005$   |
| $k_\mu = 0.1$    | $m_\mu = 0.106$  |
| $k_\tau = 2$     | $m_\tau = 1.777$ |

Table 4.7: Values of  $e, \mu, \tau$

In the first two rows of Tables 4.5, 4.6, and 4.7, the circles corresponding to quarks and leptons have identical numerical values of curvatures and masses. In the third row of these tables, there are small differences between the curvature and the mass. This is due to fluctuations in the length of the radius of the circle surrounding the geometric object due to its wave character. The following procedure is not proof but an indication of how the values of the circle's radii change due to small percentage fluctuations.

Thus, for example, we take the modified values of  $k_b^{\text{mod}}, k_t^{\text{mod}}, k_\tau^{\text{mod}}$ .  
For  $k_b$  that is in the process of shrinking, we get a 6% reduction:

$$k_b^{\text{mod}} = 5 \cdot \left(1 - \frac{6}{100}\right) = 4.7$$

For  $k_t$  that is in the process of expanding, we get a 2% increase:

$$k_t^{\text{mod}} = 166.6 \cdot \left(1 + \frac{2}{100}\right) = 170$$

For  $k_\tau$  that is a lepton (wave) in the process of shrinking, we get an 11% reduction because it shrinks more than the quarks (particles):

$$k_\tau^{\text{mod}} = 2 \cdot \left(1 - \frac{11}{100}\right) = 1.78$$

| Curvature    | Mass         |
|--------------|--------------|
| $k_d = 0.01$ | $m_d = 0.01$ |
| $k_s = 0.2$  | $m_s = 0.2$  |
| $k_b = 4.7$  | $m_b = 4.7$  |

Table 4.8: Modified  
 $d, s, b$

| Curvature     | Mass          |
|---------------|---------------|
| $k_u = 0.005$ | $m_u = 0.005$ |
| $k_c = 1.5$   | $m_c = 1.5$   |
| $k_t = 170$   | $m_t = 170$   |

Table 4.9: Modified  
 $u, c, t$

| Curvature       | Mass             |
|-----------------|------------------|
| $k_e = 0.0005$  | $m_e = 0.0005$   |
| $k_\mu = 0.1$   | $m_\mu = 0.106$  |
| $k_\tau = 1.78$ | $m_\tau = 1.777$ |

Table 4.10: Modified  
 $e, \mu, \tau$

At this point, the following questions arise:

1. Why does the Universe work this way?
2. Is it possible for particles to combine differently?
3. If not, then what binds them to follow this path?

The answer to all three of these questions arises from the definition and evolution of  $\mathbf{P}^2$ , which is due to:

- Its self-similar construction process,
- Fulfillment of its isotropy criterion, and
- Fulfillment of z.c. $\mathbf{P}^2$ .

In summary, we can say that space creates, through its definition, those simple or complex geometric objects that correspond to the simple or complex structures of matter that we perceive.

## 4.4 Ionization Energies of Hydrogen Isotopes

If the geometric model provides answers to the questions that arise in the microcosm and the macrocosm, then this constitutes the basis for developing a theory of Everything. In this section, we will find geometrically the electron ionization energies of the isotopes of Hydrogen and compare them with the experimental values. What will emerge is the equality of the theoretical with the experimental values. The isotopes of Hydrogen are Protium ( ${}^1_1\text{H}$ ), Deuterium ( ${}^2_1\text{H}$ ), and Tritium ( ${}^3_1\text{H}$ ).

The atom of  ${}^1_1\text{H}$  consists of one proton (nucleus) and one electron, the atom of  ${}^2_1\text{H}$  consists of one proton and one neutron (nucleus) and one electron, and the atom of  ${}^3_1\text{H}$  consists of one proton and two neutrons (nucleus) and one electron.

Each proton consists of two  $u$ -quarks and one  $d$ -quark, and each neutron consists of one  $u$ -quark and two  $d$ -quarks. The  $u$ -quark is represented in the space  $\mathbf{P}^2$  as a circle of radius 2 with the characteristic vector of its quadrant being  $(-2,2)$ , and the  $d$ -quark as a circle of radius 1 with the characteristic vector of its quadrant being  $(1,-1)$ .

From the characteristic vectors  $(a, b)$ , we get their charge ( $q = \frac{a+2b}{3}$ ) and their orientation ( $\hat{\mathbf{O}} = \frac{b-2a}{|b-2a|}\hat{\mathbf{e}}_3$ ), which is for the  $u$ -quark:

$$q_u = \frac{2}{3}, \quad \hat{\mathbf{O}}_u = \frac{2 - 2(-2)}{|2 - 2(-2)|}\hat{\mathbf{e}}_3 = +\hat{\mathbf{e}}_3$$

and for the  $d$ -quark:

$$q_d = -\frac{1}{3}, \quad \hat{\mathbf{O}}_d = \frac{-1 - 2(1)}{|-1 - 2(1)|}\hat{\mathbf{e}}_3 = -\hat{\mathbf{e}}_3$$

Thus, the charge and orientation of the proton are:

$$q_p = +1, \quad \hat{\mathbf{O}}_p = \hat{\mathbf{e}}_3$$

and the neutron is:

$$q_n = 0, \quad \hat{\mathbf{O}}_n = -\hat{\mathbf{e}}_3$$

The electron has a charge of  $q_e = -1$ , but its nature can be wave, wave-particle, or particle-like. We will then examine the next three cases:

1. In the case of  ${}^1_1\text{H}$ , the proton has  $q_p = +1$  and  $\hat{\mathbf{O}}_p = +\hat{\mathbf{e}}_3$ , and the electron exhibits its wave nature through the ellipse with characteristic vector (1,-2). From the characteristic vector of the quadrant of the ellipse (1,-2), we get charge  $q_e = -1$  and orientation  $\hat{\mathbf{O}}_e = -\hat{\mathbf{e}}_3$  so that the z.c.P<sup>2</sup> of the nucleus-electron cluster is satisfied.
2. In the case of  ${}^2_1\text{H}$ , the proton and neutron of the nucleus have  $q_{pn} = +1$  and  $\hat{\mathbf{O}}_{pn} = 0\hat{\mathbf{e}}_3$ . The electron exhibits its wave-particle nature through the ellipse (2,1) that surrounds the unit circle. The characteristic vector of the quadrant of the ellipse is (2,-1) and the characteristic vector of the quadrant of the circle is (-1,-1). From (2,-1) of the ellipse, we get charge  $q_E = 0$  and orientation  $\hat{\mathbf{O}}_E = 0\hat{\mathbf{e}}_3$ , and from (-1,-1) of the circle, we get charge  $q_C = -1$  and orientation  $\hat{\mathbf{O}}_C = +\hat{\mathbf{e}}_3$ . The total charge  $q_{EC}$  and the total orientation  $\hat{\mathbf{O}}_{EC}$  of this geometric object are  $q_{EC} = q_E + q_C = -1$  and  $\hat{\mathbf{O}}_{EC} = 0\hat{\mathbf{e}}_3$ . Thus, the z.c.P<sup>2</sup> nucleus-electron is satisfied overall.
3. In the case of  ${}^3_1\text{H}$ , the proton and the two neutrons of the nucleus have  $q_{pnn} = +1$  and  $\hat{\mathbf{O}}_{pnn} = -\hat{\mathbf{e}}_3$ , and the electron displays its particle nature through the quadrant of the circle of radius 1 with characteristic vector (-1,-1). From the characteristic vector (-1,-1), we get charge  $q_e = -1$  and orientation  $\hat{\mathbf{O}}_e = +\hat{\mathbf{e}}_3$  so that the z.c.P<sup>2</sup> nucleus-electron is satisfied.

We will search for the total geometric shapes that form the geometric shapes that correspond to the elementary physical entities, namely the circles of radii 1 and 2 and the ellipse with characteristic vector (1,2), by following these steps:

- A. From the geometric shapes, we will find the length of the radius  $r_0$  of the circumscribed circle that surrounds a complex geometric shape in Planck lengths  $l_p$  expressed in meters.
- B. Since the elementary physical entities are wave-particles, then each geometric object corresponding to each of them will be the complex geometric shape (circle-ellipse). Since all linear ellipses undergo the same extension of their length, then the change in their length ( $\Delta l$ ) will be the same for all ellipses. The number of these lengths will be equal to the number  $n$  of elementary particles. The resulting total length will be an order of magnitude smaller, namely  $\Delta l = 10^{-1}$ .

Thus, the total length of the number of elementary particles, namely  $n \cdot 10^{-1}$ , should be added to the radius of the circle.

- C. When a structure is made at a microscopic level, then the stage of contraction and the stage of expansion should appear. The contraction stage occurs in the creation of the nucleus and the expansion stage occurs in the removal of the electron from it and its orbit around it to fulfill the isotropy of  $\mathbf{P}^2$ .
- D. From the above procedures, we find the radius of the circle that surrounds each of these aggregates and from this its inverse which is the curvature of the circumscribed circle. Since the curvature of the circle is proportional to the energy, then free observers should find experimentally this ionization energy to remove the electron from the nucleus. Let us see in detail, through the following figures, how the curvatures of the circles corresponding to the ionization energy of the Hydrogen isotopes are calculated.

### 1. HYDROGEN

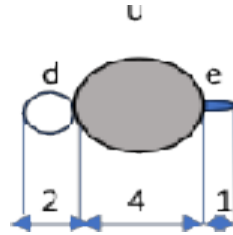


Figure 4.3: Geometric representation of hydrogen

Diameter length:

$$2 + 4 + 1 = 7$$

Rotation around the left end of the diameter.

$$\text{Number of particles: } 2(u) + 1(d) + \frac{1}{2}(e) = 3.5$$

$$r_H = (7 \cdot 10^0 + 3.5 \cdot 10^{-1}) \cdot [10^{-2} \cdot l_p \text{ (meters)}]$$

$$r_H = 0.0735 \text{ m}, \quad \kappa_H = \frac{1}{r_H} = 13.6 \cdot \text{m}^{-1}$$

The experimental value of the ionization energy of  ${}^1_1\text{H}$  is about 13.6 eV.

## 2. DEUTERIUM

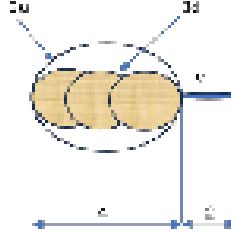


Figure 4.4: Geometric representation of Deuterium

Diameter length:

$$4 + 2 = 6$$

Rotation around the left end of the diameter. Number of particles:  $3(u) + 3(d) + 1(e) = 7$

$$r_D = (6 \cdot 10^0 + 7 \cdot 10^{-1}) \cdot [10^{-2} \cdot l_p \text{ (meters)}]$$

$$r_D = 0.067 \text{ m}, \quad \kappa_D = \frac{1}{r_D} = 14.9 \text{ m}^{-1}$$

The experimental value of the ionization energy of  ${}^2_1\text{H}$  is about 14.9 eV.

## 3. TRITIUM

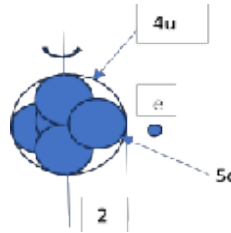


Figure 4.5: Geometric representation of Tritium

Radius length: 2 Rotation around the center of the circle. Number of particles on and inside the circle of radius 2:  $4(u) + 5(d) = 9$  and  $1(e)$  outside the circle of radius 2.

$$r_T = (2 \cdot 10^0 + 9 \cdot 10^{-1} + 1 \cdot 10^{-2}) \cdot [10^{-2} \cdot l_p \text{ (meters)}]$$

$$r_T = 0.0291 \text{ m}, \quad \kappa_T = \frac{1}{r_T} = 34.3 \text{ m}^{-1}$$

The experimental value of the ionization energy of T is approximately 34.3 eV and results from the process of  $\beta$ -radioactive decay and the conversion of Tritium into a helium nucleus. From the above, it follows that the numerical values of the curvatures of the geometric shapes calculated by the bound observers are equal to the numerical values of the ionization energies of the hydrogen isotopes found experimentally by the free observers.

## 4.5 What is Dark Matter

What free observers have observed is that all physical entities, simple and complex, consist of only three elementary particles, which are the  $d$ -quark,  $u$ -quark, and the electron  $e$ . Bound observers correspond these three elementary particles to circles of radius  $1l_p, 2l_p$  and to ellipses with semi-axis lengths  $(2l_p, 1l_p)$  or  $(1l_p, 2l_p)$ . The observations made by observers (free and bound) who are made only of these three elementary particles, as well as the observation instruments, do not allow them to investigate particles that correspond to circles with a radius equal to or greater than  $1^+l_p$  or to ellipses with semi-axis lengths  $(2^+l_p, 1^+l_p)$  or  $(1^+l_p, 2^+l_p)$ . These geometric objects arise from the circumscribed circles in the Planck patches  $P_1$  and  $P_2$  and the “circumscribed ellipses” of circles of radius  $1^+l_p$ . Thus, the observers indirectly conclude that:

**There is another type of matter that is different from the matter with which they and their observation instruments are constructed.**

We assume that the same applies to “intelligent beings” in the region with circles of radii  $1^+l_p$  and above, where they cannot determine the existence of beings in the region with circles of radii  $0 \leq r \leq 2l_p$ , according to the geometric model. A question that arises is:

**How do terrestrial observers indirectly determine the existence of dark matter?**

**The first clue** from which the existence of dark matter was determined was the rotation of galaxies. The stars and all the celestial bodies that make up a galaxy revolve around the galactic center where there is a black hole, that is, a very dense concentration of matter invisible to earthly observers, due to the Doppler effect.

This rotation is a consequence of fulfilling the criterion of isotropy of  $\mathbf{P}^2$  around an initial cluster. The same rotation also occurs for all parts of a galaxy. Astronomers initially tried to calculate the mass of galaxies by counting the number of their stars. Using this number of stars and estimating the number of other celestial bodies (planets, satellites of planets, meteorites, etc.) in the galaxy, they calculated its total visible mass. From the total visible mass of the galaxy, the astronomers were able to calculate its rotation speed.

From this value of the rotation speed, they found that the theoretical value of the mass must be much greater than the value they found from their previous calculations. If the observed rotational velocity were so large and the total visible mass so small, then stars would have to be ejected from the periphery of galaxies. This contradiction could only be eliminated if astrophysicists assumed that in each galaxy there is a huge amount of invisible or dark matter. Thus, the concept of dark matter was introduced as a basic component of the Universe.

In the following Figure, we can distinguish the two cases of non-existence of dark matter (left) and existence of dark matter (right).

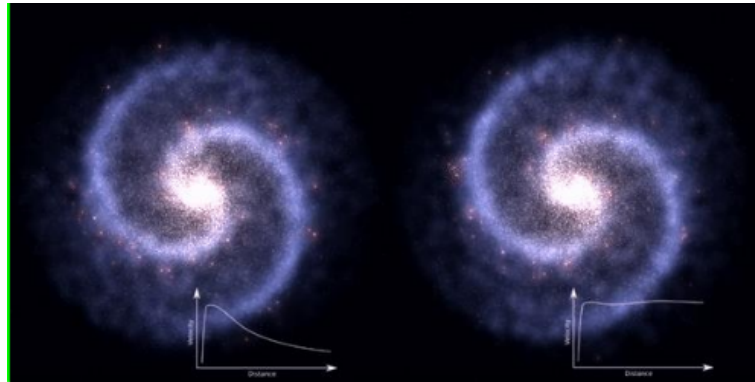


Figure 4.6: (a) Galaxy without dark matter (b) Galaxy with dark matter

The  $V_{\text{rot}}$  vs  $r$  plots below the galaxy shapes show the rotation velocities  $V_{\text{rot}}$  (vertical axis) of various points in the galaxy as a function of the distance  $r$  (horizontal axis) from its center.

- **Left:** Simulation of a rotating spiral galaxy without dark matter, where we have a continuous decrease in velocity with distance.
- **Right:** Simulation of a rotating spiral galaxy with a “flat rotation curve” (almost parallel to the  $r$  axis) expected due to the presence of dark matter, i.e., constant velocity with respect to distance.

**The second clue** that convinced astronomers of the existence of dark matter was the bending of light coming from a quasar (a quasi-star). Quasars are the initial stages of galaxy formation. Studying the sky, astronomers saw the image of a galaxy. Turning their telescope a few degrees, they saw the image of another galaxy that was very similar to the first in shape, color, and radiation coming from it. The observation of the same galaxy twice is again explained by the interference between the galaxy and the observer of a quantity of dark matter and the bending of light around it to satisfy the isotropy criterion of  $\mathbf{P}^2$ , as shown in the following Figure.



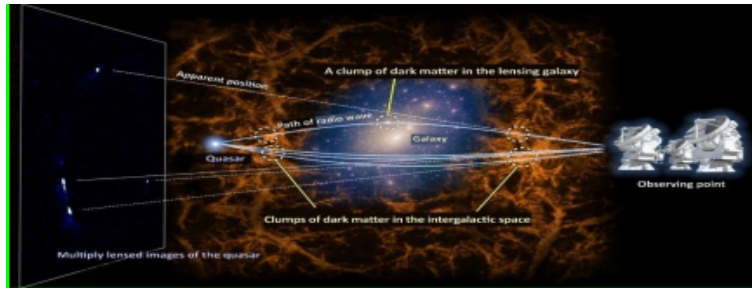


Figure 4.7: Dark matter acting as a gravitational lens

A team of Japanese researchers used the ALMA telescope to study the gravitational lens system known as MG J0414+0534, formed by an elliptical galaxy and a quasar, located in the direction of the constellation Taurus as seen from Earth. The distance of the quasar from Earth is estimated at 11 billion light-years. In Figure 4.7, this system has the foreground object forming not one, but four images of the background object, due to the satisfaction of the  $\mathbf{P}^2$  isotropy criterion for light moving around a massive galaxy. Using a new method of analyzing the data, the team was able to detect variations in the distribution of dark matter with greater resolution than ever before, on a scale of 30,000 light-years.

**The third clue** that convinced astronomers of the existence of dark matter was the collision of galaxies or galaxy clusters. One of these phenomena is the collision of two galaxy clusters and the creation of a galaxy cluster, known as the Bullet Cluster. The name comes from the shape of the red shock wave on the right of the image, suggesting a huge collision in the past.



Figure 4.8: Simulation of galaxy cluster collision

In this image, created in 2003, two colliding galaxy clusters have passed by each other. The collision occurred, according to the geometric model, during a stage of contraction of a huge Planck patch containing both galaxy clusters.

Photons originating from the red region of the cluster followed the shortest path around the blue regions. According to bound observers, the bending of light occurs to satisfy the  $\mathbf{P}^2$  isotropy criterion around any form of matter (visible or dark). Free observers attribute this path of the photons to the “gravitational pull” that the photons experience from the matter present in the blue regions. As found by the Chandra X-ray Observatory, the red coloration indicates where X-rays from high-temperature hydrogen gas were detected.

Most of the visible (or baryonic) matter in a galaxy cluster is in hydrogen gas, not in stars, black holes, or other visible or invisible objects. The electromagnetic interaction of the superheated gases from the two galaxies is through the orientations of their components, so they are concentrated in the middle of the collision point to satisfy z.c. $\mathbf{P}^2$ .

The blue color shows where the dark matter is, as inferred from the weak “gravitational lensing” that free observers consider, i.e., the distortion of the images of galaxies (taken by the Hubble telescope) behind this galaxy cluster. The lensing is weak in this image, so it is difficult to see, but a more obvious example can be seen in the image of a galaxy in the next Figure.



Figure 4.9: The central red-yellow galaxy distorts the image of the blue galaxy right behind it

From the above, we conclude that dark matter:

- Distorts the trajectories of photons
- Is invisible
- Is found in large structures such as galaxies or their clusters and
- Does not interact with visible matter or with itself.

## 4.6 What is Dark Energy

In the 1990s, scientists (free observers) were trying to answer the question:

**How fast is the expansion of the Universe slowing down?**

This question arose from the discovery of Hubble about the expansion of the Universe and from the belief that all matter in the Universe is “attracted to its center by gravity”. The general belief was that gravity would cause the Universe to contract to return to the original point from which the Big Bang and its subsequent expansion began.

## 4.7 The Big Bang model

Its essential feature is the emergence of the Universe from a state of extremely high temperature and density, the so-called Big Bang, which occurred  $\sim 13.8$  billion years ago. Although this type of Universe was proposed by the Russian mathematician Aleksandr Friedmann and the Belgian astronomer Georges Lemaitre in the 1920s, the modern version was developed by the Russian-born American physicist George Gamow and his colleagues in the 1940s.

### 4.7.1 History of the Big Bang

The Big Bang model is based on two assumptions. The first assumption is that Einstein’s general theory of relativity, rather than Newton’s theory of gravity, correctly describes the gravitational interaction of all matter.

The second assumption, called the Cosmological Principle, states that an observer’s view of the Universe depends neither on the direction in which he is looking nor on his position.

This second assumption is true only for the large-scale properties of the Universe, but it implies that the Universe has no end, so that the beginning of the Big Bang did not occur at a specific point in space but rather in all of space at once.

These two assumptions make it possible to determine the history of the Universe after a certain epoch called the Planck epoch. Scientists have not yet determined what was before the Planck epoch.

### 4.7.2 Expansion of the Universe according to Einstein

After Einstein published his theory of gravity, researchers realized that the Universe should be expanding, a finding that was confirmed by observations. According to the Big Bang model, the Universe expanded rapidly from an extremely compressed primordial state, which resulted in a significant decrease in its density and temperature.

At this stage, many types of elementary particles may have existed. After a few seconds, the Universe cooled enough to allow the formation of some H nuclei. The theory predicts that specific amounts of hydrogen, helium, and lithium were produced. Their abundances are consistent with those observed today. About a million years later, the Universe was cold enough for atoms to form. The radiation that also filled the Universe was then free to travel through space. This remnant of the early Universe is today the cosmic microwave background radiation with a temperature of  $\sim 2.728^\circ\text{K}$ .

### 4.7.3 Big Bang Model or Theory of the Evolution of the Universe

According to free observers, the Universe expanded from a state of high compression and temperature, releasing enormous amounts of energy, which soon coalesced into grains of matter. In addition to accounting for the presence of ordinary matter and radiation, the model predicts that the present Universe should also be filled with neutrinos, which are fundamental particles with almost zero mass and almost zero electric charge. According to these assumptions, scientists tried to measure the rate of deceleration of the expansion of the Universe.

However, what they found was that not only was the expansion of the Universe not slowing down, but on the contrary, it was accelerating at a great rate. Free observers, having no idea about the cause of this “behavior” of the Universe, assumed that this cause was an unknown type of energy, which they called **Dark Energy**.

### 4.7.4 Expansion of the Universe - Current Views

The velocities of stars are estimated through the Doppler effect (the faster a star moves away from an observer, the redder its light becomes, i.e., its spectrum is shifted towards the red).

The positions of specific stars (type Ia supernovae) are determined through their luminosity. The luminosity of type Ia supernovae (White Dwarfs) is the same for all stars of this type. The reason is that White Dwarfs explode when they acquire a mass slightly greater than 1.4 times the mass of our Sun (Chandrasekhar Limit). If a Type Ia supernova that is farther away from another appears fainter, then a calibration of the distances can be made and from these their ages can be found. What was revealed by observations of Type Ia supernovae was that today the stars are moving away from us faster than in the past.

For bound observers, according to the geometric model, this results from the fact that the circle surrounding one Planck patch has a smaller radius than the circle surrounding the next Planck patch. The speed of expansion of space is constantly increasing.

This means that today the Universe is expanding faster than in the past. If one examines all these issues from the perspective of bound observers of  $\mathbf{P}^2$  space, then there are answers to all these questions.

The first question is why the expansion of space occurs.

The answer to this question is to continuously decrease the curvature of the circumscribed circle in each of the subsequent Planck patches until it reaches zero.

The second question is why the Universe is expanding faster today than in the past.

The answer is that the increase in the radius of the circumscribed circle of two successive Planck patches for the continuous decrease in curvature has as a consequence an increase in the expansion speed of space, and therefore its expansion acceleration. This is the reason why the increase in the speed of space today is greater than in the past.

## 4.8 The mysteries of mass

As mentioned previously, the structure of macroscopic bodies is made by atoms which are synthesized together to give molecules. Molecules are synthesized together to give more complex compounds, etc.

Let us examine the simplest complex structural unit, the atom. As has been mentioned, the atom consists of a tiny, almost point-like nucleus surrounded by almost point-like electrons that are separated from it by enormous distances for the microscopic scale of the atom.

In other words, the nucleus and the electrons consist of point-like geometric objects, and between them there is only empty space! This observation leads us to consider that we and everything around us is an almost empty space, so the concept of mass or amount of matter is meaningless.

Nevertheless, free observers attribute to every macroscopic body the quantity they call mass. The usual interpretation" of this quantity is that the numerical value of the mass of a body is the sum of the amounts of matter" of which this body is composed. However, this definition of mass is reduced to another concept, namely the "amount of matter", so it is not an answer but a displacement of the answer to another concept that has not been defined.

All that remains is to examine whether the structure of the atom as a whole can provide an explanation for mass. The questions are:

**What do free observers measure as the mass of the quarks that make up the nuclei of atoms and the electrons that orbit around the nuclei?**

**Are these mass values sufficient to provide a macroscopic representation of the mass of a body?**

The answer is no.

Free observers try to explain this discrepancy by proposing that the binding energy that holds the quarks of the nuclei together be included in the mass value. But the following questions immediately arise:

**A. How does the electron have mass when it is a nearly point object?**

**B. The  $t$ -quark has over 75,000 times the mass of the  $u$ -quark and the same charge and spin as it. The question is: Can both participate in the same interactions?**

**C. What does it mean that a particle has zero mass, such as photon or gluon?**

These questions cannot be answered by free observers, so they pass them on to bound observers to be answered through 1-1 correspondences. Bound observers, in order to give an interpretation of the concept of mass (**connect it geometrically with the concept of curvature**). Thus, both groups of observers consider that the correlation of mass and curvature is through the relations:

$$M_H = \lambda_H \cdot \kappa_H \quad \text{with } \kappa_H = \frac{1}{r_c} \quad (\text{free observers})$$

and

$$\mu_c = \kappa_c \quad \text{with } \kappa_c = \lambda_c \cdot \frac{r_c}{l_P^2} \quad (\text{bound observers})$$

$\lambda_H$  is a constant of proportionality with units  $[\lambda_H] = [M][L] = [\lambda_c]$ .  $\lambda_c$  is a parameter whose value is determined by the radius of the circle surrounding the Planck patch and has a numerical value such that  $M_H = \mu_c$ , that is,  $\lambda_c = \lambda_H \cdot \frac{l_P^2}{r_c^2}$ . We will return to these relations later.

For free observers, the smaller the radius of the circle surrounding an aggregate of geometric objects that we correspond to physical entities, the greater its curvature and therefore its mass. A free observer can easily determine from the relation that defines the concept of energy  $E_H$  in circles. From the relation:

$$E_H = \left| -\frac{\kappa}{2r_c} \right| = \lambda \cdot \kappa_H$$

where  $\lambda = \frac{\kappa}{2}$ ,  $\kappa_H = \frac{1}{r_c}$ , the quantity  $E_H$  is proportional to the curvature or mass. Through the correlation of the geometric quantities of  $\mathbf{P}^2$  and the physical quantities defined in 3D-space, interpretations for various physical quantities can be given.

The answer to the first question is that the circle surrounding the electron has a curvature value equal to the value of the mass measured by free observers, as we saw in section 4.3.

The answer to the second question is that the curvatures of the 3-generation circles corresponding to the elementary particles are formed during the expansion-contraction process of  $\mathbf{P}^2$ , as we saw in section 3.3.

The answer to the third question is that photons or gluons, like each quark or lepton corresponding to a circle or an ellipse, has total zero charge and mass due to the pairwise opposite characteristic vectors of its four quadrants.

Let us consider the case where an aggregate of geometric objects tends to move by passing successively through different regions of space. The cause that causes the aggregate to move is the fulfillment of z.c. $\mathbf{P}^2$  for the charge or for the orientation.

In 3D-space, free observers find that every macroscopic body that tends to move experiences a resistance to the change of its kinetic state which they call **inertia**. In this case, free observers consider that the cause of the body's resistance to movement is due to a macroscopic quantity which they call **inertial mass**.

The question posed by free observers is: **Where do inertia and inertial mass come from?**

Free observers cannot give any interpretation or answer to this question, so on the one hand, they define inertial mass axiomatically and, on the other hand, they turn to bound observers to give an interpretation.

According to the structure of the space  $\mathbf{P}^2$ , the bound observers have defined that  $\mathbf{P}^2$  undergoes expansion and contraction due to self-similarity. In the following, we will use the same word "body" for the physical object in 3D-space and for the geometric object in the space  $\mathbf{P}^2$ .

Every geometric aggregate within  $\mathbf{P}^2$ , when it tends to move, increases the space it initially occupied. Since the expansion-contraction process is the same in every part of the total space  $\mathbf{P}^2$ , then at the stage of space contraction, it leads to a decrease in the speed of the geometric aggregate.

Another reason for the appearance of resistance during the movement of a body is the number of its available orientations and their direction. If during the movement of the body there are orientations that must be combined with others or moved away from others to fulfill z.c. $\mathbf{P}^2$ , then the body accelerates or decelerates.

Macroscopically, this expansion-contraction process and the fulfillment of z.c. $\mathbf{P}^2$  is interpreted by free observers as **resistance** to the change of its kinetic state.

Another question that free observers ask is: **Where does gravitational mass come from?**

The same applies when a body falls towards the Earth. If one lifts a stone from the surface of the Earth, then macroscopically the space occupied by the Earth-stone aggregate increases (expansion process). By releasing the stone, it will fall to the Earth to reduce the space previously occupied by the two bodies (contraction process). In this case, free observers call the mass

of the body **gravitational**.

## 4.9 Why are the values of inertial and gravitational mass equal?

The inertial mass has the same numerical value as the gravitational mass, as shown by experimental measurements of free observers (Einstein's experiment) and theoretically by bound observers, due to the self-similar expansion-contraction process of the space  $\mathbf{P}^2$ .

The expansion-contraction process of the space  $\mathbf{P}^2$  has nothing to do with the direction we choose, due to the isotropy of  $\mathbf{P}^2$ . Thus, in any direction the movement of a body occurs, the same curvature of the corresponding geometric object will appear. Therefore, the value of the quantity we call mass (inertial or gravitational) of a body is the same.

From the above, the answer to the question of why the values of the inertial and gravitational mass of a body are equal arises. The question that arises is:

**How are inertial and gravitational mass quantified?**

### A. Inertial mass

From observations and experiments, it has been found that every body that we attempt to change its kinetic state exhibits a resistance to this change. Observing this phenomenon macroscopically, free observers consider that the “matter” of which each body is composed reacts to this change.

For example, suppose that a free observer begins to break a very large rock into pieces of constantly decreasing size. This observer begins to push the pieces of different sizes on a flat surface of the Earth, starting from the smallest to the largest pieces in size, trying harder each time to move them. During this process, he establishes the existence of a resistance of the body during its movement, where he does not know why it exists.

Thus, he arbitrarily gives it the name inertial mass, attributing it to the “matter” that the body contains. When we say we touch, hold, or push an object, we have the feeling that our fingers are touching the object. In reality, our fingers and the object are very far apart on a microscopic scale.

The above processes occur only through electrical interactions between the directions of the spins of the electrons of the fingers and the palm of the free observers and the directions of the spins of the electrons of the materials of which the object is composed. We will examine in detail what happens in each case, guided by the interpretation of the bound observers for the fulfillment of z.c. $\mathbf{P}^2$ .



#### 4.9. WHY ARE THE VALUES OF INERTIAL AND GRAVITATIONAL MASS EQUAL? 69

As the free observers observe, as the size of the rock pieces increases, their speed decreases until it becomes zero from some piece onward. From this process, the free observers conclude that each piece of the original rock behaves in this way because the quantity they call **inertial mass**  $m_I$  increases, which is correlated with the size of the rock piece.

The bound observers interpret the process of moving the rock pieces by describing the next steps one by one, through  $\mathbf{P}^2$ . We will use the same names of the quantities used by the free observers with the corresponding geometric analogs used by the bound observers, as follows:

1. Initially, the stone is at rest on the surface of the Earth. This means for the bound observers that the orientations of the electrons of the structural materials of the stone have opposite directions from the directions of the orientations of the electrons of the materials of which the part of the Earth where the stone is located is composed in order to fulfill z.c. $\mathbf{P}^2$ .
2. A free observer “touches” the stone with his hands in order to push it. In order to move this piece of rock, the free observer’s brain gives an order (through chemical processes) to the orientations of the electrons of his fingers and palm to acquire the same direction as that of the desired direction of movement of the body on the surface of the Earth.

In the piece of rock, its available orientations take the same direction as that of the orientations of the electrons of the fingers and palm of the free observer. This means that z.c. $\mathbf{P}^2$  is not fulfilled. To fulfill z.c. $\mathbf{P}^2$ , the piece of rock moves away from the hand as it approaches it. The movement or immobility of the piece of rock is due to the number of available electron orientations of the bodies approaching or moving away from each other to fulfill z.c. $\mathbf{P}^2$ .

3. If the number of electron orientations of the hands of the free observer is smaller than the number of electron orientations of the rock materials, then the orientations cancel each other out, satisfying z.c. $\mathbf{P}^2$ . The piece of rock remains motionless because the remaining electron orientations of the rock materials satisfy z.c. $\mathbf{P}^2$  with the electron orientations of the materials that make up that particular part of the Earth.

This process gives the illusion to free observers that there is a quantity which they call **inertial mass** which resists the motion of each body. However, according to the above analysis, we make the following observations:

- Microscopically, bound observers define as inertial mass the quantity that expresses the number of available orientations for fulfilling z.c. $\mathbf{P}^2$ .

- Macroscopically, free observers define inertial mass as the scalar quantity that expresses the resistance of bodies to changes in their kinetic state.

What is true for measuring the mass of a microscopic body such as the proton  $p \rightarrow uud$  or the neutron  $n \rightarrow udd$ , which are composed of the elementary particles  $u, d$ ?

We mentioned this earlier when examining the curvature of geometric objects and their correlation with mass. Another issue that arises experimentally is that the total mass of  $uud$  or  $udd$  is roughly only 1% of the mass of  $p, n$ . The question that arises is:

**Where is the remaining 99%?**

The quarks that make up the  $p, n$  are connected to gluons which are represented by circles with the same radius as the quarks' circles. This increases the curvature of the quarks-gluons aggregates. Since the space occupied by the  $p, n$  is infinitesimal, then the macroscopic bodies that are composed of them actually express the "empty".

## B. Gravitational mass

Let's start with an example to see what happens to a body, for example a stone that we have lifted from the surface of the Earth and then let it fall to it. The interpretation of the movement of the stone towards the surface of the Earth was interpreted by Newton through an attractive force" exerted by the Earth on the stone which he called Gravity" or Gravitational attraction".

The concept of Gravity was generalized for every body on the Earth on the one hand and on the other hand in the entire Universe, thus creating a multitude of contradictions and paradoxes, such as the mass deficit" of galaxies.

Describing the previous example from the moment the lifted body is allowed to fall, we try to interpret only the falling process, that is, half the process, by arbitrarily introducing physical quantities" such as the force of Gravity".

We will examine the phenomenon from the moment we catch" the stone until the moment we let it fall from a certain height that we have raised it, above the surface of the Earth. Thus, we describe the next steps one by one:

1. Initially, the stone comes to rest on the surface of the Earth. This means for the bound observers that the orientations of the electrons of the structural materials of the stone have opposite directions from the orientations of the electrons of the materials of which the part of the Earth where the stone is located is composed, in order to fulfill z.c. $\mathbf{P}^2$ .

#### 4.9. WHY ARE THE VALUES OF INERTIAL AND GRAVITATIONAL MASS EQUAL? 71

2. We touch" the stone with our hands in order to lift it. This means that our brain instructs the orientations of the electrons in our hand to acquire the same direction as the orientations of the electrons in the stone, so that  $\text{z.c.}\mathbf{P}^2$  is not fulfilled for the stone-hand system. However, this implies that the orientations of the electrons in the stone will change direction to fulfill  $\text{z.c.}\mathbf{P}^2$  in the stone-hand system. Thus, they take on the same orientation as the electrons in the materials that make up the part of the Earth where the stone is located. This means that  $\text{z.c.}\mathbf{P}^2$  is satisfied for the stone-hand system and  $\text{z.c.}\mathbf{P}^2$  is not satisfied for the stone-Earth system, so the stone is lifted from the Earth's surface to satisfy  $\text{z.c.}\mathbf{P}^2$  for the stone-Earth system.
3. If the number of electron orientations of the hands of the free observer is smaller than the number of electron orientations of the materials of the stone, then the orientations cancel each other out, satisfying  $\text{z.c.}\mathbf{P}^2$ . The stone remains motionless because the remaining electron orientations of the stone's materials satisfy  $\text{z.c.}\mathbf{P}^2$  with the electron orientations of the materials that make up that particular part of the Earth.
4. If the stone has been lifted and our brain gives the command for the stone to fall, then the electron orientations of our fingers and palm acquire the same direction as that of the electron orientations of the stone, so our hand moves away from it to fulfill  $\text{z.c.}\mathbf{P}^2$ . To satisfy  $\text{z.c.}\mathbf{P}^2$  between the lifted stone and the Earth's surface, the orientations of the electrons of the stone take on opposite values from those of the orientations of the electrons of the materials of which that particular part of the Earth is composed, so that the stone moves towards the Earth.

Through this process, the fall of the stone is interpreted without introducing an arbitrary concept, such as the force of Gravity, but only through the interactions of the orientations of the electrons of the two bodies.

Free observers conclude that every body moving towards the Earth is "attracted" by it because it has a kind of mass called **Gravitational mass** ( $m_g$ ). According to the analysis we made for the movement of the stone on the Earth's surface or its fall towards the Earth's surface, what plays an essential role is the number of available orientations of the stone, so this determines the value of the quantity called mass. This also applies to all lifting machines that man has built.

Thus, for both masses, the number of available orientations for the fulfillment of  $\text{z.c.}\mathbf{P}^2$  for each body and for its movement is the same, so the value of the inertial mass" is equal to the value of the gravitational mass". Through this reasoning, we write the relationship  $m_I = m_g$  which expresses

the number of orientations that participate in the processes of attraction or repulsion” of bodies.

Einstein postulated the relation  $m_I = m_g$  as an axiom to formulate the General Theory of Relativity. However, as we will see, the theory of General Relativity can be derived from the definition relations of conic sections without introducing the relation  $m_I = m_g$ .

By introducing a principle or an axiom or a law into a theory to explain a phenomenon, a big why” follows, which means that the principle or axiom or law introduced into a theory was done because of ignorance of what is really happening.

## Part II

# THEORIES OF PHYSICS



## Chapter 5

# FOUNDATIONS OF CLASSICAL MECHANICS

### 5.1 Relations between the geometric quantities of conic sections

Classical Mechanics can be grounded in three-dimensional space through representations of the relations of the geometric quantities of fixed conic sections and the physical quantities of the entities of 3D-space, through 1-1 correspondences.

Let there be a unit circle in the space  $\mathbf{P}^2$ . Each point on the circumference of the unit circle is determined by the position vector  $\mathbf{r}_c(s)$  and the tangent vector  $\mathbf{u}_c(s)$ , where  $s$  is the length of the arc that characterizes this point, measured from an arbitrary point on the circumference of this circle (principle of arc length measurement). Similarly, the description of this circle can be done relative to a Cartesian coordinate system  $x_1, x_2$  with its origin at the center of this circle. If  $t$  is a real parameter such that  $t$  belongs to the interval  $I = [0, 1]$  of  $\mathbf{R}$ , where  $\mathbf{R}$  is the set of real numbers, then the coordinates of a point on this circle, with respect to bound observers, are written as:

$$x_1(\theta) = r_c(\theta) \cos(\theta) = \cos(\theta)$$

$$x_2(\theta) = r_c(\theta) \sin(\theta) = \sin(\theta)$$

where  $r_c(\theta) = 1$  and  $\theta = \theta(t)$  is the angle function, with respect to the parameter  $t$ , between the position vector and the  $x_1$ -axis, and  $x_1, x_2$  are the functions of the Euclidean coordinates.

In all subsequent relations, this notation will be used for  $\theta$ . The position vector at any point on the circumference of the circle is given by the relation:

$$\mathbf{r}_c(\theta) = x_1(\theta)\mathbf{e}_1 + x_2(\theta)\mathbf{e}_2 = \cos(\theta)\mathbf{e}_1 + \sin(\theta)\mathbf{e}_2$$

where  $\mathbf{e}_1, \mathbf{e}_2$  are the unit vectors on the  $x_1, x_2$  axes respectively. In what follows, vectors and vector fields will be denoted in bold. The unit vector  $\mathbf{e}_{r_c}(\theta)$  along  $\mathbf{r}_c(\theta)$  is given by the relation:

$$\mathbf{e}_{r_c}(\theta) = \frac{\mathbf{r}_c(\theta)}{r_c(\theta)}$$

The tangent at any point on the circumference of this circle is the vector:

$$\mathbf{u}_c(\theta) = \frac{d\mathbf{r}_c(\theta)}{dt}$$

The bound observers, substituting  $\mathbf{r}_c(\theta)$  in this relation, get:

$$\mathbf{u}_c(\theta) = \frac{d\theta}{ds}(-\sin(\theta)\mathbf{e}_1 + \cos(\theta)\mathbf{e}_2)$$

The vectors  $\mathbf{r}_c(\theta)$  and  $\mathbf{u}_c(\theta)$  are expressed in terms of the functions of the physical coordinates  $x_1, x_2$  as:

$$\mathbf{r}_c(\theta) = x_1(\theta)\mathbf{e}_1 + x_2(\theta)\mathbf{e}_2$$

$$\mathbf{u}_c(\theta) = \frac{d\theta}{ds}(-x_2(\theta)\mathbf{e}_1 + x_1(\theta)\mathbf{e}_2)$$

Bound observers define scalar quantities to determine the points of the curve, on the one hand in terms of the measure  $|\mathbf{r}_c(\theta)|$  of the position vector and on the other hand in terms of the different directions of the vector quantities at each point of the curve, so as not to be involved in their calculations. Thus, at each point of the circle, the dot product of  $\mathbf{r}_c(\theta)$  with itself and between  $\mathbf{r}_c(\theta), \mathbf{u}_c(\theta)$  is defined as follows:

$$|\mathbf{r}_c(\theta)| = \sqrt{(\mathbf{r}_c(\theta) \cdot \mathbf{r}_c(\theta))} = \sqrt{(x_1^2(\theta) + x_2^2(\theta))}$$

$$\mathbf{r}_c(\theta) \cdot \mathbf{u}_c(\theta) = x_1(\theta)u_1(\theta) + x_2(\theta)u_2(\theta)$$

where:

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$$

The vectors  $\mathbf{r}_c(\theta)$  and  $\mathbf{u}_c(\theta)$  are perpendicular to each other because their scalar product is zero, i.e.,

$$\mathbf{r}_c(\theta) \cdot \mathbf{u}_c(\theta) = 0$$

Another scalar quantity that can be determined by bound observers is the area of the rectangle formed by the vectors  $\mathbf{r}_c(\theta)$  and  $\mathbf{u}_c(\theta)$ . At a given point on the circle where  $r_c(\theta)$  is given, the expression of this area in terms of a Cartesian coordinate system is given by the relation:

$$E(\theta) = |\mathbf{r}_c(\theta) \times \mathbf{u}_c(\theta)| = x_2(\theta)u_1(\theta) - x_1(\theta)u_2(\theta)$$



if  $\mathbf{u}_c(\theta)$  has a specific direction.

If  $\mathbf{u}_c(\theta)$  has the opposite direction from the previous one, the expression of this area is given by the relation:

$$E(\theta) = -(x_2(\theta)u_1(\theta) - x_1(\theta)u_2(\theta))$$

This problem is related by the bound observers to the fulfillment of the zero criterion of  $\mathbf{P}^2$ , because the area is a scalar quantity but takes two opposite values. To answer this problem, free observers define a vector that includes both of these cases as follows:

$$\mathbf{L} = \mathbf{r}_c(\theta) \times \mathbf{u}_c(\theta)$$

The vector  $\mathbf{L}$  is the outer product (cross product) of the vectors  $\mathbf{r}_c(\theta)$  and  $\mathbf{u}_c(\theta)$ . If the vector  $\mathbf{L}$  has a direction from the surface towards free observers, then they conventionally call it positive and the measure of  $\mathbf{L}$ , that is,  $|\mathbf{L}|$ , is equal to  $E(\theta)$ , while if it has the opposite direction it is called negative and is equal to  $-E(\theta)$ .

The same image is obtained by free observers when the plane of the circle is rotated by  $\pi$  about  $\mathbf{r}_c(\theta)$ . Through the process of rotation by  $\pi$ , they obtain the same circle on both sides of the plane of the circle, where  $u(\theta)$  on one side of the circle takes the opposite value from the other side. This process defines the two faces of the plane of the circle through the unit vector  $\mathbf{L}_0 = \frac{\mathbf{L}}{L}$  called the orientation. Free observers define the vector  $\mathbf{e}_3$  as the unit vector of  $\mathbf{L}$ , i.e.,  $\mathbf{L}_0 = \mathbf{e}_3$  which is given by the outer product of  $\mathbf{e}_1$  and  $\mathbf{e}_2$  and is perpendicular to them, i.e.,

$$\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$$

and

$$-\mathbf{e}_3 = \mathbf{e}_2 \times \mathbf{e}_1$$

The vectors  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$  define a Cartesian coordinate system in three-dimensional space. Substituting the expressions for  $\mathbf{r}_c(\theta)$  and  $\mathbf{u}_c(\theta)$  into the relation given by  $L$ , we get:

$$\mathbf{L} = \dot{\theta} \mathbf{e}_3$$

If the measure of  $\mathbf{L}$  is known, then we can find the functional relation between  $\theta$  and  $t$ . If we assume that  $\mathbf{L}$  is a constant quantity with unit measure, that is,  $|\mathbf{L}| = 1$ , then the functional relation between  $\theta$  and  $t$  is defined as:  $\theta(t) = t$  with  $\theta(0) = 0$

The measure of  $\mathbf{L}$  is given by the relation:

$$L^2 = \mathbf{L} \cdot \mathbf{L} = (\mathbf{r}_c(\theta) \times \mathbf{u}_c(\theta)) \cdot \mathbf{L}$$

Due to the triple product property, where the symbols  $\times$  and  $\cdot$  are interchanged, free observers from the previous relation get:

$$\mathbf{r}_c \cdot (\mathbf{u}_c \times \mathbf{L}) - L^2 = 0$$

or

$$r_c \mathbf{e}_r \cdot (\mathbf{u}_c \times \mathbf{L}) - L^2 \mathbf{e}_r \cdot \mathbf{e}_r = 0$$

where

$$\mathbf{e}_r \cdot \mathbf{e}_r = 1$$

From this relation, dividing by  $r_c \neq 0$ , we get:

$$\mathbf{e}_r \cdot (\mathbf{u}_c \times \mathbf{L} - k \mathbf{e}_r) = 0$$

where  $k = \frac{L^2}{r_c}$ . The bound observers define the vector  $\mathbf{R}$ :

$$\mathbf{R} = \mathbf{u}_c \times \mathbf{L} - k \mathbf{e}_r$$

which is called the **Runge-Lenz vector**. In the case of the unit circle, bound observers take  $k = 1$  because  $L = 1$  and  $r = 1$ , so  $|\mathbf{R}| = 0$ .

From this procedure, free observers generally determine the vectors  $\mathbf{L}$  and  $\mathbf{R}$  by which any open or closed conic section can be clearly defined. Closed conic sections (circle, ellipse) and open conic sections (parabola, hyperbola) are created by rotating the plane of a circle that is initially perpendicular to the axis of symmetry of a cone. For free observers, the defining relations of  $\mathbf{L}$  and  $\mathbf{R}$  are written as:

$$\mathbf{L} = \mathbf{r}_c \times \mathbf{u}_c$$

$$\mathbf{R} = \mathbf{u}_c \times \mathbf{L} - \mathbf{r}_c \frac{k}{r_c}$$

From these relations, depending on the values of  $L, R$  and  $k$ , all conic sections result. In the case of a circle,  $\mathbf{r}_c, \mathbf{u}_c, \mathbf{L}$  are mutually perpendicular, so the previous relation is written as:

$$\mathbf{R} = (u_c L - k) \mathbf{e}_r$$

$\mathbf{R}$  is a vector of the plane of the circle with a measure given by the relation:

$$R^2 = (u_c L - k)^2$$

For  $L = u_c r_c$ , free observers get:

$$R^2 = 2L^2 \left( \frac{1}{2} u_c^2 - \frac{k}{r_c} \right) + k^2$$

From this expression, the quantity  $E_c$  is defined, which is called the **energy of conic sections**:

$$E_c = \frac{1}{2} u_c^2 - \frac{k}{r_c}$$

For  $u_c = \frac{L}{r_c}$  and  $k = \frac{L^2}{r_c}$ , we get:

$$E_c = -\frac{k}{2r_c} = \lambda\kappa_H$$

where  $\lambda = -\frac{k}{2}$  is a constant and  $\kappa_H = \frac{1}{r_c}$  is the curvature of the circle as defined by free observers (Holographic curvature).

## 5.2 Newtonian formulation

As we have seen and will see in detail below, every fixed conic section is constructed from the relations satisfying the fixed vectors  $\mathbf{L}$  and  $\mathbf{R}$ , which are:

$$\mathbf{L} = \mathbf{r}_c \times \mathbf{u}_c \quad (5.1)$$

$$\mathbf{R} = \mathbf{u}_c \times \mathbf{L} - \mathbf{r}_c \frac{k}{r_c} \quad (5.2)$$

where  $\mathbf{r}_c$  and  $\mathbf{u}_c$  are the position vector and the tangent vector at a point of the conic section respectively,  $k$  is a constant characterizing the given conic section, and  $r_c$  is the measure of  $\mathbf{r}_c$ . Each point of a conic section is characterized by the arc length  $s$  measured from an arbitrary origin to that point.  $\mathbf{L}$  is a vector perpendicular to the face of the plane where the conic section is located, the direction of which is from the face of the plane outwards, and  $\mathbf{R}$  is a vector in the plane of the conic section. A conic section is described by an equation in polar coordinates as follows:

$$r_c = \frac{L^2}{k + R \cos \theta} = \frac{L^2}{1 + \epsilon \cos \theta}$$

where  $\theta$  is the angle between  $\mathbf{e}_r$  and  $\mathbf{R}$ . The quantity  $\epsilon = \frac{R}{k}$  is called the eccentricity of the conic section. The change in the tangent at a point of a conic section (acceleration) is a vector defined as the derivative of the tangent with respect to the arc length  $s$  at that point, i.e.,

$$\alpha_c(s) = \mathbf{u}'_c(s)$$

where

$$\mathbf{u}'_c(s) = \frac{d\mathbf{u}_c(s)}{ds}$$

The prime ( $'$ ) denotes the derivative of a quantity with respect to  $s$ . Since  $\mathbf{L}$  and  $\mathbf{R}$  are constant vectors, then:

$$\frac{d\mathbf{L}}{ds} = 0, \quad \frac{d\mathbf{R}}{ds} = 0$$

The acceleration of a point of a conic section is a vector parallel to  $\mathbf{r}$ , given by the relation:

$$\alpha_c(s) = -\frac{k}{r_c^2(s)} \mathbf{e}_r$$

We will apply the 1-1 correspondences to determine the characteristics of the entities of the two spaces. The bound observers define a point of a circle of radius  $r_c$  in  $\mathbf{P}^2$  by the position vector  $\mathbf{r}_c(s)$  and find the tangent  $\mathbf{r}'_c(s)$  and the acceleration  $\alpha_c(s)$  of this point. All points of the circle have the same acceleration because they are equidistant from its center. The vector  $\alpha_c(s)$  for the circle is an isotropic vector of  $\mathbf{P}^2$ . According to the 1-1 correspondence of geometric objects with physical entities, bound observers find geometric quantities that correspond to quantities that free observers define for the physical entities of three-dimensional space as follows:

$$s \leftrightarrow t$$

$$r_c(s) \leftrightarrow r(t)$$

$$u_c(s) \leftrightarrow u(t)$$

The above correspondences are general, so they apply to all foundations of physical theories. The first relation is the correspondence of the arc length ( $s$ ) of a point of a conic section, measured from an arbitrary origin of it, with the parameter of  $\mathbf{P}^2$  and three-dimensional space, which bound and free observers call time ( $t$ ).

Through the parameter of time, the changes of the quantities of the physical entities of the three-dimensional space are determined. The complex geometric quantity  $\alpha_c(s)$  has as its point of application every point of the circumference of the circle of radius  $r$  in  $\mathbf{P}^2$ . The expression of  $\alpha_c(s)$  in both spaces is given by the relation of the form:

$$\alpha = -\frac{k}{r_c^2} \mathbf{e}_r$$

To preserve the isotropy and homogeneity of  $\mathbf{P}^2$  in three-dimensional space, free observers consider that each  $\alpha$  has as its point of application a point of a sphere with radius  $r$ . The direction of each  $\alpha$  lies along each radius of the sphere with measure  $r$ , and its direction is towards the center of the sphere due to the negative sign in the previous relation.

Free observers, in order to uniquely determine this construction, will have to find a scalar quantity, like bound observers, to get rid of the different directions of  $\alpha$ . Thus, they apply **Gauss's Divergence Theorem** to this sphere, integrating  $\alpha$  over the sphere. The following relation gives the solid angle element:

$$d\Omega = \frac{\mathbf{e}_r}{r_c^2} \cdot d\mathbf{S}$$

where  $d\mathbf{S}$  is a vector with a measure equal to the infinitesimal area  $dS$  of the sphere, directed from the surface of the sphere outwards.

Since  $\mathbf{e}_{r_c}$  is directed towards the center of the sphere, then

$$\mathbf{e}_r \cdot d\mathbf{S} = \cos(\pi)dS = -dS$$

. Free observers integrate the scalar quantity  $\alpha \cdot d\mathbf{S}$  over the surface of the sphere, thus obtaining the constant scalar quantity  $M_H$  which is always a positive quantity, i.e.,

$$M_H = \oint \alpha \cdot d\mathbf{S} = - \oint \frac{k}{r_c^2} \mathbf{e}_r \cdot d\mathbf{S} = k \oint d\Omega = 4\pi k$$

Free observers call the scalar quantity  $M_H$ , which is a consequence of the isotropy of the sphere in three-dimensional space and the expression of  $\alpha$ , and always takes positive values, the mass.

The previous relation expresses the fact that the cause of the appearance of the quantity  $M_H$  at some point in three-dimensional space is the isotropy of the vector field  $\alpha$  resulting from the mapping of the vector  $\alpha_c(s)$  of a circle of  $\mathbf{P}^2$  into three-dimensional space. The numerical value of  $M_H$  depends only on the constant  $k$ .

Free observers call a point in three-dimensional space a material point and identify it with a sphere that has an infinitesimal radius  $r_c$ . Free observers consider that all vectors  $\alpha$  in a sphere of any radius centered at the material point come from the quantity  $M_H$  at its center; that is, they consider  $M_H$  as the source of creation of  $\alpha$ .

Bound observers want to correspond the sphere to a geometric shape in  $\mathbf{P}^2$  that expresses its isotropy. The only geometric object that approximately satisfies this requirement is the circle. Bound observers require that the geometric quantities defined in a quadrant of a circle have a positive sign identical to the sign of the mass  $M_H$  of the sphere. The geometric quantities defined in a quadrant of a circle of radius  $r_c$  are the scalar quantity  $q$  and the vector quantity  $\hat{\mathbf{O}}$ .

The most appropriate scalar quantity for a circle that corresponds to the always positive quantity  $M_H$  is its curvature  $\kappa_c = \frac{1}{|r_c|}$  because it always takes positive values. Thus, the bound observers define in  $\mathbf{P}^2$  a corresponding quantity of  $M_H$  that they call  $\mu_c$ .

The question that arises is how the quantities  $M_H$  of 3D-space and  $\mu_c$  of  $\mathbf{P}^2$  are related. Since both of these quantities are proportional to the curvature of the same circle, then they are proportional quantities, so we get the relation:

$$M_H = \nu \mu_c$$

From the definition of  $M_H = \lambda_H \cdot \kappa_H = \frac{\lambda_H}{r_c}$ , we get the relation:

$$M_H \cdot r_c = \text{constant} \quad (5.3)$$

Relation 5.3 relates 3D-space to  $\mathbf{P}^2$ , so problems in three-dimensional space can be solved through  $\mathbf{P}^2$ .

We applied relation 5.3 when we found the values of the masses of the three generations of elementary particles  $d, u, e$  (Section 4.3) and the value of the electron ionization energy of the isotopes of  $H$  (Section 4.4).

In Appendix 17.6.1, we will apply relation 5.3 to find the limiting mass of white dwarfs (Chandrasekhar limit) and in Appendix 17.6.2 the limiting mass of neutron stars (Tolman-Oppenheimer-Volkoff limit). From the application of relation 5.3 at the microscopic and macroscopic level, the assumption we made is verified, namely that there can be a single theory to solve all problems (Theory Of Everything) and not a fragmentation through different theories for each problem separately.

The material point differs from other geometric points in three-dimensional space because it is a “point with mass”. Thus, for free observers, the material point creates a local anisotropy in its neighborhood. Free observers obtain material points in three-dimensional space by applying the procedure mentioned above.

Suppose that a bound observer considers different Planck patches in  $\mathbf{P}^2$ . This means that the circles circumscribed in them have different curvatures. If all these Planck patches are mapped onto three-dimensional space, then free observers consider these regions to be parts of a non-planar surface or space, because each of them will have a different curvature from each other. Thus, for each Planck patch of  $\mathbf{P}^2$  a neighborhood of the 3D-space will be defined that needs its own coordinate system to describe it.

In the flat space  $\mathbf{P}^2$ , as well as in any flat space, only one coordinate system covering the whole space is sufficient. But considering spaces where each region has its own coordinate system (manifolds), they will have to be examined differently, as will be done in the chapter on General Relativity. A finite region of  $\mathbf{P}^2$  (such as a region with intersecting circles in  $\mathbf{P}^2$ ) is mapped onto a finite region of three-dimensional space and is called a **body**.

The basic macroscopic features of a body are those of localization and non-separability. The foundation of Classical Mechanics to date has been made with the assumption that the mass  $M_H$  of a body exists, without its origin being interpreted. However, according to the creation of 3D-entities or holograms through the mappings of conic sections of two-dimensional space into three-dimensional space, the existence of the quantity  $M_H$  can be interpreted in three ways:

- a) The first is that  $M_H$  is a quantity in three-dimensional space produced by the vector field  $\alpha$  which is a representation of the well-defined geometric quantity  $\alpha(s)$  (acceleration of a conic section point), as considered by free observers. Free observers define a material point with mass  $M_H$  (cause) at the center of a sphere of radius  $r$  that creates an isotropic radial vector field  $\alpha(s)$  (effect) at every point on the surface of this sphere.
- b) The second is that  $M_H$  is proportional to the curvature of a circle of

radius  $r_c$  of  $\mathbf{P}^2$ .

- c) The third will be mentioned in the chapter on General Relativity and has to do with how all the geometric points of a neighborhood are connected to each other.

Let us assume that free observers consider the mass  $M_H$  at a point on the surface of the sphere with radius  $r$  and with another point mass  $M$  as its center. The mass  $M_H$  creates, in the same way as  $M$ , a vector field  $\alpha'(s)$ , where  $M = 4\pi k'$ . The mass  $M$  is acted upon by the vector field  $\alpha'(s)$  if it is located at a point on the surface of the sphere with radius  $r$  whose center is the mass  $M_H$ , and the mass  $M_H$  is acted upon by the vector field  $\alpha(s)$  if it is located at a point on the surface of the sphere with radius  $r$  whose center is the mass  $M$ . These two interactions were called the action-reaction law, as proposed by **Huygens**. The effect of the vector field  $\alpha$  ( $\alpha'$ ) on the mass  $M_H$  ( $M$ ) is proportional to this mass. Free observers denote this effect by  $\mathbf{F}$  and call it a **force**, writing the relations:

$$\mathbf{F} = \lambda m \alpha = \lambda \frac{M M_H}{r_c^2} \mathbf{e}_r$$

and

$$\mathbf{F}' = \lambda m \alpha' = -\lambda \frac{M M_H}{r_c^2} \mathbf{e}_r$$

Thus  $|\mathbf{F}| = |\mathbf{F}'|$  because the masses  $M_H$  and  $M$  are located on the surface of spheres of the same radius  $r$  where  $\mathbf{F}$  and  $\mathbf{F}'$  have opposite directions. The above relations also apply to the gravitational field of the Earth, in agreement with the experiments of Hooke and Halley. The quantity  $\lambda$  is a constant. Let us follow, as free observers, the same path that Newton followed. The relation  $\mathbf{F} = \lambda M_H \alpha$  can be divided into two relations, giving for each of them a different interpretation that results from the findings of Galileo's experimental work. Gravity is given by the relation:

$$\mathbf{F}_g = m_g \alpha_g = -G \frac{M_H m_g}{r_c^2} \mathbf{e}_r$$

where  $G$  is the universal gravitation constant and  $\alpha_g, m_g$  are called the gravitational field and gravitational mass respectively, where the subscript  $g$  refers to the supposed gravitational attraction. If we refer to the motion of a body on the surface of the Earth, we write:

$$\mathbf{F} = m_i \alpha_i$$

where the subscript  $i$  refers to the resistance or inertia (inertia) that a body presents to the change in its velocity. At this point, we will make a parenthesis that has to do with the introduction of another law for gravity that was introduced to explain the "flat curves" of the orbits of the stars of our Galaxy.

### 5.3 Modified Newtonian Dynamics

The assumption that Newton's law of gravity needs modification arose from the discrepancy between the measured values of the rotation speeds of stars and the theoretical values due to Newton's law of gravity. This discrepancy led Mordehai Milgrom to propose a different law of gravity entitled "Modified Newtonian Dynamics" (MOND). Milgrom wrote a law of gravity as:

$$\mathbf{F}_N = m_g \tilde{f}\left(\frac{\alpha}{\alpha_0}\right) \alpha$$

where  $\mathbf{F}_N$  is Newton's force,  $m_g$  is the gravitational mass of a body,  $\alpha$  is its acceleration,  $\alpha_0$  is a constant, and  $\tilde{f}(x)$  is a function with the following properties:

$$x \gg 1 \implies \tilde{f}(x) \rightarrow 1$$

$$x \ll 1 \implies \tilde{f}(x) \rightarrow x$$

Newton's law follows from the first property of  $\tilde{f}$ . From the second property of  $\tilde{f}$ , we get:

$$F_N = \frac{GMm_g}{r_c^2} = m_g \frac{\alpha^2}{\alpha_0}$$

or

$$\frac{GM}{r_c^2} = \frac{\alpha^2}{\alpha_0}$$

For a circular orbit where  $\alpha = \frac{u^2}{r_c}$ , we get:

$$u^4 = GM\alpha_0$$

From this result, the constancy of the rotation speed of stars at the edges of galaxies or their clusters follows, that is, the empirical Tully-Fisher relation, which expresses the ratio between the luminosity  $L$  and the mass of a celestial body. From the Virial theorem and the definition of surface brightness,  $u^4$  is proportional to  $L$ , so we take  $L$  to be proportional to  $M$  ( $L \propto M$ ), in agreement with the previous relation derived from MOND. However, due to the unclear foundation of the function  $\tilde{f}(x)$ , the prevailing view was that the discrepancy between observation and the results resulting from the application of Newton's law was due to the existence of dark matter, so the MOND theory did not succeed.

### 5.4 Inertial and gravitational mass

The force  $\mathbf{F}_g$  is proportional to the rate of increase of the velocity  $\alpha_g$  of a body, which is the same for each body; that is, it is independent of the



material from which the body is composed. This conclusion arose from the experiments of Galileo, who studied the movements of rolling spheres made of different materials on an inclined plane. The constant of proportionality between  $\mathbf{F}_g$  and  $\alpha_g$  is the gravitational mass  $m_g$ . If we refer to the motions of a body anywhere in the Universe, we write the equation  $\mathbf{F} = \lambda m \alpha$  and call it the law of motion of a body, which is formally similar to the law of gravity  $\mathbf{F}_g = m_g \alpha_g$ , as:

$$\mathbf{F}_I = m_I \alpha_I = \frac{d(m_I \mathbf{u})}{dt}$$

where the coefficient  $m_I$  is the inertial mass of a body, as introduced by Galileo. The inertial mass is the measure of the inertia of a body; that is, it expresses its resistance to a change in velocity, which is the same for any force or otherwise is independent of the type of force exerted on it.

The quantity  $\alpha_i$  is the acceleration of a body. The interpretation given to the masses  $m_g$  and  $m_I$  suggests that they have different characteristics. Since the gravitational mass is independent of the material of which a body is composed, then it is equal to any inertial mass, i.e.,  $m_g = m_I$ . This equality was called the **Principle of equivalence** and was verified experimentally (Eötvös experiment). Also, the force  $\mathbf{F}_I$  is independent of the type of force, therefore, due to this universal characteristic, it is considered equal to  $\mathbf{F}_g$ .

From this relation and the principle of equivalence, we get  $\alpha_I = \alpha_g$ . If this relation is suitably modified and applied to curved spacetime, then the Einstein equations of General Relativity arise. Since the relations given by  $\mathbf{F}_g$  and  $\mathbf{F}_I$  arise from a single relation, then the motions of celestial bodies have a common origin with the motions of bodies on Earth. This conclusion has been experimentally verified in “our neighborhood”, that is, in our Solar System, but not for the motions of stars at the boundaries of Galaxies. This observation forces us to consider that either the law of gravity needs modification or there is more mass than the visible mass.

By applying the equation for  $\mathbf{F}_I$  to various problems, the characteristics of the two main categories of physical entities in three-dimensional space are determined, namely those of bodies and waves. Let us first assume that three-dimensional space is Euclidean space.

The position of a body in this space is determined by its spatial coordinates, with respect to only one fixed coordinate system covering the entire three-dimensional space and the independent parameter of this coordinate system, time  $t$ . We consider only one coordinate system for the entire three-dimensional space because this is a flat space.

The movement of this body along a path in three-dimensional space is determined by the trajectory of its geometric object that exists in  $\mathbf{P}^2$ . The trajectory that a geometric object follows when it moves freely between two points, to fulfill the properties of  $\mathbf{P}^2$ , is constant and its length takes its minimum value, that is, its trajectory is a straight line. The corresponding

trajectory to the one that executes its hologram in three-dimensional space is called a **geodesic**.

For waves, it makes no sense to refer to a defined path between two points because their motion theoretically covers all points in three-dimensional space.

## 5.5 Lagrangian and Hamiltonian formalism

For an ellipse in a Cartesian coordinate system which has its origin at a focus of the ellipse, free observers write the relations for the vectors  $\mathbf{L}$  and  $\mathbf{R}$  as:

$$\begin{aligned}\mathbf{L} &= (x_1 x'_2 - x'_1 x_2) \mathbf{e}_3 \\ \mathbf{R} &= (x'_2 |\mathbf{L}| - \frac{k}{r_c} x_1) \mathbf{e}_1 - (x'_1 |\mathbf{L}| + \frac{k}{r_c} x_2) \mathbf{e}_2\end{aligned}$$

where  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$  are the unit vectors of the axes of this coordinate system. Since  $\mathbf{L}$  and  $\mathbf{R}$  are constant vectors and  $k$  is a constant quantity for a given ellipse, then taking the square of the constant vector  $\mathbf{R}$ , we find the constant quantity:

$$R^2 = \left( x'_2 L - \frac{k}{r_c} x_1 \right)^2 + \left( x'_1 L + \frac{k}{r_c} x_2 \right)^2 = 2TL^2 - 2\frac{k}{r_c}L^2 + k^2$$

where

$$T = \frac{1}{2}[(x'_1)^2 + (x'_2)^2] = \frac{1}{2}u^2$$

The prime (') denotes the derivative with respect to  $s$ . The relation for  $R^2$  is written as:

$$\frac{R^2 - k^2}{2L^2} = T - \frac{k}{r_c}$$

Thus, for  $V = -\frac{k}{r_c}$  and  $E = \frac{R^2 - k^2}{2L^2}$ , the previous relation is written as  $E = T + V$  or  $E = -\frac{k}{2a}$  where  $a = k \frac{L^2}{k^2 - R^2}$  is the length of the major semi-axis of the ellipse. For  $V = \frac{k}{r_c}$ , we get  $E = \frac{R^2 - k^2}{2L^2} = T - V$ . We will make the following correspondences between the complex quantities of an ellipse in  $\mathbf{P}^2$  and the quantities of a hologram of three-dimensional space:

$$\mathbf{L} \leftrightarrow m\mathbf{r}c \times \mathbf{u} = \mathbf{r}c \times \mathbf{p}$$

where the vector  $\mathbf{p} = m\mathbf{u}$  is called the momentum of a material point or body. Then we make the correspondences:

$$T \leftrightarrow Ek = \frac{mu^2}{2}$$

$$V \leftrightarrow Ep = m\frac{k}{r_c}$$

$$E \leftrightarrow E_0 = E_k + E_p$$

$$\Lambda \leftrightarrow \Lambda_0 = E_k - E_p$$

The quantity  $m$  is the characteristic of the holograms of closed conic sections, which was defined in the formulation of Newton. The above quantities  $\mathbf{r}_c \times \mathbf{p}$ ,  $E_k$ ,  $E_p$ ,  $E_0$ ,  $\Lambda_0$  are called angular momentum, Kinetic energy, Dynamic energy, Total energy, and Lagrangian respectively, of a body (or a material point) which has a velocity  $\mathbf{u}$  and a position vector  $\mathbf{r}$  with respect to a Cartesian coordinate system in three-dimensional space.

Due to these correspondences, Classical Mechanics is expressed in terms of kinetic and potential energy in three-dimensional space. This formulation is equivalent to the formulation of Newton. The Lagrange formulation for an  $n$ -body system starts from a function, the Lagrangian, which is a function of the position coordinates  $q$ , their changes  $q(t + \Delta t) - q(t)$  in the time interval  $\Delta t$ , and the time  $t$ . Since the Lagrangian is defined through the last equation, then that for an  $n$ -body system is written as:

$$\Lambda(q, \dot{q}, t) = \sum_{i=1}^n [E_{(k,i)} - V(q_1, q_2, \dots, q_n)]$$

If a body moving freely in three-dimensional space follows a path that is a geodesic of this space, then we require that a freely moving system of  $n$  bodies follows a path that tends to a geodesic of the space. This requirement is reflected in the fact that the quantity  $S$ , called the action and defined as:

$$S = \int_{t_1}^{t_2} \Lambda(q, \dot{q}, t) dt$$

takes an extreme value. For an infinite change of action together with the condition  $\delta S = 0$  (Principle of least action), we obtain the Euler-Lagrange equations:

$$\frac{d}{dt} \left( \frac{\partial \Lambda}{\partial \dot{q}^i} \right) - \frac{\partial \Lambda}{\partial q^i} = 0, \quad i = 1, 2, \dots, n$$

These equations are the equations of motion of an  $n$ -body system. Note that  $q$  can be curvilinear coordinates on a surface (surface parameters) within three-dimensional space or a surface independent of some space as long as its metric is known. The Hamiltonian is a function of the position and momentum of a system of  $n$  bodies at a time, and it tells us what its position and momentum will be at a later time. Going from one set of independent variables to another via the Legendre transformations, we define the Hamiltonian function as:

$$H(q, p, t) = \sum_{i=1}^n p_i \dot{q}^i - \Lambda(q, \dot{q}, t)$$

where  $p_i = m\dot{q}_i$ . The equations of motion in this case are called Hamilton equations, as follows from the principle of least action, and which are:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Thus, we obtain different formulations of Classical Mechanics, which are based on 1-1 correspondences between the relations of the geometric quantities of fixed closed conic sections and the relations of their holograms in three-dimensional space.

## Chapter 6

# FOUNDATION OF ELECTROMAGNETISM

### 6.1 Geometric quantities due to linearization of conic sections

When ellipses tend to become straight segments, then the vectors  $\mathbf{r}(s)$  and  $\mathbf{u}(s)$  at a point of the “linear” ellipse are almost parallel. The slope of the vectors  $\mathbf{r}(s)$  and  $\mathbf{u}(s)$  with respect to the major axis of this ellipse changes continuously during the linearization process, as we will see in Appendix A3. This gradient corresponds to the three-dimensional space with the gradient operator  $\nabla$  or (grad) when  $\mathbf{r}(s)$  and  $\mathbf{u}(s)$  are on the left side of each term of the conic section definition relations:

$$\begin{aligned}\mathbf{L} &= \mathbf{r}(s) \times \mathbf{u}(s) \\ \mathbf{R} &= \mathbf{u}(s) \times \mathbf{L} - \mathbf{r}(s) \frac{k}{r(s)}\end{aligned}$$

If the vectors  $\mathbf{r}(s)$  and  $\mathbf{u}(s)$  are on the right side of these terms of the above relations, then these terms are arguments of the operator  $\nabla$ , so they are defined as vector fields in three-dimensional space. The vectors  $\mathbf{r}(s)$  and  $\mathbf{u}(s)$  are defined in this way so that the operator  $\nabla$  acts on a scalar or vector quantity (via the inner or outer product). Thus, we obtain the correspondences:

$$\begin{aligned}\mathbf{r}(s) &\leftrightarrow \nabla \\ \mathbf{u}(s) &\leftrightarrow \nabla\end{aligned}$$

when  $\mathbf{r}(s)$  and  $\mathbf{u}(s)$  are on the left-hand side.

We obtain the correspondences:

$$\mathbf{r}(s) \leftrightarrow \mathbf{r}$$

$$\mathbf{u}(s) \leftrightarrow \mathbf{u}$$

when  $\mathbf{r}(s)$  and  $\mathbf{u}(s)$  are on the right-hand side of the terms in the conic section definition relations.

The quantities  $\mathbf{r}$  and  $\mathbf{u}$  are vector fields in three-dimensional space. Similar behavior is observed in the correspondences of  $L$  and  $R$  with the vector fields  $\mathbf{L}$  and  $\mathbf{R}$  of three-dimensional space, i.e.,

$$L \leftrightarrow \mathbf{L}$$

$$R \leftrightarrow \mathbf{R}$$

According to these correspondences, the definition relations of conic sections become relations that satisfy their holograms in three-dimensional space, i.e.,

$$\begin{aligned}\mathbf{L} &= \nabla \times \mathbf{u} \\ \mathbf{R} &= \nabla \times \mathbf{L} - \nabla \left( \frac{k}{r} \right)\end{aligned}$$

where  $r$  is the modulus of  $\mathbf{r}$ . The modulus  $R$  of  $\mathbf{R}$  is approximately equal to  $k$ , that is,  $R \sim k$ , due to the mapping in three-dimensional space of the relation:

$$\epsilon = \frac{R}{k} \rightarrow 1^-$$

We will examine the changes of the vector fields  $\mathbf{L}$  and  $\mathbf{R}$  in correspondence with the changes of  $L$  and  $R$  with respect to  $s$  or with respect to the coordinate  $x$  of a point of the ellipse, because  $s$  and  $x$  are almost equal during the linearization process of this ellipse.

In order to find how the vector fields  $\mathbf{L}$  and  $\mathbf{R}$  will change, we will use the correspondence between an open conic section and a light wave in a vacuum. We make this correspondence because every open conic section has two branches extending to infinity, as is the case with a wave that is divisible and extends to infinity. Thus, from the formal similarity between the equation of a wavefront of a spherical light wave and the equation of a parabola, with respect to a coordinate system which has its origin at its focus, we obtain in 3D-space the relation:

$$ct = R$$

where  $R$  is the radius of the sphere at time  $t$ . Squaring both sides, we obtain the relation:

$$c^2 t^2 - R^2 = 0$$

The equation of a parabola is written as:

$$2ax - y^2 = 0$$

or

$$x_0^2 - y^2 = 0$$

where  $x_0^2 = 2ax$ . From these two equations, we make the correspondence of the coordinate of the point  $x_0$  of a parabola with the distance  $ct$  of a point of the spherical wavefront from the center of a sphere of radius  $R$ . Thus, we get:

$$x_0 \leftrightarrow ct$$

For a linear parabola we get:

$$s \simeq x \leftrightarrow ct$$

This relation allows us to make correspondences between the changes of geometric quantities of conic sections in  $\mathbf{P}^2$  with the changes of quantities of holograms in three-dimensional space. The vector fields  $\mathbf{L}$  and  $\mathbf{R}$ , which are defined by the previous equations, will be the basis of the model which explains all the phenomena of Electromagnetism in a vacuum.

### 1. Variation of $\mathbf{L}$ with respect to $ct$

Deriving  $\mathbf{L}$  with respect to  $ct$ , we get:

$$\frac{\partial \mathbf{L}}{\partial(ct)} = \frac{\nabla \times \partial \mathbf{u}}{\partial(ct)}$$

We define the vector fields in three-dimensional space, compatible with the local isotropy of  $\mathbf{P}^2$ , as follows:

$$\mathbf{B} = \frac{\partial \mathbf{L}}{\partial(ct)}, \quad \mathbf{A} = \frac{\partial \mathbf{u}}{\partial(ct)}$$

The vector fields  $\mathbf{B}$  and  $\mathbf{A}$  are called the **magnetic field** and **vector potential**, respectively. Due to these definitions, we get:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

From this equation and the identity  $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ , we get:

$$\nabla \cdot \mathbf{B} = 0 \tag{6.1}$$

Deriving  $\mathbf{B}$  with respect to  $ct$ , we get:

$$\frac{\partial \mathbf{B}}{\partial(ct)} = \nabla \times \left( \frac{\partial \mathbf{A}}{\partial(ct)} + \nabla \Psi \right)$$

where  $\Psi$  is an arbitrary scalar function of the three-dimensional space and  $\nabla \times \nabla \Psi = 0$ . We define the vector field  $\mathbf{E}$ , compatible with the isotropy of  $\mathbf{P}^2$ , as follows:

$$\mathbf{E} = - \left( \frac{\partial \mathbf{A}}{\partial(ct)} + \nabla \Psi \right)$$

This vector field is called the **electric field**. Due to this relation, we get:

$$\frac{\partial \mathbf{B}}{\partial(ct)} = -\nabla \times \mathbf{E} \quad (6.2)$$

The quantities  $\mathbf{B}$  and  $\mathbf{A}$  are called the **magnetic field** and **vector potential** of the holograms of three-dimensional space. These relations are formally similar to the corresponding relations of Electromagnetism developed in three-dimensional space, through a systematic classification of experimental data by Maxwell.

## 2. Variation of $\mathbf{R}$ with respect to $ct$

When  $\epsilon = 1$ , then from the first of the definition equations of the conic sections, we get  $\mathbf{L} = 0$ , because  $\mathbf{r}(s)$  and  $\mathbf{u}(s)$  are parallel vectors. Representing this relation in three-dimensional space, we get:

$$\mathbf{L} = 0 \quad \text{and} \quad \mathbf{B} = 0, \quad \mathbf{R} = -\nabla \left( \frac{k}{r} \right)$$

so we get:

$$\mathbf{A} = 0 \quad \text{or} \quad \mathbf{A} = \nabla \Phi_0$$

where  $\Phi_0$  is a scalar function of three-dimensional space. From these relations and the definition of the electric field  $\mathbf{E}$ , we get:

$$\mathbf{E} = -\nabla \Phi$$

where

$$\Phi = \Psi \quad \text{or} \quad \Phi = \frac{\partial \Phi_0}{\partial(ct)} + \Psi$$

For  $\epsilon \rightarrow 1^-$ , the geometric objects of  $\mathbf{P}^2$  that correspond to the vector fields  $\mathbf{R}$  and  $\mathbf{E}$  are two parallel vectors. Due to the 1-1 correspondence between the geometric objects of  $\mathbf{P}^2$  and their holograms in three-dimensional space, as well as the preservation of the concept of parallelism between the plane space  $\mathbf{P}^2$  and the local three-dimensional space, the vector fields  $\mathbf{R}$  and  $\mathbf{E}$  are parallel in three-dimensional space. By choosing an appropriate value of  $\Phi$ , the vector fields  $\mathbf{R}$  and  $\mathbf{E}$  become equal, i.e.,

$$\mathbf{R} = \pm \mathbf{E}$$

The sign  $\pm$  means that  $\mathbf{R}$  and  $\mathbf{E}$  are parallel or anti-parallel vector fields. Thus, we get the relation:

$$\nabla \left( \frac{k}{r} \right) = \pm \nabla \Phi$$

or

$$\Phi = \pm \frac{k}{r} \quad \text{and} \quad \mathbf{E} = \frac{k}{r^3} \mathbf{r}$$



Due to the relations for  $\Phi$  and  $\mathbf{E}$ , and the fact that  $\mathbf{E}$  is an isotropic vector due to  $r$  being a mapping of  $r(s)$  of  $\mathbf{P}^2$ , the flux of  $\mathbf{E}$  through a sphere of radius  $r$  is given by the relation:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \pm 4\pi k$$

where  $\oint d\Omega = 4\pi$  and  $d\Omega$  is the solid angle element. We define the scalar quantity  $q$  proportional to the flux of  $\mathbf{E}$  through this sphere, i.e.,

$$q = \epsilon_0(\pm 4\pi k)$$

where  $\epsilon_0$  is a constant of proportionality.

We call the holographic quantity  $q$ , which is not a fundamental quantity in three-dimensional space, the **charge**. This quantity takes positive or negative values due to the fact that  $\mathbf{R}$  and  $\mathbf{E}$  are parallel or anti-parallel vector fields like their corresponding geometric objects. From this geometric interpretation, it is explained why there are positive and negative charges. Substituting for  $k$ , we get:

$$\Phi = \frac{q}{4\pi\epsilon_0 r} \quad \text{and} \quad \mathbf{E} = \frac{q}{4\pi\epsilon_0 r^3} \mathbf{r}$$

where the sign  $\pm$  has been incorporated into  $q$ . If we assume that the charge  $q$  is uniformly distributed in the volume  $V$  of the sphere with density  $\rho$ , then the relation  $q = \rho V$  holds. From the divergence theorem (Gauss's Theorem), we get:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \tag{6.3}$$

The quantity we called charge satisfies this relation called **Poisson's equation**. The identification of the name of this quantity with the physical quantity also called charge is the direct result of the correspondence between geometric and physical quantities. The existence of the charge  $q$  was initially found through observations made by Thales of Miletus and was experimentally confirmed centuries later by the English physicist J.J. Thomson. The charge was considered a fundamental, real physical quantity of three-dimensional space, but of unknown origin.

In analogy to mass, the charge can be considered by free observers as that quantity which determines the vector fields  $\mathbf{E}$  and  $\mathbf{B}$ . Through these fields and the relations they satisfy, all electromagnetic phenomena are interpreted. But through this interpretation, that is, that the charge is a fundamental quantity in three-dimensional space, the appearance of the positive or negative sign of the charge cannot be explained in a satisfactory and convincing way. However, in the context of depicting geometric objects in holograms, the existence of the two signs of the charge is interpreted. This interpretation is based on the fact that  $\mathbf{E}$  and  $\mathbf{R}$  are parallel or anti-parallel vector

fields because the geometric objects corresponding to them are parallel or anti-parallel vectors in  $\mathbf{P}^2$  during the process of linearization of an ellipse. Finally, we will examine the change of the total vector field  $\mathbf{R}$  with respect to  $ct$ . Deriving the definition equation of  $\mathbf{R}$  in terms of  $ct$ , we get:

$$\frac{\partial \mathbf{R}}{\partial(ct)} = \nabla \times \frac{\partial \mathbf{L}}{\partial(ct)} - \frac{\partial}{\partial(ct)} \nabla \left( \frac{k}{r} \right)$$

From the previous definitions, this equation is written as:

$$\frac{\partial R}{\partial(ct)} \mathbf{e}_R = \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial(ct)}$$

or

$$\frac{\partial}{\partial(ct)} \left[ \frac{q}{4\pi\epsilon_0 r^2} \mathbf{e}_R \right] - \frac{\partial \mathbf{E}}{\partial(ct)} = \nabla \times \mathbf{B}$$

Choosing the negative sign for the charge ( $q < 0$ ) in the previous relation, we get:

$$\nabla \times \mathbf{B} = \left[ \frac{1}{4\pi\epsilon_0} \left( \frac{-\partial|q|}{\partial(ct)r^2} \right) \mathbf{e}_R \right] + \frac{\partial \mathbf{E}}{\partial(ct)}$$

We define the vector field  $\mathbf{J}$  in three-dimensional space as:

$$\mathbf{J} = -\frac{1}{4\pi\epsilon_0} \frac{\partial \rho}{\partial t} \mathbf{e}_R$$

The vector field  $\mathbf{J} = \frac{\mathbf{I}}{V}$  with  $q = \rho V$ , where  $V$  denotes the unit volume, is called the **current density**, where:

$$\mathbf{I} = \frac{1}{4\pi\epsilon_0} \frac{\partial|q|}{\partial t} \mathbf{e}_R$$

Thus, we obtain:

$$\nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{\partial(ct)} + \frac{\mathbf{J}}{c} \quad (6.4)$$

The term  $\frac{\mathbf{J}}{c}$  on the right-hand side of this equation is formally identical to the “**displacement current**”, which Maxwell arbitrarily introduced into his equations so that they are compatible with the experimental data. Since the equation resulting from the change of geometric objects is formally identical to the corresponding Maxwell equation for holograms, then they are identical.

Equations 6.1, 6.2, 6.3, and 6.4 are formally identical to Maxwell equations in a vacuum. Due to the coincidence of the scalar and vector quantities obtained from the imaging process with quantities found experimentally in three-dimensional space, equations 6.1, 6.2, 6.3, and 6.4 are identical to Maxwell equations in a vacuum. All these equations are summed together, so we get:

**First pair** (Variation of  $\mathbf{L}$ ):

$$\nabla \cdot \mathbf{B} = 0, \quad \frac{\partial \mathbf{B}}{\partial(ct)} = -\nabla \times \mathbf{E}$$

**Second pair** (Variation of  $\mathbf{R}$ ):

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{\partial(ct)} + \frac{\mathbf{J}}{c}$$

with

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\left(\frac{\partial \mathbf{A}}{\partial(ct)} + \nabla \Psi\right)$$

The equations with the variations of  $\mathbf{B}$  and  $\mathbf{E}$  are the fundamental equations on which the Electromagnetic theory in a vacuum is based. The second pair of equations, produced by the variation of the vector field  $\mathbf{R}$ , contains the sources"  $\rho$  and  $\mathbf{J}$ . This observation leads us to the generalization of the definition relations, through differential forms in three or four-dimensional space, using a different type of source", such as for example a mass source which macroscopically appears solid.

## 6.2 Derivation of the wave equation in 3D-space

As already mentioned, through the linearization process of ellipses, these ellipses have almost the same eccentricity as a parabola. Thus, a linear ellipse (as a closed and finite shape) is considered as a segment of a linear parabola or linear hyperbola during the process of shrinking (linearization) of a cone. Also, the Maxwell equations are obtained by mapping the equations that arise during the linearization process of ellipses in  $\mathbf{P}^2$  into three-dimensional space. When the Maxwell equations are combined with each other, the result will be a wave equation that is formally similar to the equation of a parabola or hyperbola. This formal similarity will lead us to the discovery of new physical entities in three-dimensional space.

In general, any closed conic section that is transformed from a circle or ellipse into a linear ellipse, that is, becomes a segment of a linear hyperbola, then its hologram in three-dimensional space is a wave because it is described by the same equation as that of the linear hyperbola.

We will start from the simple case where  $\rho = 0$  and  $\mathbf{J} = 0$  in the Maxwell equations, finding the conditions under which the combination of the Maxwell equations gives a wave equation that is formally identical to the general equation of an open conic section, that is, that of a hyperbola. From:

$$\nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{\partial(ct)}$$

and the definitions:

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = - \left( \frac{\partial \mathbf{A}}{\partial(ct)} + \nabla \Psi \right)$$

we get:

$$\nabla \times (\nabla \times \mathbf{A}) = - \frac{\partial^2 \mathbf{A}}{c^2 \partial t^2} - \nabla \left( \frac{\partial \Psi}{c \partial t} \right)$$

Due to the vector identity  $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ , we get:

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{c^2 \partial t^2} = \nabla \left( \nabla \cdot \mathbf{A} + \frac{\partial \Psi}{c \partial t} \right)$$

For this equation to be formally similar to the equation of a parabola, then the right-hand side must be zero, that is, the quantity in the parentheses must be a constant. Choosing the zero value for this constant, we obtain the equation:

$$\nabla \cdot \mathbf{A} + \frac{\partial \Psi}{c \partial t} = 0$$

This equation is called the **Lorentz gauge condition**. Thus, the previous equation, due to the Lorentz gauge condition, becomes the homogeneous wave equation with group velocity  $c$ , i.e., it is the equation of a light wave:

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{c^2 \partial t^2} = 0$$

A corresponding equation is obtained for the scalar potential  $\Phi$ , which results from  $\nabla \cdot \mathbf{E} = 0$  and the Lorentz gauge condition, i.e.,

$$\nabla^2 \Phi - \frac{\partial^2 \Phi}{c^2 \partial t^2} = 0$$

These equations are formally similar to the equation of an isosceles hyperbola or the equation of a parabola with respect to a coordinate system that has its origin at a point on the axis of symmetry of the parabola between the focus and the directrix. The process of transforming an ellipse into a straight line segment (which is a segment of a linear hyperbola) on which a point circle lies leads, through correspondences, to the creation of physical entities in three-dimensional space that combine particle and wave characteristics. The existence of these physical entities in three-dimensional space has been verified experimentally. These physical entities are wave-particles due to the duality they exhibit, while simultaneously having the general characteristics of bodies and waves. Following the same procedure but with  $\rho$  and  $\mathbf{J}$  different from zero, we obtain the non-homogeneous wave equations:

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{c^2 \partial t^2} = -\mu_0 \mathbf{J}$$
$$\nabla^2 \Phi - \frac{\partial^2 \Phi}{c^2 \partial t^2} = -\frac{\rho}{\epsilon_0}$$



## Chapter 7

# FOUNDATIONS OF SPECIAL RELATIVITY

### Introduction

The foundation of Classical Mechanics in three-dimensional space was based on relations and geometric quantities that apply to fixed closed conic sections in  $\mathbf{P}^2$ . Then, these relations and their geometric quantities were depicted in three-dimensional space, through 1-1 correspondences, in relations and quantities of three-dimensional space.

This also applies to the parameter that free observers call time, which was defined through its correspondence with the arc length  $s$  of a point of a closed conic section. The principle of measurement of an arc is defined by an arbitrarily chosen point of the closed conic section. These principles apply to the study of a specific phenomenon.

However, if we consider the general concept of time, it is related to space and has a direction in the direction of the increase of space, as follows from the creation of the Planck point and the space  $\mathbf{P}^2$  through a self-similar process of expansion of the Planck patches. For free observers, time was considered a simple parameter included in the equations of motion of physical entities. The measurement principle of a time interval is determined by free observers as the moment when the observation or study of a phenomenon begins.

The transmission of information about what is happening at two different points of a closed conic section has a finite duration and “appears to happen instantaneously” for bound observers. Through this seemingly instantaneous transmission of information, the concept of simultaneity is introduced to bound observers. Thus, bound observers acquire simultaneous knowledge of the characteristics of two events occurring at two different points of a closed conic section. The concept of simultaneity implies the concept of an infinite

propagation speed of interactions.

Along with the correspondences between geometric quantities and their relations with physical quantities, the concept of simultaneity is also transferred from  $\mathbf{P}^2$  to three-dimensional space; that is, the concept of instantaneous information transmission or the transmission of information at infinite speed. From the above assumptions and correspondences, a free observer is led to the concept of universal time, where only a clock is sufficient to determine the value of the time parameter. Thus, for two events that occur “simultaneously” at two random points in our three-dimensional World, no matter how distant they are from each other, the same value of time is assigned.

The concept of simultaneous or infinite speed of information propagation, as we mentioned in Chapter 3, has as a consequence:

1. Non-locality
2. Non-causality

The previous assumptions and correspondences are based on the foundation of Classical Mechanics, taking into account only the characteristics of closed conic sections. Thus, the question that arises is whether all this also applies to open conic sections or not.

An issue that is not clear during the process of establishing Classical Mechanics is how much distance an information signal traveled from the point where an event occurred to the brain of a free observer. The non-appearance of distance in the process of transmitting information is due to the assumption of the infinite propagation speed of the interactions that free observers make. This observation leads us to examine the concept of distance, or the way of calculating distance, in relation to the type of a conic section, to see if and to what extent the process of transmitting information is affected.

Let us first examine closed conic sections. The coordinates of a point of a closed conic section, with respect to a Cartesian coordinate system in  $\mathbf{P}^2$ , satisfy the Pythagorean theorem. This is evident from the following relations:

### A) Closed conic sections

In the case of closed conic sections, the equation of any ellipse can be reduced, through a change of scale, to the equation of a circle with radius  $r_c$  corresponding to a body. Thus, we consider only the equation of a circle:

$$x_c^2 + y_c^2 = b_c^2 = r_c^2$$

where  $x_c^2$  and  $y_c^2$  are the squares of the distances  $x_c$  and  $y_c$  of a point of a circle, with respect to the axes of a Cartesian coordinate system centered at



its center, satisfying the Pythagorean theorem. To measure the infinitesimal distance  $ds_p$  traveled by a material point  $p$  in  $\mathbf{P}^2$ , we write the relation:

$$ds_p^2 = dx^2 + dy^2$$

To measure the infinitesimal distance  $ds_p$  traveled by a material point  $p$  in three-dimensional space, we will add one more dimension, obtaining a relation that satisfies the Pythagorean theorem, namely:

$$ds_p^2 = dx^2 + dy^2 + dz^2$$

This relation expresses the distance traveled by a control particle moving at a constant speed to measure the distance  $ds_p$ . This relation is called the Euclidean metric. If a body is located at a point  $p$ , different from the origin of the coordinate system  $O$ , then  $ds_p$  is always different from zero. This means that if the information about the state of the body at  $p$  reaches an observer at  $O$  in zero time, due to the instantaneous transmission of information, then the speed of information transmission is infinite.

## B) Open conic sections

In the case of an open conic section, its general equation is the equation of a hyperbola, that is:

$$\left(\frac{b_h}{a_h}\right)^2 x_h^2 - y_h^2 = b_h^2$$

For  $a_h = b_h = b_{h,e}$ , this equation becomes the equation of an isosceles hyperbola, that is:

$$x_{h,e}^2 - y_{h,e}^2 = b_{h,e}^2$$

This equation is formally similar to the equation of a parabola with respect to a Cartesian coordinate system with its origin on its axis of symmetry, that is:

$$x_p^2 - y_p^2 = b_p^2$$

where  $b_p^2 = b_{h,e}^2 = a^2 - 2ad$ ,  $y_p = y_{h,e}$ , and  $x_{h,e}^2 = 2ax_p$ . The quantity  $a$  is the focus-vertex distance of this parabola and  $d$  is the distance of the origin of a coordinate system from the perpendicular bisector of the parabola. For  $d = \frac{a}{2}$ , we obtain the equation of the parabola:

$$y_p^2 = 2ax_p$$

Thus, all equations of open conic sections are formally similar to the equation of a hyperbola. As can be seen from the equation of the hyperbola, the coordinates  $x_p, y_p$  of the point  $p$  of a parabola, or of the equivalent isosceles hyperbola, with respect to a Cartesian coordinate system, do not satisfy the Pythagorean theorem.

As we saw in the previous section and in the previous paragraphs of this section, the equation of a wave is formally identical to any equation of an open conic section, such as the equation of the hyperbola above. This observation leads us to look for a new metric when the measurement of a distance  $s_p$  in three-dimensional space is performed via waves. Since every wave in three-dimensional space has a finite group velocity, then the notion of an infinite propagation velocity of information carried via waves must be abandoned.

The distance  $s_p$  measured by a wave with group velocity  $u$  is given by the relation:

$$s_p = ut$$

or

$$u^2 t^2 - s_p^2 = 0$$

We define the left-hand side of this relation as:

$$s_w^2 = (ut)^2 - s_p^2$$

This relation is formally similar to the equation of an open conic section. Thus, a new way of measuring distances in three-dimensional space appears. The equation of an open conic section does not correspond to the relation expressing the Euclidean metric, as is the case with closed conic sections.

If the distances in three-dimensional space are infinitesimal and the wave is a light wave, where  $c$  is its constant group velocity in a vacuum, then the previous relation is written as:

$$ds_w^2 = (cdt)^2 - ds_p^2 \quad (7.1)$$

The previous relation defines a new metric in three-dimensional space, where the quantity  $ds_w$  is called the **interval**. The light wave is chosen because the maximum group velocity of a wave known to date, with which a wave can propagate, is the speed of light  $c$  in a vacuum.

In the case of an isosceles hyperbola, as in all open conic sections, the point with equal values of its coordinates  $x_0, y_0$ , i.e.,  $x_0 = y_0$ , gives  $b_0 = 0$ , as follows from the equation of the parabola. This point also belongs to the bisector of the coordinate axes, as shown in Figure 7.1.

If Figure 7.1 is rotated about the  $y$ -axis, then the bisector of the angle formed by the axes will form a cone called the light cone, because every point of each generatrix of this cone satisfies the relation:

$$ds_w^2 = 0$$

Two points inside the cone are causally connected, due to the finite speed of light. Also, at each point of the trajectory followed by a body in three-dimensional space, we construct a light cone to show that its speed is finite

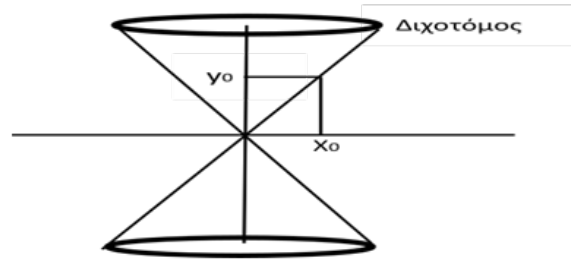


Figure 7.1: Light cone

and less than the speed of light  $c$ . The equations of closed and open conics with respect to a system of Cartesian coordinates in three-dimensional space are depicted as two different ways of measuring distances in three-dimensional space.

This depiction also defines the physical entity in three-dimensional space by which a distance is measured. If the measurement of a distance between two points in three-dimensional space is performed by a free control particle, moving along a geodesic of three-dimensional space between these two points, then this distance is calculated from the relation:

$$ds_p^2 = dx_1^2 + dx_2^2 + dx_3^2 \quad (7.2)$$

If the same distance is measured by a wave that reflects an open conic section, then this distance is given by a relationship that is formally similar to the equation of a parabola or an isosceles hyperbola. Thus, the measurement of a distance between two points in three-dimensional space is defined by equation 7.1. The measurement process by a control particle requires that this be as small as possible (such as an elementary particle), so that its presence does not affect the measurement process. However, as mentioned in the introduction and we will see below, free elementary particles also exhibit wave characteristics, so in this case, the measurement via equation 7.1 should be used in any case. For free observers, the following applies: Equation 7.1 defines a new space that has four dimensions and which is called Minkowski spacetime. The theory of three-dimensional space based on equation 7.1 is called Special Relativity. The consequences of the metric, such as length contraction and time dilation, appear as strange results for a free observer of the three-dimensional space based on the metric of equation 7.2. These two results of Special Relativity arise from comparisons of lengths or time intervals, where one refers to a non-moving" inertial frame of reference (**proper length or time**) and the other refers to an inertial frame of reference moving at a constant velocity with respect to the non-moving" inertial frame. An inertial frame of reference is defined as a frame moving at a constant velocity, and an observer moving with it is called an inertial observer. Since equation 7.1 does not contain the constant velocity of an inertial observer,

then each interval  $ds_w$  appears the same for all inertial observers. The invariant value of an interval  $ds_w$  for two inertial observers of the Minkowski spacetime leads to the Lorentz transformations, through which the space and time coordinates of these two inertial observers are related. From equation 7.1, we conclude that each inertial observer measures its own time, so the number of clocks to be used is identical to the number of inertial observers. This result invalidates the concept of **universal time**, which was introduced into the formulation of Classical Mechanics. From a further analysis of Special Relativity, the following energy-momentum relation is obtained, which will be used in the following Chapters:

$$E^2 = c^2 p^2 + m^2 c^4$$

For  $p = 0$ , we obtain a relation showing the equivalence between energy and mass, namely:

$$E = mc^2$$

Special Relativity will not be developed further, because our purpose was achieved through the correspondences between the equations of closed and open conic sections and the two different metrics (Euclidean metric and Minkowski metric) in three-dimensional space.

## Chapter 8

# FOUNDATIONS OF QUANTUM MECHANICS

### 8.1 Introduction

The geometric explanation given by bound observers for linear conic sections through the geometric model reveals the need for new physical entities in three-dimensional space.

The complex geometric shapes consist of a nearly linear ellipse and a point circle inside it. This complex geometric object (circle-ellipse) is considered as a part of a linear hyperbola (or linear parabola or linear isosceles hyperbola) during the process of shrinking a cone. This aggregate corresponds to the physical entity called a **wave-particle** in three-dimensional space.

As mentioned in the previous section, the physical entities that should exist in three-dimensional space were found from electron diffraction experiments in crystals by Davisson and Germer (1927) and independently by G.P. Thomson (1928).

In the geometric model, bound observers consider a point circle to be located at a random point within a finite, almost linear ellipse. The linear ellipse can lie on one or the other branch of a linear hyperbola. In this limit, this complex geometric object is depicted in three-dimensional space as a wave on the one hand and as a particle on the other hand, due to the point circle.

Due to the impossibility of measuring on such a microscopic scale in three-dimensional space, all one can do is create a theory based on probabilities. The probability of finding the position and momentum of the material point should “integrate” the basic quantities of a wave, which are its amplitude and intensity.

This geometric construction in  $\mathbf{P}^2$  is depicted in three-dimensional space as a physical entity, which is the wave-particle, whose existence is experi-

mentally verified. The experiment with polarized photons passing through a tourmaline crystal or the experiment with electrons passing through two slits confirms the theoretical predictions.

The integration of these quantities into the geometric model is carried out as follows:

1. We define the function  $\Psi(x, t)$  of the position  $x$  at which the geometric object corresponding to a material point can be located at time  $t$ .  $\Psi(x, t)$  is called the **probability amplitude**, corresponding to the amplitude of a wave.  $\Psi(x, t)$  expresses the pair  $(x, t)$  that characterizes the geometric object in each of the two branches of the linear hyperbola.
2. We define the square of  $\Psi(x, t)$ , that is,  $|\Psi(x, t)|^2$ , as the probability density of the geometric object being at a position  $x$  at time  $t$ , corresponding to the intensity of a wave.
3. This formalism must be compatible with the random jumps that we consider the complex geometric object “point circle-linear ellipse” to make between the two branches of the linear hyperbola, due to our ignorance of where it actually is. Thus, we obtain a linear combination of the two functions  $\Psi(x, t)$  that characterize the two branches of this hyperbola. This linear combination is called the **superposition of two states**.

In this geometric image, we have the following two reference systems whose origins are:

- A) The point where the two branches of the hyperbola begin (origin of the 1st reference system  $O_1$ ).
- B) The midpoint of the linear ellipse (origin of the 2nd reference system  $O_2$ ).

The position of the point circle with respect to  $O_1$  is determined by the tangent vector  $\mathbf{t}_g$  ( $O_1 \rightarrow O_2$ ) and by its position vector  $\mathbf{r}$  ( $O_2 \rightarrow p_c$ ). The position of the point circle with respect to  $O_2$  is determined by the position vector  $\mathbf{r}$  ( $O_1 \rightarrow O_2$ ) and the tangent vector  $\mathbf{t}_g$  ( $O_2 \rightarrow p_c$ ), where by  $p_c$  we denote the point circle, as shown in Figure 8.1.

The representation of the geometric objects of the geometric model in three-dimensional space is described as follows: The point circles are holographically represented in three-dimensional space as material points, and the linear parabolas or hyperbolas as waves. The representation of the point circle of these vectors  $\mathbf{r}$  and  $\mathbf{t}_g$  in three-dimensional space gives a material point with position vector  $\mathbf{r}$  and a momentum  $\mathbf{p} = m\mathbf{u}$  from  $O_2$ .

We map the function  $\Psi(x, t)$  and its characteristics (1), (2), and (3) into three-dimensional space, thus obtaining the function  $\Psi(\mathbf{r}, t)$  which we call

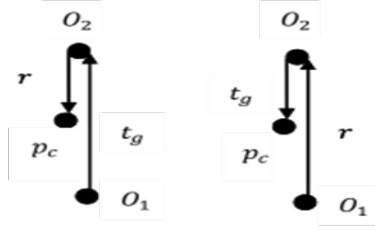


Figure 8.1: Coordinate systems in linear hyperbolas and ellipses

the wave function. The wave function  $\Psi(\mathbf{r}, t)$  is generally a complex function, in analogy to the solution of a wave equation. Through these mappings, the probabilistic model is defined in three-dimensional space.

In the geometric model of  $\mathbf{P}^2$ , the complex geometric shape should exhibit the basic properties of a wave, such as self-interference (for the interpretation of diffraction experiments) and linearity (for the interpretation of interference experiments). Thus, self-interference is the result of the divisibility of each hyperbola into two identical copies (branches of the hyperbola). Linearity is due to the possibility of constructing hyperbolas through lines parallel to their asymptotes, as we will see later. Also,  $\Psi(x, t)$  should be large in a region of space where the point circle is likely to be located and small everywhere else.

During the process of shrinking a cone, an infinitesimal linear ellipse is depicted as an infinitesimal straight line segment that is the “tangent” at a point of the “linear hyperbola”. As we mentioned above, this infinitesimal straight line segment can be located on one of the two almost straight lines, which are the two branches of a linear hyperbola degenerated into straight lines. The position of the infinitesimal linear ellipse, with respect to the point where the two branches of the hyperbola begin (beginning of the second reference system), is determined by the tangent vector  $\mathbf{t}_g$  at the point of the hyperbola where we consider the infinitesimal linear ellipse to be located.

## 8.2 Schrödinger's equation

We will try to find the equation that free wave-particles satisfy in three-dimensional space. The wave equations come from Maxwell's equations. We choose the equation:

$$\frac{\partial^2 \Phi}{c^2 \partial t^2} - \nabla^2 \Phi = 0$$

This equation is a “holographic representation” of the equation of a linear parabola, namely:

$$x_0^2 - y^2 = 0$$

From these two equations, we make the two-way correspondences between the Cartesian coordinates of a parabola point and the differential quantities

that express the changes of the quantity  $\Phi$  of the wave in three-dimensional space, namely:

$$x_0 \leftrightarrow \frac{\partial}{c\partial t} \equiv \hat{E} \quad (8.1)$$

$$y \leftrightarrow \mu \cdot \nabla \equiv \mu \cdot \hat{\mathbf{P}} \quad (8.2)$$

The members of the right-hand side of these correspondences are quantities of the holograms of the geometric quantities of the complex geometric object, which consists of a point circle, an infinitesimal linear ellipse, and a linear hyperbola.  $\mu$  is a constant vector collinear with  $\hat{\mathbf{P}}$ , ensuring that the right-hand side of equation 8.2 is a scalar quantity, as is the left-hand side.

Since the equation  $x_0^2 - y^2 = 0$  is related to the equation  $x = \frac{1}{2a}y^2$  of a parabola with  $x_0^2 = 2ax$ , then for each  $x$ , the correspondence 8.1 is written as:

$$x = \left(\frac{x_0}{2a}\right) x_0 = \mu' x_0 \leftrightarrow \mu' \frac{\partial}{c\partial t} = \mu' \hat{E} \quad (8.3)$$

The next step is to find from equations 8.1  $\rightarrow$  8.3 the equation of motion of a free hologram of a linear ellipse, that is, the equation satisfied by  $\Psi(\mathbf{r}, t)$  when there are no “interactions” between the holograms.

The correspondences 8.1 $\rightarrow$ 8.3 tell us that the coordinates of a point of a conic section can correspond to mathematical entities defined in three-dimensional space and called **operators**. The operators are linear, compatible with the linearity of  $\Psi(\mathbf{r}, t)$ , whose action on  $\Psi(\mathbf{r}, t)$  is proportional to  $\Psi(\mathbf{r}, t)$ . This requirement is made in order to point out that the result of the action of the operators on  $\Psi(\mathbf{r}, t)$  also belongs to the probabilistic model. Thus, for a general operator  $\hat{A}$ , we write the relation:

$$\hat{A}\Psi(\mathbf{r}, t) = A\Psi(\mathbf{r}, t)$$

where  $A$  is a constant called the operator eigenvalue of  $\hat{A}$ . In this case,  $\Psi(\mathbf{r}, t)$  is called the **operator eigenfunction** of  $\hat{A}$ .

The theoretical predictions of the probabilistic model must be experimentally tested in three-dimensional space (scientific method). This means that we should be able to make calculations of the theoretical values of various quantities, obtaining real numbers which we should compare with the experimental values, which are indeed real numbers. Thus, we set the following two requirements:

- **First requirement:** The operators and their properties should be defined in such a way that the result of their action on  $\Psi(\mathbf{r}, t)$  yields real physical quantities.
- **Second requirement:** A correct procedure should be defined through which the uncertainty of the values taken by a physical quantity, introduced through the probabilistic model, is managed.



The second requirement leads to the calculation of the expectation value of each operator, corresponding to a physical quantity, which is determined by the relation:

$$\langle \hat{A} \rangle = \int \Psi^*(\mathbf{r}, t) \hat{A} \Psi(\mathbf{r}, t) d\mathbf{r}$$

where  $\Psi^*(\mathbf{r}, t)$  is the complex conjugate of  $\Psi(\mathbf{r}, t)$ . The first requirement is satisfied by the group of operators called **Hermitian operators**, which are defined as follows:

**Hermitian operators are the operators whose expectation values are real numbers.**

We apply the correspondences 8.2 and 8.3 to the coordinates  $x, y$  of a point of a parabola with equation  $x = \frac{1}{2a}y^2$ , where the right-hand side of both correspondences denotes the eigenvalues of the operators when they act on  $\Psi(\mathbf{r}, t)$ . Thus, we obtain the holographic equation of a linear parabola in three-dimensional space, which is a wave equation and expresses the equation of motion of a free hologram, namely:

$$\mu' \frac{\partial \Psi(\mathbf{r}, t)}{c \partial t} = \frac{(\mu \cdot \nabla)^2}{2a} \Psi(\mathbf{r}, t) \quad (8.4)$$

or

$$\mu' \hat{E} \Psi(\mathbf{r}, t) = \frac{(\mu \cdot \hat{\mathbf{P}})^2}{2a} \Psi(\mathbf{r}, t) \quad (8.5)$$

$\Psi(\mathbf{r}, t)$  is the solution of equations 8.4 or 8.5. We make the correspondences:

$$\mu' \hat{E} \Psi(\mathbf{r}, t) \leftrightarrow E \quad (8.6)$$

$$\frac{(\mu \cdot \hat{\mathbf{P}})^2}{2a} \Psi(\mathbf{r}, t) \leftrightarrow \frac{p^2}{2m} \quad (8.7)$$

Applying equations 8.6 and 8.7 to equation 8.5, we see that this is formally similar to the energy-momentum relation of a free particle in Classical Mechanics, namely:

$$E = \frac{p^2}{2m} \quad (8.8)$$

According to what has been mentioned regarding the result of the action of the operators on  $\Psi(\mathbf{r}, t)$ , namely that this result must be incorporated into the probabilistic model, the correspondences 8.6 and 8.7 are expressed as the following equations:

$$\mu' \hat{E} \Psi(\mathbf{r}, t) = E \Psi(\mathbf{r}, t) \quad (8.9)$$

$$\frac{(\mu \cdot \hat{\mathbf{P}})^2}{2a} \Psi(\mathbf{r}, t) = \frac{p^2}{2m} \Psi(\mathbf{r}, t) \quad (8.10)$$

Equations 8.9 and 8.10 define the motion of a particle that is coupled to the particle nature of its hologram, such that this hologram behaves as a free classical particle.

What remains to be done is to find the constants  $\mu$  and  $\mu'$  that appear in equation 8.5. The solution of the homogeneous wave equation

$$\nabla^2 \Phi - \frac{\partial^2 \Phi}{c^2 \partial t^2} = 0$$

is

$$\Phi(\mathbf{r}, t) = \Phi_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$$

where  $\mathbf{k}$  and  $\omega$  denote the wavenumber vector and the angular frequency of this wave, respectively. The solution of equation 8.5, which is  $\Psi(\mathbf{r}, t)$ , should be similar to  $\Phi(\mathbf{r}, t)$ , with the difference that in the expression for  $\Psi(\mathbf{r}, t)$ , the particle and wave nature of the hologram of the complex geometric object consisting of a “linear parabola, a linear ellipse, and a point circle” should appear.

To each hologram, we attach a quantity called energy  $E$ , as we saw in Classical Mechanics. Since the hologram of the complex geometric object is an entity with a dual nature, then the energy  $E$  will be attributed to each of these two natures, namely both the particle and the wave. The energy  $E_p$  attributed to the particle nature of the hologram corresponds to the energy of a classical particle, so the energy  $E_p$  is a function of the momentum measure  $p$ . The energy  $E_w$  associated with the wave nature of this hologram corresponds to the energy of a classical wave. Thus, the energy  $E_w$  will be a function of the frequency  $\nu$  and of the group velocity measure  $u$ .

The general expression for  $\Psi(\mathbf{r}, t)$  must be the same for holograms of linear ellipses which are parts of linear hyperbolas. To find the expression for  $\Psi(\mathbf{r}, t)$ , we will consider the simplest hologram of a linear ellipse, which is the one with zero mass ( $m = 0$ ). Since this hologram also has wave characteristics, then the metric used is the one corresponding to open conics. Thus, we will use the relation:

$$E^2 = c^2 p^2 + m^2 c^4$$

For  $m = 0$ , the energy of the particle nature of this hologram is  $E_p = cp$ . In order to find the energy of the wave nature of this hologram, we use the following reasoning: The linear ellipse as a hyperbola segment corresponds to a light wave expressed through the wave equation. A light wave is characterized only by the frequency  $\nu$ , since the measure of its group velocity is the constant velocity  $c$ . Thus, the energy of a light wave will be proportional only to the frequency  $\nu$ , that is, it will satisfy the relation  $E_w = h\nu$ , where  $h$  is a proportionality constant. This relationship was confirmed experimentally by the flow of outgoing electrons in the photoelectric effect.

Since the energy  $E$  of a hologram is attributed to either its particle or wave nature, then  $E = E_p = E_w$ . From the above relations for energy, the de Broglie relations are obtained, which relate the quantities of the particle or wave nature of a hologram, namely:

$$p = \frac{E_w}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$$

where  $\lambda = \frac{c}{\nu}$ . For  $\omega = 2\pi\nu$  and  $\mathbf{k} = \frac{2\pi}{\lambda}\mathbf{e}_p$ , we get:

$$\mathbf{p} = \frac{h}{2\pi}\mathbf{k} = \hbar\mathbf{k}, \quad E = \frac{h}{2\pi}\omega = \hbar\omega$$

where  $\mathbf{e}_p$  is the unit vector in the direction of  $\mathbf{p}$  and  $\hbar = \frac{h}{2\pi}$ . Thus,  $\Psi(\mathbf{r}, t)$  is expressed as:

$$\Psi(\mathbf{r}, t) = \Psi_0 e^{\frac{i(\mathbf{p}\cdot\mathbf{r} - Et)}{\hbar}} \quad (8.11)$$

The particle and wave nature of a hologram appear in this equation through  $\mathbf{p}$  and  $E$ , respectively. Substituting equation 8.11 into equation 8.4, we get:

$$\frac{\mu'}{c} \left( \frac{-iE}{\hbar} \right) = \left( \frac{ip}{\hbar} \right)^2 \frac{\mu^2}{2a} \quad (8.12)$$

Equation 8.12 is the energy-momentum relation of the particle nature of a free hologram, so, according to what was mentioned earlier, it is identical to equation 8.8. From this identification, we obtain:

$$\mu' = ic\hbar, \quad \mu^2 = -\frac{a\hbar^2}{m} \quad (8.13)$$

To fully determine  $\mu'$  and  $\mu$  in relations 8.13, the numerical value of the constant  $h$  must be found. This was done by Max Planck in his attempt to interpret the spectrum of blackbody radiation.  $h$  is called Planck's constant. Substituting equation 8.13 into equation 8.4, we obtain the equation of motion of a free hologram, namely:

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = -\frac{\hbar^2}{2m} \nabla^2 \Psi(\mathbf{r}, t) \quad (8.14)$$

Equation 8.14 is the Schrödinger equation of a free wave-particle in three-dimensional space. Bound observers consider the general equation of a parabola expressed in terms of a coordinate system with an origin at a point other than its focus, namely the equation  $x = \frac{1}{2a}y^2 + V'$ . Following a similar procedure as in the case of a free wave-particle, the Schrödinger equation takes the form:

$$i\hbar \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = \left( -\frac{\hbar^2 \nabla^2}{2m} + V \right) \Psi(\mathbf{r}, t) \quad (8.15)$$

In this case, free observers interpret equation 8.15 in three-dimensional space as follows: the wave-particle receives the “effects of external forces” acting on it, which are reduced to a force  $\mathbf{F}$  produced by a real potential energy  $V$ , i.e.,

$$\mathbf{F}(\mathbf{r}, t) = -\nabla V(\mathbf{r}, t)$$

If the potential energy does not depend on time, i.e.,  $V(\mathbf{r}, t) = V(\mathbf{r})$ , then a partial solution of the Schrödinger equation can be written as the product  $\Psi(\mathbf{r}, t) = u(\mathbf{r})f(t)$ . Substituting this expression into 8.15, we obtain:

$$\Psi(\mathbf{r}, t) = u(\mathbf{r})e^{-\frac{iEt}{\hbar}}$$

where  $E$  is the separation constant of the two equations which result from equation 8.15. Summarizing the above procedure, we note that the derivation of the Schrödinger equation (8.14 or 8.15) was made through correspondences of the geometric quantities of a parabola in the space  $\mathbf{P}^2$  and the operators of a wave equation in three-dimensional space.

### 8.3 The Heisenberg Uncertainty Principle

As already mentioned, bound observers consider almost point-line complex geometric objects with the following structure: **point circle - infinite linear ellipse - linear hyperbola**. As mentioned previously (Figure 8.1), from this geometric construction, two coordinate systems are created where one has its origin at the middle of the major axis of the linear ellipse (linear ellipse coordinate system) and the other has its origin at the point where the two branches of the linear hyperbola “merge” (linear hyperbola coordinate system). The bound observers want to measure the value of the position vector  $\mathbf{r}$  from the ellipse coordinate system and the value of the tangent vector  $\mathbf{t}_g$  from the linear hyperbola coordinate system at the point of the linear hyperbola where the point circle is located. The uncertainty  $\Delta r$  in the measurement of the value of the magnitude of the position vector  $\mathbf{r}$  is given by the following relation:

$$(\Delta r)^2 = \langle r^2 \rangle - \langle r \rangle^2$$

In general, the symbol  $\langle w \rangle$  expresses the expectation value of the statistical quantity  $w$  given by the relation:

$$\langle w \rangle = \frac{N_1 w_1 + \cdots + N_n w_n + \cdots}{N} = \sum_n w_n f_n$$

where  $f_n = \frac{N_n}{N}$  is the frequency of occurrence of the values  $w_n, n = 1, 2, \dots$ . When  $N \rightarrow \infty$ , then  $f_n$  tends to the probability  $p_n$  of all possible values of  $w$  occurring, that is,  $f_n \rightarrow p_n$ . Thus,  $\langle w \rangle$  is written as:

$$\langle w \rangle = \sum_n w_n p_n$$

The position vector  $\mathbf{r}$  has its application point at the midpoint of the linear ellipse and its endpoint at any point on it. On this straight line segment, the probabilities  $p_i$  for the vector  $\mathbf{r}$  to take the positive value  $r_i$  or the negative value  $-r_i$  are equal, so we set  $\langle r \rangle = 0$ . In this case, the endpoint of the vector  $\mathbf{r}$  is located at two symmetrical points of the straight line segment with respect to the origin of the coordinate system. Thus, the previous relation takes the form:

$$(\Delta r)^2 = \langle r^2 \rangle \implies \Delta r = \sqrt{\langle r^2 \rangle}$$

Since this line segment is infinitesimal, then  $r \rightarrow 0$ , so we get  $\Delta r \rightarrow 0$ . This result tells us that the uncertainty in the measurement of the value of  $r$  is zero or, in other words, this measurement gives us a very precise value for  $r$  (complete knowledge). To estimate the uncertainty of the value of  $\mathbf{t}_g$ , the bound observers consider that the vector  $\mathbf{t}_g$  starts from the origin of the coordinate system of the linear hyperbola, but its end is at any of the infinite points of the two branches of the linear hyperbola. In this case, the uncertainty regarding the length of  $\mathbf{t}_g$  is also given by the relation:

$$(\Delta t_g)^2 = \langle t_g^2 \rangle - \langle t_g \rangle^2$$

On the line formed by the two branches of the hyperbola,  $\mathbf{t}_g$  takes positive and negative values of the same magnitude at two symmetrical points with respect to the origin of this coordinate system. Thus, we also obtain  $\langle t_g \rangle = 0$ . Since  $(\pm t_g)^2 > 0$ , then the bound observers, to find the average value of  $t_g^2$ , add the infinite values of  $t_g^2$  which are always positive, so they get  $t_g^2 \rightarrow \infty$ . Thus, the uncertainty  $\Delta t_g$  tends to an infinite value, that is,  $\Delta t_g \rightarrow \infty$ , as follows from the relation:

$$(\Delta t_g)^2 = \langle t_g^2 \rangle \implies \Delta t_g = \sqrt{\langle t_g^2 \rangle}$$

In other words, this relation gives us an infinite uncertainty (or complete ignorance) about the value of the modulus of  $\mathbf{t}_g$ . Thus, for the uncertainties of the measures of  $\mathbf{r}$  and  $\mathbf{t}_g$ , we get:

$$(\Delta r) \rightarrow 0 \quad \text{and} \quad (\Delta t_g) \rightarrow +\infty$$

A bound observer measures exactly  $\mathbf{r}$  at a time with respect to the coordinate system of the linear ellipse, but he cannot simultaneously measure the length of the position vector  $\mathbf{t}_g$  with respect to the coordinate system of the linear hyperbola. This is because he can only be at one of the origins of the two coordinate systems and not simultaneously in both. Similar conclusions arise if a bound observer measures  $\mathbf{t}_g$  in the coordinate system of the linear ellipse and  $\mathbf{r}$  in the coordinate system of the linear hyperbola. In this case, the bound observers get the relation:

$$(\Delta t_g) \rightarrow 0 \quad \text{and} \quad (\Delta r) \rightarrow +\infty$$

The above relations for  $\Delta r$  and  $\Delta t_g$  are unified into one relation by taking their product so that it equals a non-zero constant, i.e.,

$$(\Delta r) \cdot (\Delta t_g) = \text{constant}$$

If we consider the quantities  $\Delta r, \Delta t_g$  as measures of the vectors  $\Delta \mathbf{r}, \Delta \mathbf{t}_g$  with respect to a coordinate system of  $\mathbf{R}^2$ , then the previous relation gives their constant area. Due to the isotropy of  $\mathbf{R}^2$ , in each of its directions, the dispersion of the values of  $\mathbf{r}$  and  $\mathbf{t}_g$  must be the same. Thus, the components of  $\Delta \mathbf{r}, \Delta \mathbf{t}_g$  with respect to the coordinate system of  $\mathbf{R}^2$  are considered equal.

If we call  $t_1 = t_2 = \tau$  and  $r_1 = r_2 = \rho$  the components of  $\Delta \mathbf{t}_g$  and  $\Delta \mathbf{r}$  respectively on the two axes of the coordinate system in  $\mathbf{R}^2$ , then from the relation that gives the area of the parallelogram formed by these vectors, we get:

$$(\Delta r) \cdot (\Delta t_g) = t_1 r_2 - r_1 t_2 = \tau \rho - \rho \tau \equiv [\tau, \rho] = h$$

where  $h$  is a constant. The symbol  $[\tau, \rho] = \tau \rho - \rho \tau$  is called the **commutator** of  $\tau, \rho$ . In  $\mathbf{R}^2$ , the bound observers face the paradox that, on the one hand, the product of  $\Delta r$  and  $\Delta t_g$  is a constant different from zero and, on the other hand, zero if  $\tau \rho = \rho \tau$ .

To reconcile these two results, the bound observers consider that in such processes, where the components of one vector tend to infinity and the components of the other tend to zero, the results of the products  $\tau \rho$  and  $\rho \tau$  are not equal to each other. In other words, the interchange of  $\rho, \tau$  does not give the same result when the bound observers want to find the area of  $\Delta \mathbf{r}, \Delta \mathbf{t}_g$ . Since  $\tau \rho \neq \rho \tau$ , then one of the two products is greater than the other, let  $\tau \rho > \rho \tau$ . Comparing the area  $\tau \rho$  with half of the area  $[\tau, \rho]$  consisting of the parts  $\tau \rho, \rho \tau$ , the absolute value of  $\tau \rho$  will be greater than or equal to the absolute value of  $\frac{1}{2}[\tau, \rho]$ , i.e.,

$$|\tau \rho| \geq \frac{1}{2} |[\tau, \rho]|$$

Thus, we get:

$$(\Delta r) \cdot (\Delta t_g) \geq |\tau \rho| \geq \frac{1}{2} |[\tau, \rho]|$$

A free observer in three-dimensional space, through the 1-1 correspondences  $\mathbf{r} \leftrightarrow \mathbf{r}$  and  $\mathbf{t}_g \leftrightarrow \mathbf{p}$ , gets the relation:

$$(\Delta r) \cdot (\Delta p) = h$$

where  $h$  is a constant. Since  $\Delta \mathbf{r}, \Delta \mathbf{p}$  are mappings of  $\Delta \mathbf{r}, \Delta \mathbf{t}_g$ , then their definition relations are:

$$(\Delta r)^2 = \langle r^2 \rangle \quad \text{and} \quad (\Delta p)^2 = \langle p^2 \rangle$$

where the vectors  $\mathbf{r}, \mathbf{p}$  are now the operators  $\hat{\mathbf{r}}, \hat{\mathbf{p}}$ , as we saw in the previous paragraph. Through these relations, the probabilistic model, and Schwartz's inequality, a free observer obtains the relation:

$$(\Delta \hat{r}) \cdot (\Delta \hat{p}) \geq |\langle \hat{r} \hat{p} \rangle| \geq \frac{1}{2} |\langle [\hat{r}, \hat{p}] \rangle|$$

This relation illustrates the relation found by the bound observers. The symbol  $[\hat{r}, \hat{p}]$  is called the **commutator of the operators  $\hat{\mathbf{r}}, \hat{\mathbf{p}}$**  and is defined as:

$$[\hat{r}, \hat{p}] = \hat{r} \hat{p} - \hat{p} \hat{r}$$

The components of the vector operators  $\hat{\mathbf{r}}, \hat{\mathbf{p}}$  along the same axis are related to each other as in  $\mathbf{P}^2$ . If we make the correspondences 8.1 and 8.2 and the commutator acts on  $\Psi(\mathbf{r}, t)$  given by equation 8.11, then in three-dimensional space free observers get the following relations between the components:

$$[\hat{x}, \hat{p}_x] \Psi(\mathbf{r}, t) = i\hbar \Psi(\mathbf{r}, t) \quad \text{and} \quad [\hat{y}, \hat{p}_x] \Psi(\mathbf{r}, t) = 0$$

These relations show that the constant that expresses the product of the uncertainties is the Planck constant. Thus, the relations between the uncertainties take the form:

$$(\Delta x) \cdot (\Delta p_x) \geq \frac{\hbar}{2}$$

These relations are called the **Heisenberg uncertainty relations**.

## 8.4 Klein-Gordon-Dirac equations

The equation  $x = \frac{1}{2a}y^2 + V'$  of a parabola can be viewed as the equation of an isosceles hyperbola, which is:

$$x_0^2 - y^2 = \tilde{V}^2 \tag{8.16}$$

where  $x_0^2 = 2ax$ ,  $\tilde{V}^2 = 2aV'$ . As we have seen, this isosceles hyperbola is depicted in a hologram in three-dimensional space through the correspondences:

$$x_0 \leftrightarrow \frac{\partial}{c\partial t} \equiv \frac{\hat{E}}{c} \tag{8.17}$$

$$y \leftrightarrow \mu \cdot \nabla \equiv \frac{\mu \cdot \hat{\mathbf{P}}}{i\hbar} \tag{8.18}$$

Applying equations 8.17 and 8.18 to equation 8.16, we obtain the equation of the hologram of an isosceles hyperbola in the context of the probabilistic model, namely:

$$\left( \left( \frac{\partial}{c\partial t} \right)^2 - \mu^2 \nabla^2 \right) \Psi(\mathbf{r}, t) = \tilde{V}^2 \Psi(\mathbf{r}, t) \quad (8.19)$$

or

$$\left( \frac{\hat{E}}{c} \right)^2 \Psi(\mathbf{r}, t) = \mu^2 \hat{\mathbf{P}}^2 \Psi(\mathbf{r}, t) + \tilde{V}^2 \Psi(\mathbf{r}, t) \quad (8.20)$$

The hologram of an isosceles hyperbola on which a point circle lies in three-dimensional space is a wave-particle satisfying equation 8.19. Considering that relation 8.11 is a solution of equation 8.20, we get:

$$-\frac{E^2}{c^2} = -\mu^2 p^2 + \tilde{V}^2 \quad (8.21)$$

The relation 8.21 is formally similar to the energy-momentum relation of a relativistic wave-particle, when  $\mu$  and  $\tilde{V}$  take the following values:

$$\mu^2 = 1, \quad \tilde{V}^2 = -\frac{m^2 c^2}{\hbar^2} \quad (8.22)$$

Substituting the relations 8.22 into equation 8.19, we get the equation of a relativistic wave-particle of mass  $m$ , which is formally identical to the **Klein-Gordon equation** [7]:

$$\left( \frac{\partial}{c\partial t} \right)^2 \Psi(\mathbf{r}, t) - \nabla^2 \Psi(\mathbf{r}, t) + \frac{m^2 c^2}{\hbar^2} \Psi(\mathbf{r}, t) = 0 \quad (8.23)$$

Equation 8.23 describes a relativistic wave-particle, but does not specify whether or not its existence can be confirmed experimentally. Direct experimental confirmation relies on whether the observer can see" this wave-particle (*visible wave-particle*), through some measurable quantities of it. In other words, for an observer to be able to see" a wave-particle, the wave-particle must have the wave characteristic of a light wave, or else it must be a composite wave-particle consisting of a "dark wave-particle" and at least one photon. For example, in such a composite wave-particle, its vector quantity called *spin* is observationally revealed.

Also, in this case, it is explained in a simple way why the wave nature of the photon of visible wave-particles is destroyed in the two-slit experiment, when we try to reveal the trajectory of a visible wave-particle by illuminating it[9]. The wave nature of a dark wave-particle does not appear in our visible world . Special Relativity is embodied in equation 8.23. However, equation 8.23 has the disadvantage that the norm of a state is not generally independent of time, so probability is not conserved. In summary, the Klein-Gordon wave equation satisfies Special Relativity, but not Quantum Mechanics. To reconcile these two theories of three-dimensional space, we will reduce this problem to a geometric problem in  $\mathbf{P}^2$  and transfer the solution of this problem to three-dimensional space through the 1-1



correspondences. The geometric problem to be solved is the following: We will find two suitable linear equations that are structurally related to Quantum Mechanics. These two equations should be combined appropriately to produce the equation of an isosceles hyperbola, which is related to Special Relativity. In order to satisfy the above requirements, we proceed with the following reasoning: The isosceles hyperbola expressed by equation 8.16 has asymptotes with the following equations:

$$y = \pm x_0 \quad (8.24)$$

We write the equations of the lines that are parallel to the asymptotes as:

$$y + x_0 = \tilde{V} \quad (8.25)$$

$$y - x_0 = -\tilde{V} \quad (8.26)$$

where  $\tilde{V}$  is the constant term of equation 8.16. When the relations 8.25 and 8.26 are multiplied member by member, they give the equation 8.16 of the isosceles hyperbola. Thus, in  $\mathbf{P}^2$ , the above requirements for geometric objects are met, because we have two linear equations (related to Quantum Mechanics in three-dimensional space), and from them the equation of an isosceles hyperbola (related to Special Relativity in three-dimensional space) is produced. The last step that must be done is the representation of the relations 8.25 and 8.26 in three-dimensional space, through the above correspondences 8.17 and 8.18, in order to find the equation of a hologram in which Special Relativity and Quantum Mechanics are integrated. Applying the correspondences 8.17, 8.18 and the relations 8.22 to the equations 8.25, 8.26, we obtain the desired equations of a hologram in three-dimensional space. Thus, for  $\Psi(\mathbf{r}, t) \rightarrow \Psi$ ,  $\mu = \alpha = \pm 1$  and  $\tilde{V} = \beta \frac{mc}{\hbar}$ , we obtain:

$$i\hbar c\alpha \cdot \nabla \Psi + i\hbar \frac{\partial \Psi}{\partial t} = \beta mc^2 \Psi \quad (\beta = -|\beta|) \quad (8.27)$$

$$i\hbar c\alpha \cdot \nabla \Psi - i\hbar \frac{\partial \Psi}{\partial t} = -\beta mc^2 \Psi \quad (\beta = |\beta|) \quad (8.28)$$

where  $|\beta| = 1$ . We define the Hermitian operators:

$$\hat{E} \equiv i\hbar \frac{\partial}{\partial t} \quad (8.29)$$

$$\hat{\mathbf{P}} \equiv -i\hbar \nabla \quad (8.30)$$

Due to equations 8.29 and 8.30, equations 8.27 and 8.28 are written as:

$$\hat{E}\Psi - c\alpha \cdot \hat{\mathbf{P}}\Psi - \beta mc^2 \Psi = 0 \quad (8.31)$$

$$\hat{E}\Psi + c\alpha \cdot \hat{\mathbf{P}}\Psi + \beta mc^2 \Psi = 0 \quad (8.32)$$

We make the correspondences  $x \rightarrow 1, y \rightarrow 2, z \rightarrow 3$ . Equations 8.31 and 8.32 should be combined together to obtain the equation of a hologram. The procedure for combining equations 8.31 and 8.32 is analogous to the procedure followed for geometric objects. Thus, by multiplying equations 8.31 and 8.32 while maintaining the order of the operators during the multiplication, we obtain an equation that is formally similar to the Klein-Gordon equation, so they are identical:

$$\hat{E}^2 - c^2 \left[ \sum_{i=1}^3 \alpha_i^2 \hat{P}_i^2 + \alpha_1, \alpha_2 \hat{P}_1 \hat{P}_2 + \alpha_2, \alpha_3 \hat{P}_2 \hat{P}_3 + \alpha_3, \alpha_1 \hat{P}_3 \hat{P}_1 \right] - m^2 c^4 \beta^2 \Psi = 0 \quad (8.33)$$

The quantities  $\alpha_1, \alpha_2, \alpha_3$  are the components of  $\alpha$  and  $\hat{P}_1, \hat{P}_2, \hat{P}_3$  are the components of  $\hat{\mathbf{P}}$  in the same coordinate system of three-dimensional space. Equation 8.33 is the Klein-Gordon equation, written in the form:

$$\hat{E}^2 - c^2 \hat{\mathbf{P}}^2 - m^2 c^4 \Psi = 0$$

Comparing the coefficients of equal powers, we obtain the following relations:

$$\alpha_i^2 = \beta^2 = 1 \quad (8.34)$$

$$\{\alpha_i, \alpha_j\} = \alpha_i \alpha_j + \alpha_j \alpha_i = 2\delta_{ij} \text{ for } i, j = 1, 2, 3 \quad (8.35)$$

$$\{\alpha_i, \beta\} = \alpha_i \beta + \beta \alpha_i = 0 \text{ for } i = 1, 2, 3 \quad (8.36)$$

The symbols  $\{\alpha_i, \alpha_j\}$  and  $\{\alpha_i, \beta\}$  are called *anti-commutators* of  $\alpha_i, \alpha_j$  and  $\alpha_i, \beta$  respectively, which are defined through the relations 8.35 and 8.36. Relations 8.34, 8.35, 8.36 apply to  $\mathbf{P}^2$  and to the three-dimensional space due to the 1-1 correspondences. If the quantities  $\alpha_i, \beta$  were scalar numbers, then from equation 8.36,  $\alpha_i$  would be zero ( $\alpha_i = 0$ ) for each  $i$ .

Thus, from this observation, we conclude that the quantities  $\alpha_i$  and  $\beta$  are matrices with eigenvalues  $\pm 1$ . First we take  $\mathbf{P}^2$ . The quantities  $\alpha_i, \beta$  in this space are expressed as  $2 \times 2$  matrices.

The matrix  $\beta$  is diagonal, but not the identity matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

because in that case the  $\alpha_i$  matrices would still be the zero matrices. Since the eigenvalues of  $\beta$  are  $\pm 1$ , then the form of  $\beta$  is

$$\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (8.37)$$

In order to satisfy equation 8.36, the matrices  $\alpha_1$  and  $\alpha_2$  must have **anti-diagonal form**, that is:

$$\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \alpha_1 = \begin{pmatrix} 0 & \gamma_1 \\ \gamma_2 & 0 \end{pmatrix}, \quad \alpha_2 = \begin{pmatrix} 0 & \gamma_3 \\ \gamma_4 & 0 \end{pmatrix} \quad (8.38)$$

From equation 8.35 and for  $i = j$ , we get  $\alpha_i^2 = 1, i = 1, 2$ . We write the equations:

$$\gamma_1\gamma_2 = 1, \quad \gamma_3\gamma_4 = 1$$

Choosing the eigenvalue of  $\alpha_1$  to be  $+1$ , we get  $\gamma_1 = \gamma_2 = 1$ . Also from equation 8.35, we get:

$$\gamma_3 + \gamma_4 = 0$$

Solving the two previous equations with the unknowns  $\gamma_3, \gamma_4$ , we get  $\gamma_3 = \pm i$ . If we choose, for example,  $\gamma_3 = -i$ , then we get  $\gamma_4 = i$ . Thus, the  $2 \times 2$  matrices  $\alpha_1, \alpha_2$  and  $\beta$  become the  $2 \times 2$  *Pauli matrices*:

$$\sigma_1 = \alpha_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \alpha_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_3 = \beta = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The structure of  $\alpha_i, \beta$  in a four-dimensional space (such as the Minkowski spacetime of Special Relativity) has the same form, due to the two-way correspondences, as in the two-dimensional case [21], that is:

$$\beta = \begin{bmatrix} I & \emptyset \\ \emptyset & -I \end{bmatrix}, \quad \alpha_i = \begin{bmatrix} \emptyset & \mathcal{M}_i \\ \mathcal{M}_i & \emptyset \end{bmatrix}, \quad i = 1, 2, 3$$

where  $I$  is the  $2 \times 2$  identity matrix,  $\emptyset$  is the  $2 \times 2$  zero matrix, and  $\mathcal{M}_i$  is the  $2 \times 2$  matrix:

$$\mathcal{M}_i = \begin{bmatrix} 0 & \mathcal{M}_{1i} \\ \mathcal{M}_{2i} & 0 \end{bmatrix}$$

Thus, from the above expressions of the matrices in the four-dimensional Minkowski spacetime, we obtain the  $4 \times 4$  matrices[19]:

$$\alpha_1 = \begin{bmatrix} \emptyset & \sigma_1 \\ \sigma_1 & \emptyset \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} \emptyset & \sigma_2 \\ \sigma_2 & \emptyset \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} \emptyset & \sigma_3 \\ \sigma_3 & \emptyset \end{bmatrix}, \quad \beta = \begin{bmatrix} I & \emptyset \\ \emptyset & -I \end{bmatrix} \quad (8.39)$$

If relations 8.39 are substituted into equations 8.31 and 8.32, then the equations obtained are formally identical to **Dirac's equations**. The Dirac equations are the equations of holograms in three-dimensional space derived from the correspondences with the equations of lines parallel to the asymptotes of an isosceles hyperbola. The Dirac equations determine the motion of a relativistic wave-particle in three-dimensional space.

## 8.5 Born's Rule

The persistent mathematical incompatibility between the smooth, geometric spacetime of General Relativity and the probabilistic, linear Hilbert spaces of Quantum Mechanics represents the central crisis of modern theoretical physics. The Geometric Model resolves this tension by demonstrating that both regimes emerge naturally from a single, discrete, self-replicating 2D boundary. We establish this model on three core postulates:

1. **The 2D Holographic Boundary:** The fundamental degrees of freedom of the Universe reside entirely on a flat, two-dimensional boundary; 3D space is an emergent projection.
2. **Strict Outward Self-Similarity:** The boundary evolves discretely via structural self-similarity (fractal scaling).
3. **Information Irreversibility:** Information packets (particles) are constrained to propagate exclusively along newly generated geometric states.

## 8.6 The Geometric Growth Engine ( $1 \rightarrow 4$ )

The dynamics of the holographic boundary are driven by a discrete replacement rule. Let  $t_0$  represent the initial state containing a single square cell of fundamental area  $A_0 = l_P^2 = 1$ . The length of each side is the "Planck's length  $l_P$ ". To transition to the next temporal stage  $t_1$ , the initial square produces 3 additional, identical copies of itself because of self-similarity between successive Plank areas. These 3 new copies assemble around the original unit to construct a single, perfectly bound, larger square. The total number of units at any given stage  $n$  follows a strict power law:

$$N(n) = 4^n, \text{ where } n = 0, 1, 2, \dots \quad (8.40)$$

This scaling behavior matches the geometric requirement for doubling the side length ( $L$ ) of a 2D square, where  $\text{Area} \propto L^2$ .

## 8.7 Geometric Derivation of Born's Rule

We define the border of the Universe as a flat 2D-space. In conventional quantum mechanics, the probability of finding a particle is given by Born's Rule:  $P(x) = |\psi(x)|^2$ . For a century, physics has treated the power of 2 as an empirical axiom. The Geometric Model derives this exponent directly from spatial scaling. A quantum wave function  $\psi$  represents an information field mapped onto our 2D boundary. When the boundary steps from  $t_0 \rightarrow t_1$ , the underlying geometric grid expands by a factor of 4 ( $2^2$ ).

To preserve total probability at exactly 1 (100%) across the expanding system, the probability density must distribute evenly across these 4 available quadrants:

$$\sum_{i=1}^{4^n} P_i = \sum_{i=1}^{4^n} |\psi_i|^2 = 1 \quad (8.41)$$

If probability scaled linearly (power of 1) or cubically (power of 3), the physical information would either tear or overlap, failing to cleanly carpet the newly emerged 4-part geometric surface. The quadratic power is the unique exponent that matches the 2D surface scaling of our  $1+3=4$  growth rule.



## Chapter 9

# QUANTUM MECHANICS TOPICS

### 9.1 Magnetic moment of the electron

#### A) Electron magnetic moment according to Dirac

The Klein-Gordon equation describes a wave-particle with mass. The possibility of experimental study confirms that it is a visible wave-particle, that is, it consists of a dark mass wave-particle and a photon.

Having the Dirac equations, free observers can find how and why these two particles of three-dimensional space “compose” through their corresponding geometric objects in  $\mathbf{P}^2$ . The bound observers multiply the equations of the asymptotes of the hyperbola  $y = \pm x_0$  and obtain the homogeneous wave equation with  $V = 0$ . The free observers apply the correspondences to this relation, thus obtaining the following homogeneous wave equation in three-dimensional space:

$$\left(\frac{\partial}{c\partial t}\right)^2 \Phi - \nabla^2 \Phi = 0 \quad (9.1)$$

This wave equation expresses the wave quanta, which are massless wave-particles identical to photons ( $\gamma$ ), the only massless wave-particles in the visible three-dimensional space. The Klein-Gordon equation is also a wave equation and expresses a wave whose quanta are massive wave-particles. The question is: **“How do dark mass wave-particles compose with photons to create visible wave-particles?”**

To answer this question, we will return to  $\mathbf{P}^2$  to see how the geometric objects corresponding to photons and dark mass wave-particles are composed, so that these compositions can be mapped to “visible mass wave-particles” in three-dimensional space. Dark mass wave-particles correspond to  $\mathbf{P}^2$  with isosceles hyperbolas whose equations are constructed from straight lines ex-

pressed by the following equations:

$$y + x_0 - V = 0 \quad (9.2)$$

$$y - x_0 + V = 0 \quad (9.3)$$

Photons also correspond to hyperbolas in  $\mathbf{P}^2$  whose equations are constructed from straight lines expressed by the following equations:

$$-y_\gamma - x_\gamma = 0 \quad (9.4)$$

$$-y_\gamma + x_\gamma = 0 \quad (9.5)$$

Equations 9.4 and 9.5 represent the equations of the bisectors of the right angles of a Cartesian coordinate system (dashed lines) and equations 9.2, 9.3 represent the equations of the lines parallel to them (solid lines), as shown in Figure 9.1.

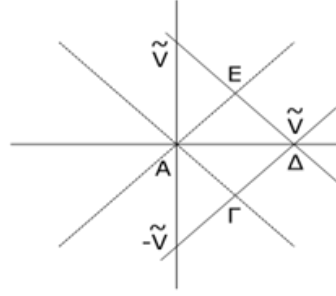


Figure 9.1: Composition of geometric objects corresponding to a wave-particle and a photon

The intersection of all parallel lines (dashed and solid) forms the square  $A\tilde{E}\Delta\Gamma$ , as is easily demonstrated by elementary geometry. The composition of a massive particle with a photon in three-dimensional space, determined by bound observers in  $\mathbf{P}^2$ , is called a “composition of lines”. This “composition” is achieved by parallel displacement of the line  $\Gamma\Delta$  to the line  $AE$  and then by parallel displacement of the line  $A\Gamma$  to the line  $E\Delta$ . Equivalently, this “composition” in  $\mathbf{P}^2$  is defined as the displacement of point  $\Gamma$  along the path  $\Gamma A + AE$  until it coincides with point  $E$ , where  $\Gamma A = AE$ .

This composition process is defined algebraically by adding member-wise the pairs of equations expressing parallel lines, namely equations 9.2, 9.3 and 9.4, 9.5. Thus, the equations of the lines are obtained, which are considered as the geometric objects corresponding to the composite visible and massive wave-particles, namely:

$$y - y_\gamma + (x_0 - x_\gamma) = V \quad (9.6)$$



$$y - y_\gamma - (x_0 - x_\gamma) = -V \quad (9.7)$$

These equations are obtained simply by adding, member by member, the equations of the geometric object corresponding to a photon and the equations of the geometric object corresponding to a dark wave-particle. In three-dimensional space, this geometric composition is depicted as a “collision” of a photon with a dark wave-particle and the creation of a composite physical entity that is a “visible wave-particle”.

In the two-slit experiment with electrons in three-dimensional space, when the electrons are illuminated to reveal which slit they pass through, their wave nature is destroyed, resulting in their distribution on the screen being the same as that of classical particles. The bound observers determine the geometric objects corresponding to the photons from the equations:

$$x_\gamma + y_\gamma = 0 \quad (9.8)$$

$$-x_\gamma + y_\gamma = 0 \quad (9.9)$$

By adding equations 9.6 and 9.8 on the one hand, and equations 9.7 and 9.9 on the other hand, we obtain equations 9.2 and 9.3, which are the equations of the geometric object corresponding to a dark wave-particle that does not display its wave properties in our visible world. This explains the behavior of the electron as a material point when one tries to reveal its trajectory in the double-slit experiment.

In three-dimensional space, free observers write the previous equations as:

$$(\hat{E} - \hat{E}_\gamma)\Psi + c\alpha \cdot (\hat{\mathbf{P}} - \hat{\mathbf{P}}_\gamma)\Psi + \beta mc^2\Psi = 0 \quad (9.10)$$

where  $\beta = -|\beta|$ , and

$$(\hat{E} - \hat{E}_\gamma)\Psi + c\alpha \cdot (\hat{\mathbf{P}} - \hat{\mathbf{P}}_\gamma)\Psi - \beta mc^2\Psi = 0 \quad (9.11)$$

where  $\beta = +|\beta|$ .

Suppose that a massive wave-particle also carries a charge  $q$ . Let there be an electromagnetic wave with photons as its quanta, which is expressed in terms of the scalar potential  $\Phi$  and the vector potential  $\mathbf{A}$ . The wave-particle interacts with a photon. During the synthesis of these particles, the energy operator  $\hat{E}_\gamma$  and the momentum operator  $\hat{\mathbf{P}}_\gamma$  of the photons act on the wave function  $\Psi$ , which includes the physical characteristics of both wave-particles, to give their eigenvalues  $q\Phi$  and  $\frac{q\mathbf{A}}{c}$  respectively, i.e.,

$$\hat{E}_\gamma\Psi = (q\Phi)\Psi \quad (9.12)$$

$$\hat{\mathbf{P}}_\gamma\Psi = \left(\frac{q\mathbf{A}}{c}\right)\Psi \quad (9.13)$$

Substituting equations 9.12 and 9.13 into equations 9.10 and 9.11, we get:

$$(\hat{E} - q\Phi)\Psi + \alpha \cdot (c\hat{\mathbf{P}} - q\mathbf{A})\Psi + \beta mc^2\Psi = 0 \quad (9.14)$$

where  $\beta = -|\beta|$ , and

$$(\hat{E} - q\Phi)\Psi - \alpha \cdot (c\hat{\mathbf{P}} - q\mathbf{A})\Psi - \beta mc^2\Psi = 0 \quad (9.15)$$

where  $\beta = +|\beta|$ .

The equation of motion of a “visible wave-particle” in three-dimensional space is again a Klein-Gordon equation obtained from equations 9.14 and 9.15. Thus, following the well-known procedure, we obtain, after some approximations, a Schrödinger equation of the form:

$$E'\Psi = \left\{ \frac{1}{2m} \left( \hat{\mathbf{P}} - \frac{q}{c}\mathbf{A} \right)^2 + q\Phi - \frac{q\hbar}{2mc}\boldsymbol{\sigma}' \cdot \mathbf{B} + \frac{ic\hbar}{2mc}\boldsymbol{\alpha} \cdot \mathbf{E} \right\} \Psi \quad (9.16)$$

where  $\mathbf{E}, \mathbf{B}$  are the electric and magnetic fields of the electromagnetic wave respectively, and  $\boldsymbol{\sigma}'$  is given by the relation:

$$\boldsymbol{\sigma}' = \begin{pmatrix} \sigma & \emptyset \\ \emptyset & \sigma \end{pmatrix} = \boldsymbol{\sigma} \cdot I$$

The term of equation 9.16 which contains  $\mathbf{B}$  is related to the energy of a magnetic dipole with the dipole moment:

$$\mu_s = \frac{q\hbar}{2mc}\sigma = \mu_B\sigma \quad (9.17)$$

where  $\mu_B = \frac{q\hbar}{2mc}$  is the Bohr magneton. The new physical quantity that appears in all visible wave-particles is called spin. From the general definition of the magnetic moment of spin, we get the relation:

$$\mu_s = \frac{g}{2}\mu_B\sigma \quad (9.18)$$

When equation 9.17 is compared with equation 9.18 for an electron, we get  $g = 2$ . We consider the above interaction process in  $\mathbf{P}^2$  as follows: In Figure 9.1, the point  $\Gamma$  moves on the path  $l = \Gamma A + AE$  until it coincides with the point  $E$ . The point  $\Gamma$  moves on this path which consists of two steps that have unit length in some units of length, that is, we have  $\Gamma A = AE = 1$ , so  $l = 2$ .

If  $g$  is the sum of the number of steps, then we get  $g = 2$ , so  $\frac{g}{2} = 1$ . In this case, the sequence of steps is denoted by  $\{1, 1\}$  and coincides with the first two members of the Fibonacci sequence. We will see in the next section how this observation is used, namely the use of the complete Fibonacci sequence, to find the numerical value of the fine-structure constant.

The spin in every visible wave-particle was considered by free observers to be an intrinsic characteristic of the wave-particle whose existence is confirmed experimentally. However, in our analysis, the appearance of spin is a result of the synthesis of a dark massive wave-particle with a massless photon. Thus, spin can be considered as a characteristic only of a composite visible wave-particle.

## 9.2 Numerical value of the fine-structure constant

**1. Approach to the numerical value of the fine structure** The need to determine the magnetic moment of the electron with greater accuracy than that of the Dirac method arose from the need to explain the energy differences found experimentally between the  $2S_{\frac{1}{2}}$  and  $2P_{\frac{1}{2}}$  orbitals of the hydrogen atom (Lamb shift), according to free observers.

According to Dirac, these levels have the same energy. The interpretation of the Lamb shift led to the development of Quantum Electrodynamics (QED), within which a method was developed by Feynman. This method is the first and, so far, the only one that provides a highly accurate value for the magnetic moment of the electron, in agreement with experimental data. As previously mentioned, in the method developed in  $\mathbf{P}^2$  for particle interactions in three-dimensional space, the sequence of steps from point C to point E is 1,1.

This sequence consists of terms that are identical to the first two terms of the Fibonacci sequence. Thus, for a more detailed analysis, the full Fibonacci sequence will be tested.

For this purpose, we must first determine the scale at which we will calculate the numerical value of a physical quantity (distance scale). For different distance scales, different numerical values of physical quantities appear, as happens macroscopically when measuring the length of a coastline from an airplane using aerial photographs. In the first aerial photograph, the length between two points of the coastline is roughly estimated.

As the altitude decreases, capes and bays of the coastline appear in the aerial photograph. In another measurement of the same initial length, the lengths of the coastline's details (capes and bays) must be calculated in the same units of length as before, and therefore must be added to the initial length.

In the case of the coastline, as with the majority of physical entity quantities, the same structure appears at all scales down to an infinitesimal scale. Thus, self-similarity emerges between structures at different scales (fractal structure [6]).

To apply the measurement process, the following question must be answered:

**“How will the distance scale  $S_k$ , at which we calculate the numerical values of various quantities, be determined?”**

This question can be answered as follows: Each distance scale is characterized by a parameter, so a way must be found to define this parameter. This distance scale constant should include in its expression only constants derived from or containing the full Fibonacci sequence. One constant of the sequence is the ratio  $\frac{F_p}{F_{p-1}}$ , where  $F_p$  is the term of the Fibonacci sequence at position  $p$ , and its limit is the golden ratio.

$$\phi = \lim_{p \rightarrow \infty} \left( \frac{F_p}{F_{p-1}} \right) = \frac{(1 + \sqrt{5})}{2} = 1.618 \dots$$

or its reciprocal  $\phi^{-1}$

$$\phi^{-1} = 0.618 \dots$$

A second constant containing an expression of the full Fibonacci sequence is the reciprocal of the number 89, which is the eleventh term of this sequence, namely

$$\psi^{-1} = 89^{-1} = \sum_{p=1}^{\infty} F_p \cdot 10^{-(p+1)} = 0.01123 \dots \quad (9.19)$$

The distance scale constant,  $c_{\mu,k}$ , is defined as a linear combination of these two constants, i.e., as

$$\mu = k + 1 \quad (9.20)$$

For this value of  $\mu$ , equation (9.20) is written as

$$c_k = 10^{-(k-1)}(\tilde{\phi} + \tilde{\psi}) \quad (9.21)$$

where

$$\tilde{\phi} = (10^2 \phi)^{-1}, \quad \tilde{\psi} = (10 \psi)^{-1}$$

The distance scale constant  $c_1$  in the Dirac method is determined by the integer value  $k = 1$ , which yields

$$c_1 = \tilde{\phi} + \tilde{\psi} = (10^2 \phi)^{-1} + (89 \cdot 10)^{-1} \simeq (137)^{-1} \quad (9.22)$$

The constant  $c_1$  has the same numerical value, to a very good approximation, as the experimentally determined value of the "fine-structure constant."

Since any distance can be converted into momentum or energy via Planck's constant and the speed of light, it follows that any distance can be transformed into momentum or energy. For low energy values, i.e., for values  $k \simeq 1$  or  $k \simeq 1.0004$ , we get  $c_{\text{low}} \simeq (137)^{-1}$ , and for high energy values ( $\simeq 200$  GeV) or  $k = 0.96736$ , we get  $c_{\text{high}} \simeq 127^{-1}$ , as was theoretically

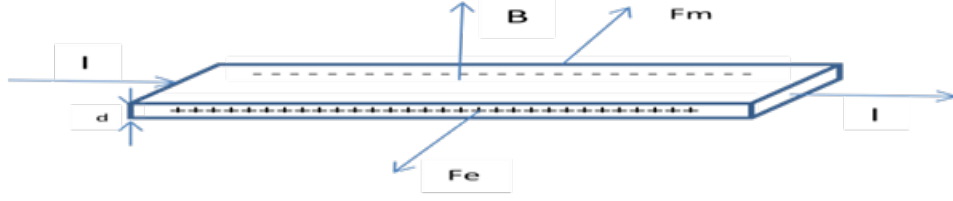
predicted by Gell-Mann and Low and subsequently verified experimentally. This confirms the fact that the distance scale constant is not a universal constant, as previously mentioned.

**2. Experimental determination of the numerical value of the fine-structure constant  $\alpha$**  The fine-structure constant  $\alpha$  is a dimensionless number with a value nearly equal to  $\frac{1}{137}$ . This is the "coupling constant" or the measure of the strength of the electromagnetic force that governs how electrically charged elementary particles (e.g., electron, muon with electron charge  $q_e$ ) and light (photons ( $\gamma$ )) interact. Currently, the value of  $\alpha$  with the smallest uncertainty comes from comparing the theoretical expression of  $\alpha$  and the experimental value ( $\alpha_{\text{exp}}$ ) of the anomalous magnetic moment of the electron. Starting in the 1980s, a new and entirely different measurement approach using the Quantum Hall Effect (QHE) generated significant excitement. The value of  $\alpha$  obtained from it independently confirms the  $\alpha$  value derived from the anomalous magnetic moment of electrons. The value of  $\alpha$  obtained via the QHE does not have as small an uncertainty as the value from the electron's magnetic moment, but it provides a major independent confirmation of that value. The quantity  $\alpha$  was introduced to physics by A. Sommerfeld in 1916, at which time it was referred to as the Sommerfeld fine-structure constant. In order to explain the observed splitting or fine structure of the hydrogen atom's energy levels, Sommerfeld extended Bohr's theory to include elliptical orbits and the relativistic dependence of mass on velocity. The quantity  $\alpha$ , which is equal to the ratio  $\frac{v_1}{c}$  where  $v_1$  is the velocity of the electron in the first circular Bohr orbit and  $c$  is the speed of light in a vacuum, emerged naturally in Sommerfeld's analysis and determined the scale of the splitting or fine structure of Hydrogen's spectral lines.

$$\alpha = \frac{e^2}{4\pi\epsilon_0 c \hbar} = \frac{\mu_0 c e^2}{2h}$$

While the value of  $\alpha$  can be determined from estimates of the constants appearing in any of its definitions, the theory of Quantum Electrodynamics (QED) provides a way to measure  $\alpha$  directly using the Quantum Hall Effect or the anomalous magnetic moment of the electron. The Hall effect has been known for more than a century and is outlined in the figure below. A magnetic field  $B$  is applied perpendicularly to the direction of current flow in a metal, causing a voltage to appear in the third direction. This potential is due precisely to the deflection of positively and negatively charged carriers toward the edge of the metal. The Hall potential  $V_H$  can be easily calculated at equilibrium, i.e., when the magnetic force is balanced by the electric force, yielding:

$$V_H = \frac{I \cdot B}{e \cdot n \cdot d}$$

Figure 9.2: Experimental setup for determining the value of  $\alpha$ 

### 3. The associated Hall resistance

$$R_H = \frac{B}{e \cdot n \cdot d}$$

increases monotonically with the applied magnetic field. The Quantum Hall Effect occurs at low temperatures and is a manifestation of the so-called Landau quantization. In a two-dimensional metal (and in a semiconductor as well), the Hall resistance does not increase continuously with an increasing magnetic field, but exhibits quantum step behavior. It is thus expressed in terms of the fundamental resistance

$$R_K = \left( \frac{e^2}{h} \right)^{-1}$$

known as the von Klitzing constant, which is used to calibrate resistance with high precision. It is evident that the von Klitzing constant is directly linked to the fine-structure constant.

Measurements made with the Quantum Hall Effect are remarkably accurate and can allow the determination of the fine-structure constant. Other methods include the A.C. Josephson effect and photon recoil in atom interferometry. There is general agreement on the value of  $\alpha$  as measured by these different methods.

The preferred methods in 2019 are measurements of anomalous electron magnetic moments and photon recoil in atom interferometry. QED theory predicts a relationship between the dimensionless magnetic moment of the electron and the fine-structure constant  $\alpha$  (the electron magnetic moment is also referred to as the  $g$ -factor).

### 4. Determination of the numerical value of the fine-structure constant via the geometric model

The geometric object corresponding to a dark electron ( $e_d$ ) consists of a circle of radius 1 and its circumscribed ellipse. From this geometric structure, it follows that  $e_d$  has a non-zero charge and mass, as these quantities are defined for conic sections. The geometric object corresponding to the photon  $\gamma$  is an ellipse in  $\mathbf{P}^2$  that compensates for one quadrant of the ellipse or circle corresponding to the dark electron, as shown in Figure 2.3 of Chapter 2.

A) Microscopically, bound observers consider that the space  $\mathbf{P}^2$ , due to the infinitesimal dimensions of the Planck point, resembles an isotropic and

homogeneous space. Thus, these properties determine a constant through the rotation and translation of each part of a Planck patch. Rotation is expressed via a circle during an expansion stage. During the contraction stage, the circle transforms into an ellipse. As we have seen from the expressions of the vectors  $L, R$ , all conic sections are derived. When we set the values  $|L| = 1$  and  $|R| = 1$ , we obtain the equation:

$$\alpha^2 - \alpha - 1 = 0$$

The solution to this equation is the golden ratio:

$$\phi = \frac{(1 + \sqrt{5})}{2} = 1.618$$

The golden ratio  $\phi$  also appears in the Fibonacci sequence by taking the ratio  $\frac{F_{n+1}}{F_n}$  of two consecutive terms. This correlation between the conic sections of  $\mathbf{P}^2$  and the Fibonacci sequence leads us to find constants from the Fibonacci sequence that are also constants in  $\mathbf{P}^2$ .

Initially, we use the Fibonacci sequence constant that is the reciprocal of the number 89, which is the 11<sup>th</sup> term of the Fibonacci sequence. This number is correlated with the Fibonacci sequence through the relation:

$$\psi^{-1} = \frac{1}{89} = 0.01 + 0.001 + 0.0002 + 0.00003 + 0.000005 + \dots$$

Thus, we take the sum of these two constants, adjusted to the same order of magnitude, to obtain a new constant derived from the Fibonacci sequence, namely:

$$C_n = 10^{-n}(\psi^{-1} + 10\phi^{-1}) \sim 0.73 \cdot 10^{-n}$$

The constant  $C_n$  begins to vary slightly after the  $n^{\text{th}}$  decimal place, because the space containing the visible electron changes through the Expansion and Contraction processes of  $\mathbf{P}^2$ .

B) The parameter  $C_n$  participates in all construction processes of geometric objects in  $\mathbf{P}^2$ , enabling the formation of more complex yet identical geometric objects. The composite geometric structure consisting of a geometric object corresponding to a dark electron ( $e_d$ ) and a geometric object corresponding to a photon ( $\gamma$ ) is the geometric structure that corresponds to a visible electron. Free observers call this constant the "Fine-structure constant"  $\alpha$ , whose experimental value is:

$$\alpha = 0.0072973525693$$

or

$$\alpha \sim 0.0073$$

The theoretical value of the constant  $C_n$  found by bound observers must be nearly identical to the experimental value found by free observers. For this to happen, the initial value of  $n$  must be  $n = 2$ , yielding  $C_2 \sim 0.0073$ . However, the numerical value of a dimensionless quantity must be exactly the same in both  $\mathbf{P}^2$  and 3D space. Thus, bound observers must modify the process of determining  $C_2$  to obtain the exact experimental value of  $\alpha$ , taking the following into account:

A. The continuously repeated Expansion (E)-Contraction (C) process of the Planck patches during the construction process of  $\mathbf{P}^2$ . According to the continuous Contraction (gray color)-Expansion (yellow color) process, a composite structure must be formed by  $(e_d)$  and  $(\gamma)$ , as shown in Table 1.5 below.

B. The structure of the Fibonacci sequence, namely the way the terms of the Fibonacci sequence are formed and the fact that the terms of this sequence form triplets in the order "(Odd (O) - Odd (O) - Even (E))", as exemplified by the first 3 triplets: (1,1,2), (3,5,8), and (13,21,34).

C. The equal number of Expansion (E)-Contraction (C) processes, the equal number of exponents, and their frequency of occurrence.

D. The creation of regions when the order of magnitude of the exponents changes. Thus, the first region  $R_1$  contains exponents of the order of magnitude of  $10^0$ , i.e., exponents in the interval [5–9]. The second region  $R_2$  contains exponents with an order of magnitude in the interval  $[10^1–10^2]$ , and so forth. In order to preserve the properties of  $\mathbf{P}^2$ , each region of the table will be generated from its preceding one via translation and rotation, just as a Planck patch is generated from its initiator.

To satisfy the above four conditions, bound observers use the relation for determining  $C_n$ , which yields the value of the constant  $\alpha$ . We apply the preceding steps (A), (B), (C), and (D) to the relation for  $C_n$ . The initial value of this relation is obtained for  $n = 2$ . Since we initially add the terms  $\phi^{-1}$  and  $10\psi^{-1}$ , we assume that this occurs during the expansion process of a Planck patch.

To obtain a different result for the 4<sup>th</sup> decimal place than the one provided by the equation above for  $n = 2$ —in order to approach the experimental value—we change the order of magnitude of the exponent of 10 in the equation above, taking the value  $n = 5$ , thereby finding  $C_5$ . Since the expansion process must be followed by the contraction process, we subtract  $C_5$  from  $C_2$ . Accordingly, we obtain a new value for  $\alpha$ , which is:

$$\alpha = C_2 - C_5 - (C)$$

To demonstrate the structure of the Fibonacci sequence, its first 3 terms are expressed by the number of times the successive exponents of 10 appear, namely the numbers 5, 6, 7.

These exponents appear 1, 1, 2 times, following the structure of each



triplet of the Fibonacci sequence ("Odd (O) - Odd (O) - Even (E)"). This sequence of appearance of the exponents must be consistent with the Expansion-Contraction— . . . process of the Planck patches. Thus, we write the different values that  $\alpha$  takes as follows:

$$\alpha = [C_2 - C_5] + C_6 - (E)$$

$$\alpha = [C_2 - C_5 + C_6] - C_7 - (C)$$

$$\alpha = [C_2 - C_5 + C_6 - C_7] + C_7 - (E)$$

At this point, the 1<sup>st</sup> triplet of region  $R_1$  is completed. The 2<sup>nd</sup> triplet of region  $R_1$  begins with the exponent number 8, which appears 1 time as the first number of the Fibonacci sequence. The 2<sup>nd</sup> term of the 2<sup>nd</sup> triplet, which is also the last exponent of 10 within the range of  $R_1$ , will appear 3 times, as shown by the sum of the appearance times of exponents 7 and 8.

The type of the Expansion-Contraction process for this exponent will be examined at the end of the process, so that the entire structure is consistent with steps (1–4). Region  $R_2$  will be structured as follows: A parallel translation of region  $R_1$  will be performed to determine the same repeating Expansion-Contraction sequence in region  $R_2$ .

Starting with an expansion process (as in region  $R_1$ ) and a rotation around the imaginary line located at the boundary of the two regions to ensure that the exponents appear the same number of times in both regions.

The rotation of region  $R_1$  is performed from exponent 5 to exponent 9 to yield an equal number of exponents in both regions. Thus, region  $R_2$  begins with exponent 10 appearing 3 times (the same as exponent 9) and within an expansion process, identical to the first operation of region  $R_1$ .

Exponent 11 appears 1 time (the same as exponent 8) in the contraction process. Exponent 12 appears 2 times (the same as exponent 7) in the expansion process.

Exponent 13 appears 1 time (the same as exponent 6) in the contraction process, and finally, exponent 14 appears 1 time (the same as exponent 5) in the expansion process.

In region  $R_2$ , the number of expansion processes is  $N_{E_2} = 6$  and the number of contraction processes is  $N_{S_2} = 2$ . In region  $R_1$ , the number of expansion processes is  $N_{E_1} = 2$  and the number of contraction processes is  $N_{S_1} = 3$ .

To match the number of contraction processes in region  $R_1$  with the expansion processes in region  $R_2$ , the three occurrences of exponent 9 must reside within a contraction process, which gives  $N_{S_1} = 6$ . Consequently, condition (3) is satisfied. All of this is displayed in the table of the following figure, where the theoretical value of the fine-structure constant found

by bound observers is identical to the experimental value found by free observers. The need to determine the magnetic moment of the electron with greater accuracy than that of the Dirac method arose from the need to explain the energy differences found experimentally between the  $2S_{1/2}$  and  $2P_{1/2}$  orbitals of the hydrogen atom (*Lamb shift*).

According to Dirac, these levels have the same energy. The interpretation of the Lamb shift led to the development of Quantum Electrodynamics (QED), within which a method was developed by Feynman. This method is the first and, so far, the only one that provides a highly accurate value for the magnetic moment of the electron, in agreement with experimental data. As previously mentioned, in the method developed in  $\mathbf{P}^2$  for particle interactions in three-dimensional space, the sequence of steps from point C to point E is  $\{1, 1\}$ . This sequence consists of terms that are identical to the first two terms of the Fibonacci sequence. Thus, for a more detailed analysis, the full Fibonacci sequence will be tested. For this purpose, we must first determine the scale at which we will calculate the numerical value of a physical quantity (*distance scale*). For different distance scales, different numerical values of physical quantities appear, as happens macroscopically when measuring the length of a coastline from an airplane using aerial photographs; in the first aerial photograph, the length between two points of the coastline is roughly estimated. As the altitude decreases, capes and bays of the coastline appear in the aerial photograph. In another measurement of the same initial length, the lengths of the coastline's details (capes) must be calculated in the same units of length as before, and they must be added to the initial length. In the case of the coastline, as with the majority of physical entity quantities, the same structure appears at all scales down to an infinitesimal scale. Thus, self-similarity emerges between structures at different scales (**fractal structure**)[6]. To apply the measurement process, the following question must be answered: "How will the distance scale  $S_k$ , at which we calculate the numerical values of various quantities, be determined?" This question can be answered as follows: Each distance scale is characterized by a parameter, so a way must be found to define this parameter.

As mentioned above, we will test the full Fibonacci sequence, so we will use it in order to find a distance scale constant. This parameter should include in its expression only constants derived from or containing the full Fibonacci sequence. One constant of the sequence  $\{\frac{F_p}{F_{p-1}}\}$ , where  $F_p$  is the term of the Fibonacci sequence at position  $p$ , is its limit, namely the *golden ratio*

$$\phi = \lim_{p \rightarrow \infty} \left( \frac{F_p}{F_{p-1}} \right) = \frac{1 + \sqrt{5}}{2} = 1.618 \dots$$

or its reciprocal

$$\phi^{-1} = 0.618\dots$$

A second constant containing an expression of the full Fibonacci sequence is the reciprocal of the number 89, which is the eleventh term of this sequence[15], namely:

$$\psi^{-1} = 89^{-1} = \sum_{p=1}^{\infty} F_p 10^{-(p+1)} = 0.01123\dots \quad (9.23)$$

The distance scale constant,  $c_{\mu,k}$ , is defined as a linear combination of these two constants, i.e., as:

$$c_{\mu,k} = 10^{-\mu} \phi^{-1} + 10^{-k} \psi^{-1} \quad (9.24)$$

where  $\mu, k$  are real numbers. The numbers  $\phi^{-1}, \psi^{-1}$  differ by one order of magnitude, as shown by their values. In order for both terms of equation (9.25) to have the same order of magnitude, we set:

$$\mu = k + 1 \quad (9.25)$$

For this value of  $\mu$ , equation (9.25) is written as:

$$c_k = 10^{-(k-1)} \{ \tilde{\phi} + \tilde{\psi} \} \quad (9.26)$$

where:

$$\tilde{\phi} = (10^2 \phi)^{-1}, \quad \tilde{\psi} = (10 \psi)^{-1}$$

The distance scale constant  $c_1$  in the Dirac method is determined by the integer value  $k = 1$ , which yields:

$$c_1 = \tilde{\phi} + \tilde{\psi} = (\phi \cdot 10^2)^{-1} + (89 \cdot 10^1)^{-1} \cong (137)^{-1} \quad (9.27)$$

The constant  $c_1$  has the same numerical value, to a very good approximation, as the numerical value of the “*fine-structure constant*”

Since any distance can be converted into momentum or energy via Planck’s constant and the speed of light, it follows that any distance can be transformed into momentum or energy. For low energy values, i.e.,  $k \cong 1$  or  $k \cong 1.0004$ , we get  $c_{\text{low}} \cong (137)^{-1}$ , and for high energy values ( $\cong 200$  GeV) or  $k = 0.96736$ , we get  $c_{\text{high}} \cong (127)^{-1}$ , as was theoretically predicted by Gell-Mann and Low and subsequently verified experimentally [10]. This confirms the fact that the distance scale constant is not a universal constant, as previously mentioned.

| Exponents | Power of 10 | Number of times | Inverse of Golden Ratio | Inverse of 89 | Sum of "D" and "E" columns | REMARKS             |                                    |                                                   |
|-----------|-------------|-----------------|-------------------------|---------------|----------------------------|---------------------|------------------------------------|---------------------------------------------------|
| 2         | 1.00E-02    | 0               | 6.18E-03                | 1.12E-03      | 0.0073009353931            | Crude Approximation |                                    |                                                   |
| 5         | 1.00E-05    | 1(O)            | 6.18E-06                | 1.12E-06      | 0.0072966314577            | S                   | 1st group of                       | 1st REGION EXONENTS WITH ZERO ORDER OF MAGNITUDE  |
| 6         | 1.00E-06    | 1(O)            | 6.18E-07                | 1.12E-07      | 0.0072973618513            | E                   |                                    |                                                   |
| 7         | 1.00E-07    |                 | 6.18E-08                | 1.12E-08      | 0.0072972888119            | S                   |                                    |                                                   |
| 7         | 1.00E-07    | 2(E)            | 6.18E-08                | 1.12E-08      | 0.0072973618513            | E                   | 1st region                         |                                                   |
| 8         | 1.00E-08    | 1(O)            | 6.18E-09                | 1.12E-09      | 0.0072973545473            | S                   | 2nd                                |                                                   |
| 9         | 1.00E-09    |                 | 6.18E-10                | 1.12E-10      | 0.0072973538169            | S                   | semi-group                         |                                                   |
| 9         | 1.00E-09    |                 | 6.18E-10                | 1.12E-10      | 0.0072973538865            | S                   | of                                 |                                                   |
| 9         | 1.00E-09    | 3(O)            | 6.18E-10                | 1.12E-10      | 0.0072973523561            | S                   | 1st region                         | CHANGE OF ORDER OF MAGNITUDE-SYMMETRY             |
| 10        | 1.00E-10    |                 | 6.18E-11                | 1.12E-11      | 0.0072973524292            | E                   | 1st group of                       | 2nd REGION EXONENTS WITH FIRST ORDER OF MAGNITUDE |
| 10        | 1.00E-10    |                 | 6.18E-11                | 1.12E-11      | 0.0072973525022            | E                   |                                    |                                                   |
| 10        | 1.00E-10    | 3(O)            | 6.18E-11                | 1.12E-11      | 0.0072973525753            | E                   |                                    |                                                   |
| 11        | 1.00E-11    | 1(O)            | 6.18E-12                | 1.12E-12      | 0.0072973525680            | S                   |                                    |                                                   |
| 12        | 1.00E-12    |                 | 6.18E-13                | 1.12E-13      | 0.0072973525687            | E                   | 2nd region                         |                                                   |
| 12        | 1.00E-12    | 2(E)            | 6.18E-13                | 1.12E-13      | 0.0072973525694            | E                   |                                    |                                                   |
| 13        | 1.00E-13    | 1(O)            | 6.18E-14                | 1.12E-14      | 0.0072973525693            | S                   |                                    |                                                   |
| 14        | 1.00E-14    | 1(O)            | 6.18E-15                | 1.12E-15      | 0.0072973525694            | E                   | Experimental Value=0.0072973525693 |                                                   |

Figure 9.3: Geometric calculation of the value of the fine-structure constant ( $\alpha$ )

As follows from the above process, we can now provide an answer to Feynman's famous conjecture, which states:

"There is a most profound and beautiful question associated with the observed coupling constant,  $\alpha$ , the amplitude for a real electron to emit or absorb a real photon. It is a simple number that has been experimentally determined to be close to  $-0.085424$ ." "My physicist friends won't recognize this number, because they like to remember it as the inverse of its square: about 137.03597 with an uncertainty of about 2 in the last decimal place." "It has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it. Immediately you would like to know where this number for a coupling comes from: is it related to  $\pi$  or perhaps to the base of natural logarithms?" "Nobody knows. It's one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding by man. You might say the 'hand of God' wrote that number, and we don't know how He pushed His pencil. We know what kind of a dance to do experimentally to measure this number very accurately, but we don't know what kind of dance to do on the computer to make this number come out, without putting it in secretly!"

### 9.3 The EPR paradox

#### A) quantum entanglement

In this section, we will study the **Einstein-Podolsky-Rosen (EPR) paradox** and interpret it in the context of the representation of conic sections in three-dimensional space. The EPR paradox is described through the following thought experiment: Suppose that at a point in three-dimensional space there is a source that emits electron ( $e$ )-positron ( $e^+$ ) pairs along a straight line segment. Each electron is directed towards one end of this segment, where observer A is, and each positron is directed towards the other end of this segment, where observer B is.

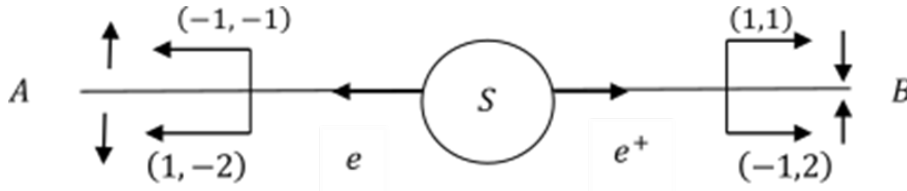


Figure 9.4: Thought experiment for the EPR paradox

According to quantum mechanics, the source is arranged in such a way that each pair ( $e, e^+$ ) it emits forms a state called a **spin singlet state**. The particles in a pair are called “**entangled particles**”. This state is considered to be a quantum superposition of two states, which we call state I and state II. In state I, the electron moves along the line segment  $AB$  towards A with spin  $+1$ , and the positron moves along the same line segment towards B with spin  $-1$ . In state II, the electron has spin  $-1$  and the positron has spin  $+1$ , and they both move along the same line segment as before. Thus, it is impossible to know, without measurement, the value of the spin of one or the other particle in the spin singlet state. Through this thought experiment, the EPR paradox is formulated as follows: Observer A measures the spin of the electron, which belongs to an electron-positron pair, along a specific direction (e.g., perpendicular to  $AB$ ) and finds a specific value for this spin. Observer B measures the spin of the positron along the same direction and finds that it has an opposite value from the value of the spin of the electron. Thus, the following questions arise:

1. How does the positron know that the spin of the electron has this particular value, in order to take the opposite value?
2. If observer A decides to measure the spin of an electron along a random direction, then how does the spin of the positron take an opposite value from that of the electron along the same direction?

The above paradox can be expressed equally well for photon-antiphoton pairs, which belong to the same category, which were studied experimentally by A. Aspect and his collaborators[3], [4],[1],[2].

To explain the above paradox, we will return to  $\mathbf{P}^2$  and conic sections. The electron is associated with a conic section that has a specific characteristic vector from which its reduced charge and orientation are derived. The positron is associated with the conjugate conic section of the electron, that is, with the one that has the opposite characteristic vector to that of the electron. The spin of electrons and the spin of positrons correspond to the orientations of these conic sections. The measurement by observer A of the value and sign of the spin of the electrons along a random axis in three-dimensional space determines the conic section and its orientation in  $\mathbf{P}^2$  to which the electron corresponds. The conjugate conic section to which the positron corresponds always has the opposite orientation to the orientation of the conic section to which the electron corresponds. This means that the measurement of a vector quantity of an elementary particle determines its value as well as automatically determines the value of the same vector quantity of its antiparticle.

Suppose that the free observer A measures the  $x$ -component of the spin of an electron in a coordinate system in three-dimensional space and finds the value -1. The mapping of this measurement to  $\mathbf{P}^2$  is determined by the value -1 of the orientation of the conic section with the characteristic vector  $(1, -2)$ . Through this representation, observer A immediately knows that the characteristic vector of the conjugate conic section is  $(-1, 2)$  with orientation +1, i.e., the source emits a positron which corresponds to this conic section. A future measurement of the spin of the positron along the same axis by observer B will indeed give the value +1.

Measuring the spin of the electron in three-dimensional space eliminates the bound observer's dilemma regarding the type of conic section (i.e., whether it is the unit circle or the ellipse with characteristic vectors  $(-1, -1)$  or  $(1, -2)$  respectively) that should be chosen in  $\mathbf{P}^2$  to correspond to the electron. Through this single measurement, observer A finds both conic sections corresponding to an electron-positron pair in three-dimensional space. If observer B measures the spin of a positron along the  $z$ -axis and at the same time is informed of the result of the spin measurement that observer A has made along the  $x$ -axis, then one can claim that observer B simultaneously knows the exact values of the  $x, z$  components of the spin of the positron. This means that the Heisenberg uncertainty principle does not apply, because observer B knows the values of two different components of the positron's spin simultaneously.

Let us see with an example how ignorance is introduced about the type of conic section or orientation that the bound observers are asked to choose. The free observer  $E_A$  measures the spin of the electron on the  $x$ -axis and finds, for example, that it is in the positive direction (up).  $E_A$  notes this

result as  $e^\uparrow$ . The bound observer  $\Delta_A$ , associated with  $E_A$ , has two choices for the geometric object corresponding to the electron, such that both have the same reduced charge -1. One is the unit circle  $C$  with characteristic vector  $(-1, -1)$  and the other is the ellipse  $E$  with characteristic vector  $(1, -2)$ . From the circle we get an up orientation while from the ellipse we get a down orientation. From the correspondences of geometric objects and their holograms, we choose the circle with an up orientation to correspond to the electron with spin up.

The free observer  $E_B$  measures the spin of the positron on the  $z$ -axis and finds, for example, that it is in the positive direction (up).  $E_B$  notes this result as  $p^\uparrow$ . The bound observer  $\Delta_B$ , associated with  $E_B$ , has two choices for the geometric object corresponding to the positron, such that both have the same reduced charge +1. One is the unit circle  $C$  with characteristic vector  $(1, 1)$  and the other is the ellipse  $E$  with characteristic vector  $(-1, 2)$ . From the circle we get the down orientation while from the ellipse we get the up orientation. From the correspondences of geometric objects and their holograms, we choose the ellipse with the up orientation to correspond to the positron with spin up. In the above example, the bound observer  $\Delta_B$  receives the information  $C \uparrow$  from  $E_A$  and  $E \uparrow$  from  $E_B$ . All possible cases that  $\Delta_B$  faces are the following four cases, as illustrated in the following arrangement:

$$\uparrow\uparrow \rightarrow CE \quad \uparrow\downarrow \rightarrow CC \quad \downarrow\uparrow \rightarrow EE \quad \downarrow\downarrow \rightarrow EC$$

From this arrangement, the observer  $\Delta_B$  at the same time has complete ignorance about whether the conic section is  $C$  or  $E$  if the orientations are the same, that is,  $\uparrow\uparrow$  or  $\downarrow\downarrow$ , and also has complete ignorance about whether the orientations are  $\uparrow\downarrow$  or  $\downarrow\uparrow$  if the conic sections are the same. Due to the 1-1 correspondences, this ignorance is also transmitted to the free observer  $E_B$ , so the Heisenberg uncertainty principle still holds as does complementarity.

The measurement procedure performed above is consistent with the quantum mechanical formalism, which is expressed in terms of the spin singlet state as follows:

$$\Psi = \frac{1}{\sqrt{2}}(\psi_+(e)\psi_-(p) - \psi_-(e)\psi_+(p))$$

The  $+$ ,  $-$  indices refer to the spin directions along an axis and correspond to the reduced  $\pm 1$  values of spin. Each term of the spin singlet state shows the components of the spin of an electron-positron pair along this axis, that is, all possible directions of the spins of this pair along each axis.

The term **collapse of the wave function**  $\Psi$ , used in quantum mechanics during a measurement, means that only one term of the wave function  $\Psi$  survives to give the correct result, without the explanation of why this collapse occurs. The “collapse of the wave function  $\Psi$ ”, according to the correspondences of conic sections with physical entities, means the following:

When a free observer makes a measurement of a quantity, then the bound observer associated with him chooses a specific conic section from all possible conic sections, which is compatible with the measurement made by the free observer.

In the above example, it was assumed that an electron can be associated on the one hand with a circle with characteristic vector  $(-1, -1)$  and on the other hand with an ellipse with characteristic vector  $(1, -2)$ . This means that the electron in the first case displays only its particle nature (due to its correspondence with a circle) and in the second case it displays only its wave nature (due to its correspondence with a linear ellipse which is considered a segment of a linear hyperbola). This behavior of the electron exists and is evident during the creation of the isotopes of Hydrogen[13]. This interpretation gives an answer to the so-called **measurement paradox**; that is, how and why the collapse of a wave function occurs.

During the measurement, one of the two conic sections is chosen by the bound observer in agreement with the measurement or observation, so the wave function “collapsed” into a single part of it. In other words, the choice of the appropriate conic section by the bound observer is interpreted as a collapse of the total wave function into a part of it by the free observer. The modeling of the observations or measurements of a free observer in three-dimensional space, carried out by the bound observer on a two-dimensional surface or on a very small part of it that resembles  $\mathbf{R}^2$  (such as the cerebral cortex), also leads us to understand how the human brain works, by constructing simple models for the interpretation of physical phenomena.

The construction of the spin singlet state for a quantum mechanical interpretation of the EPR paradox, although formally correct, implies a **spooky action at a distance**” (or *spukhafte Fernwirkung*” in German) according to Einstein. During each measurement, according to quantum mechanics, it is believed that between the members of an electron-positron pair there is a mysterious coupling, called **quantum entanglement**, which is arbitrarily introduced to complete the quantum mechanical interpretation of the EPR paradox in three-dimensional space. However, through correspondences between conic sections and elementary particles and the “cooperation of bound and free observers”, the nature of this mysterious coupling is revealed. Thus, the above process becomes clear. We summarize this process as follows:

An observer measures the value of a vector quantity of an elementary particle in three-dimensional space and identifies the value of this vector quantity with the value of the orientation of a conic section corresponding to the elementary particle.

The observer selects only the conic section with a characteristic vector, from the pair of possible characteristic vectors, that is compatible with the measurement. A single measurement of a vector quantity of a particle in three-dimensional space automatically determines the conjugate conic section corresponding to its antiparticle. Through this process, the interpreta-



tion of the measurements is immediately understandable.

We assume that each member of the positron-electron pair emitted by the source moves along a straight line with a specific direction (*Order*), that is, the electron always moves in one direction (to the left) and the positron in the opposite direction (to the right). At some points on both sides of the source, there are two observers A (left) and B (right) who are equipped with instruments that measure the spin of the electron. The two observers are clearly separated from each other by measuring the same amount of two different particles in two different directions (*Asymmetry*), as shown in Figure 9.4. In this version of the EPR paradox, we have the following arrangement:

$$Order \rightarrow Asymmetry \rightarrow Quantum\ entanglement$$

At this point, we will give the definition of **locality** and the definition of a **realistic theory**, to see how the theory of conic sections in  $\mathbf{R}^2$  and the theory of their holograms (Quantum Mechanics) in three-dimensional space can be characterized.

We define a **local theory** as one where the results of an experiment on one system are independent of the actions performed on a different system that has no causal relationship with the first.

We define a theory as **realistic** if experiments reveal properties that already exist. In other words, an existing property of a system is independent of whether or not this property is measured.

The result of the experiments measuring the spin in the electron-positron system shows that there is a causal connection between them in three-dimensional space (quantum entanglement). Also, there is a causal connection between the orientations of geometric objects in  $\mathbf{R}^2$  that correspond to the spin, through the 1-1 mappings, because these geometric objects are conjugate conic sections. This observation confirms the fact that our theory is *non-local* in both spaces. Also, in our theory, the values of the orientations of the conic sections pre-exist in their creation space, i.e., in  $\mathbf{R}^2$ , as well as the values of the spins, due to the 1-1 correspondence, so the experiment simply reveals them. According to the definition of a realistic theory, our theory is *realistic*. Thus, the features of our theory are compatible with **Bell's theorem**, which is formulated as follows[16]:

$$A\ quantum\ theory\ cannot\ be\ both\ local\ and\ realistic''$$

or

$$A\ quantum\ theory\ is\ either\ non-local\ or\ non-realistic''$$

Through this theory, quantum entanglement is interpreted through geometric arguments, such as the correlation of vector quantities of geometric

objects, which through 1-1 correspondences appear as correlations of vector quantities of physical entities. The correlation that exists between the members of an electron-positron pair is determined from the principle and is precisely determined by a single measurement of the physical vector quantity of one member of this pair in three-dimensional space. This “correlation” holds everywhere in three-dimensional space and does not depend on the distance between the members of an electron-positron pair, thus defining the principle of information conservation in  $\mathbf{R}^2$ , as well as in three-dimensional space. The question that immediately arises for the principle of information conservation in three-dimensional space is the following: **“Will the information that an elementary particle once existed (free or as a component of a complex structure) be lost when this particle is absorbed by a Black Hole and then the Black Hole evaporates, via thermal radiation, due to the Hawking phenomenon?”**

We can give a simple interpretation of this paradox, according to what we have said so far. A free elementary particle or a member of a complex structure in three-dimensional space is a representation of some geometric object on the flat boundary of the Universe ( $\mathbf{P}^2$  space), which does not undergo any change. If we follow the course of intercommunication and cooperation between the two observers by transferring information from the bound to the free observer and vice versa to extract the final conclusions, as has been done so far, then **the principle of conservation of information applies as well as Quantum Mechanics.**

## B) Complementarity

Let us consider a second version of the EPR paradox, where a positron or an electron is emitted randomly in every direction from the source S. In this case, the free observers do not know in advance the type of particle emitted towards them. This randomness in the emission of particles makes the two observers equivalent and not distinct, as in the previous section. In other words, the **randomness** in the emission of one or the other particle creates a *symmetry* with respect to the measurements for the two free observers. Thus, we will examine the results of measurements by a single observer (e.g., B) through the following Figure 9.5. If observer B measures the spin of a particle without knowing whether it is an electron ( $e$ ) or a positron ( $e^+$ ) and determines that it is up, then he does not know whether this spin value belongs to  $e$  or  $e^+$ .

If observer B measures the charge of a particle and determines that it is  $e$  or  $e^+$ , then he does not know whether its spin is up or down. A single measurement (of spin or charge) leads to uncertainty about the value of the other quantity, or in other words, **complementarity** is introduced according to the Copenhagen interpretation, which says that two conjugate

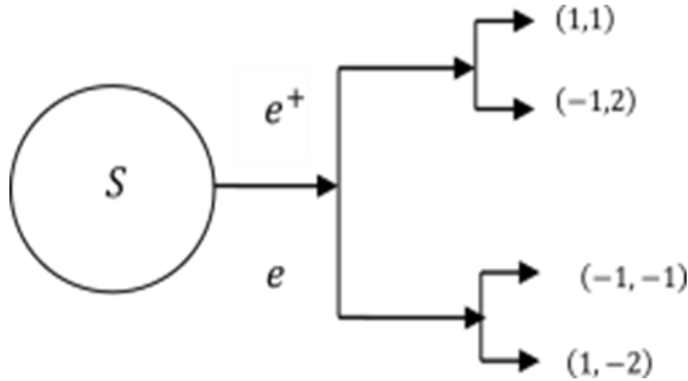


Figure 9.5: Experiment for complementarity in the EPR paradox

variables cannot be determined simultaneously[23]. In this version of the EPR paradox, we have the following arrangement:

$$Randomness \rightarrow Symmetry \rightarrow Complementarity$$

## 9.4 Born's Rule

The persistent mathematical incompatibility between the smooth, geometric spacetime of General Relativity and the probabilistic, linear Hilbert spaces of Quantum Mechanics represents the central crisis of modern theoretical physics. The Square Screen Model resolves this tension by demonstrating that both regimes emerge naturally from a single, discrete, self-replicating 2D boundary screen. We establish this model on three core postulates:

1. **The 2D Holographic Boundary:** The fundamental degrees of freedom of the Universe reside entirely on a flat, two-dimensional screen; 3D space is an emergent projection.
2. **Strict Outward Self-Similarity:** The screen evolves discretely via structural self-similarity (fractal scaling).
3. **Information Irreversibility:** Information packets (particles) are constrained to propagate exclusively along newly generated geometric states.

### 9.4.1 The Geometric Growth Engine ( $1 \rightarrow 4$ )

The dynamics of the holographic screen are driven by a discrete replacement rule. Let  $t_0$  represent the initial state containing a single square cell of fundamental area  $A_0 = l_P^2 = 1$ . The length of each side is the "Planck's length

$l_P$ ". To transition to the next temporal stage  $t_1$ , the initial square produces 3 additional, identical copies of itself because of self-similarity between successive Plank areas. These 3 new copies assemble around the original unit to construct a single, perfectly bound, larger square. The total number of units at any given stage  $n$  follows a strict power law:

$$N(n) = 4^n, \text{ where } n = 0, 1, 2, \dots \quad (9.28)$$

This scaling behavior matches the geometric requirement for doubling the side length ( $L$ ) of a 2D square, where  $\text{Area} \propto L^2$ .

### 9.4.2 Geometric Derivation of Born's Rule

We define the border of the Universe as a flat 2D-space. In conventional quantum mechanics, the probability of finding a particle is given by Born's Rule:  $P(x) = |\psi(x)|^2$ . For a century, physics has treated the power of 2 as an empirical axiom. The **Square Screen Model (Planck patch Model)** derives this exponent directly from spatial scaling. A quantum wave function  $\psi$  represents an information field mapped onto our 2D screen. When the screen steps from  $t_0 \rightarrow t_1$ , the underlying geometric grid expands by a factor of 4 ( $2^2$ ).

To preserve total probability at exactly 1 (100%) across the expanding system, the probability density must distribute evenly across these 4 available quadrants:

$$\sum_{i=1}^{4^n} P_i = \sum_{i=1}^{4^n} |\psi_i|^2 = 1 \quad (9.29)$$

If probability scaled linearly (power of 1) or cubically (power of 3), the physical information would either tear or overlap, failing to cleanly carpet the newly emerged 4-part geometric surface. The quadratic power is the unique exponent that matches the 2D surface scaling of our  $1 + 3 = 4$  growth rule.

## Chapter 10

# TOPICS OF OPTICS

### 10.1 Partial Reflection of Light by Glass

Let us assume that monochromatic light (e.g., red) is emitted by a light source and falls on a piece of glass in the shape of a rectangular block. In front of the glass at position A, there is a photomultiplier that collects the light (photons) reflected from the first surface of the glass. To measure the photons that pass through the glass, we place a second photomultiplier at position B inside the glass, as shown in Figure 10.1.

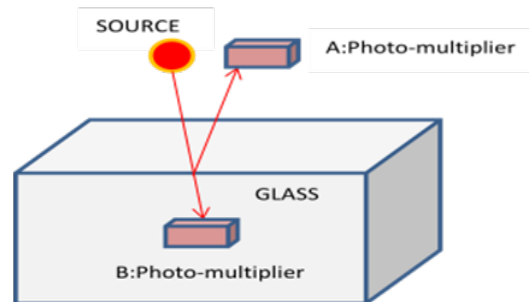


Figure 10.1: Partial reflection from the 1st surface of the glass

What the photomultiplier at position A measures is 4% of the photons emitted by the source and reflected at the 1st surface of the glass. What the photomultiplier at position B measures is 96% of the photons emitted by the source and passing through the glass.

If the photomultiplier at position B is placed outside the thin glass plate and in alignment with the source, then we expect that the 2nd surface of the glass will reflect 4% of the photons that penetrate the glass and fall on it, so we would again get  $96\% \times 4\% = 3.84\% \sim 4\%$ , as shown in Figure 10.2.

According to the above, approximately 8 out of 100 photons should reach the photomultiplier at position A. However, what we measure is that rarely

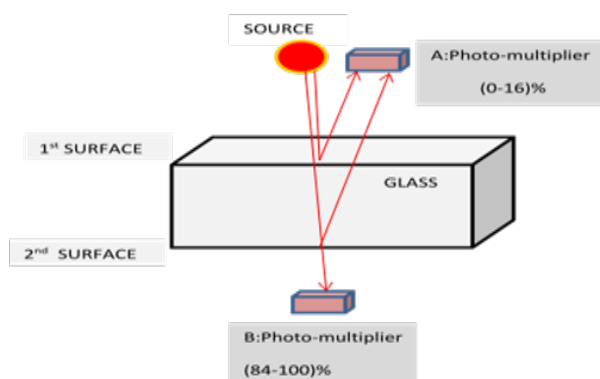


Figure 10.2: Partial reflection from the 1st and 2nd glass surfaces

8 photons reach it, and most of the time 15 or 16 photons do; that is, we get twice the number of photons than we expect.

The 1st question is: **Why does the numerical value of the percentage of partial reflection take this specific value and not a random value, given that photons have no mass or charge, therefore the reflection or passage of the initial 100 photons from or to the glass respectively should be a random number in the interval  $[0 - 100]$ ?**

The 2nd question is: **What happens when we reduce or increase the thickness of the glass?**

The only conclusion that can be drawn from such experiments is that the thickness of the glass plays an important role in the partial reflection of light from both surfaces of a piece of glass in the shape of a rectangular block. So, starting from a very thin plate of glass, we will measure the percentage of photons that reach the photomultiplier at position A.

First, we see that with the minimum possible thickness of glass, the number of photons that reach the photomultiplier at position A is almost always zero and only a few times 1. By gradually increasing the thickness of the glass, we find that the number of reflected photons ranges in the interval  $[0, 16]$  from the total of 100 photons. With slightly thicker plates of glass, we find that the number of photons that reach A approaches 8%. With even thicker plates of glass, we find that the number of photons that reach A exceeds 8%.

Increasing the thickness of the glass further, that is, taking a thickness of 0.0127 cm, we find that the number of photons reaching A takes on a maximum value of 16% and then follows the reverse course; that is, as the thickness of the glass increases, the percentage of reflected photons decreases continuously until it reaches the value of zero.

At this point, the first cycle of fluctuations in the percentages of reflected photons from both surfaces of the glass closes. We continue to increase the thickness of the glass in order to successively obtain the same circles as the

first. Thus, the periodic circles are determined, which today have reached the number  $10^8$  for glass 50 m thick, where laser beams that provide purely monochromatic light are used as the photon source.

These findings lead us to think that these rates, which are relatively constant, should be due to the structure of space followed by the photons emitted by a light source. Also, this process indirectly confirms that the space  $\mathbf{P}^2$  consists of Planck patches, through the rates of reflected photons from the two surfaces of the glass.

The holographic representation of the structure of the space  $\mathbf{P}^2$  in three-dimensional space for any number of elementary particles should give us answers for the numerical values of various sizes. Thus, we expect answers and interpretations for the numerical values of the rates of reflected photons. The same procedure that we examine here will be used for the rates of the three generations of solar neutrinos, which are particles with negligible mass and charge that resemble photons (Appendix 6, Section 6.3).

As we will see in Appendix 5, after the initial expansion of the initial Planck point, there was a sharp increase in space (inflation) where, after the structural unit  $P_1$  (Planck point) of  $\mathbf{P}^2$ , the Planck structural unit  $P_9$  was created.  $P_9$  consists of  $9 \times 9$  Planck points that have  $10 \times 10 = 100$  geometric points corresponding to photons.

When such a loose structure of photons collides with a solid structure (such as any glass surface), then the four photons of the vertices of  $P_9$  will be removed from the 1st and 2nd surfaces. This happens because only these have greater freedom to move, because they are free in two directions (one along the  $\pm x$  axis and one along the  $\pm y$  axis). Thus, the four “cornerstones” of  $P_9$  are removed from this structure.

If the glass thickness is as small as possible, then all the photons removed from the structure  $P_9$ , due to the collision with the two glass surfaces, can almost immediately reunite with this structure because in  $P_9$  the structure of the structural unit of  $\mathbf{P}^2$  should reappear. But during this process, one of the photons of  $P_9$  can be randomly absorbed by the photomultiplier A, so we get the value 1 from it. This is extremely rare under these conditions, because the dominant process is the reappearance of the structure of  $\mathbf{P}^2$ .

By increasing the thickness of the glass, the structure of  $P_9$  evolves for a longer time inside the glass, so not only do the four photons not reconnect with  $P_9$ , but also its perimeter dissolves, allowing greater freedom of movement for its peripheral photons. This results in the Planck patch  $P_7$  remaining as a “compact” structure with  $8 \times 8 = 64$  geometric points.

With a continuous increase in the thickness of the glass, we will also have a dissolution of the next Planck patches, namely  $P_7$ ,  $P_5$ , and  $P_3$ . The Planck patch  $P_1$  does not dissolve further because this is the initial building block of  $\mathbf{P}^2$ . Taking successively for each of the Planck patches  $P_9$ ,  $P_7$ ,  $P_5$ , and  $P_3$  the four photons of its four vertices that are successively removed, we interpret the percentages of 0%, 4%, 8%, and 16% of the reflected photons

as the thickness of the glass increases.

According to this view, on the one hand, the successive percentages from 0% to 16% of the reflected photons and, on the other hand, the structure of space through the Planck patches are verified.



Figure 10.3: Photon grid in a  $P_9$  patch before the collision with a glass surface

Since there is a continuous flow of photons from the source and a continuous increase in the thickness of the glass, then there is a continuous repetition of the partial reflection process, because the free photons reunite to form the structural unit of  $\mathbf{P}^2$  again, that is, the  $P_9$  Planck patch. From a table similar to the one above, we will explain the mystery of the lost neutrinos coming to Earth from the Sun's core.

## 10.2 Reflection of monochromatic light from a mirror

From the light source, photons are emitted in all directions towards the mirror from its left to right end and are reflected by it. To prevent the photomultiplier from detecting photons directly from the source, we place a black screen (curtain) between them, as shown in Figure 10.4.

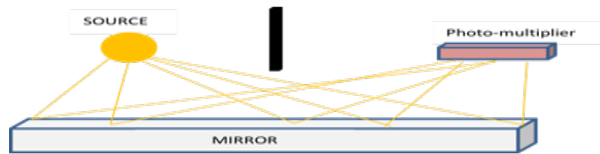


Figure 10.4: Reflection of light in a mirror

For each photon in three-dimensional space, whose point of incidence is any point on the mirror, the equation that characterizes its propagation is the wave equation. The wave equation corresponds to  $\mathbf{P}^2$  with the equation of a parabola. Since this parabola must pass through the source and the photomultiplier and is by definition a symmetric curve, then its axis of symmetry must be perpendicular to the middle of the mirror. Thus, the total path (incidence-reflection path) of each photon at each point on the mirror must be modified to become a parabola. This does not mean that all parabolas are the same, but that they share a common axis of symmetry.



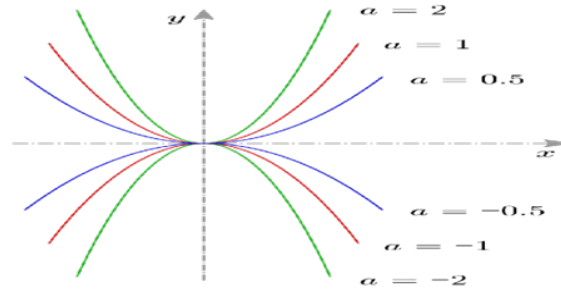


Figure 10.5: Parabolas with a common vertex - Average parabola

In three-dimensional space, the adjustment of each such path to form a symmetrical curve is achieved through the wave nature of the photon, where the perpendicular to the middle of the mirror becomes the axis of symmetry of each parabola originating from each point of the mirror. The equation of the parabola is:

$$y_0 = \frac{x^2}{2a}$$

The parameter  $a$  has units of length, therefore the equation of the isosceles hyperbola can be derived, which is:

$$y^2 - x^2 = 0$$

where  $2ay_0 = y^2$ . The equation of this isosceles hyperbola is given by the pair of equations:

$$y = \pm x$$

of the right-angle bisectors of the Cartesian reference system. Due to the above observations, we can study the parabolas through these bisectors. The set of parabolas in the above figure can be reduced to a single parabola, which expresses an “average parabola” of all of them. The average parabola in this figure is denoted by the color red. The construction of the average parabola can be done through the average value of the sum of each member of the equations expressing the asymptotes of the isosceles hyperbolas that have equations:

$$y = \pm x$$

These isosceles hyperbolas are equivalent to these parabolas through equations of the form:

$$\sqrt{2ay_0} = \pm x$$

Each parabola of the set of these parabolas differs from the others in the value of the parameter  $a$ . The concept of the average parabola is consistent with the empirical rules that tell us that:

- The angle of incidence is equal to the angle of reflection.
- Reflection occurs in the middle of the mirror.
- The path followed by the light is the shortest possible, therefore the time it takes is the shortest possible due to the constant speed of light (Fermat's principle).

### 10.3 Diffraction of white light by a mirror

Let's divide the mirror into parts of equal length starting from the left end. We carefully etch the mirror starting from the first part. This part of the mirror is called the diffraction grating. The second part remains unchanged. In the third part of the mirror, we etch it so that the distance between two adjacent slits is smaller. We continue in this way by constructing different diffraction gratings from their etched neighboring parts. This process in three-dimensional space is interpreted in  $\mathbf{P}^2$  by parabolas that have almost the same  $a$  for each diffraction grating. When this section is illuminated with white light, it will give a color corresponding to its wavelength, which is equal to the distance between two adjacent slits. When the entire mirror is illuminated with white light, it will give all the colors of the visible spectrum from red to blue. Another way to obtain all the colors of the visible spectrum with a single drawing (e.g., with the one that takes the red color) of all the diffraction gratings of a mirror is to successively change the angle of observation of the reflected beam, as shown in Figure 10.6. Through this

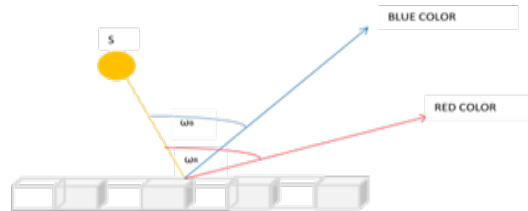


Figure 10.6: Diffraction Grating That Works for All Colors

process, we change the wavelength, resulting in shorter wavelengths than that of the red color, changing the angle of observation, so the spectrum moves towards the blue color. We apply the same procedure as in the case of reflection. From the equation of the parabola:

$$x_0 = \frac{1}{2a}y^2$$

we obtain the equation of the isosceles hyperbola:

$$x^2 - y^2 = 0$$

where  $x = \sqrt{2ax_0}$ . The equation of the isosceles hyperbola results from the equations of the right-angle bisectors of the rectangular reference system, which are:

$$y \pm x = 0 \quad \text{or} \quad y \pm \sqrt{2ax_0} = 0$$

Thus, we obtain a set of hyperbolas, each of which depends on the value of the parameter  $a$ . On the other hand, the wave equation for a photon is of the form:

$$\left(\frac{\partial}{c\partial t}\right)^2 \Psi - \mu^2 \nabla^2 \Psi = 0$$

From the similarity of these two equations in  $\mathbf{P}^2$  and in three-dimensional space, we make the correspondences as follows:

$$\begin{aligned} \sqrt{2ax_0} &\leftrightarrow \frac{\partial}{c\partial t} \\ y &\leftrightarrow \mu \cdot \nabla \end{aligned}$$

From the first correspondence, we make the correspondence:

$$a \leftrightarrow \frac{1}{c^2} = \frac{1}{(\lambda\nu)^2}$$

where  $\lambda$  is the wavelength and  $\nu$  is the frequency of the wave. Initially, we see red color from the diffraction grating, so we make the correspondence:

$$a \leftrightarrow \frac{1}{c^2} = \frac{1}{(\lambda_{\text{red}} \cdot \nu_{\text{red}})^2}$$

When we change the observation angle  $\omega_{\text{red}} \rightarrow \omega_{\text{blue}}$  gradually, where  $\omega_{\text{blue}} < \omega_{\text{red}}$ , the above relation becomes:

$$a \leftrightarrow \frac{1}{c^2} = \frac{1}{(\lambda_{\text{blue}} \nu_{\text{red}})^2}$$

Thus, by changing the observation angle, we change the wavelength so that it takes on smaller values. After a critical angle  $\omega_{\text{cr,red}}$  and due to the relationship  $c = \lambda\nu = \text{constant}$ , we have:

$$\nu_{\text{red}} \leftrightarrow \nu_{\text{blue}}$$

and the color of the reflected beam now becomes blue.

## 10.4 Refraction of light between two media

When a photon passes from air to water, its speed decreases due to its interaction with an electron. The equations that define the photon-electron interaction in  $\mathbf{P}^2$  are the sum of the equations of the asymptotes:

$$y_\gamma = \pm x_\gamma \implies (y_\gamma)^2 - (x_\gamma)^2 = 0$$

of the hyperbola defining the photon ( $\gamma$ ) and the equations of the asymptotes:

$$y_e \pm x_e = V_e \implies y_e^2 - x_e^2 = V_e^2$$

of the hyperbola defining the electron ( $e$ ). In three-dimensional space, the hyperbolas correspond to waves, so the photon-electron interaction is due to the contribution of these two waves. As we know from reflection, light follows the minimum path that is a consequence of the structure of the parabola, which is a symmetric curve, as shown in Figure 10.7. Due to

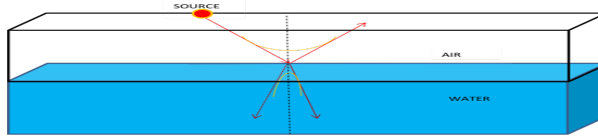


Figure 10.7: Light refraction between air and water

the fact that light moves at a constant speed in any given medium, then the time interval during which the photon moves in two media takes its minimum value (principle of minimum time or Fermat's principle). Snell's law can be derived from Fermat's principle, which states that light travels the path that takes the least time. Taking the derivative of the optical path length, the stationary point is found to give the path that light follows. Because the refractive indices of air and water are different, the fastest way for a rescuer to reach a drowning person at sea is to run along a path that follows Snell's law. Let us assume that the refractive indices of medium 1 and medium 2

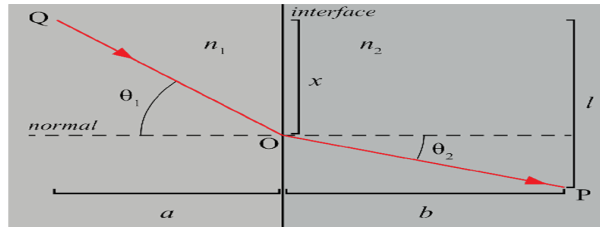


Figure 10.8: Refraction of light

are  $n_1$  and  $n_2$  respectively. Light enters medium 2 from medium 1 through point  $O$ .  $\theta_1$  is the angle of incidence,  $\theta_2$  is the angle of refraction with respect to the normal. The phase velocities of light in medium 1 and medium 2 are  $v_1 = \frac{c}{n_1}$  and  $v_2 = \frac{c}{n_2}$  respectively, where  $c$  is the speed of light in a vacuum. Let  $t$  be the time required for light to travel from point  $Q$  through point  $O$  to point  $P$ :

$$t = \frac{(x^2 + a^2)^{\frac{1}{2}}}{v_1} + \frac{((l - x)^2 + b^2)^{\frac{1}{2}}}{v_2}$$

$a, b, l$  and  $x$  are denoted in the previous figure, with  $x$  being the variable parameter. To minimize this, we can take the derivative of this relation,

namely:

$$\frac{dt}{dx} = \frac{x}{v_1(x^2 + a^2)^{\frac{1}{2}}} - \frac{l - x}{v_2((l - x)^2 + b^2)^{\frac{1}{2}}} = \frac{\sin \theta_1}{v_1} - \frac{\sin \theta_2}{v_2} = 0$$



## Chapter 11

# FOUNDATION OF GENERAL RELATIVITY

### 11.1 Definition of a differential manifold

In this Chapter, we will see how the definition relations of conic sections apply to a *differentiable manifold*  $M$  by free observers of three-dimensional space. From this process, free observers find equations of the same form as the equations of General Relativity of Einstein, *without using the equivalence principle*. Thus, these equations, free from the equivalence principle related to “gravity”, can be used at all scales of Physics, that is, in the microcosm and in the macrocosm.

Let us consider a space constructed of “points”, each of which consists of a random set of Planck points (a Planck random patch). This means that the “points” of this space are different from each other. The differentiation of the points of this space lies in the fact that the inscribed or circumscribed circles in the random Planck patches are different from each other, so their curvatures (or correspondingly their masses) are different.

From this structure, it follows that the space cannot be described by a single coordinate system, as is the case with Euclidean spaces (such as  $\mathbf{P}^2$ ) whose all points (geometric or Planck points) are considered equivalent.

In the chapter Classical Mechanics, we considered that a point circle in  $\mathbf{P}^2$  is depicted as a material point in three-dimensional space, creating a local anisotropy in its neighborhood due to the mass attributed to it. In other words, the neighborhood of this material point exhibits curvature. In order to calculate various physical quantities, a curvilinear coordinate system must be defined in this neighborhood.

Suppose that a second material point is located in the neighborhood of the first, with a different mass than that of the first. We assume that these two points are correlated with each other. In the case of Classical Mechanics, it was assumed that the three-dimensional space does not change its

basic characteristics, that is, that it continues to be isotropic and homogeneous. However, in curved spaces, the second material point creates its own neighborhood in which curvature also appears, different from that of the neighborhood of the first material point. Therefore, a second, different coordinate system is required in addition to the coordinate system of the first material point, in order to calculate various physical quantities in the neighborhood of the second material point. Since these two material points are considered to be correlated, then this correlation must be expressed as a correlation of both coordinate systems.

When the space consists of geometric points, the concept of a point's neighborhood is ambiguous because there are infinitely many other geometric points nearby. However, if we consider the space to consist of "points", each of which is a random few-member Planck patch, then we can have a clear idea of the neighborhood of each Planck patch. The question that arises in this construction is: *How are two adjacent random Planck patches related?*

We will answer this question through an example. We consider a Planck square consisting of four Planck points. Within this Planck square, the following Planck patches can be defined:

1. Planck patch with one Planck point
2. Planck patch with two Planck points
3. Planck patch with three Planck points
4. Planck patch with four Planck points

Macroscopically, the entire Planck square can be considered as a flat two-dimensional surface. Microscopically, each of the above Planck patches can be considered to exhibit curvature, which is the curvature of the inscribed or circumscribed circle in it. According to this microscopic view, the four aforementioned Planck patches have four different curvatures and common overlapping regions.

In general, any collection of such random Planck patches is called a **differentiable manifold**  $M$ . The basic characteristics of a differentiable manifold are that it is covered by a set of coordinate neighborhoods where each neighborhood has the same number of coordinates as every other. This is because all these neighborhoods are parts of a Planck patch that macroscopically belongs to a flat space with a certain number of dimensions. Thus, its parts will have the same number of dimensions as the total space. Also, a basic condition that must be satisfied by two different neighboring coordinate systems is that in a common region, they are related by a differentiable transformation of class at least  $C^1$ . The construction of the differentiable manifold from a flat space is in agreement with the observational data from which we conclude that the Universe as a whole is a flat space.



## 11.2 Definition of mass

Since the neighborhood of a “point” (which is a random Planck patch) of a differentiable manifold  $M$  can also be considered as a neighborhood of Euclidean space according to the previous construction, then a conic section can be defined in this neighborhood. In the considered flat neighborhood of this point of the differentiable manifold  $M$ , we write the general equation of conic sections. Thus, we get:

$$\mathbf{R} = dV + \tilde{\mathbf{D}} \quad (11.1)$$

In the case of applying this relation to a circle in  $\mathbf{P}^2$ , the quantity  $dV$  expresses the association of the center of the circle with a point on its circumference. This correlation results from the change of the tangent at two infinitely close points on the circumference of the circle, defining the “acceleration” of the point. The quantity  $\tilde{\mathbf{D}}$  gives the measure of this correlation for bound observers, because it is the mapping of the term  $\mathbf{r}_c \left( \frac{k}{r_c} \right)$  containing the  $k$  from which the mass or charge is derived. Based on these observations, free observers on a differentiable manifold follow a corresponding procedure as follows:

At each point of a differentiable manifold, as we defined it previously, that is, through a few-member Planck patch, a matrix  $V$  is attached, whose elements are expressed by the basis  $(dx^i)$  of 1-forms, resulting from the coordinate system  $(x^i)$  associated with this point. The matrix  $V$  of a point of a differentiable manifold curve is called the *affine connection*. The elements  $V_j^i$  are expressed as[24]:

$$V_j^i = \Gamma_{jk}^i dx^k \quad (11.2)$$

The coefficients  $\Gamma_{jk}^i$  belong to the set  $\Lambda^0(M)$ , where the set  $\Lambda^0(M)$  includes all functions of  $M$  called the *connection coefficients*. At each point of a curve of a neighborhood, there is attached a matrix  $V$ , whose elements are homogeneous of first degree with respect to  $dx^k$ , as follows from the isomorphic mapping between the tangent spaces of two points belonging to this curve. The matrix  $V$  is the mapping of an “intrinsic connection” of a point of  $\mathbf{R}^2$  to an intrinsic connection attached to a point of the differentiable manifold  $M$ . The elements  $dV_j^i$  of  $dV$  in equation 11.1 are 2-forms, because they are expressed as a product of two differentials (here of  $dx^p \wedge dx^k$ ), that is,

$$dV_j^i = d\Gamma_{jk}^i \wedge dx^k = (\partial_p \Gamma_{jk}^i) dx^p \wedge dx^k \quad (11.3)$$

The quantity  $dV$  corresponds to the change in the tangent, that is, to the acceleration that relates the center of a circle and a point on its circumference and characterizes this point. We will now examine how  $\tilde{\mathbf{D}}$ , on the right-hand

side of equation 11.1, is expressed in each neighborhood of a differentiable manifold.

Free observers call this neighborhood of a differentiable manifold the *central neighborhood*, in correspondence with the circle of  $\mathbf{P}^2$ . At each point of the central neighborhood, the intrinsic connection  $V$  and its variation  $dV$  are attached. We consider all the curves passing through each point of this neighborhood that connect it to all other points of this neighborhood. Since all points of each neighborhood of the differentiable manifold  $M$  are characterized by a matrix  $V$  and are correlated with each other, then the intrinsic connections of the points of the central neighborhood are correlated with each other.

In order to express these correlations mathematically, we construct a space with basis elements  $dx^r \wedge dx^q$ , in which the 2-forms are defined. This basis of 2-forms results from the composition of the dual spaces with bases of the 1-forms of the tangent spaces of the neighboring points of a central neighborhood. Consequently, the elements of the matrix  $\tilde{\mathbf{D}}$  express all possible interactions of any two points of a central neighborhood. Thus, the elements of the matrix  $\tilde{\mathbf{D}}$  can be written as the sum of all intrinsic connections  $V_s^i$  of all points of this neighborhood, as follows:

$$\tilde{\mathbf{D}}_j^i = V_s^i \wedge V_j^s = \Gamma_{sr}^i dx^r \wedge \Gamma_{jq}^s dx^q \quad (11.4)$$

where the summation is done with respect to  $s$ . Thus, to the elements of the array  $dV_j^i$ , the elements of the array  $\tilde{\mathbf{D}}$  are added. Through equation 11.1, the concept of the *source-mass* is defined, which is expressed by  $\tilde{\mathbf{D}}_j^i$  in a central neighborhood. This procedure is similar to that followed in the derivation of Maxwell's equations, where  $\tilde{\mathbf{D}}$  was interpreted there as a source of charge[13].

### 11.3 The general equations

Having determined the elements of the matrices  $dV_j^i$  and  $\tilde{\mathbf{D}}_j^i$  of the terms  $dV$  and  $\tilde{\mathbf{D}}$  respectively of the right-hand side of equation 11.1, then the question that arises is what the elements of the matrix  $R_j^i$  of  $\mathbf{R}$  express on the left-hand side of this relation. We write the relation 11.1 in terms of the components of a mixed tensor as follows:

$$R_j^i = dV_j^i + V_s^i \wedge V_j^s \quad (11.5)$$

Substituting  $dV_j^i$  and  $V_s^i \wedge V_j^s$  from equations 11.3 and 11.4 into equation 11.5, we get:

$$R_j^i = d\Gamma_{jk}^i \wedge dx^k + \Gamma_{sr}^i dx^r \wedge \Gamma_{jq}^s dx^q$$

Due to the **antisymmetry rule**, that is:

$$dx^r \wedge dx^q = -dx^q \wedge dx^r$$

we get:

$$R_j^i = (\partial_p \Gamma_{jk}^i - \partial_k \Gamma_{jp}^i) dx^p \wedge dx^k + (\Gamma_{sr}^i \Gamma_{jq}^s - \Gamma_{rs}^i \Gamma_{qj}^s) dx^r \wedge dx^q$$

By appropriately changing the dummy indices of the above equation, we get the relation:

$$R_j^i = (\partial_p \Gamma_{jk}^i - \partial_k \Gamma_{jp}^i + \Gamma_{sp}^i \Gamma_{jk}^s - \Gamma_{ps}^i \Gamma_{kj}^s) dx^p \wedge dx^k \quad (11.6)$$

The expression in parentheses represents the components of the **curvature tensor**, that is:

$$R_{jkp}^i = \partial_p \Gamma_{jk}^i - \partial_k \Gamma_{jp}^i + \Gamma_{sp}^i \Gamma_{jk}^s - \Gamma_{ps}^i \Gamma_{kj}^s \quad (11.7)$$

We can use a symmetric intrinsic connection for a central neighborhood point (**central point**), where the coefficients are the **Christoffel symbols**, i.e., the symbols  $\Gamma_{jk}^i$ , which are defined as symmetric functions with respect to their lower indices:

$$\Gamma_{jk}^i = \Gamma_{kj}^i \quad (11.8)$$

The **torsion tensor** is defined as:

$$T_{jk}^i = \frac{1}{2}(L_{jk}^i - L_{kj}^i) \quad (11.9)$$

For a symmetric affine connection, the torsion tensor is zero because

$$\Gamma_{jk}^i = \Gamma_{kj}^i$$

Acting with the exterior derivative operator  $d$  on both sides of equation 11.1 and using the Poincaré lemma, i.e., the relation  $ddV = 0$ , we get the relation between the elements of the matrices of  $\mathbf{R}$  and  $\tilde{\mathbf{D}}$ , i.e.,

$$dR_j^i = d\tilde{\mathbf{D}}_j^i \quad (11.10)$$

To have the same physical units on both sides of equation 11.10, we write the relation:

$$\tilde{\mathbf{D}} = k\mathbf{T}$$

where  $\mathbf{T}$  is the energy-momentum tensor and  $k$  is a constant with appropriate units. If  $\tilde{\mathbf{D}}$  is a conserved quantity, then  $d\tilde{\mathbf{D}}_j^i = 0$ , so from equation 11.10 we get:

$$dR_j^i = 0$$

The above relation leads us to the **Bianchi identities**. We write equation 11.6 as:

$$R_j^i = R_{j k p}^i dx^p \wedge dx^k \quad (11.11)$$

where  $R_{j k p}^i$  are the components of the Riemann curvature tensor. Acting with the operator  $d$  on equation 11.11, we get:

$$dR_j^i = \nabla_q R_{j k p}^i dx^q \wedge dx^p \wedge dx^k = 0 \quad (11.12)$$

By making a cyclic permutation of the indices  $k, p, q$ , then, due to the wedge product rules, we obtain the relation:

$$(\nabla_m R_{j k l}^i + \nabla_k R_{j l m}^i + \nabla_l R_{j m k}^i) dx^m \wedge dx^k \wedge dx^l = 0$$

From these relations, the **Bianchi identities** are obtained, namely:

$$\nabla_m R_{j k l}^i + \nabla_k R_{j l m}^i + \nabla_l R_{j m k}^i = 0 \quad (11.13)$$

Initially, the left-hand side of equation 11.13 is contracted with respect to the indices  $i, l$ , and after this expression, the index  $j$  is raised, so we get[18]:

$$\nabla_m R_k^j - \nabla_k R_m^j + \nabla_l R_{m k}^{j l} = 0$$

In this relation, the indices  $j, k$  are contracted, so we get:

$$\nabla_j \left( R_m^j - \frac{1}{2} \delta_m^j R \right) = 0$$

where  $R$  is the scalar curvature. From this relation, we define the **Einstein tensor** as:

$$G_j^i = R_j^i - \frac{1}{2} \delta_j^i R \quad (11.14)$$

or for the covariant components we get the relation:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \quad (11.15)$$

Thus, if  $\tilde{\mathbf{D}}$  is a conserved quantity, then the resulting equations are identical to Einstein's field equations for gravity:

$$G_{\mu\nu} = k T_{\mu\nu}$$

or

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = k T_{\mu\nu} \quad (11.16)$$

From equation 11.10, we find Einstein's equations or the general equations. We write equation 11.10 as:

$$d(R_j^i - kT_j^i) = 0$$

The quantity in parentheses must be constant, i.e.,

$$R_j^i - kT_j^i = \tilde{\Lambda}\delta_j^i \quad (11.17)$$

For the covariant components of the tensors of equation 11.18, we obtain the field equations:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = kT_{\mu\nu} + g_{\mu\nu}\Lambda \quad (11.18)$$

where  $\Lambda$  is the **Cosmological constant**. Equations 11.18 are called the **General field equations** because they apply to all physical scales and not just gravity.

Equation 11.18 was derived from a member of the Planck patch, namely from a conic section, so due to this self-similarity, it will also express the evolution equation of the Planck patch at each stage of its expansion. The self-similarity mentioned earlier should also occur in the present era, where almost all large structures, such as galaxies or clusters of galaxies, have formed. Indeed, on a galactic scale, galaxies form a "homogeneous and isotropic grid" within the Visible Universe.

The so-called **Cosmological principle** or **Copernican principle** no longer makes sense as an axiom on the galactic scale, because it is a direct mathematical consequence of the isotropy of  $\mathbf{P}^2$ , as it results from the evolution of a Planck patch. What follows is that, due to self-similarity, the equation that determines the evolution of any geometric object of a Planck patch can be used in a different form and with a different meaning by free observers. Thus, we get the following cases:

- a) The equation is used to describe macroscopically the evolution of the total Planck patch (as is done with the general equations), and
- b) The equation is used to determine the three-dimensional entities and their properties that fill this space.

What follows as a conclusion is that through the equations of conic sections, we can describe the whole and the part macroscopically and microscopically respectively. Let us write the Einstein equations with the cosmological constant as follows:

Adding and subtracting the quantity  $g_{\mu\nu}(\frac{R}{2})$  to the

$$R_{\mu\nu} = kT_{\mu\nu} + g_{\mu\nu}(\frac{R}{2}) + g_{\mu\nu}(\tilde{\Lambda} - \frac{R}{2}) \quad (11.19)$$

or

$$R_{\mu\nu} = kS_{\mu\nu} + g_{\mu\nu}\Lambda \quad (11.20)$$

where from (11.19) and for  $\Lambda = \tilde{\Lambda} - \frac{R}{2} = 0$  the quantity  $kS_{\mu\nu}$  is written as [22]

$$k(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T)$$

For the derivation of this relation, the relation  $R = -kT$  was used, which resulted from (11.19) and the definitions  $R = g^{\mu\nu}R_{\mu\nu}$  and  $T = g^{\mu\nu}T_{\mu\nu}$ . Thus, we define the quantity

$$kS_{\mu\nu} = k\left(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T\right) \quad (11.21)$$

Equations (11.20) are called **General equations** because they hold for all physical scales and not just for gravity, and the constant  $\Lambda$  is called the **Cosmological constant**.

If  $\tilde{\Lambda} = \frac{R}{2}$  then we obtain Einstein's equations with  $\Lambda = 0$ , namely

$$R_{\mu\nu} = kS_{\mu\nu} \quad (11.22)$$

If  $\tilde{\Lambda} > \frac{R}{2}$  then we obtain the general equations with  $\Lambda \neq 0$ , namely equation (11.22)

$$R_{\mu\nu} = kS_{\mu\nu} + g_{\mu\nu}\Lambda \quad (11.23)$$

Equation (11.23) originated from a member of the Planck patch, i.e., from a conic section, so due to self-similarity, it will also express the evolution equation of the Planck patch at every stage of its expansion. The self-similarity mentioned earlier should also occur in the present era, where almost all large structures, such as galaxies or clusters of galaxies, have formed. Indeed, on a galactic scale, galaxies form a “homogeneous and isotropic lattice” within the Observable Universe.

Let us write Einstein's equations with the cosmological constant as follows

$$R_{\mu\nu} - g_{\mu\nu}\left(\frac{R}{2}\right) - g_{\mu\nu}\Lambda = kT_{\mu\nu} \quad (11.24)$$

For free observers, the right side of this equation states that the source of the gravitational field is the total energy-momentum tensor. For  $\Lambda = 0$ ,  $T_{\mu\nu}$  expresses the sum of visible and non-visible mass ( $M$ ), meaning that  $T_{\mu\nu}^{(M)}$  determines the curvature  $R_{\mu\nu} - g_{\mu\nu}(\frac{R}{2})$ . Correspondingly, the term  $-g_{\mu\nu}\Lambda$  should be determined by dark energy through a tensor  $T_{\mu\nu}^{(e)}$ . Since the propagation of dark energy is isotropic in space, the tensor  $T_{\mu\nu}^{(e)}$  must be Lorentz-invariant in a local inertial frame. The only Lorentz-invariant tensor

in this case is the metric tensor  $\eta_{\mu\nu}$  of flat space, which generalizes directly locally to  $g_{\mu\nu}$  in an arbitrary coordinate system. Thus, we get

$$T_{\mu\nu}^{(e)} = -\rho_e g_{\mu\nu} \quad (11.25)$$

Comparing these two relations, we get

$$p_e = -\rho_e \quad (11.26)$$

The isotropic pressure  $p_e$  of this perfect fluid is constant, so the energy density  $\rho_e$  is also constant. We write Einstein's equations as

$$R_{\mu\nu} - g_{\mu\nu} \left( \frac{R}{2} \right) - g_{\mu\nu} \Lambda = 8\pi G \left( T_{\mu\nu}^{(M)} + T_{\mu\nu}^{(e)} \right) \quad (11.27)$$

Comparing the last terms on both sides of (11.26) and for  $T_{\mu\nu}^{(e)} = -\rho_e g_{\mu\nu}$ , we get

$$\rho_e = \frac{\Lambda}{8\pi G} \quad (11.28)$$

From this relation, it follows that the energy density of the vacuum equals the cosmological constant. Astronomical measurements gave a value for the vacuum energy density on the order of  $\sim 10^{-8} \text{ erg/cm}^3$ .

If Quantum Field Theory (QFT) is applied to a Planck point, then the vacuum energy density yields a value of

$$\rho_{\Lambda}^{\text{QFT}} = \frac{E_p}{l_p^3} \sim 10^{112} \frac{\text{erg}}{\text{cm}^3} \quad (11.29)$$

This result leads to a massive discrepancy, greater than 120 orders of magnitude, between these two values (**Cosmological Constant Problem**). Alternatively, we can calculate the energy density of dark energy using the acceleration of the Planck patch. In this manner, the theoretical results match the observational data





## Chapter 12

# INTERACTIONS

### 12.1 The structure of geometric objects of the simplest composite particles

#### 12.1.1 Relationships between geometric quantities of conic sections

##### A. The structure of geometric objects corresponding to the simplest composite particles.

In a Planck patch, concentric circles are created. Let us consider the circles of radii  $r = 1, r = 2$  in units  $l_p$  that correspond to the particles of visible matter. From the appropriate characteristic vectors of the quadrants of these circles we find their charge and orientation as mentioned in Chapter 3.

Thus, the quadrants of these circles, through their characteristic vectors, correspond to all the elementary particles of  $3D$ -space and their antiparticles, that is, to the leptons-antileptons and *quarks – antiquarks*.

Leptons and *quarks* belong to the first physical scale of the structure of matter.

**The circle with radius  $r = 1$  of the first physical scale gives charge and orientation**

A1.  $q_r = 1, \hat{\mathbf{O}} = \hat{\mathbf{O}}^\downarrow$  from the  $1^o$  quadrant and  
 $q_r = -1, \hat{\mathbf{O}} = \hat{\mathbf{O}}^\uparrow$  from the  $3^o$  quadrant.

B1.  $q_r = \frac{1}{3}, \hat{\mathbf{O}} = \hat{\mathbf{O}}^\uparrow$  from the  $2^o$  quadrant and  
 $q_r = -\frac{1}{3}, \hat{\mathbf{O}} = \hat{\mathbf{O}}^\downarrow$  from the  $4^o$  quadrant

**The circle with radius  $r = 2$  of the first physical scale gives charge and orientation**

A2.  $q_r = 2, \hat{\mathbf{O}} = \hat{\mathbf{O}}^\downarrow$  from the  $1^o$  quadrant and  
 $q_r = -2, \hat{\mathbf{O}} = \hat{\mathbf{O}}^\uparrow$  from the  $3^o$  quadrant

B2.  $q_r = -\frac{2}{3}, \hat{\mathbf{O}} = \hat{\mathbf{O}}^\downarrow$  from  $4^o$  quadrant and  $q_r = \frac{2}{3}, \hat{\mathbf{O}} = \hat{\mathbf{O}}^\uparrow$  from  $2^o$  quadrant

To construct the second physical scale of matter, a procedure similar to that of the first physical scale of matter should be followed. Thus, according to what was mentioned earlier regarding the comparison of similar geometric objects, the stable structures of the second physical scale that have “mass” (curvature) and reduced charge with integer values will be four, that is, as many as exist in the first physical scale.

This means that we will have to find two different particles as well as their antiparticles in  $3D$ -space. These particles will correspond to circles with radii such that they give through their characteristic vectors the same value  $q_r$  as that of the geometric objects that correspond to the elementary particles.

Thus, we will consider the cases  $q_r = \pm 1, q_r = 0$  or  $0^\pm$  by finding circles corresponding to composite particles of the second physical scale of  $3D$ -space. These circles should have radii  $r = 3$  and  $r = 0$  respectively.

The characteristic vectors of these circles that give the above values of  $q_r$ , are

$$A = \left\{ \begin{array}{l} b_3 = (-3, 3) \\ b_{-3} = (3, -3) \\ b_0 = (0, -0) \end{array} \right\} \quad (12.1)$$

To find the components of the characteristic vectors of the members of A, we will decompose them into two, three or more elementary characteristic vectors, using the first members only of the families of circles that correspond to the first members of the families of elementary particles.

To construct the second physical scale of matter, a procedure similar to that of the first physical scale of matter should be followed. Thus, according to what was mentioned earlier regarding the comparison of similar geometric objects, the stable structures of the second physical scale that have “mass” (curvature) and reduced charge with integer values will be four, that is, as many as exist in the first physical scale.

This means that we will have to find two different particles as well as their antiparticles in  $3D$ -space. These particles will correspond to circles with radii such that they give through their characteristic vectors the same value  $q_r$  as that of the geometric objects that correspond to the elementary particles.

Thus, we will consider the cases  $q_r = \pm 1, q_r = 0$  or  $0^\pm$  by finding circles corresponding to composite particles of the second physical scale of  $3D$ -space. These circles should have radii  $r = 3$  and  $r = 0$  respectively.

Initially, the characteristic vectors in relation (12.1) are decomposed into 2 components. The only allowed combinations, with the restriction that we must only obtain feature vectors of the form  $(\pm 1, \mp 1)$ ,  $(\pm 2, \mp 2)$ , are the following:

$$(0, 0) = \left\{ \begin{array}{l} (1, -1) + (-1, 1) \rightarrow d\bar{d}, q_r(d\bar{d}) = 0, \hat{\mathbf{O}}(d\bar{d}) = 0\hat{\mathbf{e}}_3 \\ (2, -2) + (-2, 2) \rightarrow \bar{u}u, q_r(\bar{u}u) = 0, \hat{\mathbf{O}}(\bar{u}u) = 0\hat{\mathbf{e}}_3 \end{array} \right\} \quad (12.2)$$

$$(-3, 3) = (-2, 2) + (-1, 1) \rightarrow u\bar{d}, q_r(u\bar{d}) = 1, \hat{\mathbf{O}}(u\bar{d}) = 2\hat{\mathbf{e}}_3 \quad (12.3)$$

$$(3, -3) = (2, -2) + (1, -1) \rightarrow d\bar{u}, q_r(d\bar{u}) = 1, \hat{\mathbf{O}}(d\bar{u}) = -2\hat{\mathbf{e}}_3 \quad (12.4)$$

The symbols  $q_r(q\bar{q}), \hat{\mathbf{O}}(q\bar{q})$  express the sum of the reduced charge values and the vector sum of the orientation of the geometric objects of  $(q, \bar{q})$  *quarks* respectively, i.e.

$$q_r(q\bar{q}) = q_r(q) + q_r(\bar{q})$$

$$\hat{\mathbf{O}}(q\bar{q}) = \hat{\mathbf{O}}(q) + \hat{\mathbf{O}}(\bar{q})$$

The characteristic vectors of relation 12.1 consist of the characteristic vector of a quadrant of a circle corresponding to a *quark* and the characteristic vector of a quadrant of a circle corresponding to its anti-*quark*. These have the same  $q_r$  as the geometric object of the neutrino (antineutrino), but a different orientation value from it. The same arguments apply to relations 12.3, 12.4 which compare the orientations of the geometric objects of these composite entities with those of the positron (electron).

The observation that the orientations of the conic sections that are the geometric objects of these composite entities do not match the orientations of the conic sections that are the geometric objects of the neutrino (antineutrino) or the electron (positron) leads us to the conclusion that these structures either have not been created at all or, if they are created, will immediately disintegrate.

Composite structures of this type decay rapidly, as has been experimentally verified. In general, composite particles consisting of a *quark* and an *antiquark*, regardless of the member of their families, are called mesons (*mesons*). When mesons consist only of the first member of their family, they are called pions (*pions*). The pions are of the form

$$\pi^+ = u\bar{d}, \pi^- = d\bar{u}, \pi^0 = \left\{ \begin{matrix} \bar{u}u \\ d\bar{d} \end{matrix} \right\}$$

The meson  $\pi^-$  is considered to be the antiparticle of  $\pi^+$ , while  $\pi^0$  coincides with its antiparticle.

The analysis of the characteristic vectors of the circles of relation 12.1 into three components in a unique way, due to the restriction of the values of the characteristic vectors mentioned above, is as follows:

$$(0,0)= \left\{ \begin{matrix} (-2,2) + (1,-1) + (1,-1) \rightarrow udd, \\ q_r(udd) = 0, \hat{\mathbf{O}}(udd) = -\hat{\mathbf{e}}_3 \\ (2,-2) + (-1,1) + (-1,1) \rightarrow \bar{u}\bar{d}\bar{d}, \\ q_r(\bar{u}\bar{d}\bar{d}) = 0, \hat{\mathbf{O}}(\bar{u}\bar{d}\bar{d}) = \hat{\mathbf{e}}_3 \end{matrix} \right\} \quad (12.5)$$

$$(-3,3)= \left\{ \begin{matrix} (-2,2) + (-2,2) + (1,-1) \rightarrow uud, \\ q_r(uud) = 1, \hat{\mathbf{O}}(uud) = \hat{\mathbf{e}}_3 \\ (-1,1) + (-1,1) + (-1,1) \rightarrow \bar{d}\bar{d}\bar{d}, \\ q_r(\bar{d}\bar{d}\bar{d}) = 1, \hat{\mathbf{O}}(\bar{d}\bar{d}\bar{d}) = 3\hat{\mathbf{e}}_3 \end{matrix} \right\} \quad (12.6)$$

$$(3,-3)= \left\{ \begin{matrix} (2,-2) + (2,-2) + (-1,1) \rightarrow \bar{u}\bar{d}\bar{d}, \\ q_r(\bar{u}\bar{d}\bar{d}) = -1, \hat{\mathbf{O}}(\bar{u}\bar{d}\bar{d}) = -\hat{\mathbf{e}}_3 \\ (-1,1) + (-1,1) + (-1,1) \rightarrow ddd, \\ q_r(ddd) = -1, \hat{\mathbf{O}}(ddd) = -3\hat{\mathbf{e}}_3 \end{matrix} \right\} \quad (12.7)$$

From relation (12.5) the quantities  $q_r, \hat{\mathbf{O}}$  of the circles, which are the geometric objects of the composite particles  $udd, \bar{u}\bar{d}\bar{d}$  are compared with the  $q_r$  of the circles and the orientations  $\hat{\mathbf{O}}$  of the ellipses which are the geometric objects of  $\nu, \bar{\nu}$ .

The comparison of these quantities shows that the quantities  $q_r, \hat{\mathbf{O}}$  of the circles corresponding to  $udd, \bar{u}\bar{d}\bar{d}$  correspond to the quantities of circles and ellipses  $q_r, \hat{\mathbf{O}}$  corresponding to  $\nu, \bar{\nu}$ . Therefore, we make the correspondences

$$n = udd \rightarrow \nu \quad (12.8)$$

$$\bar{n} = \bar{u}\bar{d}\bar{d} \rightarrow \bar{\nu} \quad (12.9)$$

The composite entity  $n \equiv udd$  called *neutron* corresponds to the neutrino of the elementary particle, while the composite entity  $\bar{n} \equiv \bar{u}\bar{d}\bar{d}$  called *antineutron* corresponds to the antineutrino of the elementary particle, after comparing the quantities  $q_r, \mathbf{O}$  of their geometric object. According to the correspondences (12.8),(12.9) the entities  $n, \bar{n}$  can exist as stable and isolated particles, as confirmed by experiments.

From relations (12.6) and (12.7), the only composite entities that correspond to elementary particles through their corresponding circles are the structures  $uud, \bar{u}\bar{u}\bar{d}$ . Comparing  $q_r, \mathbf{O}$  of the cycles that are g.o of  $uud(\bar{u}\bar{u}\bar{d})$  with  $q_r, \mathbf{O}$  of the cycles and ellipses that are geometric object of  $e^+(e)$  ensures that they are identical. Thus, we make the correspondences

$$p \equiv uud \rightarrow e^+ \quad (12.10)$$

$$\bar{p} \equiv \bar{u}\bar{u}\bar{d} \rightarrow e \quad (12.11)$$

The complex entity  $p \equiv uud$  called *proton* corresponds to positron. The complex entity  $\bar{p} \equiv \bar{u}\bar{u}\bar{d}$  called *antiproton* corresponds to electron. The reduced charge of  $p, e^+$  fits, therefore, we consider the geometric object of  $p$  as the unit circle of the second physical scale.

According to the correspondences (12.10),(12.11)  $p, \bar{p}$  can exist as stable isolated entities in nature, as confirmed by experiments. Protons and neutrons (*nucleons*) constitute the next scale of organization of matter, which is the *nuclei*.

The other complex entities resulting from equations (12.6) and (12.7) are unstable (or rapidly degradable), because they do not correspond to a conic section depicting an isolated elementary particle. Composite natural particles consist of three quarks or three antiquarks, regardless of the member of the three-member family, and are called *baryons*.

In nature, the only stable baryons are the proton and the neutron. The neutron is a relatively stable particle, because if it is free it decays in about 14.7 minutes, while the proton does not decay at all. We will discuss in detail the reason for the non-decay of the proton in the next section.

**Matter** is defined as the set of all elementary natural entities and the simplest composite stable structures, that is, the set  $\{u, d, e, \nu, \gamma, p, n\}$ .

**Antimatter** is defined by the set  $\{\bar{u}, \bar{d}, e^+, \bar{\nu}, \bar{\gamma}, \bar{p}, \bar{n}\}$  of elementary and simplest complex structures. The study of nature shows that only matter structures appear at all scales of organization of matter, while antimatter is completely absent.

Thus, the first question arises:

**Why does antimatter not appear in complex structures of the third scale of organization of matter (such as nuclei), while all members of the set satisfy the stability criteria according to the correspondences (12.8), (12.9),(12.10),(12.11)?**

To answer this question, we will consider the third scale of organization of matter, namely the nucleus. The nucleus consists only of stable particles, namely protons and neutrons. The nucleus is depicted in the generator space as a complex structure of geometric objects of protons and neutrons. The geometric object of a proton corresponds to the geometric object. of a positron, i.e. to a particle characterized as an antimatter particle in  $3D$ .

The gamma of a neutron corresponds to the gamma of a neutrino, that is, to a particle characterized as a matter particle in 3D. Thus, in the generator space  $\mathbf{P}^2$  the gammas of the components of a nucleus represent particles and antiparticles in 3D. According to the correspondence procedure we have followed so far, the previous question is meaningless.

A similar answer to the above question could also be given by correspondences of equations (12.8),(12.9) and (12.10),(12.11). The complex geometric entities that are the geometric object. of protons and neutrons are expressed, by analyzing their characteristic vectors, as follows:

### 1. Proton

$$\mathbf{b}_p = (-3, 3) = \begin{bmatrix} (-1, 2) + (-2, 1) \rightarrow \nu + e^+ \\ (-2, 2) + (-1, 1) \rightarrow u + \bar{d} \end{bmatrix} \quad (12.12)$$

From this analysis of the characteristic vector of the complex geometric entity, which is the geometric object of a proton, we see that it consists of the characteristic vector of a conic section corresponding to a matter particle (neutrino ( $\nu$ ) or  $u$ -quark) and the characteristic vector of a conic section corresponding to an antimatter particle (positron  $e^+$  or  $\bar{d}$  - antiquark).

### 2. Neutron

$$\mathbf{b}_n = (0, 0) = \begin{bmatrix} (-1, 2) + (1, 2) \rightarrow e^+ + e \\ (2, -1) + (-2, 1) \rightarrow \bar{\gamma} + \gamma \\ (2, -2) + (-2, 2) \rightarrow \bar{u} + u \\ (-1, 1) + (1, -1) \rightarrow \bar{d} + d \end{bmatrix} \quad (12.13)$$

According to the above analysis, antimatter appears in complex structures with equal probability to matter, therefore through correspondences for conic sections and physical entities, the above paradox does not apply.

However, both members of relations (12.12) and (12.13) are compatible with  $q_r$  but are not compatible with the orientations, so the analysis of the eigenvector of the geometric object. of  $p, n$  in relations (12.12) and (12.13) is incomplete. In order to make relations (12.12) and (12.13) fully compatible, they must be modified. Thus, for example, we write the first relation of (12.12) as

$$p \rightarrow (-3, 3) = (0, 0) + (-1, 2) + (-2, 1) \rightarrow n + e^+ + \nu = n + W^+ \quad (12.14)$$

and correspondingly for  $n$  the relation

$$n \rightarrow (0, 0) = (-3, 3) + (1, -2) + (2, -1) \rightarrow p + e + \bar{\nu} = p + W^- \quad (12.15)$$

Given that  $p, n$  correspond to elementary particles of antimatter (A) and matter (M) respectively, then the general expressions of (12.14),(12.15) are

$$A \rightarrow M + A + M \quad (12.16)$$

$$M \rightarrow A + M + A \quad (12.17)$$

As can be seen from both sides of Equations (12.16) and (12.17), anti-matter appears in the  $p, n$  relations with equal probability as matter.

The nature and properties of the entities  $W^+, W^-$  will be discussed in the next section. Each of these two relations is interpreted as a spontaneous decay of protons and neutrons respectively into components that are matter and antimatter particles. The pairs of particles  $e^+, \nu$  and  $e, \bar{\nu}$  move away rapidly from the point of each decay, so that only one particle of matter remains, which gives the impression that there is only matter. We will see in the next section whether these decays are permissible or not. A second question is the following:

*"Why do the nuclei and atoms that constitute the third and fourth scales of organization of matter have this structure in 3D ?"*

Each nucleon corresponds to an elementary particle, comparing the quantities  $q_r, \mathbf{O}$  of their geometric object. The geometric object. of this elementary particle directly defines the conjugate of geometric object, so that this pair of conjugate conic sections is characterized by zero  $q_r$  and zero total orientation.

*The complex geometric object. of each nucleon must have these characteristics to complete this correspondence.*

Since there are only definite conjugate geometric object. for elementary particles and no conjugate geometric object. for composite particles, then another procedure must be applied to construct the geometric object. of composite particles that satisfy the above characteristic, namely that of zero  $q_r$  and zero total orientation. An obvious solution in  $\mathbf{P}^2$  is the following:

*Any composite structure with components the geometric object of the nucleons to be formed, such that this structure has a total zero  $q_r$  and a total zero orientation  $\mathbf{O}$ .*

Based on these arguments, the units of the third scale of organization of matter are constructed, namely the nuclei, which are representations in 3D of structures with components the geometric object of nucleons. Since these structures always have positive  $q_r$ , they should be attached to them by geometric object. with negative  $q_r$ , in such a way that this composite geometric object. has a total zero  $q_r$ .

Since a nucleon has a geometric object. that has positive or zero  $q_r$ , then the only option is to attach to it geometric object. of elementary particles with negative  $q_r$ , as many as there are geometric object. of protons. Through

this process, the geometric object. of composite structures called *atoms* can be constructed.

The geometric object of an atom should also have a zero orientation. The geometric object of a proton and a neutron have orientations  $+\mathbf{e}_3, -\mathbf{e}_3$  respectively. The geometric object of an electron can be a circle with orientation  $+\mathbf{e}_3$ , or a circle and an ellipse with orientation  $0\mathbf{e}_3$ , or an ellipse with orientation  $-\mathbf{e}_3$  corresponding to  $3\mathbf{D}$  in a classical particle, a wave particle, or a wave, respectively.

Through appropriate combinations of electron geometric object. orientations in this structure, zero total orientation appears. The parameters involved in these combinations are the number of neutron geometric object. that do not affect the total  $q_r$  but do affect the total orientation, and the freedom to choose one of the three possible *electron geometric object. orientations*. Through these arguments, we will examine the stable isotopes of the hydrogen atom.

The hydrogen atom  ${}_1^1H$  consists of one proton and one electron, so its geometric object has  $q_r = 0$ . The geometric object of the proton has an orientation of  $+1$ . For the geometric object of the hydrogen atom to have a zero orientation, then the geometric object of the electron should be the ellipse  $(1, 2)$  with characteristic vector  $(1, -2)$ , which gives an orientation of  $-1$ . The electron is considered as a wave that easily moves away from the nucleus.

The ionization energy of  ${}_1^1H$  is about  $13.6 \text{ eV}$ .

The deuterium atom  ${}_1^2H$  consists of one proton, one neutron, and one electron. The vector sum of the orientations of the geometric objects of the proton and neutron is the zero vector. For the geometric object of Deuterium to have zero orientation, the geometric object of the electron of  ${}_1^2H$  should be represented by a circle and an ellipse together, that is, it should be a wave-particle that gives zero overall orientation. It is more difficult to remove the electron from the nucleus, compared to the previous case, because the particle nature appears. The ionization energy of  ${}_1^2H$  is about  $14.9 \text{ eV}$ .

The tritium atom  ${}_1^3H$  consists of one proton, two neutrons, and one electron. The geometric object of the nucleus of  ${}_1^3H$  has an overall orientation of  $-1$ . For the geometric object of the tritium atom to have zero orientation, the geometric object of the electron should be a circle of radius 1 that gives an orientation of  $+1$ . Thus, this electron is considered a classical particle that is more difficult to remove from the nucleus than the two previous cases, because it behaves like a classical particle. The ionization energy of  ${}_1^3H$ , as it results from its  $\beta$ -radioactive decay when it transforms into a helium nucleus, is about  $34.3 \text{ eV}$ , i.e. more than twice the sum of the ionization energy values of the two previous cases.

To continue with the analysis of the characteristic vectors

$$\mathbf{b}_0 = (0, 0), \mathbf{b}_3 = (-3, 3), \mathbf{b}_{-3} = (3, -3)$$



in more than three components, we analyze the above characteristic vectors in an even or odd number of terms.

The analysis of the feature vectors results in structures that are isolated and non-isolated. Isolated structures are preserved, while non-isolated structures are transformed into isolated structures. As an example, the analysis of  $b_0 = (0, 0)$  into the following five terms

$$\begin{aligned} \mathbf{b}_0 = (0, 0) &= (-2, 2) + (1, -1) + (1, -1) + (1, -1) + (-1, 1) \\ &\rightarrow udd + d\bar{d} = n + \pi^0 \end{aligned}$$

The quantity  $q_r$  and the orientation are preserved. The first three terms give a neutron and the next two terms a  $\pi^0$ -pion. Since the  $\pi^0$ -pion decays to the isolated structures  $e^+ + e^-$  or  $\bar{\nu} + \nu$ , only one neutron remains, so, according to this analysis, a new complex structure is not produced.

## 12.2 Allowed and disallowed decays

The simplest complex structures in  $3D$  (nucleons) are examples of structures in  $\mathbf{P}^2$  consisting of geometric object that are only circles. The geometric object of nucleons, as circles of finite radius, have non-zero curvature, destroying in their neighborhood the zero curvature of  $\mathbf{P}^2$ . This also happens with their mappings, which are physical entities that destroy the zero curvature of  $3D$  in their neighborhoods.

The eigenvector of the geometric object of each nucleon is decomposed into the eigenvectors of their components. Each of these eigenvectors belongs to a geometric object that is a circle, which also destroys the zero curvature of  $\mathbf{P}^2$ . To restore the zero curvature of  $\mathbf{P}^2$  as much as possible, the geometric object of the nucleon components must behave as follows:

1. *If the geometric object is an ellipse, then it is converted into a linear ellipse, so that the elementary particle corresponding to it becomes a wave particle whose curvature is almost zero, because it spreads throughout  $3D$ , due to its wave nature.*
2. *If the geometric object is a circle, then its radius will tend to take on as large a value as possible, so that its curvature takes on as small a value as possible.*

The tendency of  $\mathbf{P}^2$  to restore its unperturbed state is illustrated in  $3D$  by the spontaneous decay of the simplest composite particles into wave particles and classical particles. Thus, these elementary wave particles that have ellipses as geometric object appear as waves in  $3D$ , restoring its zero curvature. Classical particles for which geometric object is a circle try to

increase the radius of this circle, in order to have less curvature than the curvature of the geometric object of the composite initial particles in  $3D$ .

This tendency of the geometric object composite particles of  $\mathbf{P}^2$  leads to permissible and impermissible decompositions of the simplest composite particles in  $3D$ , as verified experimentally. We will examine the decompositions of  $p, n$  in  $3D$  theoretically, based on previous observations and the relations (12.14), (12.15). We will start from relation (12.15), which depicts the decay of a neutron into a proton, an electron, and an antineutrino in  $3D$ , as verified experimentally. We will consider the resolution of the characteristic vector of the geometric object of a neutron into characteristic vectors of the geometric objects of a proton, an electron, and an antineutrino under the condition that the total charge and the total orientation are conserved. Since the electron and antineutrino have two geometric objects, we have the following four possible cases

$$n \rightarrow (0, 0) = \left[ \begin{array}{l} (-3, 3) + (-1, -1) + \left[ \begin{array}{l} (0, 0) \\ (2, -1) \end{array} \right] \\ (-3, 3) + (1, -2) + \left[ \begin{array}{l} (0, 0) \\ (2, -1) \end{array} \right] \end{array} \right] \quad (12.18)$$

In the two subbranches of the upper main branch as well as in the first subbranch of the lower main branch, at least two characteristic geometric objects vectors of elementary particles appear that are circles with curvature different from zero. Also, in these three cases, the characteristic vector of the neutron's geometric object, as well as the quantity  $q_r$ , are not conserved. Only in the last case are the geometric objects of two elementary particles ellipses. This means that these elementary particles can be transformed into wave-particles, occupying the entire  $3D$  while preserving the neutron's characteristic vector. Thus, only the last case in  $\mathbf{P}^2$  will be discussed in detail, namely

$$(0, 0) = (-3, 3) + (1, -2) + (2, -1) \quad (12.19)$$

The mapping of equation (2.19) in  $3D$  is given by the relation

$$n = p + e + \bar{\nu} \quad (12.20)$$

We write (12.20) as

$$n = p + W^- \quad (12.21)$$

where

$$W^- \rightarrow e + \bar{\nu} \quad (12.22)$$

The composite particle  $W^-$  decays immediately into an electron and an electron antineutrino. By analogy with quantum electrodynamics, the particle  $W^-$  plays the role of an intermediate carrier of this interaction, therefore the unified theory of electroweak interactions is based on this characteristic of  $W^-$ . The geometric object of the particle  $W^-$  has  $q_r = -1$  as shown by the characteristic vector  $(3, -3)$ .

We add the value of  $-1$  of charge, to the particle  $W^-$  in  $3D$ . As long as the particles  $p$  and  $W^-$  coexist (because the particle  $W^-$  decays rapidly), the geometric object of  $W^-$  can be considered as the mirror geometric object (or conjugate geometric object) of  $p$ , because it has an opposite characteristic vector compared to that of the geometric object of  $p$ . Thus, before the decay of  $W^-$  we have

$$p^\uparrow + (W^-)^\downarrow = p^\uparrow + p^{\downarrow(k)} \quad (12.23)$$

where the symbol  $p^{\downarrow(k)} = (W^-)^\downarrow$  denotes the mirror image of  $p^\uparrow$ . Through a mirror image of  $p^\uparrow$ ,  $p^{\downarrow(k)}$  we obtain

$$p^\uparrow \rightarrow^k (W^-)^\downarrow \text{ and } p^{\downarrow(k)} \rightarrow^k p^\uparrow \quad (12.24)$$

where the symbol  $\rightarrow^k$  denotes the mirror image. Thus, successively from equation (12.23), we get

$$p^\uparrow + (W^-)^\downarrow = p^\uparrow + p^{\downarrow(k)} \rightarrow^k (W^-)^\downarrow + p^\uparrow = p^\uparrow \quad (12.25)$$

The image of  $p$  through a plane mirror, **has the same and not the opposite orientation from that of the object  $p$** . The relation (12.25) expresses **the violation of parity (parity)**. The violation of parity in  $3D$  is interpreted by reducing this phenomenon to the antineutrino, in a simple and understandable way. We write equation (12.19) as

$$-[(1, -2) + (2, -1)] + (0, 0) = (-3, 3) \quad (12.26)$$

From relation (12.26) a new particle is defined in  $\mathbf{P}^2$  called  $W^+$ , whose geometric object has a characteristic vector  $(-3, 3)$  that matches that of a proton. The geometric object of particle  $W^+$  has  $q_r = +1$ , so we assign this value to particle  $W^+$  in  $3D$ . Relation (12.26) can be interpreted as follows: if particle  $W^+$  acts on a neutron, then it converts it into a proton, i.e..

$$W^+ + n = p$$

Writing the particles  $n, p$  in terms of quarks, omitting the common pair  $ud$  of the particles  $n, p$ , we get

$$W^+ + d = u \quad (12.27)$$

Illustrating the action of  $W^+$  with the symbol  $\circ$  on the  $d$ -quark, we get

$$(W^+) \circ d = u \quad (12.28)$$

The conversion of  $n$  to  $p$  or of a  $d$ -quark to a  $u$ -quark, due to the action of the  $W^+$  particle, *is compatible* with the tendency of the geometric object of each *composite structure to minimize its curvature because the geometric object of a  $u$ -quark has less curvature than the geometric object of a  $d$ -quark* ( $k_u = \frac{k_d}{2}$ ). An example of an allowed decay (reaction) is relation (12.20). Thus, relation (12.26) is an example *of a **allowed decay (reaction)***. Let us consider a possible decay of a proton. We follow a procedure similar to that for finding relation (12.18), where the only decay of a proton that is allowed is given by relation (12.14). The characteristic vector of the geometric object of a proton can be analyzed as follows

$$(-3, 3) = (0, 0) + (-1, 2) + (-2, 1) \quad (12.29)$$

The representation of equation (2.29) in 3D-space is given by the relation

$$p = n + W^+ \quad (12.30)$$

where

$$W^+ = e^+ + \nu$$

Writing Equation (12.29) as

$$-[-1, 2) + (-2, 1)] + (-3, 3) = (0, 0) \quad (12.31)$$

The representation of relation (12.31) in 3D is

$$W^- + p = n \quad (12.32)$$

The relation (12.32) is interpreted as follows: if the particle  $W^-$  acts on a proton, then the proton is converted into a neutron. Writing  $n, p$  in terms of quark and omitting the common quark pair  $ud$  of  $n$  and  $p$ , we get

$$W^- + u = d$$

or

$$W^- \circ u = d \quad (12.33)$$

The conversion of  $p$  to  $n$  or of a  $u$ -quark to  $d$ -quark, due to the action of the  $W^-$  particle, *is not compatible* in this case, *with the tendency of the geometric object. of any complex structure to reduce its curvature.* The incompatibility is due to the fact that the curvature of the geometric object of a  $d$ -quark (circle of radius 1 with  $k_d = 1$ ) is greater than that of a  $u$ -quark (circle of radius 2 with  $k_u = 0.5$ ). The relation (12.30) or (12.33) is

an example of *an impermissible decay (reaction) that expresses the fact that the proton does not decay!*

The definition of the particles  $W^+, W^-$  with characteristic vectors  $(-3, 3)$ ,  $(3, -3)$ , correspond to the proton and the antiproton and belong to the second scale of “construction” of matter. These correspondences lead us to the search for a third particle with zero charge. Thus, in correspondence with the neutron, the characteristic vector of the geometric object. of such a particle must be  $(0,0)$ . We define a particle called  $Z^0$  in  $D$  whose geometric object. has the characteristic vector  $(0,0)$ . In interactions through the  $Z^0$  particle in  $D$ , a quark or a muon emits or absorbs a  $Z^0$  particle, i.e.

$$e \rightarrow e + Z^0$$

By analogy with the definition of the  $W^+, W^-$  particles, the  $Z^0$  particle can be decomposed, in general, as

$$Z^0 \rightarrow b + \bar{b}$$

The set of this type of interactions between particles in  $3D$  forms the set of particle interactions whose geometric object tend to minimize the local curvature of  $\mathbf{P}^2$ . These interactions are called *weak interactions*. This name comes from the comparison of the “strong forces” between the quarks that make up nucleons and the “weak forces” that convert quarks between the same or different generations.



## Chapter 13

# NUCLEON STRUCTURES-STRONG INTERACTIONS

According to the analysis in Chapter 12, nucleons consist of three quarks. As already mentioned, each of these quarks is the representation of a circle where the scalar quantity  $q_r$  and the vector quantities are defined as charge orientation, mass orientation, and total orientation, i.e., the vector sum of the first two orientations. Through the terms charge orientation, mass orientation, and total orientation, we briefly characterize the respective vector quantities of the quarks. The experimental search for nucleon vector quantities showed that every nucleon exhibits a vector quantity called *magnetic moment*, which for protons and neutrons has the following values (per unit of *nuclear magneton*  $\mu_N$ )

$$\mu_p = +2.792847$$

$$\mu_n = -1.91304273$$

The subscripts  $p, n$  refer to the proton and the neutron, respectively. The values of the magnetic moments of the proton and the neutron do not correspond to the vector sum of the charge orientations, nor to the vector sum of the mass orientations, nor to the vector sum of the total orientations of the circles with radii 3 and 0, which correspond to the proton and the neutron. This means that a proton and a neutron are not planar structures embedded in  $3D$ . Thus, the next step is to assume that they form a three-dimensional structure in  $3D$ . To study nucleons, we define a right-handed Cartesian coordinate system  $x, y, z$  with its origin at the center of mass of each nucleon, where initially its quarks are concentrated.

### 13.1 Kinematics of Strong Interaction

#### Proton

Initially, we will study the proton, where the charge orientations, mass orientations, and total orientations of the three quarks lie on the  $x, y, z$  axes of the aforementioned Cartesian coordinate system. These vector quantities will appear in three different figures from which certain characteristic properties of the proton emerge. The same procedure applies to the neutron.

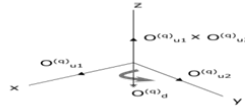


Figure 13.1: Charge orientation of the proton quark

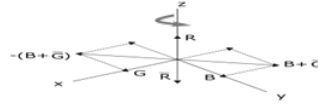


Figure 13.2: Mass orientation of the proton quark

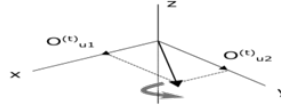


Figure 13.3: Total orientation of the proton quark

We will begin with figure (13.3) which gives the total orientation of a proton consisting of  $2u$ -quarks and  $1d$ -quark. The following Table 13.1 gives us the characteristic vector, the charge, the orientation due to charge, the mass (curvature), the orientation due to mass, and the total orientation of the  $u$ -quark and the  $d$ -quark. From Table 13.1, the mass relation of the  $u, d$  quarks emerges as  $m_d = 2m_u$ , which has been experimentally verified, namely  $m_{d,\text{exp.}} = 5$  and  $m_{u,\text{exp.}} = 2.5$  in units of  $\frac{MeV}{c^2}$ . The radii of the circles of the  $u, d$  quarks are  $r_u = 2, r_d = 1$ .



The curvatures of the circles of the  $u, d$  quarks are  $\kappa_u = 0.5, \kappa_d = 1$ . Also, the total orientation is  $O(t) = O(q) + O(m)$ . The column with the Quarks is denoted by  $Q$ .

| Q | Ch.v.charge | charge         | O(q)   | Ch.v.mass                | $m(\kappa)$              | $O(m)$ | $O(t)$  |
|---|-------------|----------------|--------|--------------------------|--------------------------|--------|---------|
| u | (-2,2)      | $\frac{2}{3}$  | $e_3$  | $3(-\kappa_u, \kappa_u)$ | $\kappa_u = \frac{1}{2}$ | $e_3$  | $2e_3$  |
| d | (1,-1)      | $-\frac{1}{3}$ | $-e_3$ | $3(-\kappa_d, \kappa_d)$ | $\kappa_d = \frac{1}{1}$ | $e_3$  | $0 e_3$ |

Table 13.1: Geometric quantities of the  $u, d$ -quarks

The  $u$ -quarks have a total orientation with a positive sign, therefore they are placed along the positive semi-axes  $x$  and  $y$ . The total orientation of a  $u$ -quark is  $\mathbf{O}_u^{(t)} = 2\mathbf{e}_3$  and the total orientation of a  $d$ -quark is  $\mathbf{O}_d^{(t)} = 0\mathbf{e}_3$ , as shown in Table (13.1). The magnitude of the total orientation of the proton  $|\mathbf{O}_p^{(t)}|$  is the vector sum of the total orientations of the  $u$ -quarks on the  $x, y$  axes.

Thus, the magnitude of  $\mathbf{O}_p^{(t)}$  is

$$|\mathbf{O}_p^{(t)}| = \sqrt{|\mathbf{O}_{u1}^{(t)}|^2 + |\mathbf{O}_{u2}^{(t)}|^2} = 2\sqrt{2} \approx 2.828427 \quad (13.1)$$

This value is close to the experimental value of the magnetic moment of the proton, so this fact reinforces the hypothesis we made regarding the three-dimensional structure of the nucleon quarks. The direction of  $\mathbf{O}_p^{(t)}$  is the positive direction of the axis on which the total orientation of the proton lies. This axis can be any of the three axes of the coordinate system, so it is chosen in such a way that it coincides with the axis of the total orientation of the neutron, in order to compare the values of their magnetic moments. Figures (13.1), (13.2), (13.3) are not correlated with each other, and the only vector quantity we measure in a proton is its magnetic moment. This means that the total charge orientation and the total mass orientation of the proton should be zero.

To find the other features of the proton structure, we will study figure (13.1), which gives the charge orientations of the proton quarks. For the total charge orientation of a proton to be zero, the charge orientation along each axis of the coordinate system should be zero. Since we have positive and negative charge orientations, then the charge orientation along each axis of a coordinate system is defined by the following relations

$$\mathbf{O}_x^{(q)} = \mathbf{O}_{u1}^{(q)} \times \mathbf{O}_{u2}^{(q)} + \mathbf{O}_d^{(q)} = \mathbf{e}_3 - \mathbf{e}_3 = 0 \quad (13.2)$$

$$\mathbf{O}_y^{(q)} = \mathbf{O}_{u2}^{(q)} \times \mathbf{O}_d^{(q)} + \mathbf{O}_{u1}^{(q)} = -\mathbf{e}_1 + \mathbf{e}_1 = 0 \quad (13.3)$$

$$\mathbf{O}_z^{(q)} = \mathbf{O}_d^{(q)} \times \mathbf{O}_{u1}^{(q)} + \mathbf{O}_{u2}^{(q)} = -\mathbf{e}_2 + \mathbf{e}_2 = 0 \quad (13.4)$$

In the cross product of these relations, the *right-hand rule* is used. These relations show that the total charge orientation along each axis of this coordinate system is zero, so that the total charge orientation of the proton is zero. The total mass orientation of protons is found through the following procedure: Mass orientations are positive unit vectors, which lie on the positive semi-axes of the coordinate system and are called **colors**, namely

$$\mathbf{O}_{u1}^{(m)} = \mathbf{G}, \mathbf{O}_{u2}^{(m)} = \mathbf{B}, \mathbf{O}_d^{(m)} = \mathbf{R}$$

The opposite unit vectors of these vectors are denoted by  $\bar{\mathbf{G}}, \bar{\mathbf{B}}, \bar{\mathbf{R}}$  and are called **anti-colors**. The property of the zero total mass orientation of protons is called **white color** ( $W$ ). This property is defined by analogy with the total charge orientation as follows:

Since mass orientations are only positive unit vectors, then for any two colors, the **left-hand rule** is used for their vector product in order to obtain the anti-color of the third color. By adding color and anti-color along the axis of the third color, the zero mass orientation for that axis is obtained. Thus, for example, we get

$$\mathbf{G} \times \mathbf{B} = \bar{\mathbf{R}}, \quad \mathbf{O}_z^{(m)} = \bar{\mathbf{R}} + \mathbf{R} = 0$$

Quarks can change colors without affecting the total mass orientation, which is a zero vector. This can be done as follows: From color  $\mathbf{B}$  we subtract the vector  $\mathbf{B} + \bar{\mathbf{G}}$  and add it to color  $\mathbf{G}$  as shown in figure (13.2). The addition and subtraction of the same vector does not bring about a change in the arrangement of the quarks, but causes only a rotation of their colors or **equivalently a continuous rotation of the axes of the coordinate system**, as shown in the following relations

$$\mathbf{B} : \mathbf{B} - (\mathbf{B} + \bar{\mathbf{G}}) = -\bar{\mathbf{G}} = \mathbf{G}$$

$$\mathbf{G} : \mathbf{G} + (\mathbf{B} + \bar{\mathbf{G}}) = (\mathbf{G} + \bar{\mathbf{G}}) + \mathbf{B} = \mathbf{B}$$

To maintain the original arrangement of colors, the  $\mathbf{R}$ -color (of the original arrangement) and the  $\mathbf{B}$ -color (of the new arrangement) must alternate, i.e., the vector  $\mathbf{R} + \bar{\mathbf{B}}$  between these colors must be added and subtracted, as shown in the following relations

$$\mathbf{R} : \mathbf{R} - (\mathbf{R} + \bar{\mathbf{B}}) = -\bar{\mathbf{B}} = \mathbf{B}$$

$$\mathbf{B} : \mathbf{B} + (\mathbf{R} + \bar{\mathbf{B}}) = (\mathbf{B} + \bar{\mathbf{B}}) + \mathbf{R} = \mathbf{R}$$

#### Neutron

To obtain the right-handed cross product of two identical charge orientations of the neutron, we place both  $d$ -quarks along the positive  $x, y$ -axes and the  $u$ -quark along the negative  $z$ -axis. The same procedure as that used in

the case of the proton is repeated in such a way as to obtain zero charge and mass orientation for a neutron. Because of these relations, we say, in general, that every nucleon has a white color. The arrangements of the charge and mass orientations, as well as the total orientations of a neutron, are shown in the figures below. For the neutron we obtain

$$\mathbf{O}_u^{(t)} = \mathbf{O}_n^{(t)} = -2e_3 \quad (13.5)$$

because  $\mathbf{O}_u^{(t)}$  lies along the negative  $z$ -axis. This value is close to the value of the magnetic moment of neutrons found experimentally.

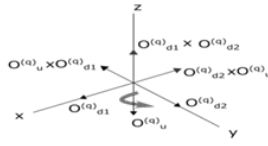


Figure 13.4: Charge orientations of neutron quarks

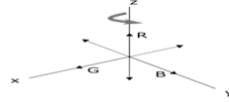


Figure 13.5: Mass orientation of neutron quarks

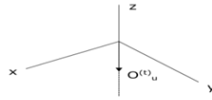


Figure 13.6: Total orientation of neutron quarks

From the process of color exchange between quarks, the idea of a different description of this process emerges. Thus, a hypothetical particle is defined that corresponds to the vector that is added and subtracted between the colors of two quarks, as examined in the case of protons. The process of adding and subtracting this vector between two colors of two quarks can be considered as emission or absorption” of this particle by a quark. Since all this happens only at the origin of a coordinate system, in order to preserve the white color of the nucleon, we consider that this particle has the property of gluing quarks together. Due to this property, this particle was named *gluon*,  $g$ .

The vectors that are added and subtracted between the colors of the quarks are eight and not nine, because if we take any two of the three vectors

$$\{(\mathbf{G} + \bar{\mathbf{G}}), (\mathbf{B} + \bar{\mathbf{B}}), (\mathbf{R} + \bar{\mathbf{R}})\} \quad (13.6)$$

then the third is automatically determined. The other six are

$$\{(\mathbf{G} + \bar{\mathbf{B}}), (\mathbf{G} + \bar{\mathbf{R}}), (\mathbf{B} + \bar{\mathbf{G}}), (\mathbf{B} + \bar{\mathbf{R}}), (\mathbf{R} + \bar{\mathbf{B}}), (\mathbf{R} + \bar{\mathbf{G}})\} \quad (13.7)$$

The process of adding and subtracting vectors is equivalent to a right-handed rotation of the coordinate system of colors. These 8 vectors are called **color states** of gluons. According to the Standard Model, the role of color states is attributed to gluons, which are responsible both for changing the color of a quark and for concentrating the quarks around the center of mass of each nucleon. This interpretation is consistent with the  $SU(3)$ -octet of gluons. Since each gluon does not change the charge or mass of a quark, i.e., it does not interact with that quark, it is considered a particle without charge and mass.

We assume that gluons must be the dark” copies of photons, which correspond to a g.o. that is an ellipse and a circle with the same characteristic vectors as those of photons. Thus, gluons must be characterized by a scalar and a vector potential just like photons and satisfy equations similar to *Maxwell's* equations.

According to the theory of interactions, when a gluon interacts with a quark, then a spin value of  $\frac{1}{2}$  is assigned to the quark. Gluons are not the cause that forces quarks to cluster. The bonds” between the quarks and their center of mass are due to the tendency to maintain the flatness of the plane in which they lie, and the tendency to preserve the isomorphisms between points in a neighborhood of a differential manifold and in their interactions through the tensor  $\tilde{D}$ .

All these issues will be studied in detail in the next section, where the dynamics of strong interactions will be examined. These hypothetical particles behave like photons, but do not appear in our visible world. Thus, we propose that these hypothetical particles be named *erevos*.

The word **Erevos** ( $E\rho\epsilon\beta\omicron s''$ ) means absolute darkness in Greek. According to Greek mythology, Erevos lived very deep underground, so no one could see him. Erevos and Night were children of Chaos.

If the three quarks of a nucleon move away from the  $x, y, z$  axes, then they must quickly return to them, so that the white color of the nucleon is preserved. In strong collisions of composite particles, a breakdown of a nucleon has never been observed. During these collisions, new composite particles with two or three members are created, which always have a white color.

In order to satisfy the condition of the white color of nucleons, each quark is simulated by a classical point particle and not by a wave-particle or a wave, which can be located anywhere in  $3D$ . Thus, the motion of quarks through a mechanical model" will be studied, for two reasons:

*First reason:* To preserve the white color of a nucleon, quarks must be indivisible and localized particles, i.e., classical particles at specific positions on the  $x, y, z$  axes which are attracted" by their center of mass.

*Second reason:* Quarks are representations of circles on  $\mathbf{P}^2$  and therefore correspond to classical particles.

## 13.2 Dynamics of Strong Interaction

### 13.2.1 The mechanical model

According to what we mentioned in section 12.1, we consider a central neighborhood of a differential manifold in which a circle is represented. The characteristic vector of a circle represented in a nucleon is the sum of three characteristic vectors of circles corresponding to three quarks, according to the analysis mentioned in Chapter 12.

We also consider a fourth point of the differential manifold, which is the center of mass of the quarks. This central neighborhood has the center of mass of the quarks as its central point. Since macroscopically (through experiments) it appears that quarks are attracted to the center of mass" while maintaining the same distances from it, then at the center of mass it is considered that there is *some kind of mass*", which causes an *attractive force*. This mass" is due to the tensor  $\bar{D}$ , as described in Chapter 11, which can be expressed through a parameter  $a_s$ , as we will see later.

Thus, a nucleon can be simulated by a *three-member rigid body*" whose members are the three quarks. Quarks are considered classical point particles, which tend to maintain the distances between them and with the center of mass, as well as all the properties of this neighborhood. If the quarks tend to move randomly away from the axes of the coordinate system  $x, y, z$ , then their center of mass  $O$  also tends to move randomly. Let us define  $Q$  as the point showing the position of a quark. If the quark moves by an infinitesimal distance  $dr_Q$ , then this motion is expressed as

$$dr_Q = dr_O + d\phi \times \mathbf{r} \quad (13.8)$$

where  $dr_O$  is the vector of an infinitesimal displacement of the center of mass  $O$  with respect to the origin of the axes of the coordinate system  $x, y, z$ ,  $r$  is the distance  $QO$  and  $d\phi \times \mathbf{r}$  is the displacement vector with respect to  $O$ , which comes from a rotation by an infinitesimal angle  $d\phi$ . Both sides of (13.8) are divided by the time interval  $dt$ , in which these displacements occur. Thus, we obtain the relation[14]

$$\mathbf{v}_Q = \mathbf{v}_O + \boldsymbol{\Omega} \times \mathbf{r} \quad (13.9)$$

where

$$\frac{\mathbf{r}_Q}{dt} = \mathbf{v}_Q, \frac{\mathbf{r}_O}{dt} = \mathbf{v}_O, \frac{d\phi}{dt} = \boldsymbol{\Omega} \quad (13.10)$$

The symbol  $\boldsymbol{\Omega}$  denotes the angular velocity of this rigid body” in the same direction as  $d\phi$  along its axis of rotation. Thus, the velocity of each quark of a nucleon relative to the origin of the coordinate system  $x, y, z$  is expressed as the sum of the translational velocity and the angular velocity of rotation of a nucleon. At the center of mass  $O$ , a moving Cartesian coordinate system is also taken into account, whose unit vectors are  $\mathbf{O}_1, \mathbf{O}_2, \mathbf{O}_3$ . Let us see what happens as the quarks of a nucleon tend to move randomly away from  $O$ , in order to transform into wave-particles. Let us assume that the quarks of this rigid body” are initially concentrated around  $O$ . The three quarks define the position of an infinitesimal neighborhood of  $D$ . The remaining space of  $D$  outside this neighborhood exhibits all its properties, i.e., isotropy and homogeneity. If the quarks begin to move randomly, then the region of  $3D$  that does not satisfy these properties becomes larger. *To minimize the region of  $3D$ -space that does not exhibit its properties, the three quarks must return to their initial position.* Free observers in  $3D$  interpret this shrinkage process under the assumption that a “spherically symmetric and strong attractive center” exists at the center of mass  $O$ . Thus, they consider that a curved spacetime manifold is created around  $O$ . The solution of the general equations (11.41) with  $\Lambda = 0$  in empty space, i.e.,  $T_{\mu\nu} = 0$ , is the Schwarzschild metric

$$ds^2 = -B(r)^{-1}dt^2 + B(r)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (13.11)$$

where  $c = 1$  and  $B(r) = (1 - \frac{a_s}{r})^{-1}$ . We consider the general equations (11.34) with  $\Lambda = \text{const}$ . If we use appropriate combinations of the Ricci tensor [17], we obtain the following differential equation

$$\frac{1}{rB^2(r)} \frac{dB(r)}{dr} + \frac{B(r) - 1}{r^2B(r)} = \Lambda$$

Since  $B(r) = (1 - \frac{a_s}{r})^{-1}$  is a partial solution of this differential equation, i.e., for  $\Lambda = 0$ , then a general solution of the above differential equation is obtained when we assume that  $a_s = a_s(r)$ . Substituting  $B(r) = (1 - \frac{a_s(r)}{r})^{-1}$  into this differential equation, we get

$$\frac{da_s(r)}{dr} = r^2\Lambda$$

The solution of this differential equation is

$$a_s(r) = \Lambda \frac{r^3}{3} + a_s$$

Substituting  $a_s(r)$  into  $B(r)$ , we get

$$B(r) = (1 - \frac{a_s}{r} - \frac{\Lambda}{3}r^2)^{-1}$$

Thus, we obtain the *Schwarzschild-De Sitter Metric*, namely

$$ds^2 = -(1 - \frac{a_s}{r} - \frac{\Lambda}{3}r^2)dt^2 + (1 - \frac{a_s}{r} - \frac{\Lambda}{3}r^2)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (13.12)$$

The Killing vectors[8] of this metric are

$$K_t = g_{tt} = -(1 - \frac{a_s}{r} - \frac{\Lambda}{3}r^2) \quad (13.13)$$

$$K_\phi = g_{\phi\phi} = r^2 \sin^2\theta \quad (13.14)$$

From these Killing vectors, we obtain the following conserved quantities

$$K_\mu \frac{dx^\mu}{d\lambda} = \text{constant}$$

where  $\lambda$  is an intrinsic parameter (*affine connection*) defined as

$$ds^2 = -\varepsilon d\lambda^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (13.15)$$

In the case of massive particles we define  $\lambda = \tau$  (proper time) and  $\varepsilon = c^2$ . In our case we get

$$E = K_t \frac{dt}{d\lambda} = -(1 - \frac{a_s}{r} - \frac{\Lambda}{3}r^2) \frac{dt}{d\lambda} \quad (13.16)$$

$$L = K_\phi \frac{d\phi}{d\lambda} = r^2 \sin^2\theta \frac{d\phi}{d\lambda} \quad (13.17)$$

$$\frac{da_s(r)}{dr} = r^2 \Lambda$$

The conservation of the direction of angular momentum  $L$  means that each quark must move in a plane perpendicular to this direction. Choosing the equatorial plane as the plane perpendicular to this direction, we set  $\theta = \frac{\pi}{2}$ . If a quark does not move in the equatorial plane, then the coordinate axes rotate in such a way that this condition is satisfied for each quark. Satisfying this condition leads to a *continuous rotation of the coordinate system that is equivalent to maintaining the white color of the nucleons*, as mentioned earlier for the process of color exchange between quarks. From equation (3.15) which gives the expression for  $\varepsilon$ , we get

$$\mathcal{E} = K_r + V(r) \quad (13.18)$$

where

$$\dot{r} = \frac{dr}{d\lambda}, \mathcal{E} = \frac{1}{2}E^2, K_r = \frac{1}{2}\dot{r}^2 \quad (13.19)$$

$$V(r) = -\frac{\Lambda\varepsilon}{6}r^2 - \frac{a_s\varepsilon}{r} + \frac{L^2}{2r^2} - \frac{L^2a_s}{r^2} + \frac{1}{2}\left(\varepsilon - \frac{\Lambda L^2}{3}\right) \quad (13.20)$$

For  $r \rightarrow 0$  we get  $V(r) \rightarrow -\infty$  and for  $r \rightarrow \infty$  we also get  $V(r) \rightarrow -\infty$ . This means that as quarks tend to move away from their center of mass, an infinite attractive potential forces them to return to their initial positions, i.e., to the region of the center of mass. Thus, quarks are confined to the region of the center of mass (*confinement*). The only motion that quarks make is a rotational motion for the reasons mentioned above, namely the preservation of the direction of angular momentum. The kinetic energy of this rigid body" is purely rotational with angular velocity  $\boldsymbol{\Omega}$  around an axis passing through the center of mass, which is written as

$$T_{rot} = \frac{1}{2}I_{ik}\Omega^i\Omega^k \quad (13.21)$$

Thus, the three quarks are free to execute only rotational and not translational motion. This means that quarks can have any color in the fixed coordinate system, with the constraint that a nucleon will always have a white color. This state is called **asymptotic freedom**.  $\Omega^i, X_i$  are components of the vectors  $\boldsymbol{\Omega}, \mathbf{r}$  with respect to the coordinate system  $\hat{\mathbf{O}}_1, \hat{\mathbf{O}}_2, \hat{\mathbf{O}}_3$ . The symmetric tensor  $I_{ik}$  is the *inertia tensor*

$$I_{ik} = \sum_l m_l \{X_l^2 \delta_{ik} - X_i X_k\} \quad (13.22)$$

Like any symmetric second-rank tensor, the inertia tensor can be expressed in diagonal form with an appropriate choice of axis directions  $\hat{\mathbf{O}}_1, \hat{\mathbf{O}}_2, \hat{\mathbf{O}}_3$ , which are the *principal axes of inertia*. The corresponding values of the diagonal components of the inertia tensor are the **principal moments of inertia**  $I_1, I_2, I_3$ . Let us assume that the coordinate systems  $x, y, z$  and  $\hat{\mathbf{O}}_1, \hat{\mathbf{O}}_2, \hat{\mathbf{O}}_3$  have the center of mass  $O$  as a common origin, but these two sets of axes do not coincide. This means that the components  $\Omega^i$  can be expressed through the **Euler angles** as follows:  $\hat{\mathbf{O}}_1$  forms an angle  $\theta$  with the  $z$ -axis. The intersection of the plane of axes  $\hat{\mathbf{O}}_1, \hat{\mathbf{O}}_2$  with the plane of axes  $x, y$  is the **line of nodes**  $ON$ . The  $x$ -axis forms an angle  $\phi$  with  $ON$ , and  $\hat{\mathbf{O}}_1$  forms an angle  $\psi$  with  $ON$ . Thus, the components of  $\Omega$  along the axes  $\hat{\mathbf{O}}_1, \hat{\mathbf{O}}_2, \hat{\mathbf{O}}_3$  are functions of the angles  $\theta, \phi, \psi$ . With this construction, we obtain the relations

$$\Omega^1 = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi \quad (13.23)$$

$$\Omega^2 = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi \quad (13.24)$$

$$\Omega^3 = \dot{\phi} \cos \theta + \dot{\psi} \quad (13.25)$$



where the derivative with respect to time is denoted by a dot. If the axes  $\hat{\mathbf{O}}_1, \hat{\mathbf{O}}_2, \hat{\mathbf{O}}_3$  are chosen as the principal axes of inertia, then for  $\Omega^i = \Omega_i$ , the kinetic energy  $T_{rot}$  takes the simple form

$$T_{rot} = \frac{1}{2}(I_1\Omega_1^2 + I_2\Omega_2^2 + I_3\Omega_3^2) \quad (13.26)$$

where

$$\begin{aligned} I_1 &= \sum_i m_i(X_{i2}^2 + X_{i3}^2), I_2 = \sum_i m_i(X_{i3}^2 + X_{i1}^2), I_3 \\ &= \sum_i m_i(X_{i1}^2 + X_{i2}^2) \end{aligned} \quad (13.27)$$

### 13.2.2 Equations of motion of quarks

The next step is to find the Lagrangian of the nucleon at the origin of a fixed coordinate system, and from it the equations of motion of the quarks. The kinetic energy is given by Equation (3.21) and the potential energy is given by Equation (3.20). Thus, we get

$$\mathcal{E}_k = \frac{1}{2}I_{ik}\Omega^i\Omega^k, (\Omega^i = \Omega_i) \quad (13.28)$$

The Lagrangian with respect to the principal inertial axes for  $\lambda = t$ ,  $c = 1$ , has the form

$$L = \frac{1}{2}(I_1\Omega_1^2 + I_2\Omega_2^2 + I_3\Omega_3^2) - \left(-\frac{\Lambda\varepsilon}{6}r^2 - \frac{a_s\varepsilon}{r} + \frac{L^2}{2r^2} - \frac{L^2a_s}{r^2}\right) \quad (13.29)$$

The constant term  $\frac{1}{2}(\varepsilon - \frac{\Lambda L^2}{3})$  has been omitted in this expression. Since at any moment the quarks define a plane, then to simplify calculations we assume that they form an equilateral triangle. At the base of this triangle are the quarks with equal masses and at the apex of this triangle is the third quark which has a different mass from the other two. Let us assume that this equilateral triangle has geometric elements determined by the assumption  $I_1 = I_2$ . Thus, due to the symmetry of the equilateral triangle, we get

$$I_3 = I_1 + I_2 = 2I_1$$

The rotational kinetic energy is written as

$$T_{rot} = \frac{1}{2}I_1(\Omega_1^2 + \Omega_2^2 + 2\Omega_3^2) \quad (13.30)$$

With equations (13.23), (13.24), (13.25), the relation (13.30) is written as

$$T_{rot} = \frac{1}{2}I_1[\dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2 + 2(\dot{\phi}^2 \cos^2 \theta + 2\dot{\phi}\dot{\psi} \cos \theta + \dot{\psi}^2)] \quad (13.31)$$

The Lagrangian of this symmetric top, with respect to the principal inertial axes for  $\lambda = \tau$ ,  $c = 1$ , is

$$L = \frac{1}{2}I_1[\dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2 + 2(\dot{\phi}^2 \cos^2 \theta + 2\dot{\phi}\dot{\psi} \cos \theta + \dot{\psi}^2)] \\ - \left(-\frac{\Lambda\varepsilon}{6}r^2 - \frac{a_s\varepsilon}{r} + \frac{L^2}{2r^2} - \frac{L^2 a_s}{r^2}\right) \quad (13.32)$$

From Equation (3.32) we can find the Euler-Lagrange equations which are the equations of motion of this rigid body”, namely

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0 \quad (13.33)$$

We will not solve these equations because this Lagrangian has a special form through the above assumptions, which restricts this method to a specific problem. But, through this method, one can appropriately choose the principal inertial axes and solve similar problems, from which one must be selected that gives results closer to experimental or observational data.

## Part III

# STAGES OF EVOLUTION OF THE UNIVERSE



## Chapter 14

# THE ERA OF RADIATION DOMINANCE

### 14.1 Big Bang or Big Expansion?

#### Questions and Answers

To further develop the geometric model, we need to analyze it properly so that we can visualize it in three-dimensional space and see its consequences through the eyes of free observers. The theory widely accepted today by free observers for the origin and evolution of our Universe is the Big Bang model. According to the Big Bang model, the Universe began as an extremely hot and superdense point about 13.7 billion years ago and evolved into the structure we see today.

**How did the Universe evolve from such a “point” to the huge and complex structure we observe today?**

Some of the questions that arise from the Big Bang model are the following:

1. *If the early Universe was only a “singularity”, then to what space does this point belong, and how is such a space constructed from all these points?*
2. *What does a “super-hot point” mean when the concept of temperature has not yet been defined?*
3. *What does a “superdense point” mean when the concept of mass and the concept of volume of a space have not yet been defined to define the concept of density?*

Since the definition and construction of a space consisting of such points are difficult to understand, we will start differently by defining an “understandable point”, and we will construct a space from these understandable

points. Thus, if instead of an incomprehensible and unique point, which is a dimensionless geometric object, we consider an infinitesimal region such as the Planck point in two dimensions, then from this we can construct the two-dimensional Planck space  $\mathbf{P}^2$  through the Planck points.

As already mentioned, this space is constructed through a self-similar process and constitutes the boundary of the Universe for the following reasons:

- **First**, we invoke the obvious observational fact that what we see and perceive around us is a three-dimensional space. If a space has  $n$  dimensions, then its boundary, if it exists, has one dimension less, that is,  $n - 1$  dimensions. Since the space we perceive around us is three-dimensional, then its boundary will be a two-dimensional space.
- **Second**, we invoke the non-obvious theoretical and observational data that come from the Friedmann equation and the fact that the energy density parameter  $\Omega$  tends to unity, which means that space is flat. From these data, space on a scale larger than that of super-clusters of galaxies, which are the boundary of the Universe, is flat, as is the Universe as a whole.

Combining these two facts, we conclude that the boundary of the three-dimensional space we perceive around us is a flat space that can be constructed as the two-dimensional Planck space  $\mathbf{P}^2$ . Having defined the boundary of three-dimensional space, we can then answer the above questions. Before doing so, let's see how the name Big Bang came to be, which is not the most appropriate term. This search is being done because there was no measure for its definition as "Big", nor was there space to be characterized as an "Explosion". This name was given by the British astrophysicist Fred Hoyle to discredit the idea that the Universe began from a single point. If this had happened, then the Universe would have had a continuous evolution and would not have been eternal and unchanging, as Hoyle believed. Due to the construction of  $\mathbf{P}^2$  through the self-similar process we mentioned, we propose to change the name Big Bang to the **Big Expansion** (*BE*).

What happens in the flat space  $\mathbf{P}^2$  can be depicted holographically in 3D-space, creating the illusion to free observers that only 3D-space exists. We will start the study of the evolution of the Planck patches, through a self-similar expansion-contraction process from a Planck point, and continue with the next two Planck patches. After these, we will continue with the Planck patch  $P_9$  because:

- a) This is the first Planck patch after inflation, i.e., a sharp increase in the initial Planck patches, as we will see below, and
- b) In it, the first simple and the simplest of the complex geometric objects are created, i.e., circles with infinitesimal radius  $0^+$ , circles with radius  $1l_p$ , circles with radius  $2l_p$ , and circles with radius of  $1^+l_p$ .

The bound observers, due to the infinitesimal area of the Planck patch  $P_9$ , namely  $A_{P_9} \sim 10^{-64} \text{ cm}^2$ , consider that the circle that surrounds it has a “huge” positive curvature, which also means that its temperature will tend to a huge value because the curvature is proportional to the temperature. This fact, for the bound observers, means that the z.c. $\mathbf{P}^2$  is not satisfied for the curvature. Thus, as we mentioned above, the process of expansion of the Planck patches that started from the Planck point should continue, so that the curvature of its initial circumscribed circle gradually begins to decrease until it reaches zero. The curvature becomes zero when the radius of the circumscribed circle in a huge Planck patch tends to an infinite value.

This process of enlargement of the Planck patches does not happen randomly, but follows a self-similar enlargement process. This means that the space  $\mathbf{P}^2$  is created in a discontinuous way, as mentioned in Chapter 3, through the Planck patches. The parameter  $t$  defines the time it takes for each Planck patch to form during the expansion process. As the beginning of the measurement of the parameter  $t$ , we define the moment when the creation of a Planck point begins. Thus, at this moment we define the value zero for the parameter  $t$ , that is,  $t_0 = 0$ . The completion of the creation of a Planck point requires the minimum Planck time, that is,  $\Delta t_p = t_p - t_0 = 5.39 \cdot 10^{-44} \text{ sec}$  (Planck time). From now on, we will use the term *BE* for the moment when the creation of the Universe began, and not the term Big Bang.

This definition frees us from the initial singularity that appears when considering that everything started from a Big Bang. As the boundary of the Universe expands, various simple and complex geometric objects are created within it. These geometric objects within the Universe are realized as 3D holographic objects by free observers. The name holographic objects arises from the technique of constructing holograms by humans today. The two-dimensional film in which the object to be projected in three-dimensional space is created and stored has a self-similar structure, as does the human cerebral cortex that is the receiver of this representation in the film. This means that the entire Universe and its boundary must have been structured through a self-similar process, as the two-dimensional film and the human cerebral cortex have been structured in order to be understandable by humans.

## 14.2 Inflation

When the Universe was very young, that is, at about the time  $\sim 10^{-35} \text{ sec}$ , it experienced an exponential expansion. During this expansion, known as **Inflation**, the Universe grew exponentially, as shown in the right-hand Table 17.9 of Appendix 17.5.

At the beginning of the expansion, these conditions, as considered by

bound observers, led to an “isothermal and adiabatic change of an ideal fluid” consisting of non-interacting geometric objects, e.g., an “ideal gas of photons”, as considered by free observers. These luminous geometric objects, after this isothermal and adiabatic expansion, moved isotropically within the Planck patch to fulfill the isotropy criterion of  $\mathbf{P}^2$ .

During this isothermal and adiabatic expansion, the area of the Planck patch increased exponentially in a short time, as follows from the relation  $TV^{\gamma-1} = \text{constant}$  for an isothermal and adiabatic expansion, without following the previous self-similar process. But the expansion of the subsequent Planck patches, after inflation, continued through a corresponding self-similar process that had as its initial Planck patch the one that resulted from the isothermal and adiabatic change, namely the Planck patch  $P_9$ .

The inflationary period began at  $10^{-34}$  sec and stopped at  $10^{-29}$  sec after  $BE$ , as shown in the right-hand Table 17.9 of Appendix 17.5, where the Planck patch grew significantly. This sharp increase in the Planck patch resulted in a dramatic decrease in the curvature of its circumscribed circle and therefore its temperature.

This entire process can be described by free observers with mathematical methods based on Quantum Field Theory and the Klein-Gordon equation. To describe the entire space occupied by a Planck patch, the Klein-Gordon equation contains an unknown quantity to describe it. This quantity was called the inflaton. Through the idea of inflation and the use of the equations of General Relativity, one can give an answer to the Horizon Problem and the Flatness Problem of the Universe. In our theory, these two problems do not arise because, since the construction of the Planck patches and their evolution are isotropic and flat as parts of the flat two-dimensional Universe on the one hand, there is a causal connection between them on the other hand.

The fact that the sharp increase in the Planck patch stopped in such a short time is due, as we mentioned, to the contraction process during which the curvature of the Planck patch increased. Thus, the Planck patch, after the inflationary period, returned to its previous expansion rates. The Universe, over time, expanded and became cooler and less dense. After inflation, the Universe continued to expand, but at a slower rate than during the inflationary period. Over time, as space expanded sufficiently, the Universe cooled further and the forms of matter, as we know them today, began to appear.

### 14.3 Elementary Geometric Objects

In this chapter and in the following chapters, when we refer to the characteristic vectors of circles or ellipses corresponding to physical entities, we will mean the characteristic vectors of their quadrants (Chapter 2). The phys-



ical entities corresponding to these luminous geometric objects, i.e., circles of radius  $r = 1l_p$  enclosed in almost linear ellipses, are called photons ( $\gamma$ ) by bound and free observers. As the Planck patches expand continuously, then the length of the semi-major axis of the linear ellipses increases continuously. This means that the linear ellipse becomes part of a linear hyperbola whose equation is identical to the wave equation. Thus, these complex linear objects exhibit wave properties.

Photons are geometric entities that have no charge. Since the mass is proportional to the curvature of the circle surrounding the linear ellipse with a continuously increasing radius, then the mass of the photon tends to zero. Photons appear mainly as waves to free observers, but in some cases, photons also exhibit their particle nature due to the circle  $c_0$  that is enclosed within the linear ellipse. Photons retain the curvature of the circle that surrounds them. Thus, in the last Planck patch of the radiation-dominated era, there are photons of various curvatures or energies, where energy is the magnitude with which free observers correspond to curvature.

1. **Photons-Gluons:** According to the geometric model, as each Planck patch expands, the value of the curvature and therefore its temperature decreases. After a certain Planck patch, all the  $\gamma$ 's that were formed in it and will be formed in its subsequent Planck patches become dark. Thus, in these Planck patches, the gluons, which are circles of radius  $r = 1$  or ellipses  $(a, b) = (1, 2)$  surrounding circles of radius  $r = 1$ , have the same structure as photons, but because the temperature has been significantly reduced, these become the “dark copies of photons”.

These geometric objects are called **gluons** by bound and free observers, due to their tendency to form, together with quarks, the complex structures (nuclei) which in turn create even more complex structures (atoms, molecules, etc.). Thus, structures are formed from gluons alone or from gluons and circles of radii  $r = 1, r = 2$  corresponding to quarks, or ellipses  $(a, b) = (1, 2)$  corresponding to electrons.

The creation of complex structures between gluons, or between gluons and other geometric objects, continued until the last Planck patch of the radiation-dominated era. This is how large and small stable dark structures were created, such as primordial black holes, the galaxy halo, and nucleons; that is, structures made of gluons only, and of gluons and quarks which are the baryons (protons and neutrons), respectively. The dark geometric structures with  $r = 1^+, r = 2^+$ , and  $(a, b) = (1^+, 2^+)$  that correspond to three-dimensional physical entities in 3D-space are called **Dark Matter** by free observers.

These dark matter structures should follow the constraints that arise from the application of the definition axioms of  $\mathbf{P}^2$ , namely isotropy and the zero criterion. For example, we mention the constant velocity

of the stars of a galaxy that are within its dark halo. This is equal to the constant rotation speed of the dark halo around the Black Hole at the center of the galaxy in order to satisfy the isotropy criterion. Correspondingly, the creation of atoms is done in order to satisfy the zero criterion of  $\mathbf{P}^2$ . More complex structures, such as molecules, are created from atoms, and so on.

2. **Neutrinos ( $\nu$ ):** In the initial Planck patches of the radiation-dominated era, geometric objects are created, except for photons, which are circles of infinitesimal radius  $r = \frac{l_p}{2}$  because they are inscribed in every Planck point. Bound and free observers call these neutrinos and their holograms **neutrinos**. Since these circles are the inscribed circles of any Planck point, bound observers consider them as Planck points or Planck circles. According to the definition of the mass on the one hand and the charge on the other hand in the conic sections, these have values proportional to  $\frac{l_p}{2}$ . From the characteristic vector  $(-\frac{l_p}{2}, \frac{l_p}{2})$  of these circles and the relation that gives the mass, we get  $m_\nu = 2.7 \cdot 10^{-34}$  g. The charge of the geometric objects corresponding to neutrinos is found from the characteristic vector  $(\frac{l_p}{2}, \frac{l_p}{2})$  and has a numerical value given by the relation:

$$q_\nu = \frac{l_p + 2l_p}{3}$$

For the mass, the experimental value found is  $\sim 10^{-34}$  g, that is, it is in very good agreement with the theoretical value that we get from the geometric model. Since no matter structure is created with an infinitesimal charge that is opposite to the neutrino charge, then neutrinos are not synthesized with any structure to compensate for its charge. The fact that neutrinos pass through matter as if it did not exist, moving at enormous speeds towards the boundaries of the Planck patches which are constantly being created, is a consequence of z.c. $\mathbf{P}^2$ .

The satisfaction of z.c. $\mathbf{P}^2$  explains why neutrinos do not interact with matter and do not merge with it, because no non-zero characteristic of them, such as the neutrino charge, is allowed to exist in any of their regions. These geometric objects are formed by circles with infinitesimal charge and mass and a specific orientation, which are surrounded by ellipses with the same charge but with an opposite orientation to that of the circles. These ellipses have semi-axes lengths  $(a, b) = (1, 2)$  in the various Planck patches.

According to this geometric structure, these particles correspond to wave-particles for free observers. The conjugate particles of these have opposite charges and are called antineutrinos. Free observers have

experimentally found an infinitesimal mass, but have not yet experimentally found an infinitesimal positive or negative charge. This representation leads to uncertainties in the measurements of their charge or orientation for determining the type of conic section, in agreement with the Complementarity Principle of Quantum Mechanics.

3. **Electrons:** In the initial Planck patches, circles of radius  $r = 1$  were formed within the ellipses with lengths of their semiaxes  $(2l_p, 1l_p)$ , as well as circles of various other radii. Due to the high curvature (energy/temperature for free observers), these geometric objects were parts of geometric aggregates of very high curvature. When the Planck patch was greatly enlarged, these geometric objects became dark, displaying the characteristics of particles, waves, and wave-particles, as determined by bound and free observers. The characteristics of these geometric objects are as follows:

- a) Unit radius circles are characterized by a vector which we called a characteristic vector, with coordinates determined by the free quadrant of the unit circle because the other three give zero values for their scalar and vector quantities. Thus, we obtain the vectors  $(1, 1), (-1, 1), (-1, -1), (1, -1)$  for the first, second, third, and fourth quadrants of the circle, respectively.
- b) The ellipses that surround unit radius circles have characteristic vectors  $(1, 2), (-1, 2), (-1, -2), (1, -2)$  for the first, second, third, and fourth quadrants of the ellipse, respectively.

From these characteristic vectors, the scalar quantity  $q_r$  arises that the bound and free observers call the reduced charge, with values  $+1$  or  $-1$  for the circle and ellipse. Also, the vector quantity  $\hat{\mathbf{O}} = \pm \hat{\mathbf{e}}_3$  arises, perpendicular to the plane of the circle or ellipse, which the bound observers call **orientation**.

Free observers give different names for this vector quantity depending on what they want to observe. As in the case of neutrinos, the above geometric objects with the same value  $q_r$  and opposite values of their orientations create uncertainties when measuring their vectors for the bound and free observers, according to the complementarity principle of quantum mechanics.

We call these geometric objects dark electrons if they have  $q_r^e = -1$  and their conjugates dark positrons if they have  $q_r^{e^+} = 1$ .

Neutrinos-antineutrinos and dark electrons-positrons have the same geometric structure but belong to different scales and are called leptons. The building block of the Planck patch in which the dark electrons-positrons are created tends to increase its curvature due to its contrac-

tion. This results in the formation of electrons that combine with photons in their appropriate quadrants, which are called visible electrons and visible positrons, or simply electrons and positrons. These visible geometric objects, which result from the synthesis of dark geometric objects with photons, have quantities that have been experimentally revealed today, such as spin, in an extended Planck patch.

4. **Quarks:** If we consider the characteristic vectors  $(-1, 1)$  and  $(1, -1)$  of the second and fourth quadrants of the unit circle, then their reduced charge is  $q_r = \pm \frac{1}{3}$ , respectively. The ellipses  $(1, 2)$  do not have a characteristic vector that gives the same  $q_r = \pm \frac{1}{3}$  as that of the unit circle with characteristic vectors  $(-1, 1)$  and  $(1, -1)$ . This means that the circles with radius  $r = 1$  and with these characteristic vectors, although surrounded by the ellipses  $(1, 2)$ , do not form permanent complex quarks with them, so they do not correspond to wave-particles but only to particles. The reason is that the reduced charge of the circle is different from the reduced charge of the ellipse.

Geometric objects with fractional charge values  $q_r$  are called quarks. Quarks with  $q_r = -\frac{1}{3}$  are called  $d$ -quarks and quarks with  $q_r = \frac{1}{3}$  are called  $\bar{d}$ -antiquarks, respectively.

The circles with radius  $r = 2$  and characteristic vectors  $(-2, 2)$ ,  $(2, -2)$ , although they are surrounded by the ellipses  $(3, 2)$ , do not correspond to wave-particles.

The reason is that the reduced charges of these circles and ellipses are different. Thus, they correspond only to particles. If we consider the characteristic vectors  $(-2, 2)$ ,  $(2, -2)$  of these circles, then their reduced charges are  $q_r = \pm \frac{2}{3}$ . The quarks with reduced charges  $q_r = \frac{2}{3}$  and  $q_r = -\frac{2}{3}$  are called  $u$ -quarks and  $\bar{u}$ -antiquarks, respectively.

#### **Why are there only these two types of quarks-antiquarks?**

The reason is that these two types correspond to the simplest geometric objects and express the particle part of the structures we have examined so far. Quarks are bound by gluons, forming specific structures for reasons we explained earlier.

As the Planck patches continue to expand, their temperature becomes lower and lower until it reaches a critical Planck patch, which is the last Planck patch of the radiation-dominated era. We call this Planck patch  $P_R^f$ .

## Chapter 15

# THE ERA OF MATTER DOMINANCE

### 15.1 Preface

In the Planck patch  $P_M^i$  created by the Planck patch  $P_R^f$ , according to the self-similar expansion-contraction process, the era of matter dominance begins. The existing geometric objects acquire a smaller curvature when they are in a larger Planck patch  $P_M^i$  which is surrounded by a circle of larger radius. The temperature of a Planck patch decreases when the curvature of the circle surrounding it decreases because these quantities are proportional (Appendix 17.4: Definition of temperature).

To satisfy the zero criterion of  $\mathbf{P}^2$ , the geometric objects are joined together, forming increasingly complex structures whose curvature value approaches the curvature value of  $P_R^f$  (during the shrinkage process). Thus, these complex structures initially have the curvature of  $P_M^i$  and then their curvature increases to approach the curvature of  $P_R^f$ , which is greater than that of  $P_M^i$ , according to the expansion-contraction process. The production of increasingly complex structures does not continue in perpetuity, because each Planck patch is constantly expanding and its curvature is decreasing.

Thus, the existing structures are spread out to acquire the same curvature as the Planck patch in which they exist. Then they are joined together to approach the curvature of the previous Planck patch. Thus, they become looser structures than the previous ones. This process continues until the Planck patch  $P_M^f$ , which defines the end of the matter-dominated era, where the maximum complex structure (a super-super-cluster of tree-shaped galaxies) is formed.

The geometric objects of  $P_9$  have the same properties as  $P_9$  itself, that is, they are “luminous geometric objects” because they have a high temperature and form a high-density aggregate because they also have a high curvature. Circles with radii  $1l_p, 2l_p, \dots$  cannot move freely within this Planck patch,

nor in the subsequent Planck patches from it. This is due to the fact that all geometric objects are concentrated together in a Planck patch in order to have a local curvature equal to the curvature of the previous Planck patch from it, according to the self-similar structuring process of  $\mathbf{P}^2$ .

The physical entities corresponding to these luminous geometric objects, i.e., the circles of radius  $1l_p$ , are called photons ( $\gamma$ ) by bound and free observers. Photons correspond to geometric entities that have no charge and mass, as follows from their characteristic vectors. Photons are considered wave-particles by free observers. Thus, photons can exhibit either particle or wave nature (as electrons do during the creation of complex structures). Photons retain the curvature of the Planck patch in which they were created, so their curvature or energy takes on distinct values. In the last Planck patch of the radiation-dominated era, there are photons of different curvatures or energies.

Free observers classify these photons by their “energy quanta”, while bound observers classify them by their “curvature quanta”. As the Universe cools, after about 47,000 years from the Big Expansion (*BE*), the Universe becomes dominated by matter. This happens because the energy density of matter begins to exceed both the energy density of radiation and the energy density of the vacuum that begins to appear due to the expansion of the Planck patches, as we will see later. As we will see in the next section, the matter in the Universe is distributed as follows:

- **Dark matter:**  $\frac{64}{76} \sim 84\%$
- **Normal (baryonic) matter:**  $\frac{12}{76} \sim 16\%$

We call baryonic matter “normal” because we are made of it, even though its percentage is the smallest of the percentages of dark matter and dark energy. Visible matter should be called “non-normal” because the terminology “normal” is more appropriate for dark matter, which has a larger percentage than visible matter.

Free observers, from this point on and for several billion years thereafter, believe that the presence of dark matter accelerates “structure formation”, i.e., the formation of galaxies, galaxy clusters, and super-superclusters of galaxies in our Universe. For bound observers, the cause for the creation of the aforementioned structures is the inherent properties of  $\mathbf{P}^2$ .

## 15.2 Recombination

The last Planck patch of the radiation-dominated era is  $P_R^f$ . The next Planck patch from  $P_R^f$  is  $P_M^i$ .  $P_M^i$  is the first Planck patch of the matter-dominated era. About 370,000 years after the *BE*, two events occurred that were related to each other:

- a) The end of recombination, and
- b) The decoupling of photons.

For bound observers, the term **recombination** describes the combination of the geometric objects corresponding to the first nuclei  $H$  and  $He$  and the geometric objects corresponding to the electrons  $e$ , so that the total charge of the complex geometric structures is zero. This is done to satisfy z.c. $\mathbf{P}^2$ . The name recombination is inaccurate (it is mentioned only for historical reasons) because, in reality, the electrons and atomic nuclei are combined together for the first time.

Most of the radiation corresponding to photons could not travel significantly long distances, so the Universe corresponding to this Planck patch was opaque or “hazy”. Although there was light in this Planck patch, it is not possible to observe it through telescopes today.

For bound observers, starting about 18,000 years after the  $BE$ , the Planck patch corresponding to the Universe had less curvature or lower temperature than its previous Planck patches. Neutral nuclei  $H$  and  $He$  then began to form about 100,000 years into this era, with hydrogen formation peaking at about 260,000 years. By the end of recombination, most of the proton-corresponding particles in the Universe had formed geometric objects corresponding to neutral atoms.

This change from charged to neutral geometric objects means that free space has increased due to the creation of complex structures. Also, due to the expansion of the Planck patch, the free space has increased even more, so that geometric objects corresponding to free photons can travel isotropically over large distances. This motion of photons has been called **photon decoupling**. Thus, for free observers, the Universe has become transparent to visible light, radio waves, and other electromagnetic radiation for the first time in its history.

The bound observers call the Planck patch where the photon decoupling occurred “the Planck transparent patch”. This occurred about 370,000 years after the  $BE$ , as estimated today from observational data and theoretically from Table 17.9 of Appendix 17.5.

After billions of years since the decoupling, and as the Universe has expanded, the frequency of these photons has shifted toward the red, which today corresponds to a temperature of about 2.73 K. The redshift describes photons that have acquired longer wavelengths (or lower frequencies) as the Universe has been continuously expanding for billions of years. Thus, this radiation gradually changed from visible light to radio waves. These photons are now detected as radio waves that make up the Cosmic Microwave Background Radiation (CMBR) and provide important information about the early Universe and its evolution.

From the transparent Planck patch on wards, the geometric objects that correspond to the large complex structures of all dark geometric objects begin

to form. Thus begins the era of matter creation, as observed in the initial anisotropies that appear in the Cosmic Microwave Background. The photons that were formed at the different stages of the expansion of a Planck patch in the era of radiation dominance have different curvatures (proportional to the expansion). This results in the quantization of the curvature (or energy) of the photons.

The cosmic microwave background radiation was discovered by chance by radio astronomers Robert Wilson and Arno Penzias of Bell Labs in 1965. The two scientists were unable to identify the “noise” recorded on the antenna as a constant intensity radiation with a temperature of about 3 Kelvin. This radiation was received by the antenna regardless of its orientation. This last observation confirms the isotropic propagation of cosmic microwave radiation.

Despite the two scientists’ persistent efforts to locate and interpret this noise, even cleaning the antenna of pigeon droppings, the noise remained. At that time, Professor Robert Dicke and his team at Princeton University had found that the radiation from the early Universe should today have a value of about 10 Kelvin and were preparing to make measurements when the results of the research of Wilson and Penzias were announced to them. This discovery showed that the Universe at the beginning of its expansion was extremely hot and that the radiation it had at each stage of its evolution was and is isotropic. Thus, these two scientists were awarded the Nobel Prize in Physics in 1978.

### 15.3 Photon decoupling

Large-scale structures corresponding to Planck patches contain geometric objects with opposite orientations and opposite reduced charges, so free observers only take into account their mass. Thus, dark large-scale structures were created, such as black holes, and small-scale structures such as nucleons (protons, neutrons). Since black holes are electrically neutral geometric objects, it is up to the nucleons to satisfy the self-similarity criterion and the zero-valued  $\mathbf{P}^2$  criterion. Through self-similarity, the transition to appropriate nuclei was created from combinations of protons and neutrons, such as the nuclei of hydrogen ( $H$ ) and helium ( $He$ ). By combining nuclei with geometric objects corresponding to electrons, the zero criterion  $\mathbf{P}^2$  is fulfilled. Thus, the creation of Visible Matter begins.

After the concentrations of geometric objects at specific points in the Planck patch to be consistent with the properties of  $\mathbf{P}^2$ , such as, for example, its isotropy (this is seen today from the positions of galaxies that form a nearly isotropic lattice and each of them has at least one black hole in its center), a lot of space remained in the Planck patch.

Photons took advantage of this empty space to move freely and isotrop-



ically, and the Universe began to become luminous. About 380,000 years after the *BE*, the Universe contained radiation from its early stages, which we now see as the Cosmic Microwave Background.

At this cosmic epoch, the Universe entered the Dark Age, so named because the bright stars and galaxies (light sources) we see today had not yet formed. Most of the matter in the Universe at this stage was dark matter, with a smaller percentage of ordinary matter that was largely neutral hydrogen and helium. Over the next hundreds of millions of years, the Universe entered a critical juncture in its evolution known as the **epoch of Reionization**.

During this period, the dominant dark matter began to coalesce into larger structures, due to self-similarity, along with photons forming a halo around them to further increase the local curvature. Thus, the first stars and galaxies began to form through turbulent flow, as we will see later. Since the corresponding galaxies that formed in the Planck patch of the entire Universe had a large curvature, then their members began to separate from each other in such a way that their curvature became equal to that of the Planck patch in which they formed. This process was called **reionization**. This resulted in the nuclei being separated from their electrons.

**Question:** Why were the quarks of nucleons not separated and only the nuclei separated from their electrons?

**Answer:** The parts of the complex structures that join together with some of the original structures in larger Planck patches are more loosely connected to the original structures, due to the smaller curvature of the Planck patch that was created; therefore, they break apart more easily. In general, the parts of a structure are more easily removed from the structure the larger the Planck patch in which this structure was formed. The quarks joined together in Planck patches of large curvature. The electrons joined with nuclei to form atoms in Planck patches of smaller curvature. In general, the reason is the conservation of the curvature that a complex structure has, whose value is the same as that of the Planck patch in which it was formed. This is the reason for the different apparent strengths of the various “interactions”.

The strong interactions reflect the fact that the formation of nucleons took place in the early Dark Ages, i.e., in a relatively small area and therefore large curvature Planck patch. The electromagnetic interactions reflect the fact that the formation of atoms took place towards the end of the Dark Ages, i.e., in a more extensive Planck patch. The weak interactions reflect the fact that only the neutron decays because the geometric object corresponding to it has a greater curvature than that of the geometric object corresponding to a proton. This shows the local tendency of the geometric object corresponding to a neutron to have the smallest possible curvature.

The reverse reaction, namely the formation of neutrons from protons, is possible during a contraction process inside stars, as we will see later.

Finally, “gravitational interactions” express the tendency for each Planck patch to grow so that the zero criterion  $\mathbf{P}^2$  is satisfied, i.e., a process in which the strength of this interaction is constantly decreasing.

The interpretation given by free observers is that after the formation of the first stars and galaxies, these physical entities in turn released large amounts of ultraviolet light. Ultraviolet light has a very high energy capable of removing electrons from the surrounding neutral matter. The period of re-ionization took place in the distant past of the Universe and is at the limit of current cosmological observations.

## 15.4 Dark Ages of the Universe

After recombination and decoupling of photons, bound observers consider that the Planck patch corresponding to the entire Universe has grown so much that it is quite sparse. Thus, its density  $\rho = \frac{N_{\text{geo}}}{S_P}$  is small, where  $N_{\text{geo}}$  is the number of geometric objects within it and  $S_P$  is its total area. Also, this Planck patch has a low temperature due to the increase in the radius of its circumscribed circle, thus a decrease in its curvature and consequently its temperature. Due to these conditions, light can travel long distances within it. Large luminous structures, such as the geometric objects corresponding to stars and galaxies, have not yet been created to “illuminate” the Planck patch corresponding to the Universe at that time.

These complex structures require a long time to form, during which repeated processes of expansion-contraction of the Planck patches occur. The geometric structures corresponding to stars and galaxies took hundreds of thousands, perhaps even millions of years after recombination to form, so the Universe remained dark for a long time. The period during which this sequence of Planck patches appeared, which followed the era of photon decoupling until the Universe became bright, is called the **Dark Age of the Universe**.

This period began about 370,000 years after the *BE*. During the dark ages, the temperature of the Universe dropped greatly. Free observers believe that there were two sources of photons during this period:

1. Photons released during the recombination-decoupling period that we can detect as the cosmic microwave background (CMB), and
2. Photons occasionally emitted by neutral hydrogen atoms.

The photon emission process known as the “**21 cm spin line**” of neutral hydrogen is described, from the perspective of bound observers, as follows: If a geometric object corresponding to the *H* atom is formed by the geometric object corresponding to the proton, which as we have seen has an orientation of  $+1\hat{e}_3$ , and a geometric object corresponding to an electron with the same orientation of  $+1\hat{e}_3$ , then:

- a) The zero criterion of  $\mathbf{P}^2$  is not met because the total orientation of the  $H$  atom will be  $+2\hat{e}_3$  and not zero.
- b) It is dark.

An electron with orientation  $+1\hat{e}_3$  has a dominant particle nature, so it is difficult to remove it and replace it with an electron with orientation  $-1\hat{e}_3$ , whose dominant nature is a wave nature. This has been clarified previously. To satisfy the zero criterion of  $\mathbf{P}^2$ , the orientation of the electron must be reversed (from positive to negative) and the electron must absorb a photon to become a wave with a negative orientation. To do this, we consider two photons each with an orientation of  $-1\hat{e}_3$ , so both are considered as waves. One of these photons nullifies the electron's initial orientation and moves away from the electron as a wave-particle, i.e., with zero orientation. The other photon binds to the electron, creating the wave nature as its dominant nature. Thus, the electron acquires an orientation opposite to its initial orientation. What happens is a reversal of the electron's orientation (spin-flip), as shown in the following Figure 15.1. Making calculations, we find that

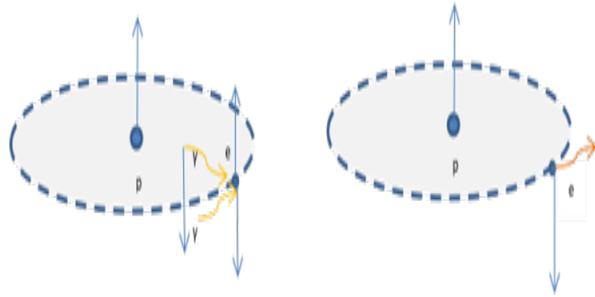


Figure 15.1: Spin-flip in a hydrogen atom

the wavelength of the photon that moves away is about 21 cm (or frequency  $f = 1420$  MHz). This process does not indicate an additional source of photon creation, as free observers believe, but a process of converting the dominant nature of photons from wave to wave-particle nature.

## 15.5 Discussion on the evolution of matter

In the last Planck patch dominated by radiation, namely  $P_R^f$ , all the simple and simpler complex geometric objects (wave-particle) and some large gluon structures that formed in the previous Planck patches of the radiation-dominated era had been created. All these formed a compact ensemble that had a large curvature and therefore a high temperature, due to the relatively small area occupied by  $P_R^f$ . In Table 17.9 of Appendix 17.5, we see that in the Planck  $P_R^f$  patch the corresponding temperature is of the order of  $10^6(^0K)$ .

We define the Planck  $P_R^f$  patch as the last Planck patch of the radiation-dominated era, because its curvature (or temperature) has decreased so much that the number of circles with radii greater than  $2l_p$  do not correspond to any hologram in three-dimensional space. In the Planck  $P_R^f$  patch, a second inflation occurs, but much milder than the first one, leading to the Dark Ages. The next Planck patch after  $P_R^f$  will be the initial Planck patch of the matter-dominated era, which we denote by  $P_M^i$  and is depicted in the following Figure 15.2.  $P_M^i$  has as its structural unit  $P_R^f$ . When we refer to the term structural unit, we mean only the specific space without the geometric objects that exist in it. At the beginning of the evolution of  $P_M^i$ , the entropy of this Planck patch takes its maximum value due to the maximum disorder of the geometric objects that exist in it.

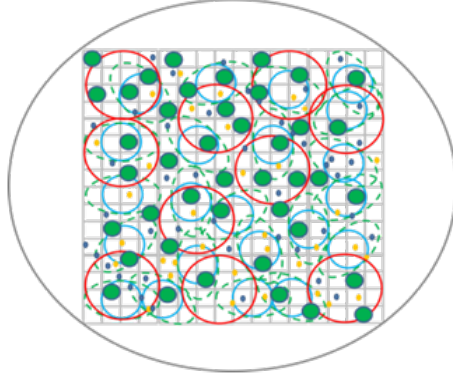


Figure 15.2: Giant molecular cloud in  $P_M^i$

All the geometric objects present in  $P_M^i$  are those that existed in  $P_R^f$ , and new geometric objects are created in  $P_M^i$ . Figure 15.2 shows the dispersion of all geometric objects of  $P_R^f$  in the Planck patch  $P_M^i$ . This happens so that the curvature of their circumscribed circle is equal to the curvature of the circumscribed circle of the Planck patch  $P_M^i$ . All the  $P_R^f$  structures start to become less and less bright until they become completely dark, as the temperature decreases in  $P_M^i$  and in the subsequent Planck patches after  $P_M^i$ .

The only geometric objects that remain luminous are the photons that were created in all previous Planck patches before  $P_R^f$ , which retain the curvature of the Planck patch in which they were created. This curvature of the photons is also called energy by free observers and takes only discrete values because the Planck patches in which the photons were created are structured through a quantum broadening, i.e., discontinuously. All the scattered geometric objects within  $P_M^i$  are concentrated around and within its building blocks, which are the Planck patch  $P_R^f$ .

According to the self-similar construction of  $P_M^i$ , it is created by  $P_R^f$ ,  $P_R^f$

by its building blocks, and so on. The geometric objects around and inside the  $P_R^f$  building blocks begin to approach each other so that the structure they will create will have, at least approximately, the curvature of  $P_R^f$ . This also applies to the building blocks of  $P_R^f$ , the building blocks of the  $P_R^f$  building blocks, and so on.

In general, the greater the curvature of a Planck patch, the more strongly the parts of the structures created in the immediately following Planck patch are connected. As the curvature of a Planck patch decreases more and more, the parts of the structures created in the next Planck patch are connected more loosely to each other. The complex structures created cover a larger area overall because they must lie within the circumscribed circle of the new Planck patch. Every large structure is composed of members, and every member is composed of parts. Thus, we get the general proposition:

**The members of each part of a complex structure are more strongly connected to each other than the parts themselves, because their curvature should tend to the greater curvature of the previous Planck patch than the present Planck patch.**

In other words, simpler structures are more compact than complex ones. The creation of a structure is due to two opposing processes, which are the increase of the distances between the most complex geometric objects of the structure and the decrease of the distances between the immediately less complex geometric objects of the structure, during the evolution of the Planck patches. These two opposing processes occur in order to satisfy the criterion of self-similarity between two successive Planck patches.

These two opposing processes are repeated continuously, resulting in the creation of increasingly complex structures in the era of matter dominance. But as the Planck patch reaches a huge magnification, then the reverse process appears to begin, namely the phenomenal removal of the members of this very large structure. At this stage of the evolution of the Universe, the dominance of the era of dark energy begins, which expresses the fact that the removal of parts of the structures formed in the era of matter dominance begins to become apparent. Free observers who observe the removal of the members of the structures are led to the wrong conclusion that:

*There is some greater repulsive force than the attractive force of gravity that leads to the gradual disintegration of the structures.*

But according to the interpretation we have given, there is only one process for the synthesis (and apparent decomposition) of matter that has as its ultimate goal the restoration of  $\mathbf{P}^2$  to its original form ( $R_0^2$ ) by zeroing any local curvature, through the fulfillment of the zero criterion of  $\mathbf{P}^2$ .

Free observers of three-dimensional space interpret these two opposite processes differently as follows: the radiation pressure of trapped photons in a region of space causes the particles to spread out and then the gravitational

attraction causes the particles to concentrate in the same region of 3D-space. But gravitational attraction without rotation makes no sense, as we have seen!

Based on the interpretation of the creation of complex geometric structures, through the geometric model, one can interpret dark matter, the creation and evolution of stars, the creation of star clusters, the formation of galaxies, the creation of galaxy clusters, and finally the formation of a huge super-cluster of galaxies at the end of the era of matter dominance.

The largest super-cluster of galaxies ever observed formed in the last part of the Planck epoch when matter dominated. However, the boundaries between two successive epochs are not clearly defined, due to the unique process of evolution of the Universe. If this super-super-cluster of galaxies begins to gradually decay, creating a new structure whose parts are further apart, then the super-clusters of galaxies will take up more space. If each of the super-clusters of galaxies gradually decays, then clusters of galaxies will be created that take up even more space. If each galaxy in a galaxy cluster gradually decays, then star clusters will appear and this process will continue in smaller and smaller structures. This process is repeated continuously until the components of the first complex structures that will become the elementary geometric objects are removed from each other.

This is the reverse course of the creation of the various forms of matter. The decomposition becomes apparent from the initial Planck patch  $P_{\text{er}}^i$  (where "er" means erebons) in the era of the dominance of dark energy. Let us now look more closely at what was mentioned in general before, about the creation of structures in the era of the dominance of matter.

## 15.6 Emergence of Early Structures

As mentioned previously and we will see in detail in a later chapter, the total matter in the Universe consists of approximately 84% dark matter and 16% ordinary matter. Free observers believe that ordinary matter is concentrated in places where dark matter is densest. In these places, ordinary matter appears as clouds of mostly hydrogen gas. For bound observers, the concentration of ordinary matter where dark matter is also present, and its collapse into dark matter, means that the part of the Planck patch being examined is in the process of shrinking to increase the curvature of that part of the Planck patch, which corresponds to its building block.

The first stars and galaxies formed from these clouds of hydrogen and helium. Where numerous galaxies have formed, galaxy clusters and super-super-clusters of galaxies will eventually form. Large gaps with few stars will develop between them, signaling that dark matter was less dense there. The exact times of formation of the first stars, quasars, galaxies, and supermassive black holes, as well as the start and end times and the evolution of the

period known as **re-ionization**, are not well-defined and are still under investigation.

As of 2019, the first confirmed galaxies date from about 380–400 million years after the *BE*, suggesting surprisingly rapid condensation of gas clouds and star birth rates. The formation of structure in the *BE* model proceeds hierarchically with smaller structures forming before larger ones. The first large structures to form are the first stars known as Population III. These stars are currently a hypothetical population because they have not been observed in the distant past.

Also among the first structures are dwarf galaxies and quasars, which are believed to be the bright, first active galaxies containing a supermassive black hole surrounded by a disk of gas. Population III stars will also be responsible for converting the few light elements that were initially formed (hydrogen, helium, and small amounts of lithium) into heavier elements. The largest stars of this era had very short lifetimes compared to most of the Main Sequence stars we see today. They usually complete their hydrogen burning and explode as supernovae after millions of years, scattering the heavier elements that had been created within them into the Universe, as we will see later. Observations of the Cosmic Microwave Background Radiation can now be used to determine when the two youngest populations began forming stars.

An analysis of such observations by the Planck Space Telescope in 2016 concluded that the first generation of stars may have formed about 300 million years after the *BE*. In December 2012, the first candidate galaxies were discovered that date back to before re-ionization, i.e., 13.4 billion years ago. Quasars provide some additional clues to the formation of early structures. Their light shows that they contain chemical elements such as carbon, magnesium, iron, and oxygen. This is evidence that by the time the quasars formed, a long period of star formation had already taken place, involving a large number of generations of Population III stars, to produce these elements.

## 15.7 Gravitational collapse

Let us consider a dust ball, where by the term “dust” we mean an aggregate of  $H_2$  and  $He$  molecules that are in a state of equilibrium. Given that the initial chemical elements formed in the era of the dominance of radiation were 75% hydrogen and 25% helium, then the dust ball will consist of these chemical elements and other elementary particles such as photons and electrons.

Let  $P_\sigma$  (where  $\sigma$  means dust) be the Planck patch in which the geometric object corresponding to the dust ball exists. When it is in the process of expansion, then its parts move away from each other. This separation results in the creation of circular layers consisting mainly of  $H_2$  and  $He$  molecules

as well as other elementary particles to satisfy the isotropy criterion of  $\mathbf{P}^2$ .

In the next stage of the shrinkage process of the Planck patch due to self-similarity, the dust sphere returns almost to its original state. Depending on the shrinkage process in each circular layer of the dust cloud, its condensation rate will differ from one circular layer to each successive circular layer. The inner circular layers condense faster because the distances between them are smaller, so the z.c. $\mathbf{P}^2$  is satisfied faster. Thus, in a short time, the initial spherical cloud becomes an almost solid rotating ball. Free observers call this phenomenon **gravitational collapse**. The question that arises is the following:

**How do free observers on Earth detect the gravitational collapse of a dust ball?**

During the process of shrinking  $\mathbf{P}^2$ , the curvature of the circle surrounding the Planck patch whose particles correspond to the spherical dust cloud increases. This results in an increase in temperature (or curvature) and the emission of photons by the electrons of the  $H_2$  and  $He$  molecules to increase the curvature of this circle. Photons do not easily escape the spherical dust cloud because they follow the process of shrinking the Planck patch on the one hand and fulfilling the z.c. $\mathbf{P}^2$  on the other hand during an accelerating contraction of the circular layers towards the center of the cloud.

Due to the contraction of the circular layers with an initially constant velocity towards the center of the cloud, a decrease in the frequency of the photons occurs, called the **redshift** due to the Doppler effect. There is also another decrease in the frequency of the photons, which is due to the acceleration of the circular layers towards their common center. In this case, observers on Earth call this decrease in frequency the **gravitational redshift**.

Observers on Earth always see a decrease in their frequency through the redshift of the emission spectrum, due to the summation of frequencies in both cases, as we will see later. Let us consider two observers A and B, where A is on Earth and B is near the spherical dust cloud. The two observers agree to send light signals at equal intervals of time before the cloud begins to shrink. This process is transferred and studied in the space  $\mathbf{P}^2$  according to the holographic correspondence. The initial distance between A and B is  $z$ , as shown in the following Figure 15.3.

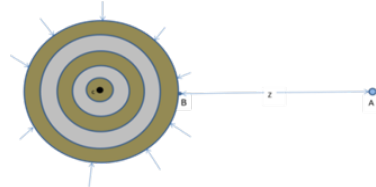


Figure 15.3: Shrinkage process during gravitational collapse



The first neutral particles created in  $\mathbf{P}^2$ , that is, those that satisfy z.c. $\mathbf{P}^2$ , correspond to the atoms  $H$  and  $He$ . These gather to form a circular shell centered at a point  $C$  in order to satisfy the isotropy criterion of  $\mathbf{P}^2$ . The subsequent particles that approach the initial shell and correspond to the atoms  $H$  and  $He$  gather together to form a new circular shell that surrounds the previous one and is distant from the first one. This process continues until all the geometric objects in this neighborhood of  $\mathbf{P}^2$  are exhausted, continuously forming new circular shells, as shown in Figure 15.3.

Observers A and B, in order to study what is happening from their own perspective, agree to send a photon every 1 second to each other. At the beginning of the creation of this circular disk, the time interval for the transition of a photon from A to B is  $\Delta t_{A \rightarrow B} = \frac{z}{c}$ . The same time passes for the transition of a photon from B to A, that is,  $\Delta t_{B \rightarrow A} = \frac{z}{c}$ , so  $\Delta t_{A \rightarrow B} = \Delta t_{B \rightarrow A}$ . This situation cannot remain the same forever because the Planck patches that make up  $\mathbf{P}^2$  follow the self-similar process:

### Shrink - Expand - Shrink ...

In the first stage of shrinkage of the Planck patch that contains the last circular disk, the two observers find that  $\Delta t_{A \rightarrow B}$  and  $\Delta t_{B \rightarrow A}$  are no longer equal, finding that  $\Delta t_{A \rightarrow B} < \Delta t_{B \rightarrow A}$ . The following question is therefore born:

**Why does the contraction of the circular disk create this asymmetry in the time intervals  $\Delta t_{A \rightarrow B}$  and  $\Delta t_{B \rightarrow A}$  given that A is considered a fixed point?**

To answer this question, we will see how the layer of the circular disk shrinks. Let us assume that the velocity of the outer circular layer at the beginning of the shrinkage process is the constant velocity  $v_{B_0}$  directed towards the center of the circular layers, which we define as the positive direction. If the observer at point B emits a photon towards the observer at point A, then the velocity of the photon will be  $v_B^{\text{ph}} = c - v_{B_0}$ . The time it takes for the photon to reach point A is:

$$\Delta t_{B \rightarrow A} = \frac{z}{c - v_{B_0}} > \frac{z}{c}$$

We assume that at the moment of emission of the photon from B, a photon from A reaches B. The observer at B considers that the velocity of this photon is collinear and in the same direction as the velocity  $v_{B_0}$ , having a value of:

$$v_A^{\text{ph}} = c + v_{B_0}$$

The observer at B calculates the time elapsed from its emission from A until it reaches him from the relation:

$$\Delta t_{A \rightarrow B} = \frac{z}{c + v_{B_0}} < \frac{z}{c}$$

From these relations, we get:

$$\Delta t_{A \rightarrow B} < \Delta t_{B \rightarrow A}$$

This process explains why the time intervals for the photon motion differ between the two observers. Next, we will examine how the frequency (or wavelength) of the photons emitted by observer B changes when:

- A) The shrinkage velocity of spherical shells is almost constant.
- B) The shrinkage acceleration of spherical shells is almost constant.

We will first consider the case where the velocity of observer B is almost constant, i.e.,  $v_B \simeq \text{constant}$ , given that the velocity of observer A is zero ( $v_A = 0$ ). In this case, from the relation for the change in frequency when the source ( $s$ ) and the receiver ( $r$ ) move with velocities  $\pm v_s, \pm v_r$  respectively, we obtain from the Doppler effect the relation:

$$f = f_0 \cdot \frac{c \pm v_r}{c \pm v_s} \implies \frac{\Delta f}{f_0} = \frac{\pm v_r - (\pm v_s)}{c \pm v_s}$$

The first case is the one where B sends a photon to A. A's velocity is always  $v_{A,r} = 0$  and B's velocity is always  $v_{B,s} = v_B$ . From the previous relation, we get:

$$\frac{\Delta f}{f_0} = -\frac{v_B}{c}$$

where  $v_B \ll c$ . From this relation, it follows that  $f < f_0$ , which means that the frequency of the photon emitted by B decreases, or else the frequency spectrum moves towards the red (redshift). Applying the Doppler relation for a photon emitted from A to B, we find that its frequency increases ( $f > f_0$ ), or else the frequency spectrum moves towards the blue (blueshift).

As the shrinkage of the Planck patches continues into smaller and smaller Planck patches, the time interval between two successive Planck patches becomes shorter. This results in a change in the distance between two successive layers and therefore in the velocity magnitude. In a first approximation, we consider that the change in velocity is:

$$\Delta v = a \Delta t$$

where  $a = \text{constant}$  and  $\Delta t = \frac{z}{c}$ . From the relation for the quotient  $\frac{\Delta f}{f_0}$  and for  $v_{A,r} = 0, v_{B,s} = +v_B$ , we get:

$$\frac{\Delta f}{f_0} = -\frac{\Delta v}{c}$$

where  $\Delta v = v_{A,r} - v_{B,s} = -v_B$  and  $c + v_B \simeq c$  because  $v_B \ll c$ . Substituting  $\Delta v$  in the previous relation, we get:

$$\frac{\Delta f}{f_0} = -\frac{a\Delta t}{c} = -\frac{az}{c^2}$$

where  $\Delta t = \frac{z}{c}$  was used, giving  $\Delta f < 0$ .

From this relation, it follows that as the circular layers of the circular disk shrink with constant acceleration, then we have an even greater decrease in the frequency of the photons emitted by it. Since during the process this value of acceleration can take on increasingly larger values, then the frequency will decrease more and more, resulting in the spherical body not being visible to observer A.

Thus, the free observer A calls the process of contraction of the spherical body **gravitational collapse** and the now invisible spherical body, due to its shift beyond the redshift, is called a **Black Hole**.

The Universe is accelerating therefore, its galaxies or clusters of galaxies are moving away from us at the present time with an acceleration:

$$a_{\text{univ}} = 1.345 \cdot 10^{-8} \text{ cm} \cdot \text{s}^{-2}$$

as shown in Table 17.9 of Appendix 17.5. Thus, an additional redshift appears in the emission spectrum of a distant galaxy, according to the previous formula, which is:

$$\frac{\Delta f}{f_0} = -\frac{a_{\text{univ}} \cdot z}{c^2}$$

where  $z$  is the distance between us and this galaxy.

## 15.8 Black Holes

According to what was mentioned in the previous section, a spherical dust cloud can collapse into a superdense structure beyond the Schwarzschild radius. This results in the light from this object being redshifted so that it does not reach an observer on Earth. The question is:

### What is the core of this spherical dust cloud?

In a previous Planck patch, a huge number of scattered dark geometric objects, namely gluons, electrons, and quarks, had been created. Due to the contraction process of this Planck patch, these merged together to form a huge core. Around this core, layers of  $H_2$  and  $He$  molecules gathered in a state of equilibrium. This was the initial state of this aggregate. The next stage is the gravitational collapse, as we described it in the previous section. The celestial objects that correspond to these aggregations of geometric objects are called **black holes**. Since a black hole is not visible to be determined through its radiation spectrum, it can be detected indirectly as follows:

1. Due to the closed path performed by a visible star or visible star clusters around an invisible center, such as, for example, the rotational motion of our solar system around the invisible center of our galaxy (Black hole) to fulfill the isotropy criterion of  $\mathbf{P}^2$ .
2. Due to the radiation emitted by a binary star system where one star is invisible, as we will see in Chapter 17.

**How can one detect the existence of an invisible star such as a black hole?**

The first clue, as we mentioned, is the orbit of a visible star in a circular or elliptical orbit, meaning that this motion is around some other invisible star within the orbit, to satisfy the isotropy criterion of  $\mathbf{P}^2$ .

The second clue comes from the radiation that is probably emitted by such binary star systems. The interpretation given by bound observers is as follows: Bound observers examine the process of expansion of the Planck patch containing the geometric objects that make up the binary star system.

The geometric objects that make up the visible star will start to move away from it and will start to surround the dark geometric object (invisible star) due to the criterion of isotropy of  $\mathbf{P}^2$ . Thus, an accretion disk is formed around the invisible star. Since the next process is that of contraction, then this Planck patch will acquire a greater curvature (temperature) and so will the accretion disk, as a result of which it will start to radiate at a certain frequency.

One such case is the source Cygnus X-1 that emits X-rays as detected by X-ray telescopes. The visible star was found to be a spectral type O or B star, according to the spectral classification of stars. Type B stars have a high mass and luminosity. The mass of this star has been estimated to be at least 20 solar masses, and its orbital period is 5.6 days. Its radial velocity can also be calculated, and the mass of its unseen companion is roughly estimated at 15 solar masses. Making various assumptions about the nature of the unseen star, such as that it may be a neutron star, we conclude that its limiting mass is much larger than the limit of 2-3 solar masses for neutron stars (Appendix 17.6.2). We therefore conclude that the unseen star must be a black hole.

The question that arises is how and when black holes were created. We have previously examined this issue from the perspective of bound observers. According to the self-similar expansion-contraction process of the various Planck patches and the collapse of spherical dust clouds, dark structures corresponding to such celestial bodies merged. The creation of these began towards the end of the radiation-dominated era, when the curvature (or temperature) of the Planck patches had significantly decreased.

At this point, we can briefly say that the cause of black hole creation is the tendency of restoring the large curvature of one Planck patch to the

next Planck patch due to self-similarity. Another consequence of black hole creation is the emission of waves.

In two successive Planck patches, the black hole tends to shrink in one Planck patch and expand in the other Planck patch; that is, the geometric object corresponding to it tends to become a circle in one and an ellipse in the other Planck patch. These successively repeated processes result in the creation of the so-called gravitational waves, as mentioned in the next section.

As the Planck patches continue to expand, the repeated contraction-expansion processes begin to be spaced at increasingly greater intervals. This results in the frequency of gravitational waves continuously decreasing, increasing the difficulty of detecting them. Finally, we will mention the mass-temperature correlation (Hawking effect) in a black hole (Appendix 17.4.4) and the information loss paradox.

## 15.9 Creation of gravitational waves through the Geometric model

The precession of the perihelion of a planet around the Sun or the precession of the periastron of a star around another, as happens, for example, with the binary system of a neutron star and a pulsar (such as the Hulse–Taylor pulsar, Appendix 17.8.1), is a combination of two motions.

In the first case, the planet moves in an elliptical orbit around the Sun.

In the second case, the planet moves in a circular orbit with its radius being the semi-major axis of the elliptical orbit.

Combining these two motions into one motion, we consider that the circle shrinks to an almost linear ellipse and then the ellipse expands to a circle. This is a repeating self-similar process that is found in all stages of the evolution of the Universe.

From this process, we obtain equations similar to Maxwell's equations, and from Maxwell's equations, we obtain the wave equations. The difference of this wave equation from the corresponding one of Electromagnetism has to do with the nature of the sources  $\rho$  and  $\mathbf{J}$ . Thus, the scalar quantity  $\rho$  expressing the charge density is replaced by the quantity  $\rho_m$  expressing the mass density, and the charge current density  $\mathbf{J}$  is replaced by the mass current density  $\mathbf{J}_m$ . The corresponding equations with Maxwell's equations for mass are written as:

**Variation of L:**

$$\nabla \cdot \mathbf{B} = 0, \quad \frac{\partial \mathbf{B}}{\partial(ct)} = -\nabla \times \mathbf{E}$$

**Variation of R:**

$$\nabla \cdot \mathbf{E} = \frac{\rho_m}{\epsilon_0}, \quad \nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{\partial(ct)} + \frac{\mathbf{J}_m}{c}$$

### 15.10 Derivation of the wave equations for mass in 3D

As already mentioned, through the linearization process of ellipses, linear ellipses have almost the same eccentricity as a parabola. Thus, a linear ellipse is considered as a segment of a linear parabola or hyperbola in the process of linearizing a cone.

Also, Maxwell's equations arise, through their representation in three-dimensional space, from the process of linearizing ellipses in  $\mathbf{P}^2$ . When Maxwell's equations are combined, then the result is a wave equation that is formally similar to the equation of a parabola. This formal similarity will lead us to the discovery of new physical entities in three-dimensional space. We will start from the simple case where  $\rho_m = 0$  and  $\mathbf{J}_m = 0$  by finding the conditions under which the combination of Maxwell's equations gives a wave equation that is formally similar to the parabola equation.

From the equation  $\nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{c \partial t}$  and the definitions  $\mathbf{B} = \nabla \times \mathbf{A}$  and  $\mathbf{E} = -\left(\frac{\partial \mathbf{A}}{c \partial t} + \nabla \Psi\right)$ , we get:

$$\nabla \times (\nabla \times \mathbf{A}) = -\frac{\partial^2 \mathbf{A}}{c^2 \partial t^2} - \nabla \left( \frac{\partial \Psi}{c \partial t} \right)$$

Due to the vector identity  $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ , we get:

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{c^2 \partial t^2} = \nabla \left( \nabla \cdot \mathbf{A} + \frac{\partial \Psi}{c \partial t} \right) \quad (15.1)$$

To be formally similar, equation 15.1 with the equation of a parabola, then the right-hand side must be zero, that is, the quantity in the parentheses must be a constant. Choosing the zero value for this constant, we obtain the equation:

$$\nabla \cdot \mathbf{A} + \frac{\partial \Psi}{c \partial t} = 0 \quad (15.2)$$

Equation 15.2 is called the **Lorentz gauge condition**. Thus, equation 15.1, due to equation 15.2, is the homogeneous wave equation with group velocity  $c$ , that is, it is the equation of a light-like gravitational wave:

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{c^2 \partial t^2} = 0 \quad (15.3)$$

A corresponding equation is obtained for the scalar potential  $\Phi$ , which results from  $\nabla \cdot \mathbf{E} = 0$  and the Lorentz gauge condition, i.e.,

$$\nabla^2 \Phi - \frac{\partial^2 \Phi}{c^2 \partial t^2} = 0 \quad (15.4)$$

Following the same procedure but with  $\rho_m \neq 0$  and  $\mathbf{J}_m \neq 0$ , we get the equations of inhomogeneous waves:

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{c^2 \partial t^2} = -\mu_0 \mathbf{J}_m \quad (15.5)$$

$$\nabla^2\Phi - \frac{\partial^2\Phi}{c^2\partial t^2} = \frac{\rho_m}{\epsilon_0} \quad (15.6)$$

The solutions of wave equations of this type result in the creation of space waves, which for historical reasons are called **gravitational waves**.

## 15.11 Star Formation

Stars, like our Sun, have not always been around. Stars are born and die over millions or billions of years. The explanation of star formation by free observers was given through the concept of gravity, which means that stars form when regions of dust and gas in the Universe collapse due to gravity. The expression “collapse due to gravity” means that the gas and dust form huge clumps of matter due to a force that free observers called gravity. However, there is another process of combining hydrogen molecules without the introduction of the concept of gravity, as we will see later. Without this dust and gas, stars would not form.

### Dust and Gas

A galaxy contains not only billions of stars, but also vast amounts of gas and dust. These regions of gas and dust in a galaxy are found in interstellar space. The space between stars in a galaxy is called the **interstellar medium** or interstellar space. Interstellar space is the space inside galaxies where there are no stars, but a lot of gas and dust. The regions of gas and dust are called **molecular clouds**, which are found within the interstellar medium. Molecular clouds are made up of a mixture of atoms, molecules, and dust. Atoms are the small building blocks of everything around us. Molecules are made up of two or more atoms bonded together. The molecules in molecular clouds are usually molecular hydrogen  $H_2$ , but they can also be more complex molecules, such as methanol, which consists of six atoms, or water, which consists of three atoms. Dust grains are even larger clumps of matter and can be up to a few millimeters in size, which is huge compared to the sizes of atoms or molecules.

Molecular clouds in the interstellar medium are large. In fact, a single molecular cloud can contain hundreds or even thousands of times more matter than our Sun. Their volumes also vary: a molecular cloud can be the same size or many times larger than our entire solar system. These huge molecular clouds undergo turbulent flow. This means that the gas and dust within the clouds do not stay in the same place over time.

## 15.12 What is turbulent flow and what causes it?

Bound observers consider a Planck patch in which the Planck patches corresponding to gas and dust components are located. During a contraction stage of the Planck patch, covalent bonds develop between the hydrogen atoms so that  $z.c.P^2$  is satisfied. According to these requirements, hydrogen molecules are created which, due to the contraction stage, form even larger aggregates of hydrogen molecules.

Simultaneously with the creation of the initial accumulations of hydrogen molecules, a rotation of them around the center of this cluster occurs, as well as a rotation of each small cluster around a larger cluster that is close to it. This happens in order to satisfy the isotropy criterion of the Planck patch within which these clusters are located. This process of forming clusters of hydrogen molecules is called “gravitational collapse” by free observers, which they attribute to a mysterious force called gravity that forces the hydrogen molecules to form in different regions large or small clusters of hydrogen molecules and other chemical elements.

Through the process shown in the following Figure 15.4 and explained below, all the formations that we observe today in the Universe are created, from stars and their planetary systems to clusters of galaxies. In a neighborhood of three-dimensional space, free observers examine all small clusters located around a large cluster at various distances from it.

Let us consider a line with a large cluster at one end and smaller clusters along it. The cluster closest to the large cluster begins to rotate around it in order to satisfy the isotropy criterion. The closer a small cluster is to a larger cluster, the earlier it begins to rotate around it, relative to other similar or nearly similar clusters. If this process is described mathematically (as will be done at the end of this paragraph), then the locus defined by the positions of the smaller clusters is found to be the Archimedean spiral, which expresses a form of turbulent flow. Figure 15.4 shows the creation of a spiral galaxy from four stars at positions A, B, C, D. The same process is followed for the formation of stars.

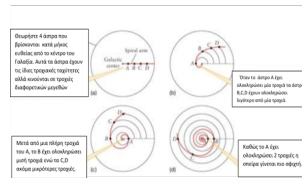


Figure 15.4: Formation of a spiral Galaxy from 4 stars located in an Archimedean spiral at positions A, B, C, D



This turbulent flow of gas and dust distributes the atoms and molecules unevenly, so that some regions of the molecular cloud have more matter than other regions (Figure 15.5A). If the gas and dust accumulate to a very high level in a particular region, then the matter in that region begins to condense into a further stage of contraction. Free observers interpret the behavior of matter in terms of its “collapse due to its own gravity”. The region that is created is within the molecular cloud, so it takes up less space than the entire molecular cloud. Initially, the largest distances within this region are only a few hundred Astronomical Units ( $AU$ ), where 1  $AU$  is the distance from Earth to the Sun.

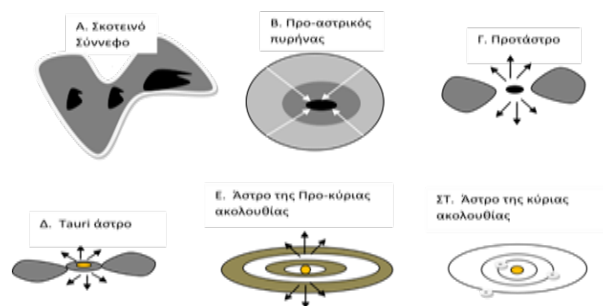


Figure 15.5: Formation of stars and planets around them

The process begins in Figure 15.5A, where gas and dust are located in some region of space. The formation of different stars that follows leads to the name of the space as the **Interstellar Medium**. The gas and dust, through the turbulent flow described earlier, form a dense ball of gas called the **protostellar core** (Figure 15.5B) that will eventually evolve into a star. During the contraction process and due to the fulfillment of the isotropy criterion, the matter orbiting the core follows the contraction process and the rotation process. The contraction process leads to the formation of a disk around the core and the fulfillment of the isotropy criterion in its rotation (Figure 15.5C).

The question is how this disk of rotating gas and dust is created. Bound observers consider a relatively small Planck patch where there are geometric objects forming a relatively compact circle of geometric objects (core) and geometric objects rotating loosely around the compact core. The particles outside the core, in the next stage of contraction of the Planck patch, will be compressed towards the core, creating an ellipse in the center of which are the accumulated particles that constitute the core of the star, as in the following Figure 15.6. This process in three-dimensional space resembles the compression of the top and bottom of a spherical balloon simultaneously to obtain a flat, almost circular shape, as shown in Figure 15.6.

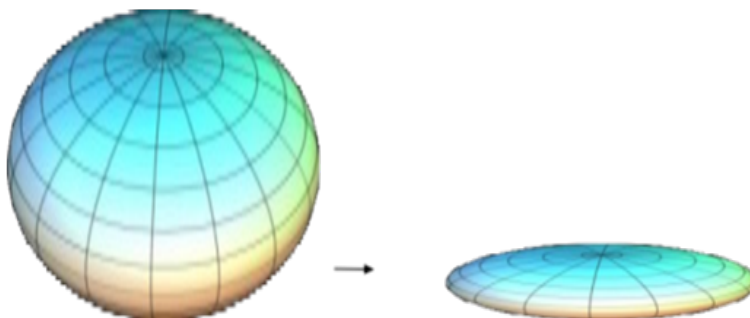


Figure 15.6: Formation of the accretion disk

The two jets of matter emitted towards the poles determine the axis of rotation of the system and the removal of excess matter in the expansion stage. At some point, the star stops growing, but gas continues to fall into the disk (Figure 15.5D) due to the expansion-contraction process. After a few million years, this process stops. Thus, a star is born (Figure 15.5E), while planets form in the disk from the remaining material, which will eventually become the star's planetary system (Figure 15.5F). The fact that the planets revolve in the same plane around the star results from the fact that they are created from the accretion disk. A solar system usually lives 10 billion years after its formation process.

### 15.13 The Collapsing Molecular Cloud

A molecular cloud is initially very cold, only a few degrees above absolute zero (0 K). But as gas and dust begin to accumulate in a region within the molecular cloud, it slowly warms up. For bound observers, the increase in temperature is a consequence of the increase in curvature of the Planck patch to which the molecular cloud belongs, which is proportional to temperature. The outer edge of a collapsing region will have a temperature of about 10 K, and the inner region will slowly warm up to 300 K, which is about room temperature.

When the collapsing region reaches a size of about 10,000  $AU$ , it is called a **protostellar core** (Figure 15.5B) and is officially a star in progress. The word core refers to the gas and dust, which are now so dense that the term core is more accurate than the terms cloud or region. This protostellar core will also later become the inner core of the star.

Over the next 50,000 years or so, the protostellar core contracts due to successive contractions of the corresponding Planck patch containing the core. This may sound like a long time, but on an astronomical scale it is considered a fairly rapid process compared, for example, to the age of the Universe, which is about 14 billion years.

The core contracts until its radius is about 1,000  $AU$  (Figure 15.5C) and

it is still composed of the same gas and dust, so this means that the density of this material increases as the diameter shrinks to  $\frac{1}{10}$  the original size of the condensation region. After 50,000 years, the system will have formed a disk around the central core and the excess material will have been ejected from the poles of the star. The poles of a star define the axis around which the star rotates. In Figure 15.5C, two fountain-like jets are visible where this excess material is ejected. This system is called a **protostar**, meaning that it is in its first stage of becoming a real star.

### A) From protostar core to star

The disk is crucial for protostars to evolve into stars of suitable size. The disk consists mainly of gas, which rotates with the disk and slowly approaches the surface of a protostar. When the gas gets close enough to the star, it falls to the surface of the star due to the contraction process and the star grows. This growth process is called accretion. Over the next 1,000 years, matter from the disk is either accumulated by the star or ejected from the disk (Figure 15.5D).

At this stage, the star has grown in size and density such that nuclear reactions begin in the central region, which are responsible for the brightness of stars, as is the case with our Sun. From this point on, the star is called a T Tauri star, and this is the first time the star can be observed optically. The star eventually stops accreting matter from the disk, but the remaining material around the star still has a disk shape (Figure 15.5E).

The disk no longer serves the purpose of feeding the star with matter to grow. Thus, the disk becomes a circularly rotating plane of matter, which will slowly begin to gather and rotate around the star. **These small clusters, made of the material left over from the formation of the star, will form the planets.** This means that the planets in our solar system are made of the material left over from the formation of the Sun. This is also why all the planets in the solar system are in the same plane, as mentioned earlier! The final solar system (Figure 15.5F) is complete when the disk is completely exhausted and all the planets have formed. Over the next 10 billion years, the star will burn nuclear fuel in its core and emit energy, i.e., radiation, which we call sunlight.

### B) Observing Molecular Clouds

Molecular clouds that host and form newborn stars are dark regions in the night sky. It is not possible for a human to see a molecular cloud, not even with an optical telescope. The reason we cannot see a molecular cloud is that dust particles are scattered throughout the cloud and absorb light from the surrounding stars. This prevents starlight from traveling through space and reaching us here on Earth, which is why a molecular cloud looks like a

dark region in the sky.

Fortunately for astronomers a molecular cloud is transparent to radio waves. This means that the radio waves are not absorbed by the dust particles in the cloud and can therefore travel freely towards Earth. Radio waves are not visible to the eye, but, using large radio telescopes, it is possible to receive signals from dark molecular clouds. These radio waves carry information about the contents of the dark molecular cloud. If a star is born, the cloud will send out different radio waves than if no stars were born in it. Using radio waves allows astronomers to see when stars are being born, even though the molecular cloud is dark. Although astronomers now know a lot about how stars like the Sun form, in reality there is still a big mystery to be solved.

### C) Stars with a mass greater than $6M_{\text{Sun}}$

The big mystery for free observers arises when very large stars form. Astronomers know that stars with a mass up to six times that of our Sun are born in the way we described earlier. More massive stars require a different process because, as free observers claim, the radiation pressure from the stars will push the disk away, preventing the stars from growing larger than six times the size of our Sun. Astronomers have seen these large stars, so they know that large stars exist and must have been born somehow. But how they are born remains a big question for astronomers around the world.

One answer to this question, according to bound observers, is that massive stars form in much larger Planck patches, during the expansion stage, which contain a much larger number of geometric objects than the previous Planck patches. In the next stage of contraction of these Planck patches, all these geometric objects create a much larger circle without the accretion disk, because the larger the Planck patch, the smaller its curvature and the contraction of the geometric objects within it. This process has to do with the larger radius of the circumscribed circle of this Planck patch (and therefore its minimum curvature) where all these geometric objects are located.

This process preserves the circular shape of the cluster of all geometric objects without shrinking the peripheral geometric objects into a disk, as happens for relatively small Planck patches. The limit of 6 solar masses for the formation of small stars is related to the size of the Planck patches (Figure 15.6). As the Universe continues to expand, the process of star formation changes with it. The stars that have formed in various regions of the Universe are considered as building blocks for even more complex structures. This is how star clusters, galaxies, galaxy clusters, and super-clusters of galaxies are created. In each such cluster, its building block follows a similar process to that of star formation, namely through turbulent flow.

Thus, for the construction of a star, its building block is the aggregate of hydrogen molecules; for the construction of a galaxy, its building block

is the star; for the construction of a galaxy cluster, its building block is the galaxy; and for the construction of the Universe, its building block is the galaxy cluster.

## 15.14 Archimedes' spiral explains turbulent flow

We will search for the trajectory traced by a geometric object corresponding to a star ( $S$ ) of a galaxy in  $\mathbf{P}^2$ . At the center of this galaxy (**Galactic Center** ( $GC$ )) there is a black hole. We will use the same names for the geometric objects and their holograms. In  $\mathbf{P}^2$ , we define the origin  $GC$  of an  $x, y$  coordinate system at the point where the black hole is located. Let us assume that a star  $S$  has formed at a distance  $d_1$  from the point  $GC$  on the  $x$ -axis. We assume that the Planck patch is in an expansion process that takes place with a speed  $u$  from the  $GC$  and in all directions of the radii of the circumscribed circle in the Planck patch to which this galaxy belongs. Due to the expansion process,  $S$  moves with a constant speed  $u$  along the  $x$ -axis away from the  $GC$ . Also,  $S$  must satisfy the isotropy criterion of  $\mathbf{P}^2$ , so that it rotates simultaneously around  $GC$  with a constant angular velocity  $\omega$ . Following the process described in the following Figure 15.7 for the formation of a galaxy, spirals are created and the turbulent flow begins.

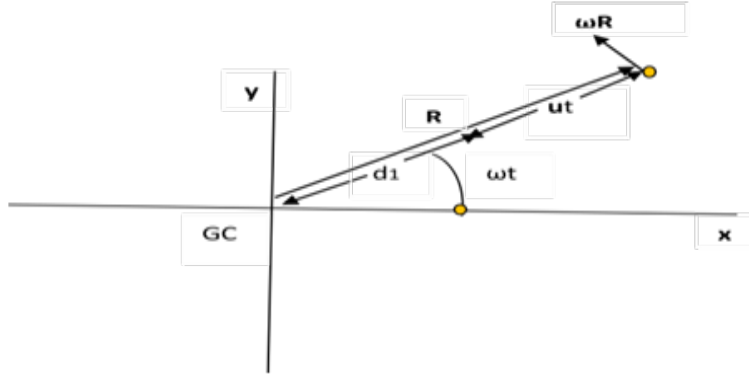


Figure 15.7: Spiral formation for the formation of stars, star clusters, galaxies and galaxy clusters

The total velocity vector of the star is the vector sum of the rotation velocity  $\omega R$  and the displacement velocity  $u$ . The components of the total velocity are:

$$u_x = \frac{dx}{dt} = u \cos(\omega t) - \omega(d_1 + ut) \sin(\omega t)$$

$$u_y = \frac{dy}{dt} = u \sin(\omega t) + \omega(d_1 + ut) \cos(\omega t)$$

Integrating these two relations member by member, we get:

$$x = (d_1 + ut) \cos(\omega t)$$

$$y = (d_1 + ut) \sin(\omega t)$$

giving

$$r^2 = x^2 + y^2 = (d_1 + ut)^2 \implies r = d_1 + ut$$

and

$$\theta = \omega t = \frac{\omega}{u}(r - d_1)$$

Rearranging this equation, we get:

$$r = \frac{u}{\omega}\theta + d_1$$

This equation is the equation that describes a spiral (Archimedes' spiral). If on the  $x$ -axis there are  $n$  stars that are initially at distances  $d_i, i = 1, 2, \dots, n$  from the  $GC$ , then a spiral will be created from them. If in some radial directions there are stars, then a bundle of spirals will be created from them which creates a spiral arm of a galaxy with a certain thickness. Figure 15.4 also shows the same process for the creation of a spiral arm with  $n = 4$  stars, which is the following:

- a) Imagine four stars that are initially located along a line extending from the Galactic center and are in the process of expanding away from it. The stars, in order to fulfill the  $\mathbf{P}^2$  isotropy criterion, begin to revolve around the Galactic center and have approximately the same orbital velocities, but they travel in orbits of different sizes.
- b) When star A has completed half of its orbit, stars B, C, D have completed only a quarter or less of an orbit.
- c) After one full orbit of star A, star B has completed only half of its orbit and stars C, D are further behind.
- d) As star A completes its second orbit, the spiral continues to move more tightly due to the contraction process.

If the stars are star clusters, then the same process again gives a spiral arm of the galaxy but with a greater thickness than the previous case. Through this methodology, the turbulent flow encountered in various fluids due to self-similarity is interpreted. The elementary particles that initially exist in each structural unit of  $P_M^i$  correspond to photons ( $\gamma$ ) or gluons ( $g$ ), electrons ( $e$ ), and quarks ( $d, u$ ). In the following, we will use the same name for the elementary particles and their holograms in three-dimensional space. To these particles are added the large structures of dark matter, that is,

clusters of particles mainly of circles with radii  $r = 1^+$ , as we will examine below.

As we have seen, the scattered particles in the Planck patch  $P_M^i$  are concentrated in each of the  $P_R^f$  building blocks of  $P_M^i$  in order to satisfy self-similarity. The set of all geometric objects of  $P_M^i$  is called the giant molecular cloud" (Figure 15.2). Each building block  $P_R^f$  of  $P_M^i$  has particles in and around it. These particles belong to each of the building blocks of  $P_R^f$  of  $P_M^i$ , so the giant molecular cloud appears to be breaking into pieces", each of which belongs to a  $P_R^f$ . This process appears macroscopically as a process of contraction and decomposition of the original giant molecular cloud. The geometric objects in each  $P_R^f$  follow the same shrinkage process, where the building blocks of  $P_R^f$  are combined, creating even more compact structures. This process continues in smaller and smaller Planck patches until the simplest composite geometric objects are synthesized (self-similarity downwards). Initially, the composite geometric objects are created, which are hydrogen atoms ( $H$ ) at a rate of 75%, helium atoms ( $He$ ) at a rate of 25%, and a few lithium atoms ( $Li$ ), for reasons mentioned earlier.

During the downward self-similarity process, the isotropy criterion and the zero criterion of  $\mathbf{P}^2$  must be incorporated into each new structure. Isotropy is expressed by rotating all layers of geometric objects around the center of each piece (like planets around our Sun). The hologram of the set of geometric objects in each of the  $P_R^f$  in three-dimensional space is called a **protostar**. The geometric objects corresponding to protostars have a different number of geometric objects corresponding to electrons and a different number of their orientations; therefore, their holograms have different masses. In what follows, we present a brief history of the evolution of stars.

## 15.15 Populations of stars

The first generation of stars, also known as Population III stars, are thought to have formed within a few hundred million years of the  $BE$ . These stars were the first source of visible light in the Universe after recombination. Structures may have begun to appear about 150 million years ago, and the first galaxies appeared from about 380 to 700 million years ago. We have no separate observations of very early individual stars. The first stars observed are parts of very early galaxies.

During 1944, astronomer Walter Baade categorized the groups of stars in our Milky Way into stellar populations. Baade observed that blue stars were strongly associated with spiral arms, and yellow stars predominated near the central galactic bulge and within globular star clusters. These two main subsets of the total number of stars were designated Population I and Population II. Then, in 1978, another subset of stars was added, which was called Population III.

These three populations are denoted by the abbreviations Pop I, Pop II, and Pop III, respectively. Between the population types, significant differences were found with the individual observed stellar spectra. These later turned out to be very important and are probably related to star formation, their observed kinematics, their stellar age, and even the evolution of galaxies in both spiral and elliptical galaxies. These three population types are used to divide stars according to their chemical composition or metallicity, i.e., their metal content. By definition, each population group exhibits a decreasing metallicity trend that is related to the increasing stellar age trend.

Therefore, the first stars formed in the Universe, i.e., those with the oldest age, as determined by our observations, and with very low metal content are Pop III stars. The next population of stars are those with low metal content and younger than Pop III stars, i.e., Pop II stars. Finally, the most recent stars formed in the Universe, i.e., those with the youngest age and relatively higher metal content, are the stars of the Pop I population. We consider our Sun to belong to the Pop I population, i.e., it is a young star with relatively high metallicity. Note that in astrophysical nomenclature, any element heavier than helium is a “metal”, including non-metals from a chemical point of view, such as oxygen, which is a non-metal.

## 15.16 Formation of early stars

Bound and free observers find that the geometric objects and their holograms, i.e., all dark and ordinary matter, were created in the era of the dominance of radiation. The large structures of gluons and the small structures of gluons-quarks (nucleons) are composed of normal matter. Normal matter consists mainly of 75% hydrogen and 25% helium, with only a very small fraction consisting of other chemical elements such as lithium and beryllium. When the Universe had cooled sufficiently (in the era of the Dark Ages), the first stars were born as Pop III stars, without heavier metals.

This is thought to have affected their structure, so that their stellar mass became hundreds of times greater than that of our Sun. In turn, these massive stars also evolved very rapidly, and their core synthesis processes created the first 26 elements (up to iron ( $Fe$ )), as shown in Appendix 15.19. It is likely that most of the high-mass Pop III stars quickly exhausted their fuel and probably exploded as supernovae. These explosions scattered their material, ejecting metals into the interstellar medium that was incorporated into subsequent generations of stars.

This catastrophe suggests that massive Pop III stars should not be observed in local galaxies. However, some Pop III stars can be observed in galaxies at high redshift, i.e., at the beginning of the galaxy formation era. However, no such star has been discovered to date.



Astronomers have found evidence of an extremely small, metal-poor star, slightly smaller than the Sun, found in a binary system in the spiral arms of our Milky Way. This discovery paves the way for the observation of even older stars. Protostars with masses less than about  $0.08M_{\text{Sun}}$ , where  $M_{\text{Sun}}$  is the mass of our Sun, never acquire a high curvature (temperature) during a contraction process to begin the conversion of hydrogen into helium. The holograms of these geometric objects are known as **Brown Dwarfs**.

## 15.17 Some Features of Structures

After the expansion process of  $P_M^i$  in each of its structural units, i.e., in each  $P_R^f$  patch, the contraction process of all  $P_M^i$  will begin. Thus, a repeated expansion-contraction-expansion ... sequence occurs for the construction of geometric objects, following the general procedure of alternating processes in all subsequent Planck patches  $P_M^j$ , where  $j = 2, 3, \dots$ .

Thus, new, more complex but also looser structures are created in the next Planck  $P_M^j$  patches of the matter-dominance era. Every complex geometric object resulting from the coupling of simpler geometric objects should meet the criteria of  $\mathbf{P}^2$ , namely the isotropy criterion, the zero criterion, as well as the self-similarity criterion. Initially, the large structures formed due to gravitational collapse are Black holes and the dark halos of galaxies. Also, geometric objects corresponding to nucleons (protons, neutrons) are created from three selected quarks and gluons. The geometric objects corresponding to nucleons have the same reduced charge and the same orientation as some elementary geometric objects so that there is self-similarity between elementary and complex geometric structures.

The geometric objects corresponding to protons are joined by an equal number of geometric objects corresponding to electrons to satisfy the z.c. $\mathbf{P}^2$  in terms of charge. Thus, the atoms  $H$  and  $He$  and some  $Li$  atoms are initially formed. The  $Li$  atom can be formed from a deuterium atom  $D$  (which is an isotope of  $H$ ), a  $He$  atom, and a neutron.

The isotropy of  $\mathbf{P}^2$  appears in the complex geometric object corresponding to the hydrogen atom. Thus, the geometric object corresponding to the electron moves around the nucleus in a nearly circular orbit in order to satisfy the isotropy of  $\mathbf{P}^2$ . In three-dimensional space, this orbit is not a perfect geometric shape because free observers consider this orbit to be composed of segments each with a different curvature, as mentioned earlier. This orbit does not have the structure of a specific particle orbit that we perceive macroscopically, such as a circle or an ellipse, for example, which is why it was called an **orbital**.

The zero orientation of each complex structure is determined by the number of neutrons in the nucleus, which also determines the behavior of the electron, that is, whether the electron exhibits its particle or wave or wave-

particle nature. This behavior of the electron is experimentally verified by free observers through the ionization energy required to remove the electron from the hydrogen atom or its isotopes.

For free observers, the hydrogen atom displays a “halo” which is a cluster of ellipses (orbitals) surrounding the nucleus, within which is a circle corresponding to the particle nature of the electron. The geometric analogue of the hydrogen atom is a composite geometric object corresponding to the nucleus (proton) and a geometric object corresponding to the electron, moving in a fixed circular orbit to satisfy the isotropy criterion of  $\mathbf{P}^2$ . If this orbit is not fixed, then the isotropy criterion of  $\mathbf{P}^2$  is not satisfied. Thus, the Bohr rules for the stability of atoms are consequences of satisfying the  $\mathbf{P}^2$  isotropy criterion.

This structure is characterized on the one hand by the curvature  $k_c^e$  of the circular orbit in which the geometric object corresponding to the electron moves around the geometric object corresponding to the nucleus, and on the other hand by the constant holographic mass of the electron  $M_h^e$ . We will then interpret the jumps of electrons between such orbits through the Fraunhofer lines.

## 15.18 Physical characteristics of stars

In order to study stars, astronomers must classify them according to those characteristics of theirs that can be measured. Two such characteristics are the star’s luminosity, which is the rate at which the star emits light, and its surface temperature. We will see how these two quantities are determined over the great distances that separate astronomers from these stars.

Having these two quantities for the stars we are studying, we place them on a diagram with the star’s luminosity on the vertical axis and its surface temperature on the horizontal axis. This diagram is called a Hertzsprung-Russell diagram or **H-R diagram**.

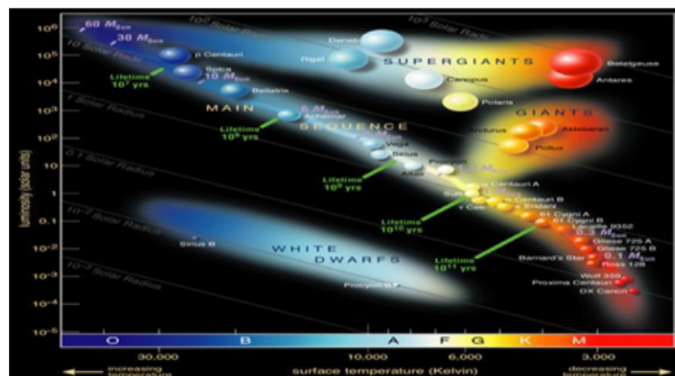


Figure 15.8: H-R diagram

What we find is that a large number of stars, including our Sun, are located in a zone that extends from the upper left corner to the lower right corner of the  $H - R$  diagram. As we move from left to right, the temperature decreases. The fact that the color changes from blue to red has to do with the frequency of the radiation they emit, as we will see below. So, the stars on the bottom right are relatively cool ( $\sim 4,000$  K) while the stars on the top left are hot ( $\sim 12,000$  K). Our Sun is in the middle of this sequence, so its surface temperature is  $\sim 5,500$  K. This zone is called the **Main Sequence**. In the  $H - R$  diagram, in the top right corner, there are stars called **giant stars** which are cold but bright, and in the bottom left corner there are stars that are hot but very dim and are called **dwarf stars**.

### 1. Brightness–Apparent Brightness

The first question is how do we define the brightness of stars, and then the second question is how do we define the scale for brightness. From each star, a quantity of photons reaches our telescopes. This quantity of photons coming from a region of the star falls on a device called a **Charge-Coupled Device** (CCD, Figure 15.9) .



Figure 15.9: CCD array

When this number of photons falls on the CCD device, it releases an equal number of electrons which are recorded by special counters. If a region of the star's surface is observed, then the part of the region where the most or fewest photons came from is determined, so we get more or fewer electrons. Thus, by means of a computer, parts of the region are converted into artificial images depending on the number of electrons, putting different colors for each part of the region.

By collecting photons from a star, the astronomer can measure the **apparent brightness** of the star. This brightness is called “apparent” because the image does not contain complete information about the star's true brightness, that is, the effective rate at which the star emits photons.

Consider a light source that emits  $L_R$  photons (considered as particles) and the same number of photons in each direction (isotropic source). Let us assume that the light source is located at the center of a sphere of radius  $r$  and an observer is located at a point on the surface of this sphere that has an area of  $4\pi r^2$ . We define the apparent luminosity  $L_F$  of this light source as the number of photons reaching the observer per unit time  $t_u$  and per unit area as:

$$L_F = \frac{L_R}{t_u \cdot 4\pi \cdot r^2}$$

If we can calculate the distance of the star from the Earth, then we can find the luminosity  $L_R$  of the source through this relationship. The calculation of the distance of a star from the Earth is done by various methods depending on their estimated distance, such as spectroscopic parallax or trigonometric parallax, as shown in Figure 15.10 .

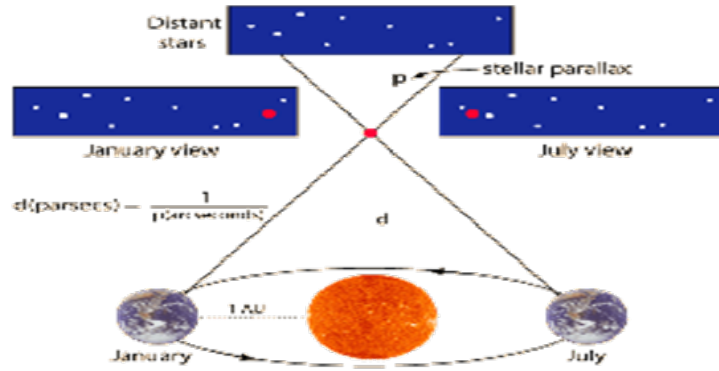


Figure 15.10: Trigonometric parallax method for measuring distance

## 2. Spectrum of stars

Sunlight can be analyzed using a spectrograph into colors from violet (at the shortest wavelengths) to red (at the longest wavelengths). In this spectrum, we observe a series of dark lines, as shown in Figure 15.11. These lines were discovered in 1814 by **Josef von Fraunhofer** and named in his honor **Fraunhofer lines**.

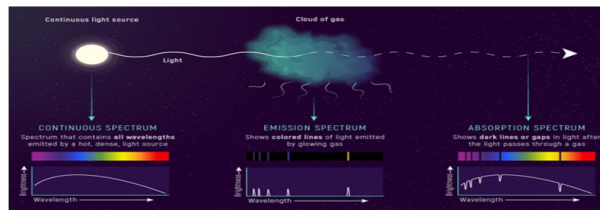


Figure 15.11: Spectrum of a star

The question that arises is: Where do the dark Fraunhofer lines come from? We will answer this question through the interpretation given by the bound observers. The product of the holographic mass  $M_h^e$  of the electron times the radius  $r_c^e$  of the circle surrounding the electron in  $\mathbf{P}^2$  is a constant, that is, the relation:

$$M_h^e \cdot r_c^e = \text{constant}$$

holds. Let us assume that a geometric object corresponding to a photon of curvature  $k_c^\gamma$  or of corresponding energy “collides” with a geometric object corresponding to an electron and unites with it. From this interaction, the total mass (or total energy) of the aggregate increases and becomes  $M_h^{e-\gamma} > M_h^e$ . For the relation  $M_h^e \cdot r_c^e = \text{constant}$  to hold, the radius  $r_c^{e-\gamma}$  of the circle surrounding the aggregate should be reduced so that the product of mass and radius remains constant, that is,  $M_h^{e-\gamma} \cdot r_c^{e-\gamma} = M_h^e \cdot r_c^e = \text{constant}$ . Thus, we get:

$$r_c^e > r_c^{e-\gamma}$$

As the radius of the circle decreases, the complex geometric structure  $e - \gamma$  moves to an orbit of smaller radius. In this case, a black line appears in the radiation spectrum due to the creation of the  $e - \gamma$  cluster where the photon is bound to the electron. If the photon  $\gamma$  is detached from the  $e - \gamma$  cluster, then  $e$  moves to an orbit of larger radius and a bright line appears in the radiation spectrum with a color corresponding to the frequency of the photon.

Free observers interpret these processes through the concept of a “quantum of energy” carried by a photon, which is added (or subtracted) from an elementary particle to increase (or decrease) its energy accordingly. Through this process, the Fraunhofer lines of the emission spectrum of light sources, such as stars, are interpreted.

### 3. Finding the surface temperature of stars

When radiation from a star encounters, for example, a hydrogen cloud, then it is almost certain that coupling of the cloud’s electrons with photons of different curvatures (or energies) of the radiation will occur. Since the photons were created at different stages of the expansion or contraction of the Universe, they also have different curvatures (or energies). Thus, the spectrum of the star’s radiation will lack all the photons that were coupled to the cloud’s electrons, showing the dark Fraunhofer lines (absorption lines).

Since each chemical element creates Fraunhofer lines at different positions and with different intensities in the spectrum of the light source, we can infer the type and content of the chemical element on the surface of the light source. The density is related to the number of atoms whose electrons cause the absorption of photons.

Another element that we can estimate from the degree of absorption is the temperature of the region of the source where the absorption took place. Theoretical work on very hot gases was achieved by the Indian astrophysicist Meghnad Saha. Saha found a mathematical relationship, using the Boltzmann equation, between the proportions of atoms, their ions, and free electrons as a function of temperature (Appendix 17.8.4).

Therefore, from the radiation spectrum of the source, one can draw conclusions about the abundance of its constituents and, from them, about the temperature of the source. Applying Saha's relation to stars, we find that they cover a wide range of surface temperatures. Thus, stars are classified into the spectral types O, B, A, F, G, K, M, R, and N, from the hottest to the coldest stars, as shown in the  $H - R$  diagram (Figure 15.8).

From these letters, Russell formed the mnemonic rule: **Oh, Be A Fine Girl/Guy, Kiss Me Right Now**. The range of surface temperatures of the above spectral types of stars is from over  $30,000^\circ\text{C}$  down to  $3,500^\circ\text{C}$ , starting from O-type stars and ending with N-type stars.

#### 4. Color of stars

As is known from the study of blackbody radiation, such as a well-insulated heated oven that does not allow significant heat loss, the electromagnetic radiation trapped inside it is distributed over various frequencies (or wavelengths). The intensity of the radiation at the various wavelengths has a well-defined profile. As the frequency increases, the intensity of the radiation also increases up to a certain limit, beyond which it decreases sharply. The radiation intensity-frequency curve peaks higher for higher temperatures. Also, as the temperature ( $T$ ) increases, the maximum of the curve shifts to the right, that is, to higher frequencies ( $\nu$ ), as shown in Figure 15.12.

This means that there must be a relationship between frequency and temperature. W. Wien found that this frequency-temperature relationship is given by:

$$T = b \cdot \nu$$

where  $b$  is a constant. Wien's law also follows from the geometric definition of temperature, as we will see in Appendix 17.4.

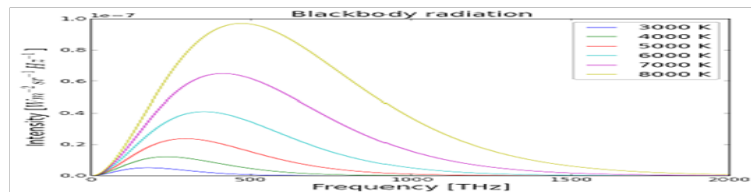


Figure 15.12: Blackbody radiation at many temperatures

In the case of a radiating star, the flux of its surface radiation is relatively small, so it does not disturb the equilibrium state of its interior. From these findings, it follows that stars behave as blackbodies (at least approximately). Thus, one can fit the continuous spectrum of a star to a blackbody radiation curve and estimate its temperature. This method is in satisfactory agreement with the results from the study of absorption spectra. Consequently: **Blue stars are hotter and red stars are cooler.**

## 5. Where does the brightness of stars come from?

Stars in the initial stage of their creation in three-dimensional space were a huge holographic cloud consisting mainly of Hydrogen (H) and possibly other elementary particles. This cloud formed as a self-similar holographic entity in three-dimensional space during the process of shrinking the geometric objects of a Planck patch. During this process, the individual  $\gamma$ 's of the Planck patch corresponding to H atoms came even closer together, creating  $H_2$  molecules, thus increasing the curvature of this geometric object. In the next expansion process of the Planck patch, the  $\gamma$ 's corresponding to the electrons were removed from the nuclei, creating a set of geometric objects of two kinds, namely H ions and electrons. Photons of various curvatures were released from the electrons, creating the emission spectrum of the star. According to Appendix 17.4, a Planck patch containing two kinds of geometric objects—such as the set we consider to consist of H nuclei and electrons—is characterized by the quantity we call temperature.

The temperature is proportional to the curvature of the Planck patch. The same applies to its hologram. The initial temperature (or curvature) of the Planck patch was very high, but as the Planck patch expands, it continuously decreases. Macroscopically, there is a relationship between the mass  $M$  and the flux of a certain number of photons (luminosity  $L_s$ ) of a star given by the relation:

$$M \propto L_s$$

Since the number of electrons is equal to the number of photons, the above relation holds. Since the temperature is equal to the curvature of a Planck patch, when the cloud is in the process of shrinking, only a few photons can escape. In other words, **as the curvature increases, the luminosity decreases.** Following the same procedure for all the stars we observe, we theoretically find the  $H - R$  diagram that coincides with the observed  $H - R$  diagram. In addition to the process we described earlier in the contraction stage, we also have the creation of the next most complex chemical element, which is He.

The most abundant isotope of He consists of 2 protons ( $p$ ), 2 neutrons ( $n$ ), and 2 electrons ( $e$ ). The reason this isotope of He appears in abundance

is its stability compared to all its other possible isotopes. This stability is due to the nature of its electrons, namely whether they behave as waves, wave-particles, or classical particles.

Since the process of expansion-contraction of the Planck patches is continuous in ever-smaller Planck patches, the flux of photons from the star is continuous. Since the interior of the star develops enormous temperatures due to the enormous curvature, the production of He continues.

## 15.19 Classification of stars

### 1. Pre-main sequence stars

If the curvature of the circle surrounding the geometric objects corresponding to the protostars is approximately equal to the curvature of the circle surrounding the geometric objects corresponding to our Sun, then these become pre-main sequence stars. We will describe the evolution of these stars from the perspective of free observers, keeping in mind the correspondences with their geometric objects studied by bound observers.

Let us return to the point where the synthesis of complex geometric objects, corresponding to H and He atoms, has been completed. We draw a circle containing the H and He atoms, photons, gluons, neutrinos, and the remaining electrons and  $d$ -quarks left after the creation of the H and He atoms. The  $u$ -quarks have been exhausted during the formation of the H and He atoms. At the end of this process, the radius of the circle that initially included all these elementary geometric objects increases. This increase occurs because the composite geometric objects (H and He) take up more space than their components, thus reducing the curvature of this circle.

The next step is the initiation of the reverse process, that is, the shrinking of the circle, so that its curvature again approaches the curvature of the Planck patch in which all these geometric objects are located. The shrinking process results in an increase in the curvature of a region around the center of the Planck patch. Thus, the existing H atoms are stripped of their electrons, reducing the surface area they occupy and consequently increasing the curvature of this entire region. The electron of the hydrogen atom H behaves like a wave, so it easily moves away from the nucleus.

When this happens, the reverse process begins, namely the decrease in curvature. This is achieved by combining an H atom, a deuterium atom D, and a neutron  $n$ , as well as two electrons to form a He atom, which has a smaller curvature than its components. The large complex geometric object in which all these processes take place corresponds to its hologram in three-dimensional space, called a **star**.



## 2. Main sequence stars

Depending on its curvature (or mass and temperature), a star is more or less luminous; therefore, it occupies a specific position on the absolute luminosity-temperature diagram (Hertzsprung-Russell ( $H - R$ ) diagram). All stars observed in the stage of converting hydrogen (H) into helium (He) are classified in a zone called the **Main Sequence**.

The brightest of these stars are at the top left and the coolest are at the bottom right, as shown in the  $H - R$  diagram. The Sun is near the middle of the Main Sequence. Stars outside the Main Sequence, at the top right of the  $H - R$  diagram, are cooler but much brighter and are called **red giants**. Stars outside the Main Sequence, at the bottom left of the  $H - R$  diagram, are hotter but much dimmer and are called **white dwarfs**.

When all the hydrogen in the core of the star has been converted to helium, then helium production stops. The hydrogen layer surrounding the circle of He atoms is divided into sublayers, each of which has a lower curvature (or temperature) as we move from the inner to the outer layer. Thus, the curvature (temperature) is constantly decreasing, so it is not enough to convert the H atoms into He atoms. At the end of the contraction process, which coincides with the exhaustion of hydrogen in the core, there is a core filled with He atoms and an outer layer of H atoms, as shown in Figure 15.13.

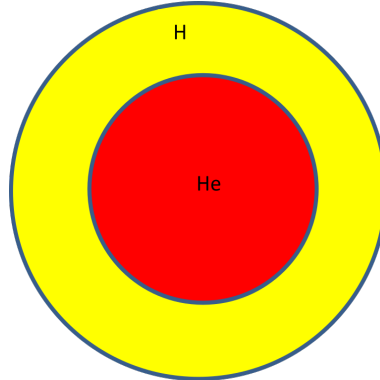


Figure 15.13: Onion structure with two layers of He – H

Looking at the core of the star, its area has increased due to the creation of He atoms, which are larger than H atoms. Thus, according to the alternation of expansion-contraction processes, a period of core contraction will follow. This means that the curvature of the core increases, resulting in the creation of larger geometric objects.

The question is: **What is the rule for the formation of new complex geometric objects in the core of a star?**

The answer to this question is given by the self-similar process of constructing new structures, similar to the construction of the first three Planck

patches. The first three Planck patches are constructed from the three solutions of the physical equation of a circle, which give three circles with three different curvatures. Thus, in addition to the initial circle, we get two additional circles, each of which has a different curvature from the other two. The He atoms are all identical in terms of the zero charge criterion and the isotropy criterion of  $\mathbf{P}^2$ , but differ in their curvatures depending on the layer they are in. Thus, it is possible to create a new structure from three He atoms.

We follow the inverse process of a single circle from which three circles originate through their physical equation. Thus, having three new circles with different curvatures, we will find which circle they originate from. The physical equation of this circle has as solutions the three curvatures of the three circles. Thus, we can determine a new circle corresponding to a new chemical element in three-dimensional space. Through the above geometric process, carbon atoms can be created. The geometric object corresponding to the carbon atom (C) is formed by three helium atoms of different curvatures from three consecutive inner layers of the nucleus. The nucleus of a carbon atom has 6 protons and 6 neutrons from the union of three He atoms.

This is the geometric rule for creating complex geometric structures and therefore their holograms. From the previous procedure, we get a star with an onion structure with three main subsequent layers, C – He – H from the inside out, in which the chemical elements Li, Be, B are contained in smaller quantities. After the creation of carbon atoms, the size of the nucleus increases, as shown in Figure 15.14.

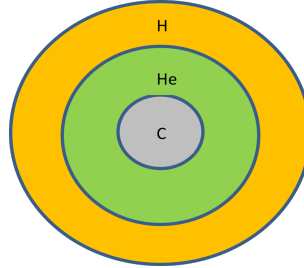


Figure 15.14: Onion structure with three layers of C – He – H

In the chemical element C, the protons are divided into three groups of different curvatures, depending on the curvature of the He atoms from which they came. The same applies to neutrons and electrons. Thus, three groups are formed for each of these elementary geometric objects that present the same characteristics. This has the consequence that the zero criterion of  $\mathbf{P}^2$  is not satisfied in these regions. Thus, a violent expansion of the space they occupy occurs for the satisfaction of z.c. $\mathbf{P}^2$ . Free observers interpret this phenomenon through the Pauli exclusion principle of Quantum Mechanics. Using the Pauli principle, free observers can arrive at a limit value for the

mass that this star must have before it explodes. Bound observers can arrive at the same limit value using only geometric arguments (Appendices 17.6.1, 17.6.2).

### 3. Red giants

The expansion of the star is not limited only to the core, but also extends to the hydrogen layer. If the growth of the core coincides with the expansion of the Planck patch, then the H layer grows significantly and stops when the expansion of the Planck patch is complete. Thus, the curvature (temperature) of the H layer gradually decreases, from the outer boundary of the He layer to the outer boundary of the H layer. Consequently, the star will become a red giant, due to the decrease in its surface temperature. Stars of this type occupy positions in the upper right part of the  $H - R$  diagram.

This process can be repeated for other initial conditions, that is, when their initial mass is greater than  $8M_{\text{Sun}}$ , in the same way in the subsequent stages of the red giant's evolution. After the core has grown, it will shrink, creating a new complex structure resulting from the union of two carbon atoms of the last two outer layers of the core and the first inner layer of the He shell. Figure 15.15 shows the onion structure of a red giant up to the formation of iron (Fe). The evolution of this star into a neutron star is different from its evolution into a white dwarf, as we will see later.

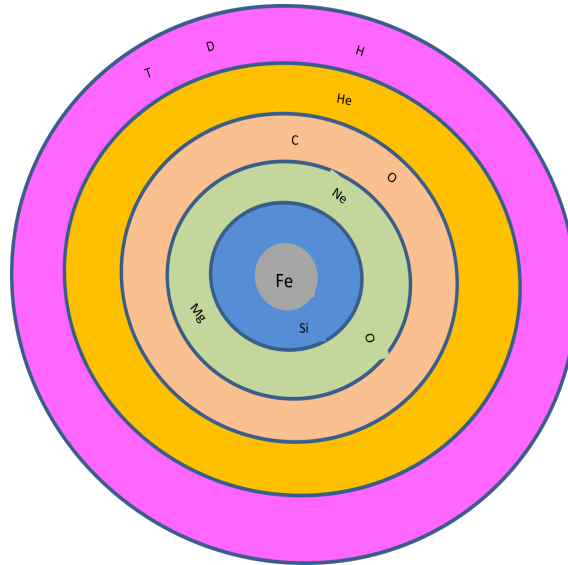


Figure 15.15: Onion structure of a red giant

The next table (Figure 15.16) shows the creation of chemical elements through this process up to the complex geometric object corresponding to the iron atom (Fe).

| ΑΡΙΘΜΟΣ ΤΩΝ $p$ | ΣΤΟΙΧΕΙΟ | ΑΝΤΙΔΡΑΣΗ                                        | ΑΡΙΘΜΟΣ ΤΩΝ $p, n$                          | ΑΡΙΘΜΟΣ ΤΩΝ $e$                                                |
|-----------------|----------|--------------------------------------------------|---------------------------------------------|----------------------------------------------------------------|
| 1               | H        | H                                                | $p^\uparrow$                                | $e_w^\uparrow$                                                 |
| 1               | D        | H + n                                            | $p^\uparrow + n^\uparrow$                   | $e_{wp}^\uparrow$                                              |
| 1               | T        | D + n                                            | $p^\uparrow + 2n^\uparrow$                  | $e_p^\uparrow$                                                 |
| 2               | He       | H + D + n                                        | $2p^\uparrow + 2n^\uparrow$                 | $e_p e_w^\uparrow$                                             |
| 3               | Li       | D + H <sub>e</sub> + n                           | $3p^\uparrow + 4n^\uparrow$                 | $e_p^\uparrow e_w^\uparrow + e_p^\uparrow$                     |
| 4               | Be       | H <sub>e</sub> + H <sub>e</sub> + n              | $4p^\uparrow + 5n^\uparrow$                 | $e_p^\uparrow e_w^\uparrow + e_{wp}^\uparrow + e_p^\uparrow$   |
| 5               | B        | H <sub>e</sub> + H <sub>e</sub> + T              | $5p^\uparrow + 6n^\uparrow$                 | $2e_p^\uparrow e_w^\uparrow + e_p^\uparrow$                    |
| 6               | C        | H <sub>e</sub> + H <sub>e</sub> + H <sub>e</sub> | $6p^\uparrow + 6n^\uparrow$                 | $3e_p^\uparrow e_w^\uparrow$                                   |
| 7               | N        | C + H + n                                        | $7p^\uparrow + 7n^\uparrow$                 | $3e_p^\uparrow e_w^\uparrow + e_{wp}^\uparrow$                 |
| 8               | O        | C + D + D                                        | $8p^\uparrow + 8n^\uparrow$                 | $4e_p^\uparrow e_w^\uparrow$                                   |
| 9               | F        | C + H <sub>e</sub> + T                           | $9p^\uparrow + 10n^\uparrow$                | $4e_p^\uparrow e_w^\uparrow + e_p^\uparrow$                    |
| 10              | Ne       | C + H <sub>e</sub> + H <sub>e</sub>              | $10p^\uparrow + 10n^\uparrow$               | $5e_p^\uparrow e_w^\uparrow$                                   |
| 11              | Na       | O + H <sub>e</sub> + T                           | $11p^\uparrow + 12n^\uparrow$               | $5e_p^\uparrow e_w^\uparrow + e_p^\uparrow$                    |
| 12              | Mg       | O + H <sub>e</sub> + H <sub>e</sub>              | $12p^\uparrow + 12n^\uparrow$               | $6e_p^\uparrow e_w^\uparrow$                                   |
| 13              | Al       | C + C + T                                        | $13p^\uparrow + 14n^\uparrow$               | $6e_p^\uparrow e_w^\uparrow + e_p^\uparrow$                    |
| 14              | Si       | C + C + H <sub>e</sub>                           | $14p^\uparrow + 14n^\uparrow$               | $7e_p^\uparrow e_w^\uparrow$                                   |
| 15              | P        | O + C + T                                        | $15p^\uparrow + 16n^\uparrow$               | $7e_p^\uparrow e_w^\uparrow + e_p^\uparrow$                    |
| 16              | S        | O + C + H <sub>e</sub>                           | $16p^\uparrow + 16n^\uparrow$               | $8e_p^\uparrow e_w^\uparrow$                                   |
| 17              | Cl       | O + O + T                                        | $17p^\uparrow + 18n^\uparrow$               | $8e_p^\uparrow e_w^\uparrow + e_p^\uparrow$                    |
| 18              | Ar       | B <sub>e</sub> + N + F                           | $18p^\uparrow + 18n^\uparrow + 4n^\uparrow$ | $7e_p^\uparrow e_w^\uparrow + 4e_p^\uparrow$                   |
| 19              | K        | L <sub>i</sub> + N + F                           | $19p^\uparrow + 19n^\uparrow + 2n^\uparrow$ | $8e_p^\uparrow e_w^\uparrow + e_{wp}^\uparrow + 2e_p^\uparrow$ |
| 20              | Ca       | C + C + O                                        | $20p^\uparrow + 20n^\uparrow$               | $10e_p^\uparrow e_w^\uparrow$                                  |
| 21              | Sc       | K + D + T                                        | $21p^\uparrow + 21n^\uparrow + 3n^\uparrow$ | $9e_p^\uparrow e_w^\uparrow + 3e_p^\uparrow$                   |
| 22              | Ti       | A <sub>e</sub> + H <sub>e</sub> + H              | $22p^\uparrow + 22n^\uparrow + 4n^\uparrow$ | $9e_p^\uparrow e_w^\uparrow + 4e_p^\uparrow$                   |
| 23              | Va       | S <sub>c</sub> + T + T                           | $23p^\uparrow + 23n^\uparrow + 5n^\uparrow$ | $9e_p^\uparrow e_w^\uparrow + 5e_p^\uparrow$                   |
| 24              | Cr       | T <sub>i</sub> + D + D                           | $24p^\uparrow + 24n^\uparrow + 4n^\uparrow$ | $10e_p^\uparrow e_w^\uparrow + 4e_p^\uparrow$                  |
| 25              | Mn       | V <sub>a</sub> + D + D                           | $25p^\uparrow + 25n^\uparrow + 5n^\uparrow$ | $10e_p^\uparrow e_w^\uparrow + 5e_p^\uparrow$                  |
| 26              | Fe       | Cr + D + D                                       | $26p^\uparrow + 26n^\uparrow + 4n^\uparrow$ | $11e_p^\uparrow e_w^\uparrow + 4e_p^\uparrow$                  |

Figure 15.16: Creation of chemical elements in stars

The arrow at the top of each symbol indicates the orientation of the geometric object  $(p, n, e)$ . In the case of electrons ( $e$ ), the lower letter indicates the nature it exhibits, namely  $p$  for particle nature,  $w$  for wave nature, and  $wp$  for wave-particle nature.

As shown in Figure 15.16, from the chemical element Ar to the chemical element Fe, the number of positively oriented electrons is equal to the number of negatively oriented neutrons per atom. The population of the group of identical electrons and the population of the group of identical neutrons begin to grow. Thus, in order to satisfy the zero criterion of  $\mathbf{P}^2$  within these groups, the members of the two groups must begin to move away from each other. Free observers interpret this phenomenon in terms of the Pauli exclusion principle for fermions, and bound observers in terms of the satisfaction of z.c. $\mathbf{P}^2$ .

The Pauli exclusion principle is used to explain the repulsion of identical electrons and neutrons confined to a region of three-dimensional space. The Pauli exclusion principle is a fundamental principle in three-dimensional space because it does not follow from any other more fundamental law. For bound observers, the Pauli exclusion principle is simply a mapping of z.c. $\mathbf{P}^2$  into three-dimensional space. Free observers call the group of identical electrons or neutrons during this process **degenerate fermions**. There are two cases that can determine the future evolution of a red giant, namely:

1. Their initial mass is less than  $8M_{\text{Sun}}$
2. Their initial mass is greater than  $8M_{\text{Sun}}$

## Red giant remnants

### 1. White dwarfs

In case (1), there are two sub-cases:

- a) Gradual reduction of the radius of the red giant that occurs by the emission of layers of light chemical elements, in such a way that the criterion of isotropy of  $\mathbf{P}^2$  is met. Thus, planetary nebulae are created around the red giant. This name arises from the fact that the orbits of the nebulae have the same shape as the orbits of the planets of our solar system and, on the other hand, the aggregates of chemical elements have the form of clouds. If the curvature of the red giant increases significantly, removing planetary nebulae, then the star can survive for a long time as a white dwarf. The upper limit of the mass of a white dwarf is discussed in Appendix 17.6.1.
- b) Suppose there is a two-star system consisting of a white dwarf and a red giant. We will consider this case from the perspective of bound observers. In the expansion phase of the Planck patch to which these two geometric objects belong, the curvature of the circle corresponding to the red giant begins to decrease because its radius increases. The outer layers of the geometric object corresponding to the red giant consist of the geometric objects corresponding to hydrogen (H) (Figure 15.15). In this phase, these geometric objects are constantly moving away from the red giant, reducing its overall curvature. A part of these geometric objects surrounds the white dwarf to satisfy the isotropy criterion of  $\mathbf{P}^2$ . Thus, in the expansion phase of the Planck patch, the curvature of the circle surrounding the white dwarf also decreases because its radius increases.

But let's see what happens in the next phase of the contraction of the Planck patch in which these two geometric objects are located. The radius of the circle containing the layers of geometric objects corresponding to H and located around the original circle corresponding to the white dwarf shrinks to form another circle of smaller radius and greater curvature. The question that arises is the following:

**How much can the curvature of the circle corresponding to the white dwarf increase, and what will happen if it increases beyond a certain value?**

First, the electrons of the hydrogen atoms H move away from their nuclei to reduce the curvature of each *Hatom* and accumulate in small areas.

Since these individual and identical electrons have the same orientation, as shown in the previous table, the zero criterion of  $\mathbf{P}^2$  is not satisfied for this region. This situation leads to a sudden and violent expansion of this region of  $\mathbf{P}^2$  that has a high density of identical electrons, which corresponds to a Type Ia supernova explosion in three-dimensional space. The evolution of this type of red giant is shown in Figure 15.21. In the second case, we will consider the evolution of red giants with masses greater than  $8M_{\text{Sun}}$ . These stars may end their lives as black holes or neutron stars.

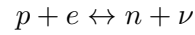
## 2. Neutron stars

Let us assume that a geometric object corresponding to a giant star has not removed a sufficient number of geometric objects (planetary nebula) to increase its curvature in the various stages of its contraction, ending up as a white dwarf. In this case, the geometric objects of its core continue to transform until the geometric objects corresponding to iron nuclei (Fe) are created. The reason why they stop at the nuclei of iron (Fe) and not at any other chemical element will appear in the following: The geometric object corresponding to an Fe atom has:

- a) 26 geometric objects oriented upwards corresponding to the protons of the Fe nucleus.
- b) 26 downward-oriented geometric objects corresponding to a portion of the neutrons of the Fe nucleus that compensate for the upward orientation of the protons.
- c) 11 upward-oriented geometric objects corresponding to electrons and 11 downward-oriented geometric objects also corresponding to electrons. The orientations of these electrons give zero orientation.
- d) 4 neutrons oriented downwards and 4 electrons oriented upwards.

Overall, for each Fe atom, the zero criterion of  $\mathbf{P}^2$  holds for charge and orientation, but two regions with identical particles are formed (Figure 15.16). One is the region of 4 neutrons oriented downwards and the other is the region of 4 electrons oriented upwards per Fe atom. The same applies to the previous chemical elements of iron, namely Ar, Sc, Ti, V, Cr, Mn. Thus, from all these chemical elements, the two groups of  $e$  and  $n$  form a sufficient number of identical geometric objects, i.e., electrons and neutrons, confined in a small space, so that they can be separated from each other for z.c. $\mathbf{P}^2$  to hold. Furthermore, during the contraction process, some of the geometric objects corresponding to protons are combined with some of the geometric objects corresponding to electrons, giving geometric objects corresponding to neutrons and neutrinos.

This happens because the circles corresponding to  $p, e$  have larger radii than the circles corresponding to  $n, \nu$ , so the contraction process is further enhanced. The reaction of the conversion of  $p, e$  to  $n, \nu$  is:



Thus, there are enlarged regions of similar geometric objects, concentrated together at very close distances, corresponding to neutrons and electrons. To satisfy z.c. $\mathbf{P}^2$ , a violent expansion of the core is caused, which is transmitted to the outer layers of the star (supernova explosion in three-dimensional space). Thus, the initial red giant appears as a remaining core and a nebula around it. The upper limit of the mass of this core (neutron star) is examined in Appendix 17.6.2.

After the process of expansion of the core, where the planetary nebula is removed, a contraction process follows. Thus, the remnant of the red giant is a star that has almost only neutrons (**neutron star**). The distances between the neutrons of the neutron star are very small. This has the consequence that the space they occupy is very small and their density is very high.

Another consequence of this explosion is the leakage of neutrinos, a few moments before the explosion. These neutrinos were formed during the formation of neutrons during the process of contraction of the core of the red giant, according to the previous reaction. If the mass of the core of the contracted neutron star is between about 1 and 3 solar masses, these newly created neutrons can stop the contraction process, leaving behind a neutron star. (Stars with a higher mass will continue the contraction process, forming black holes).

### 3. Pulsars

#### Questions:

1. How do neutron stars produce electromagnetic waves since neutrons have zero charge?
2. Is the reason for the existence of electromagnetic fields hidden in other processes inside these stars?

**Answers:** After the supernova explosion of a massive red giant, a neutron star is created from its core. In the neutron star, considered as an autonomous entity, the following processes have occurred according to the definition criteria of  $\mathbf{P}^2$ :

1. In order to satisfy the isotropy criterion of  $\mathbf{P}^2$ , a rotation around one of the neutron star axes has begun.

2. In order to satisfy the self-similarity criterion in  $\mathbf{P}^2$ , a repeated expansion-contraction process of the neutron star has begun.

Let us examine the application of the self-similarity criterion to a neutron star. A neutron consists of one  $u$ -quark and two  $d$ -quarks, as shown in Figure 15.17.

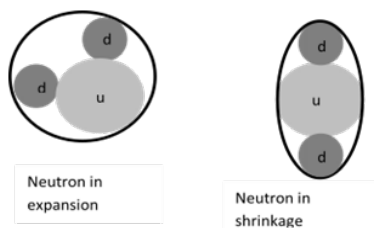


Figure 15.17: Neutron expansion-contraction process

The left sub-figure shows the neutron in the expansion stage, while the right sub-figure shows the neutron in the contraction stage of a neutron star.

During the expansion process, the circle surrounding the neutron has a finite radius, and so does the circle surrounding all neutrons. During the contraction process, the circle surrounding the neutron in the expansion phase becomes an almost linear ellipse, giving rise to Maxwell's equations. Combining the Maxwell equations with each other, as we have shown, we obtain the wave equation of the electromagnetic (EM) wave emitted by the neutron star. Because of the periodic transmission of EM waves in all directions of three-dimensional space, the neutron star is called a **pulsar** (pulsating star).

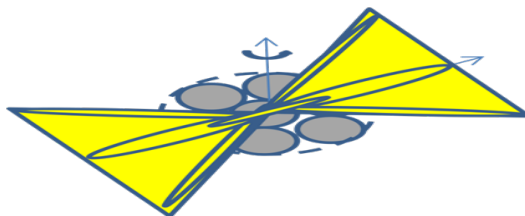


Figure 15.18: Pulsating star

From Maxwell's equations, and due to the huge number of neutrons, a very strong magnetic as well as an electric field arises along an axis with a different direction from that of the rotation axis, as seen by free observers. The frequency of EM waves emitted by a neutron star decreases over time due to the expansion of the circle in  $\mathbf{P}^2$  which corresponds to the neutron star. This expansion is due to the expansion of the Planck patches, which gradually increases the radius of the circle corresponding to the neutron star.

The circle corresponding to a neutron star oscillates between its two limiting states, that is, that of the circle and that of the ellipse where the source is the mass (curvature) of the neutron star.



Thus, a non-homogeneous wave equation with a mass source is produced, by the same method as that of EM waves, to which the name gravitational waves is attributed. Just before a supernova explodes, its core consists mainly of neutrons in a super-dense state. This is due to the process of shrinking the Planck patch (according to bound observers). Also, all neutrons have the same spin directions and are concentrated in the core, which is a small region, as shown in the table of Figure 15.16.

After the contraction process, the expansion process of the Planck patch follows, and at the same time, the z.c. $\mathbf{P}^2$  for this region of neutrons with the same directions of their spin is fulfilled. From this simultaneous enlargement of the Planck patch, a sharp expansion of the star in all directions (supernova explosion) follows. This results in the disintegration of the outer shells of the star and the ejection of the core of the star into space, creating a neutron star. To satisfy the isotropy criterion of  $\mathbf{P}^2$ , the neutron star starts to rotate around one of its axes. To satisfy z.c. $\mathbf{P}^2$  for neutrons in the nucleus with the same spin, these neutrons must coexist with particles of opposite orientation (spin).

The electrically charged particles released after the explosion into the surrounding space, which are mainly electrons, form clouds around the nucleus, rearranging their spin so that they have opposite orientations from the orientations of the spin of the neutrons. These clouds rotate around the neutron star to satisfy the isotropy criterion of  $\mathbf{P}^2$  together with the neutron star. Since the expansion-contraction process is most pronounced in the small Planck patches, then the neutrons themselves contract and expand sequentially, as in Figure 15.17.

The process of expansion-contraction in successive Planck patches for each neutron is described by the Maxwell equations. From Maxwell's equations, and due to the huge number of neutrons, a very strong magnetic field (as well as an electric field) arises along an axis with a different direction from that of the rotation axis, as seen by free observers. Free observers know that fast-moving charged particles near magnetic fields emit electromagnetic waves. This radiation is well-focused into a beam like that of a searchlight, as shown in Figure 15.18.

The bound observers believe that as the Planck patch to which the neutron star belongs expands, then all the members of the geometric structure corresponding to it move away from each other. This results in the creation, on the one hand, of layers of neutrons with different rotation speeds and, on the other hand, the process of expansion-contraction of each neutron takes place at increasingly longer intervals of time.

The process of expansion-contraction of the geometric object corresponding to the neutron is expressed by the change of shape from an ellipse to a circle and vice versa, within which the quarks  $udd$  are located. As the expansion of the Planck patch increases, the process of expansion-contraction of each neutron becomes slower and slower. This results in a decrease in the

frequency of the emitted radiation.

Using the same arguments and methodology used for white dwarfs, the limit on the mass of neutron stars is found. Due to the extreme population density of core neutrons, the limit on their mass is about twice the mass of the Sun, as demonstrated by arguments from bound observers (Appendix 17.6.2). These red giant remnants are maintained in equilibrium only if their mass is less than this limit.

According to the description of the supernova explosion, the search for neutron stars or pulsars should be done near the supernova remnants. One such example is the Crab Nebula (Figure 15.19) which formed almost 1000 years ago, where a pulsar was found within it.

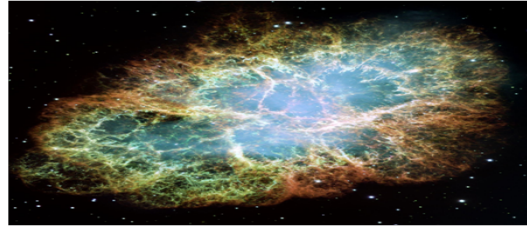


Figure 15.19: Crab Nebula

Next, we will discuss the radiation emitted by neutron stars and pulsars.

#### 4. Radiation emission spectrum from supernova explosions

Fast-moving charged particles, after their release, emit radiation whose wavelength or frequency varies depending on their speed. Since there is a layer of charged particles around the neutron star as mentioned earlier, the particles in the outer layers will have higher speeds. This means that the frequency of the emitted radiation is not one but many, thus creating a spectrum of frequencies from very high, corresponding to gamma rays, to very low, corresponding to radio waves. The spectrum of frequencies is shown in Figure 15.20.

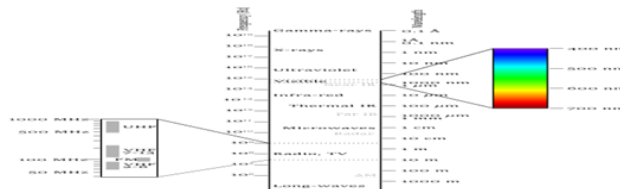


Figure 15.20: Spectrum of emission radiation from supernova explosions

Regions of different frequencies of EM waves around the Crab Nebula are the reason why this radiation spectrum is still observed today. Historical

data show that a supernova explosion occurred about 1000 years ago. This raises the question:

**Why did the charged particles not slow down so that they reduced their speeds, so that gamma radiation and X-rays do not appear in the electromagnetic spectrum that we observe today?**

- A first answer to this question is that 1000 years is an infinitesimal period of time from the point of view of Universal times.
- A second answer is that the action of dark energy is not yet evident in these dimensions to begin to break up the pulsar, thus reducing the speed of the charged particles.

The evolution of stars is shown in Figure 15.21.

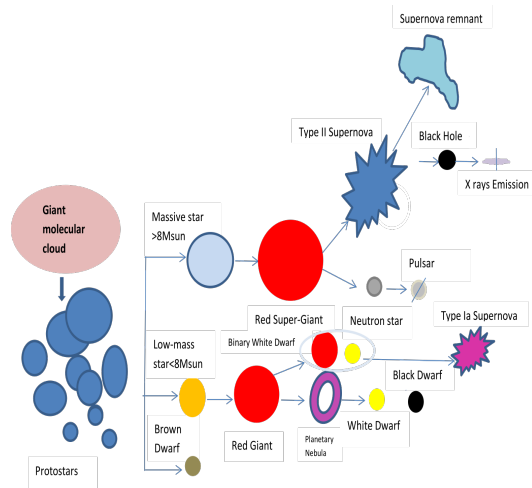


Figure 15.21: Stages of star evolution

## 5. Radio sources

In 1946, astronomers J.S. Hey, S.J. Parsons, and J.W. Philips discovered radio waves from the constellation Cygnus but without locating their source. After 5 years, F.G. Smith managed to locate the radio source, which was named Cygnus A. What happened in Cygnus A was an explosion in its central region during which electrically charged particles (free electrons, atomic nuclei) and photons were ejected in two opposite directions, forming two lobes as shown in Figure 15.22.

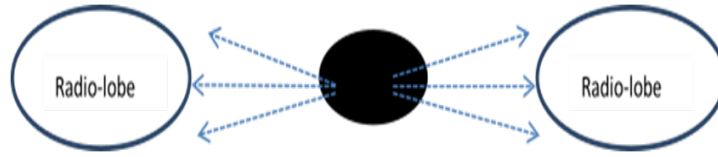


Figure 15.22: The Radio Galaxy Hercules

Let us examine, through the bound observers, the interpretation of the explosion of the star's center and the creation of the two lobes along an axis. First of all, for such an explosion to occur, the star must be in a phase of contraction and, on the other hand, large quantities of very hot electrons with the same spin orientations and the same charge, as well as atomic nuclei with corresponding characteristics, must have accumulated in a small region of its center. Due to the subsequent expansion process of the star, its overall geometric object in  $\mathbf{P}^2$  is a circle. The photons of the electrons move away from them independently of the electrons. In order to satisfy  $\text{z.c.}\mathbf{P}^2$ , the identical particles must be separated from each other.

These two conditions (expansion of the Planck patch and satisfaction of  $\text{z.c.}\mathbf{P}^2$ ) therefore lead to a violent expansion of the star's core (explosion). In the next phase of contraction, the central region of the star at  $\mathbf{P}^2$  acquires an ellipse shape, as faintly visible in Figure 15.22. This determines the two emission directions of charged particles and photons. After the explosion of the star's core, clusters of charged particles and photons are removed from each of the two directions along the major axis of the ellipse.

Since both clusters of particles and photons belong to a Planck patch, during the contraction process, they take the same shape at  $\mathbf{P}^2$  as that of the central region of the star, that is, they take the shape of an ellipse. This forms two lobes of charged particles and moving photons. The photon lobes move away from the charged particle clusters because they move at the speed of light, while the charged particles move at much slower speeds. At this stage of contraction of the Planck patch, the wavelength of the photons becomes shorter and their frequency higher, resulting in the emission of gamma radiation. At some point, the Planck patch begins the process of expansion, resulting in the wavelength of the photons increasing and their frequency continuously decreasing, so the emission of X-rays begins at a frequency lower than that of gamma radiation (Figure 15.22).

Because the Planck patches are constantly expanding due to dark energy (or the tendency for the initial curvature to vanish), the emitted radiation will have increasingly longer wavelengths or lower frequencies, reaching radio wave frequencies. For the same reason, the temperature in these two lobes will also have decreased. The photon lobes observed today at distances of hundreds of thousands of light-years from the central core are in these positions due to the expansion of the Planck patches, that is, due to the

expansion of the Universe. Thus, there is no mysterious source of energy.

## 6. Quasars

A quasar is an extremely luminous Active Galactic Nucleus (AGN). It is sometimes known as a Quasi-Stellar Object (QSO). The formation of the core is interpreted through the “gravitational collapse” of a huge spherical cloud composed mainly of hydrogen. Following the interpretation of bound observers, the hydrogen layers rotate in order to satisfy the isotropy criterion of  $\mathbf{P}^2$ . During a contraction process, the curvature of the Planck patch containing this cluster increases, as does its temperature. Thus, a supermassive black hole with a mass ranging from millions to tens of billions of solar masses is created, surrounded by a gaseous accretion disk in  $\mathbf{P}^2$ .

Through the contraction-expansion process in  $\mathbf{P}^2$ , the circle surrounding this cluster of geometric objects begins to change shape from a circle to an ellipse, in successive Planck patches, thus generating electromagnetic radiation as shown in Figure 15.23.

Despite their brightness, due to their great distance from Earth, no quasars can be seen with an unaided eye. Energy from quasars takes billions of years to reach the Earth's

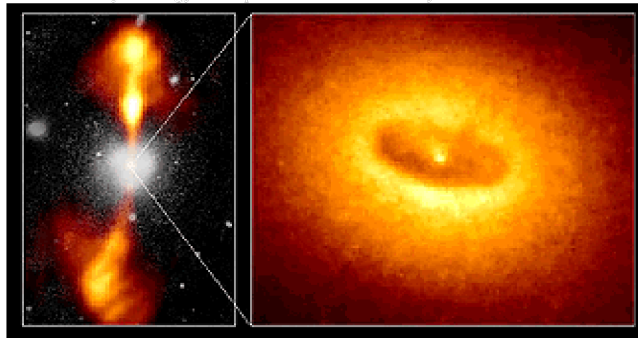


Figure 15.23: A quasar (NASA)

The emitted radiation is enormous. The most powerful quasars have a luminosity thousands of times greater than that of a galaxy such as our Milky Way. Quasars are usually categorized as a subclass of the more general class of AGN. The redshifts of quasars are of cosmological origin due to the continuous enlargement of the Planck patches. The term quasar was derived from the phrase *quasi-stellar radio sources*, because quasars were first recognized in the 1950s as sources of radio emission of unknown origin that, when detected in photographs at visible wavelengths, resembled faint, bright points of stars.

High-resolution images of quasars, particularly from the Hubble Space Telescope, have shown that quasars occur in the centers of galaxies and that some host galaxies interact strongly with or merge with neighbor galaxies. As with other classes of AGN, the observed properties of a quasar depend on

many factors, such as the mass of the black hole, the rate of gas accretion, the orientation of the accretion disk relative to the observer, the presence or absence of a jet, and the degree of obscuration by gas and dust within the host galaxy.

More than a million quasars have been found, with the nearest known one located about 600 million light-years from Earth. The record for the most distant known quasar continues to change. In 2017, the quasar ULAS J1342+0928 was detected at a redshift of  $z = 7.54$ . The light observed from this quasar, with a mass of 800 million solar masses, was emitted when the universe was only 690 million years old. In 2020, the quasar J1007+2115 was detected from an era just 700 million years after the *BE*, and with an estimated mass of 1.5 billion times the mass of the Sun. In early 2021, the quasar QSO J0313–1806, with a black hole with a mass of 1.6 billion times the mass of the Sun, was reported at  $z = 7.64$ , 670 million years after *BE*.

Using a telescope and a spectrometer, the variation of the intensity of the star's light with frequency can be measured. The resulting spectrum can be compared to the spectrum of hot gases expected in stars, such as hydrogen, in a laboratory on Earth. As shown in the ideal spectrum, to determine the redshift, features in the two spectra, such as absorption lines, emission lines, or other variations in light intensity, can be matched. Redshift (and blueshift) can be characterized by the relative difference between the observed and emitted wavelengths (or frequencies) of an object. In astronomy, it is common to refer to this change using a dimensionless quantity called  $z$ .  $z$  is given by the relationship:

$$z = \frac{f_{\text{emit}} - f_{\text{obs}}}{f_{\text{obs}}}$$

Quasar discovery surveys have shown that quasar activity was more common in the distant past. The peak era was about 10 billion years ago. Concentrations of multiple quasars that are clustering together are known as large quasar groups and are some of the largest known structures in the universe. Since the distances are enormous, the emitted radiation is also enormous and covers almost the entire electromagnetic spectrum, from X-rays to the far infrared with a peak in the ultraviolet-optical bands, while some quasars are also strong sources of radio waves and gamma rays. These observations lead us to the conclusion that these objects indicate the formation of galaxies at the beginning of the Dark Age period. The conjecture that a new galaxy is about to form, and not a star, comes from the following considerations:

1. Since the amount of radiation emitted by these objects reaches the current era, it means that this amount of radiation is enormous.
2. The enormous amount of radiation emitted means that in its previous phase, that is, in the process of shrinking the geometric object corre-

sponding to this celestial body, there would have been a huge number of scattered geometric objects in this Planck patch. These geometric objects are dark, that is, gluons, neutrinos, electrons, quarks, as well as photons. All these dark geometric objects, when gathered together, form a black hole.

We have never encountered such a huge black hole in the process of star formation, nor does a star occupy such a large area, which is  $1 \text{ Kpc} = 3.26 \cdot 10^3 \text{ light-years}$ , in the early stages of its formation, that is, as a star of the pre-main sequence state. This state is rather similar to the formation of a proto-galaxy.

In the next phase of the expansion of the Planck patch, the lighter geometric objects, namely photons and neutrinos, will be removed first, followed by the fragmentation of this large geometric object into smaller pieces. These pieces are assembled into the Planck patches resulting from the enlargement of the previous Planck patch, through the self-similar process. Each of these pieces can evolve, depending on the number of geometric objects it contains, mainly into stars and possibly into dwarf galaxies.

## 7. Creation of early galaxies

Keeping in mind the two-way correspondences of geometric objects and their physical entities (their holograms), we will develop our reasoning for the creation of early galaxies without explicitly stating these correspondences. Since the beginning of the era when the Universe is dominated by matter, dark matter gradually accumulates in specific parts of two successive Planck patches. Ordinary matter (clouds of mainly hydrogen gas) accumulates rapidly in the places where there is a high concentration of dark matter. The reason is the preservation of curvature, due to self-similarity, between two successive Planck patches. The first stars and galaxies are formed from these clouds. Where numerous galaxies have formed, galaxy clusters and super-clusters will eventually form, due to the rules of formation. Large gaps with few stars will develop between these formations, signaling that dark matter is less dense there.

The exact times of formation of the first stars, galaxies, supermassive black holes, and quasars, as well as the start and end times and evolution of geometric objects during the period known as **re-ionization**, are still being investigated. As of 2019, the earliest confirmed galaxies date back to about 380–400 million years ago (such as the galaxy GN-z11), suggesting surprisingly rapid condensation of gas clouds.

Structure formation in the *BE* model proceeds hierarchically, due to the alternating contraction-expansion process of the Planck patches, with smaller structures forming before larger ones. The first large structures of ordinary matter to form are Population III stars, dwarf galaxies, and quasars. Quasars

are believed to be the first active galaxies containing a supermassive black hole surrounded by a disk of gas.

Population III stars are responsible for the conversion of light elements (hydrogen, helium, and small amounts of lithium), according to the *BE* model, into heavier elements. These stars can be massive (or possibly small) that are non-metallic, meaning they contain no elements other than hydrogen and helium. The largest stars have very short lifetimes compared to most Main Sequence stars that we see today. They complete their hydrogen burning and explode as supernovae after millions of years, scattering the heavier elements that they have produced into the Universe, in repeated generations.

The next stage in the organization of matter, after the formation of stars ( $3 \cdot 10^6$  years after the *BE*), was the formation of galaxies from them. The process of galaxy formation follows the same general path of structure creation in the Universe, namely through self-similarity and turbulent flow. Thus, depending on the curvature of the Planck patch in which they formed and the curvature of the previous Planck patch, stars accumulate to form aggregates (star clusters). The accumulation of stars in the halo of dark matter orbiting a black hole occurs, on the one hand, to increase the curvature of the cluster and, on the other hand, to achieve the isotropy of  $\mathbf{P}^2$ . This results in a greater expansion of the space originally occupied by the galaxy, reaching a thousandfold and possibly more.

## 8. Re-ionization

Over the next few hundred million years, the Universe entered a critical turning point in its evolution, known as the **Epoch of Re-ionization** (Re-ionization). During this period, the dominant dark matter began to coalesce with stars, through turbulent flow, and this entire halo cluster began to revolve around a black hole, creating the structure we call a galaxy. Thus, the first galaxies began to form. Since the geometric objects that formed in the Planck patch of the boundary of the Universe had a large curvature, then their members began to separate from each other, in such a way that on the one hand their curvature approached the curvature of the Planck patch in which they formed and on the other hand the isotropy of  $\mathbf{P}^2$  was preserved. This process was called Re-ionization and resulted in the separation of the nuclei from their electrons.

**Question:** *Why were the quarks of nucleons not separated and only the nuclei separated from their electrons?*

**Answer:** The parts of the complex structures that were joined together with some of the original structures in larger Planck patches are more loosely



connected to the original structures, so they have less curvature and are therefore easier to remove from them.

In general, the parts of a structure are easier to remove from the structure the larger the Planck patch in which this structure was formed. So for example, quarks-gluons joined together in Planck patches of large curvature. Electrons joined with the nuclei to form atoms in Planck patches of smaller curvature.

In general, the strong interactions reflect the fact that the formation of nucleons took place in the era of radiation dominance in a relatively small area and therefore large curvature Planck patch. The electromagnetic interactions reflect the fact that the formation of atoms took place in the era of matter dominance, that is, in a relatively larger area and therefore smaller curvature Planck patch.

The interpretation given by free observers is that after the formation of the first stars and galaxies, these structures in turn released large amounts of ultraviolet light. Ultraviolet light has a very high energy capable of removing electrons from neutral matter. The period of re-ionization took place in the distant past of the Universe and is at the limit of current cosmological observations.

The electrons of neutral hydrogen have specific ways of absorbing photons, related to the energy levels (curvatures) of the photons, called the “Lyman series”. Ionized hydrogen has no such electron energy levels. Therefore, light traveling through ionized hydrogen and neutral hydrogen exhibits different absorption lines. Furthermore, the light will have traveled for billions of years to reach us, so any absorption by neutral hydrogen will have shifted in different ways, rather than in a specific way, that indicate when the absorption occurred. These features make it possible to study the state of ionization at many different times in the past.

For observers, these observations have narrowed down the time period over which re-ionization occurred, but the source of the photons that caused the ionization is still not completely certain. According to free observers, protons and electrons will recombine if energy is not continuously supplied to keep them apart, which also places limits on the number of sources and their lifetimes. With these constraints, free observers assume that quasars, stars, and early galaxies were the main sources of ionizing energy. Candidate sources for ionization are thought to be Population III stars (the first stars), dwarf galaxies (very early small galaxies), and a contribution from quasars (active galactic nuclei). For bound observers, another type of explanation is the expansion-contraction process of Planck patches, as mentioned earlier.

## 9. Evolution of galaxies

By studying the shapes of distant galaxies and carefully studying the celestial objects we see in our Milky Way, we have come to the conclusion that our Milky Way is a spiral galaxy. Our Milky Way consists of a central core in the shape of a bar which is formed by stars and hosts an extremely large black hole at its center. The core is surrounded by spiral arms, which are also formed by stars as well as gas and dust, as shown in Figure 15.24.



Figure 15.24: Our Milky Way (Universe Today)

We are in a branch that extends between the large spiral arms. The exact configuration of the spiral arms is still debated by astronomers. To the astronomers, the spiral arms of our Milky Way are thought to be a kind of density wave traveling around the flat disk. Material accumulates and stars form along the arms. Everything in our Milky Way revolves around its center (confirmation of the isotropy of  $\mathbf{P}^2$ ). Our Solar System travels in and out of the spiral arms as it orbits the center of the Milky Way.

In studying the rotation of the stars in galaxies, it was observed that they do not rotate following Newton's theory of gravity, as one would expect. Swiss astronomer Fritz Zwicky proposed in 1934 that there must be a large amount of invisible matter or "Dark Matter", making spiral galaxies more massive than they appear. The original motivation for the dark matter hypothesis was to explain the velocities of stars at the edges of galaxies, which have higher velocities than those found by Newtonian mechanics (an application of the Virial theorem). It was found by independent observations that the visible mass of galaxies and galaxy clusters is not sufficient to calculate the speed of celestial bodies at their boundaries. This deviation, which became known as the **mass deficiency**, was systematically studied in 1933 by Zwicky by applying the Virial theorem to the Coma galaxy cluster.

In the 1960s and 1970s, astronomers began to believe that there could be more matter in the Universe than the visible matter. **Vera Rubin**, an astronomer at the Carnegie Institution of Washington, studied the velocities of stars at various locations in galaxies. Newton's basic physics suggests that stars at the edges of a galaxy would rotate more slowly than stars at its

center, but Rubin found no difference in the velocities of the most distant stars compared to those closer to the center. In fact, she found that all the stars in a galaxy appear to move in circular orbits around its center at the same speed. This finding suggests that parts of a galaxy, visible or invisible, rotate around its center, confirming the conservation of the isotropy of  $\mathbf{P}^2$ .

If the gravity on the stars of galaxies were the only force holding them in their orbits, then the galaxies should have been torn apart by such rapid rotation. Since this was not the case, there should be enormous amounts of invisible matter, the pull of which kept them in their orbits, according to free observers. This became known as the **galaxy rotation problem**. Rubin's calculations showed that galaxies must contain at least ten times as much dark matter as their stars. These were the first strong results to suggest that:

- a) either Newtonian gravity does not hold throughout the Universe,
- b) or that gravity was not the only force responsible for maintaining the concentrations of matter we call galaxies,
- c) or that there was a vast halo of invisible matter in most galaxies or galaxy clusters.

Rubin's research results were confirmed in the following decades, and efforts to solve the problem of galaxy rotation built the theory of dark matter, which starts from the idea of a mysterious and invisible matter surrounding galaxies or their clusters, as shown in Figure 15.25.



Figure 15.25: Dark halo around the galaxy cluster CL 0024+17 (NASA, ESA)

The existence of dark matter, according to the theory of free observer gravity, is indirectly inferred from the gravitational attraction it exerts on

normal matter. This mystery could be solved if we assumed that dark matter consists of some exotic particles that do not interact with light or normal matter, so they are very difficult to impossible to detect. The discrepancy between theoretical and observational data, as we have seen, is due to the fact that the rotational speeds of stars should decrease with distance from the center of the galaxy, according to Newtonian mechanics. However, it was found from observations that these speeds are almost constant. Thus, in the following velocity-distance graph, the curves are almost straight lines parallel to the distance axis. The rotation curves of stars are called **plane curves** (or flat rotation curves).

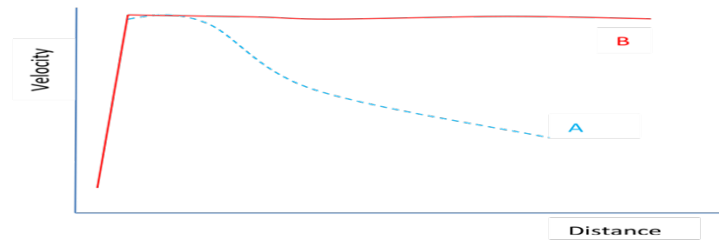


Figure 15.26: Observational evidence for the existence of Dark Matter

The dashed blue curve (A) results from Newtonian mechanics, while the solid red curve results from observational data, as shown in Figure 15.26. This inconsistency leads to the following two hypotheses:

1. Around galaxies, there is a large amount of dark matter in which there are stars. This dark cloud with the scattered stars within it rotates at a nearly constant speed around a Black Hole at the center of the galaxy, to maintain the isotropy of  $\mathbf{P}^2$ . According to this hypothesis, it is explained why stars move at almost constant speeds that are much larger than those that astronomers expected to see taking into account only visible matter.
2. Newton's universal law of gravity needs modification.

We examined the second hypothesis earlier. Dark matter is currently found from observations to be about 26% of all the components of the Universe. In the next Chapter, we will find theoretically, through the geometric model, the same result. By comparison, we mention that 5% of all the components of the Universe consists of normal or visible matter, which includes planets, stars, star clusters, galaxies, and galaxy clusters.

Based on these observational data, we will examine which dark geometric objects are created, where here by the term dark we do not mean that their temperature or curvature have very small values.

These dark objects have a different structure than the geometric objects of ordinary matter, so the detection instruments of free observers cannot detect them directly.

Summarizing the above, we make the following findings:

- **A)** Large-scale geometric structures contain geometric objects that have opposite orientations and opposite reduced charges. Thus, data show that bound observers take into account only their curvature and, correspondingly, free observers take into account only their mass. In three-dimensional space, large-scale dark structures, such as black holes and the halo of galaxies, were created, as well as small-scale dark structures, which are nucleons in which their quarks are bound by gluons. Since the geometric objects that make up black holes are electrically neutral (gluons), then it is up to the geometric objects corresponding to the nucleons to satisfy the criterion of self-similarity and  $z.c.P^2$ . Through self-similarity, the appropriate geometric objects corresponding to the nuclei will be created, from combinations of geometric objects corresponding to protons and neutrons, such as the original hydrogen (H) and helium (He) nuclei.
  
- B)** The concentrations of geometric objects at specific points in some of the initial Planck patches of the matter-dominant era were made to show the isotropy of  $P^2$  in this and later Planck patches. The isotropy of  $P^2$  is illustrated today by the positions of galaxies that form a nearly isotropic lattice where each of them has at least one black hole at its center. Spiral galaxies and dwarf satellite galaxies in larger galaxies are a strong indication of the tendency to simulate the isotropy of  $P^2$ . Nearing the end of the matter-dominated era and the beginning of the dark energy-dominated era, clusters and superclusters of galaxies are formed with characteristics similar to those of all matter structures at any scale.
  
- C)** The beginning of the dark energy dominance means that the curvature of the considered Planck patches ( $P_{er}^i$ ) is constantly decreasing. The excessive accumulation of matter stops and the curvature of the last supercluster of the matter-dominated era decreases, approaching the curvature of the first Planck patch of the dark energy-dominated era. This means that the last supercluster of galaxies or clusters of galaxies of the era of matter dominance begins to spread continuously in the successive Planck patches of this era, that is, the apparent decomposition of the structures of matter begins. This state can be described equivalently by entities expressing the empty space that we have called **erebons**, that is, circles of ever-increasing radius.

This results in the gradual removal of the parts of the last supercluster of

matter in the subsequent Planck patches, so that from an enlarged Planck patch onwards, its parts will be grains of an almost invisible cosmic dust.

## 15.20 Experimental Signatures and Observational Verifications

Because the Square Screen Model relies on discrete  $1 \rightarrow 4$  cell replication steps rather than a perfectly smooth continuous fabric, it makes distinct experimental predictions:

### 15.20.1 Cosmic Microwave Background (CMB) Pixelation Limits

At extremely early stages of cosmic expansion ( $t_n$  where  $n$  is small), the total number of available geometric screen units  $4^n$  was finite. This predicts a fundamental cutoff or “pixelation” threshold in primordial gravitational waves. Modern CMB observations should detect a discrete quantization in temperature anisotropy boundaries matching the cellular  $4^n$  power law.

### 15.20.2 High-Energy Photon Dispersion Anomalies

If the boundary screen is discrete, high-energy gamma-ray photons traveling across billions of light-years must cross a finite number of square junctions. Photons with wavelengths approaching the discrete boundaries of these squares will experience a tiny geometric refraction, predicting a wavelength-dependent speed variation over cosmological distances.

## Chapter 16

# THE DARK ENERGY ERA

### 16.1 Decomposition of the structures of the Universe

The Planck patch  $P_{\text{De}}^i$  marks the beginning of the dark energy (*DE*) era, where the reverse course from the creation of matter structures begins. Thus, the decomposition of the structures formed in the matter-dominated era initiates. This process starts from the maximum structure that has formed in the Planck patch  $P_M^f$ . What will be produced from  $P_{\text{De}}^i$  and beyond will be geometric objects of almost zero curvature that will start to dominate the next Planck patches. We call these geometric objects erebons, and we call the population of these geometric objects Dark Energy.

The cosmic moment when the energy density of matter (visible and dark) becomes equal to the energy density of dark energy cannot be explained by the methods used by free observers today. This is the so-called Coincidence Problem, which remains an unanswerable question for free observers. The other unanswered problem for free observers is the so-called Cosmological Constant Problem in the context of the study of the Universe by free observers.

What has been shown so far during the process of evolution of the Universe is that one can interpret all observed phenomena at every scale, through representations of two-dimensional geometric objects in three-dimensional entities, starting from the space  $\mathbf{P}^2$ .

As the expansion of the Planck patches continues, the structures created within them occupy more and more space. This has the result that the parts of which these structures are composed are increasingly “loosely” connected to each other. In the subsequent Planck patches, the parts of these structures move further and further away from each other. When this begins to become apparent on a large scale, we say that the Universe has entered the era of dark energy. We have already mentioned the expansion ratio of the Planck patches, which is nothing more than their tendency to satisfy on a Universal

scale both  $z.c.P^2$  and its self-similar structure process.

The photons and gluons in the Planck patches at the beginning of the matter-dominated era combine with other geometric objects to form complex structures. Thus, they can be described as geometric objects that increase the local curvature of the parts of the Planck patch. In a sufficiently expanded Planck patch whose curvature tends to zero, the complex geometric objects also have almost zero curvature. This means that these structures are spread out so that their curvature becomes equal to the curvature of this Planck patch.

For bound observers, the explanation of the dissolution of the structures is obvious for the reasons mentioned earlier, namely the satisfaction of  $z.c.P^2$  and the self-similarity criterion. For free observers, who do not connect the processes taking place at the boundary of the Universe with their observations, there is the question of why all this is happening. So, the explanation they give for the removal of the parts of the galaxies is that there is a mysterious dark energy that acts against gravity.

Einstein's original idea of the existence of a force opposite to gravity, so that the Universe remains static, led him to introduce the Cosmological Constant. As we saw earlier, the Cosmological Constant was discarded from Einstein's equations for various reasons. But since a force acting opposite to gravity is now required, the Cosmological Constant was reintroduced to play another role, namely the force that accelerates the expansion of the Universe. As we will see at the end of this chapter, the introduction of the Cosmological Constant leads us again to a dead end if we use only the methods of free observers in three-dimensional space. Based on these observations, free observers ask the question:

**“What is the reason that the members of a structure separate from each other?”**

According to the current understanding that everything can be explained in terms of particles, an effort has begun to find exotic particles that decompose complex structures. According to the theory of bound observers, what will be produced from the first Planck patch of the Dark Energy dominance era,  $P_{De}^1$  and beyond, will be geometric objects of almost zero curvature that begin to dominate the subsequent Planck patches. Free observers call these geometric objects erebons, and the population of these geometric objects is called Dark Energy.

The name “erebons” comes from the word Erebus, meaning absolute darkness. This was the name of the son of Chaos who lived deep beneath the Earth according to Hesiod's *Theogony*, as mentioned in the work of Jean Richepin, *Greek Mythology*.

To distinguish the erebons from matter particles, bound observers consider that circles of large radius (much larger than  $2l_p$ ) which have almost zero curvature correspond to the erebons. In this era, it seems that an opposite process (decomposition of matter) is followed from the process followed



in previous eras (synthesis of matter), leading the three-dimensional Universe to its initial state corresponding to the space  $\mathbf{R}^2$ , that is, to Nothingness.

Initially, the population of erebons in  $\mathbf{P}^2$  was many orders of magnitude smaller than the population of geometric objects that made up dark and visible matter. As cosmic time passed, after the era of matter dominance, the only geometric objects that continuously formed were erebons, because they fulfilled all the conditions for further reduction of the curvature of the Planck patch. Thus, at some point in time, the population of erebons becomes equal to the population of geometric objects corresponding to the physical entities (holograms of geometric objects) that have created dark and visible matter. This happened shortly before the present epoch (on a Universal time scale).

Due to the increase in the population of erebons, it is predicted that in the future they will exceed the number of all other geometric objects (and their holograms) by several orders of magnitude. This new situation will lead to the gradual separation of the parts of the dark and visible matter structures. Thus, the members of these structures will be so far apart that they will resemble “cosmic dust” or like geometric points of  $\mathbf{P}^2$  when the Planck patch expands so much that it becomes the original space  $\mathbf{R}^2$ .

## 16.2 Percentages of Visible, Dark Matter and Dark Energy

In a Planck  $P_9$  patch of the present era with  $N = 10$  geometric points per side, inscribed circles are formed that have radii and populations:

$$r_1 = 1l_p, \quad N_1 = 64$$

$$r_2 = 2l_p, \quad N_2 = 36$$

$$r_3 = 3l_p, \quad N_3 = 16$$

$$r_4 = 4l_p, \quad N_4 = 4$$

The sum of these circles is  $N_t = N_1 + N_2 + N_3 + N_4 = 120$ . Visible matter arises only from the inscribed circles of radii  $r_1 = 1l_p$  and  $r_2 = 2l_p$ , which are  $64 + 36 = 100$  in total. From the circles  $N_1, N_2$ , we obtain 12 geometric aggregates corresponding to 9 H atoms and 3 He atoms, i.e., a total of 12 H and He atoms, as mentioned in Section 3.8.

The atoms H and He belong to the Visible Matter (VM) from which free and bound observers, their observation instruments, planets, stars, galaxies, etc., are constructed. In the same Planck patch  $P_9$ , we also get 120 circumscribed circles with radii being the hypotenuses of right triangles with length  $r_{j+} = j^+l_p$ , where  $j^+ = j\sqrt{2}$  and  $j = 1, 2, 3, 4$ . The sum of all these circles is  $120 + 120 = 240$ , so the percentage of Visible Matter is:

$$VM: \frac{12}{240} = 0.05 \quad \text{or} \quad 5\%$$

The Planck patch  $P_9$  is covered by circles of radius  $r_1^+ = 1 \cdot \sqrt{2}l_p$ , which are  $N_1 = 64$  in total. The circles of radius  $r_1^+$  are not visible to observers (free and bound) because they “see” only structures made of the circles of radius  $r_1 = 1l_p$  and  $r_2 = 2l_p$ . Thus, the bound and free observers:

- A) consider the circles of radius  $r_1^+$  as building circles from which all circles with radii  $r_{j+} = j\sqrt{2}l_p$ , where  $j = 1, 2, 3, 4 \dots$ , are constructed.
- B) include the  $r_1^+$  radius circles in a type of matter they call Dark Matter ( $DM$ ).

The percentage of Dark Matter is:

$$DM: \frac{64}{240} = 0.266 \quad \text{or} \quad 26.6\% \simeq 27\%$$

As we have seen, each Planck patch, after the contraction stage, enters an expansion stage, so it quadruples according to the bound observers. Free observers believe that a mysterious energy increases space, which they call Dark Energy ( $DE$ ). Bound observers consider that the space occupied by the new Planck patch increases due to self-similarity, and the reason for its continuous increase is the decrease in the curvature of the circle surrounding each new Planck patch to satisfy z.c. $\mathbf{P}^2$ .

The bound observers consider the circles of radii  $r_1 = 1l_p$  and  $r_2 = 2l_p$  of  $VM$ , which are a total of 100, and the circles of radii  $r_1^+$  of  $DM$ , which are a total of 64. Since the space containing the total number of  $100 + 64 = 164$  circles will expand, then their percentage is:

$$DE: \frac{164}{240} = 0.6833 \quad \text{or} \quad 68.33\% \simeq 68\%$$

Thus, the percentages of  $VM$ ,  $DM$ , and  $DE$  are 5%, 27%, and 68%, respectively. The proportions of geometric objects that make up the population of  $VM$ ,  $DM$ , and  $DE$  in the Planck patch  $P_9$  of this era are the same as those recorded by the Planck satellite measurements in the Universe. From the above proportions, the conclusion follows that the geometric model works equally well at the macroscopic level, giving theoretical results that are identical to the results from observations. The surface density  $\rho$  of a Planck patch is equal to the acceleration  $a$  when we set  $c = 1$  and  $G = 1$ , as follows from dimensional analysis.

From  $c = 1$ , we get  $[L] = [T]$ , and from  $G = 1$  and Newton's law  $\frac{[M][L]}{[T]^2} = G \frac{[M]^2}{[L]^2}$ , we get  $[L] = [M]$ . Thus,  $\rho = \frac{[M]}{[L]^2} = \frac{1}{[L]}$  and  $a = \frac{[L]}{[T]^2} = \frac{1}{[L]}$ .

Having calculated the acceleration of a Planck patch, we immediately know the value of its surface density. The energy density of  $DE$  is found as

follows: The numerical value of the acceleration of  $P_9$  today is  $a = 1.345 \cdot 10^{-8}$  in the c.g.s. system of units, as shown in Table 17.9 of Appendix 17.5.

Since dark energy is responsible for the acceleration of  $P_9$ , then the density of dark energy  $\rho_{De}$  has the same numerical value as the acceleration of  $P_9$ . This result explains the paradox of the Cosmological Constant through the expansion of the Planck patches.

For free observers, this paradox arises from the discrepancy between the theoretically calculated value via Quantum Field Theory (QFT), which gives a value of  $\sim 10^{112} \text{ erg} \cdot \text{cm}^{-3}$ , and the value from measurements, which is  $\sim 10^{-8} \text{ erg} \cdot \text{cm}^{-3}$ .

From the previous result, it follows that the value of the Cosmological Constant that is theoretically obtained is not compatible with the observational data; therefore, once again, it plays no role in the birth and evolution of the Universe. From this fact, we are again led to the rejection of the Cosmological Constant from the field equations of the General Theory of Relativity.

Furthermore, the geometric model we have developed provides answers to old and new problems in Cosmology without the mathematical complexity of the General Theory of Relativity. This in no way diminishes the value of the equations of General Relativity, whose findings have been verified by various methods, as well as their application to the interpretation of strong interactions. Through this process, the boundary of the Universe will once again become  $\mathbf{R}^2$  space, and the holographic Universe will return to Nothingness.



Part IV

APPENDICES



## Chapter 17

# APPENDICES

### 17.1 THE DELIAN PROBLEM

The Delian problem was one, and perhaps the most significant, of the three classic problems of Greek geometry (the other two being the trisection of an angle and the squaring of the circle).

This problem was named Delian because, according to a myth recounted by Theon of Smyrna in a dialogue titled *Platonic*, the inhabitants of Delos were suffering from a plague. Thus, they asked the Oracle of Delphi what they should do to be saved. The Pythia replied that they had to double the volume of the altar of Apollo, which had a cubical shape, while preserving its geometric form. The Delians initially believed that this problem could be solved simply by doubling the sides of the cube.

However, when they discovered that the volume became eight times larger instead of double, they sent ambassadors to Plato's Academy and requested a solution to the problem. Plato replied that the God gave this prophecy not because he needed an altar double in volume, but firstly to criticize and secondly to rebuke the Greeks for neglecting mathematics and despising geometry, and to emphasize the importance of Mathematics for humanity.

Plato's answer was indeed a wise one, which was vindicated over the following centuries, because many applications at both theoretical and technological levels emerged from the solution to the Delian problem. The solution to the Delian problem concerned finding the edge of a cube that has twice the volume of a given cube using only a straightedge and a compass. Expressing the above proposition with a mathematical formula, we obtain:

$$x^3 = 2a^3$$

where  $x$  is the edge of the new cube. Hippocrates of Chios was the first to provide a solution to this problem by reducing it to the equivalent problem of constructing two line segments  $x$  and  $y$  in  $R^2$ .

The segments  $x$  and  $y$  are called mean proportionals between two straight line segments  $a$  and  $\beta$ , such that they satisfy the relation:

$$\frac{a}{x} = \frac{x}{y} = \frac{y}{\beta}$$

If  $\beta = 2a$  then:

$$xy = 2a^2$$

$$x^2 = ay$$

$$y^2 = 2ax$$

Thus, we obtain:

$$x^3 = 2a^3$$

With the proof of this result, ancient Greek mathematicians attempted to

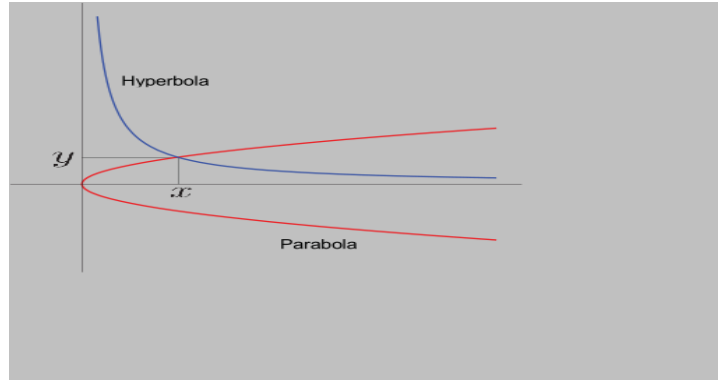


Figure 17.1: The construction of Menaechmus

find two mean proportionals toward the final solution of the Delian problem. Menaechmus, initially a student and later an associate of Plato, provided a highly interesting solution that initiated a new type of geometric approach, ultimately leading to the formulation of the theory of conic sections. According to the geometric construction of Menaechmus, illustrated in Figure 17.1, the segments  $x$  and  $y$  are constructed in such a way that they form the coordinates of the intersection point of a hyperbola and a parabola. The axes of the coordinate system are asymptotes of a hyperbola, one of which is the axis of symmetry of a parabola. From the preceding relations, we obtain:

Hyperbola:

$$2a^2 = xy$$

Parabola:

$$2ax = y^2$$



The terms ellipse, parabola, hyperbola, and conic sections were established after Menaechmus. These terms were introduced for the first time around 300 B.C., and subsequently, these curves were identified with the curves generated by intersecting a cone with a plane, hence the name conic sections. Other terms, such as the coordinate system and the asymptotes of the hyperbola, are much newer. Archimedes (287–212 B.C.) used the following terminology:

1. "Conic section of an acute angle" for the ellipse
2. "Rectangular conic section" for the parabola
3. "Conic section of an obtuse angle" for the hyperbola

Figure 17.2: Definition of conic sections by Apollonius

These names derived from comparing the side length of a rectangle to a fixed length corresponding to the section of a cone by a plane, which was called the **parameter**. Thus, if the section of a cone by a plane corresponds to the parameter  $\epsilon$ , which is the side length of a rectangle whose area is equal in magnitude to the area of a square with side  $y$ , then the parameter  $\epsilon$  is equal to the side AB of the rectangle whose other side is  $x$  (in Greek, “παράβállεται” means “to be placed alongside/equal”), hence the curve is a

parabola. Through this process, a parabola is defined as shown in Figure 17.2. If the curve is such that the side AB is less than the parameter  $\epsilon$  (i.e., it falls short, “ελλείπει”, then the curve is an ellipse. Finally, if this curve is such that the side AB is greater than the parameter  $\epsilon$  (i.e., it exceeds, “υπερβάλλει”, then the curve is a hyperbola. The parameter  $\epsilon$  is known today as the **eccentricity of conic sections**.

In his work, Apollonius emphasized the theoretical aspect rather than the applications of conic sections, as he points out in the prefaces of the fourth and fifth volumes of his work. Conic sections as a unified mathematical theory appeared to be completed by the work of Apollonius. In the centuries that followed, physical phenomena were not associated with conic sections due to the philosophical view of the ancient Greeks that mathematics is a “pure science” that should not be linked to practical constructions, even though the properties of conic sections were used in several applications, such as the acoustics of theaters. Such an effort to connect them to physics was initiated later by Galileo and Kepler, and continued by Newton.

The Delian problem was the first problem in the history of mathematics to be formulated in three-dimensional space ( $3D$ ) and solved in two-dimensional space ( $R^2$ ). This solution was transferred back to three-dimensional space for practical applications, according to Plato’s method, as mentioned in the Preface. Based on this general method, a similar approach will be developed for mapping conic sections onto physical entities.

## 17.2 MAGNITUDES OF CONIC SECTIONS

### 17.2.1 Definition of the Cartan Matrix

The Cartan matrix of a non-singular, differentiable square matrix  $A(u)$  is defined as:

$$C(A(u)) = \frac{dA(u)}{du} A(u)^{-1} = \dot{A}(u) A(u)^{-1} \quad (17.1)$$

From this definition, we obtain the following relationship:

$$C(AB) = C(A) + AC(B)A^{-1} \quad (17.2)$$

The rotation matrix in a plane is:

$$P(\phi) = \begin{pmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{pmatrix}$$

Its transpose matrix is:

$$P^t(\phi) = \begin{pmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{pmatrix}$$

From these definitions, we get:

$$P(\phi)P^t(\phi) = U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

or:

$$P^t(\phi) = P^{-1}(\phi) \quad (17.3)$$

Any matrix that satisfies Equation 17.3 is called an **orthogonal matrix** and has a determinant value of  $\pm 1$ . If its value is  $+1$ , it is called a rotation matrix. A specific rotation in the plane is given by the matrix:

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

where  $\phi = \frac{\pi}{2}$ . The Cartan matrix of a function of an orthogonal matrix is skew-symmetric, i.e.,

$$C(A)^t = -C(A)$$

### 17.2.2 The Moving Frame in a Plane

A frame is defined by the vector pair  $(\mathbf{e}_1, \mathbf{e}_2)$ , where  $\mathbf{e}_1, \mathbf{e}_2$  are mutually perpendicular unit vectors such that  $\mathbf{e}_2$  is obtained from  $\mathbf{e}_1$  by a rotation through an angle of  $\frac{\pi}{2}$ . Let there be a planar curve parameterized by its arc length  $s$ . The position vector of a point corresponding to the arc length  $s$  is measured from an arbitrary origin. This is given by:

$$\mathbf{x}(s) = x_1(s)\hat{\mathbf{e}}_1 + x_2(s)\hat{\mathbf{e}}_2 \quad (17.4)$$

The tangent to this curve is the unit vector:

$$\mathbf{t}(s) = x_1'(s)\hat{\mathbf{e}}_1 + x_2'(s)\hat{\mathbf{e}}_2 \quad (17.5)$$

Rotating  $\mathbf{t}(s)$  by  $\frac{\pi}{2}$ , we obtain the normal vector:

$$\mathbf{n}(s) = -x_2'(s)\hat{\mathbf{e}}_1 + x_1'(s)\hat{\mathbf{e}}_2 \quad (17.6)$$

where the prime denotes differentiation with respect to  $s$ . We define the frame  $(\mathbf{t}(s), \mathbf{n}(s))$  as the moving frame of a planar curve. A moving frame in an Euclidean plane defines a Cartesian coordinate system at each point of this curve, such that the point  $\mathbf{x}(s)$  is its origin and the  $+x$  axis lies along the tangent of this curve in the direction of increasing  $s$ . The moving frame is derived from the fixed frame  $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2)$  through a rotation by an angle  $\phi$ . Thus, we have:

$$(\mathbf{t}(s), \mathbf{n}(s)) = \mathbf{P}(\phi)(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2) \quad (17.7)$$

The matrix  $\mathbf{P}(\phi)$  is called the **frame matrix of the planar curve**. The variation of the moving frame along a curve is defined from Equation 17.7 as follows:

$$(\mathbf{t}(s), \mathbf{n}(s))' = \mathbf{P}'(\phi)(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2) = P'(\phi)P^{-1}(\phi)(\mathbf{t}(s), \mathbf{n}(s)) = C(P)(\mathbf{t}(s), \mathbf{n}(s)) \quad (17.8)$$

The equations resulting from Equation 17.8 are the Frenet equations of planar geometry. Since  $C(P)$  is skew-symmetric, it takes the form:

$$C(P) = \begin{pmatrix} 0 & \kappa(s) & -\kappa(s) & 0 \end{pmatrix} = \kappa(s) \begin{pmatrix} 0 & 1 & -1 & 0 \end{pmatrix} = \kappa(s)J \quad (17.9)$$

### 17.2.3 The Moving Plane Method

A complete description of the moving plane method can be found in reference [11]. Here, we present some useful results utilized in this text. The moving plane  $\Pi_f$  is the plane defined by the moving frame  $(\mathbf{t}(s), \mathbf{n}(s))$  on a curve  $f$  of the fixed plane  $\Pi$ , which is defined by the fixed frame  $(\mathbf{e}_1, \mathbf{e}_2)$ . Let  $g(s_f)$  be another curve in  $\Pi$  using the arc length  $s_f$  of curve  $f$  as a parameter, as shown in Figure 17.3 below. The arc length of this curve  $g$  does not correspond directly to  $s_f$ .

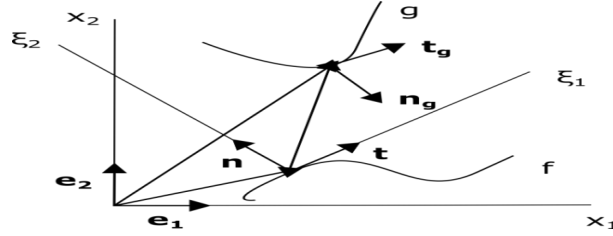


Figure 17.3: The moving plane

We consider a point moving along curve  $g$ . This point “traces” a curve  $\gamma$  in  $\Pi_f$ . The question is:

**”What is the curve  $\gamma$ , given the curves  $f$  and  $g$ ?”**

For simplicity, we denote both the curve and the position vector of a point on that curve by the same letter. Every point  $(g_1, g_2)$  of curve  $g$  in plane  $\Pi$  coincides with a point  $(\gamma_1, \gamma_2)$  of curve  $\gamma$  in  $\Pi_f$ . The curve  $f$  defines  $\Pi_f$  via the moving frame  $(\mathbf{t}(s), \mathbf{n}(s))$  at the point  $(f_1, f_2)$ , and the same point in  $(\mathbf{t}(s), \mathbf{n}(s))$  is  $(0,0)$ , which is the origin of the moving frame. The arc length and curvature of curve  $\gamma$  are indicated by  $s_\gamma$  and  $\kappa_\gamma$ , respectively. For the sake of simplicity, we will omit the argument  $(s)$  across various quantities in the following relationships where it is not strictly necessary.

The frame matrices of the position vectors  $f$  and  $g$  of curves  $f$  and  $g$  are given by:

$$(\mathbf{t}(s_f), \mathbf{n}(s_f)) = F(s_f)(\mathbf{e}_1, \mathbf{e}_2) \quad (17.10)$$

$$(\mathbf{ng}, \mathbf{tg}) = (\mathbf{n}\gamma, \mathbf{t}\gamma) = G(s_g)(\mathbf{e}_1, \mathbf{e}_2) \quad (17.11)$$

Relation 17.11 is reflected in the geometric configuration of Figure 17.3, namely the fact that the moving frames  $(\mathbf{ng}, \mathbf{tg})$  and  $(\mathbf{n}\gamma, \mathbf{t}\gamma)$  are mapped to each point of  $g$  or  $\gamma$  for each value of  $s_f$ , yielding:

$$G(s_g) = \Gamma(s_f)F(s_f) \quad (17.12)$$

The frame matrix of curve  $\gamma$  with respect to  $(\mathbf{t}, \mathbf{n})$  is given by the relation:

$$(\mathbf{n}\gamma, \mathbf{t}\gamma) = \Gamma(s_\gamma)(\mathbf{t}, \mathbf{n}) = \Gamma(s_\gamma)F(s_f)(\mathbf{e}_1, \mathbf{e}_2) \quad (17.13)$$

From Equations 17.10, 17.11, and 17.13, we obtain:

$$G(s_g)(\mathbf{e}_1, \mathbf{e}_2) = \Gamma(s_\gamma)F(s_f)(\mathbf{e}_1, \mathbf{e}_2)$$

or:

$$G(s_g) = \Gamma(s_\gamma)F(s_f) \quad (17.14)$$

We evaluate the Cartan matrix from Equation 17.14 by differentiating with respect to  $s_f$ . Thus, we obtain:

$$C(G(s_g))\frac{ds_g}{ds_f} = C(\Gamma(s_\gamma))\frac{ds_\gamma}{ds_f} + \Gamma(s_f)C(F(s_f))\Gamma^{-1}(s_f) \quad (17.15)$$

In the second term of the right-hand side of Equation 17.15, Equation 17.12 is utilized. Since all matrices in Equation 17.15 represent rotations, by applying Equation 17.9 we can write:

$$\kappa_g \frac{ds_g}{ds_f} = \kappa_\gamma \frac{ds_\gamma}{ds_f} + \kappa_f \quad (17.16)$$

For an arc element  $s_f$ , the position vectors of  $f, g$ , and  $\gamma$ , as illustrated in Figure 17.3, satisfy the relation:

$$g(s_f)(\mathbf{e}_1, \mathbf{e}_2) = f(s_f)(\mathbf{e}_1, \mathbf{e}_2) + \gamma(s_f)(\mathbf{t}, \mathbf{n})$$

Due to Equation 17.11, the preceding relation is written as:

$$g(s_f) = f(s_f) + \gamma(s_f)F(s_f) \quad (17.17)$$

Differentiating Equation 17.17 with respect to  $s_f$ , we obtain:

$$\frac{dg}{ds_f} = \frac{df}{ds_f} + \gamma(s_f)C(F(s_f))F(s_f) + \frac{d\gamma}{ds_f}F(s_f) \quad (17.18)$$

We choose a point on curve  $f$  characterized by the arc length  $s_0$  for which the moving frame matches the fixed frame, namely:

$$(\mathbf{e}_1, \mathbf{e}_2) = (\mathbf{t}(s_0), \mathbf{n}(s_0))$$

As  $s_f \rightarrow s_0$ , we have  $F(s_f) = U$  (the identity matrix) and  $(f_1, f_2) = (s_f, 0)$ ; therefore, Equation 17.18 simplifies to:

$$\frac{dg}{ds_f} = \frac{d\gamma}{ds_f} U + \gamma \kappa_f(s_f) \cdot J + \frac{df}{ds_f} \quad (17.19)$$

From Equation 17.19, we extract the following two equations:

$$\frac{dg_1}{ds_f} = \frac{d\gamma_1}{ds_f} - \kappa_f \gamma_2 + 1 \quad (17.20)$$

$$\frac{dg_2}{ds_f} = \frac{d\gamma_2}{ds_f} + \kappa_f \gamma_1 \quad (17.21)$$

If curve  $g$  collapses into a single point, then  $\frac{dg_1}{ds_f} = \frac{dg_2}{ds_f} = 0$ . Equations 17.20 and 17.21 then take the form:

$$\frac{d\gamma_1}{ds_f} - \kappa_f \gamma_2 + 1 = 0 \quad (17.22)$$

$$\frac{d\gamma_2}{ds_f} + \kappa_f \gamma_1 = 0 \quad (17.23)$$

In the coordinate system of the moving frame  $(\mathbf{t}, \mathbf{n})$ , we set  $(\gamma_1, \gamma_2) = (\xi_1, \xi_2)$ , so equations 17.22 and 17.23 are written as:

$$\xi_1' = \kappa_f \xi_2 - 1 \quad (17.24)$$

$$\xi_2' = -\kappa_f \xi_1 \quad (17.25)$$

#### 17.2.4 The Natural Equation of Conic Sections

Let us assume that the point curve  $g$  coincides with the origin of the coordinate system in  $\Pi$  and curve  $\gamma$  is a conic section in  $\Pi_f$ , when the frames in both planes coincide for some  $s = s_0$ . The equation of this conic section is:

$$\xi_2 = \frac{1}{2}(a(s)\xi_1^2 + 2b(s)\xi_1\xi_2 + c(s)\xi_2^2) = \frac{1}{2} \begin{bmatrix} \xi_1 & \xi_2 \end{bmatrix}^t \begin{bmatrix} a(s) & b(s) \\ b(s) & c(s) \end{bmatrix} \begin{bmatrix} \xi_1 & \xi_2 \end{bmatrix} \quad (17.26)$$

where  $a(s), b(s), c(s)$  are functions of the arc length  $s = sf$  of this conic section. The coordinates  $\xi_1, \xi_2$  satisfy equations 17.24, 17.25, and 17.26. Differentiating Equation 17.26 with respect to  $s$  and using equations 17.24 and 17.25 to eliminate  $\xi_1'$  and  $\xi_2'$ , we obtain:

$$(a - \kappa_f)\xi_1 + b\xi_2 = \frac{1}{2}\{(a' - 2b\kappa_f)\xi_1^2 + 2(\kappa_fa + b' - \kappa_fc)\xi_1\xi_2 + (2\kappa_fb + c')\xi_2^2\}$$

Since we can evaluate this at any value of  $s$ , this expression must represent the same conic section as Equation 17.26. Thus, comparing the coefficients that represent the conic sections, we get the relations:

$$a - \kappa_f = 0 \quad (17.27)$$

$$a' - 2b\kappa_f = ab \quad (17.28)$$

$$\kappa_fa + b' - \kappa_fc = b^2 \quad (17.29)$$

$$2\kappa_fb + c' = cb \quad (17.30)$$

From equations 17.27 and 17.28, we find:

$$b = \frac{1}{3} \frac{\kappa'_f}{\kappa_f} \quad (17.31)$$

From Equation 17.26, it follows that the invariant quantities of the conic section are the trace  $T$  and the determinant  $D$  of the matrix  $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ , namely:

$$T = a + c \quad (17.32)$$

$$D = ac - b^2 \quad (17.33)$$

From equations 17.27, 17.28, 17.29, 17.30, 17.32, and 17.33, we obtain:

$$\frac{T'}{T} = (\ln T)' = \frac{1}{3} \frac{\kappa'_f}{\kappa_f} = \frac{1}{3} (\ln \kappa_f)' \quad (17.34)$$

$$\frac{D'}{DKennel} = (\ln D)' = \frac{2}{3} \frac{\kappa'_f}{\kappa_f} = \frac{2}{3} (\ln \kappa_f)' \quad (17.35)$$

Solving equations 17.34 and 17.35, we obtain the expressions for  $T$  and  $D$ :

$$T = A\kappa f^{\frac{1}{3}}, \quad D = B\kappa f^{2/3} \quad (17.36)$$

where  $A, B$  are constants. From equations 17.34 and 17.35, we establish the **invariant quantities of conic sections**:

$$I_1 = \frac{(\ln T)'}{(\ln \kappa_f)'} = \frac{1}{3} \quad (17.37)$$

$$I_2 = \frac{(\ln D)'}{(\ln \kappa_f)'} = \frac{2}{3} \quad (17.38)$$

From equations 17.34 and 17.35, we derive the identity:

$$b^2 = aT - D - a^2$$

Substituting the values of  $a, b, T$ , and  $D$  from equations 17.27, 17.31, and 17.36 into the relation above yields an equation that is independent of coordinate systems, containing only the curvature of a point on the conic section as an unknown. This equation is called the **natural equation of conic sections**, and its expression is:

$$\left( \frac{1}{3} \frac{\kappa_f'}{\kappa_f} \right)^2 = A\kappa_f^{4/3} - B\kappa_f^{2/3} - \kappa_f^2 \quad (17.39)$$

### 17.3 CONSTRUCTION OF LINEAR ELLIPSES

The linearization of ellipses in  $\mathbf{P}^2$  occurs in two ways:

**The first** is as follows: during the collapse phase of a Planck patch, a circle shrinks along one of its diameters, thereby generating an ellipse.

**The second** follows this procedure: between two circles in a random Planck patch, there exist points belonging to an ellipse.

If the inner circle is the circumscribed circle at a Planck point with radius  $r_p \propto l_p$  and the other is the circumscribed circle around a Planck patch (outer circle) with radius  $r_p \propto 2l_p$ , then the eccentricity of the ellipse is nearly equal to the eccentricity of a circle, meaning we have  $\epsilon \rightarrow 0^+$ . If, for the same inner circle, the radius of the outer circle is, for example,  $r_p \propto 10^{14}l_p$ , then the ellipse formed between these circles “appears” as a linear ellipse with an eccentricity tending toward 1, namely  $\epsilon \rightarrow 1^-$ , as shown in Figure 17.4. These nearly linear ellipses constitute segments of linear open conic sections (parabola, hyperbola), as derived from the process of collapsing the circular bases of a cone into points.

Since the equations of open conic sections are formally identical to wave equations, the equations of their segments will also express waves, as will be seen below. Through this process, the generation of wave-packets or waves is interpreted geometrically.



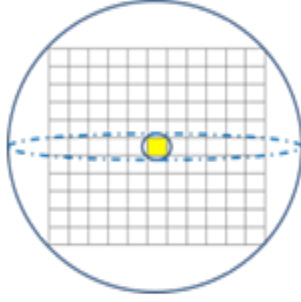


Figure 17.4: Linearization of an ellipse between the circumscribed circles of two Planck patches

Let there be the position vector of a point on an ellipse. The origin of this vector is located at one focus of the ellipse. The expression for this position vector is given by:

$$\mathbf{r}_c(\theta) = \mathbf{r}_x + \mathbf{r}_y$$

where:

$$\mathbf{r}_x = r_c(\theta(s)) \cos(\theta(s)) \mathbf{e}_1$$

$$\mathbf{r}_y = r_c(\theta(s)) \sin(\theta(s)) \mathbf{e}_2$$

The vectors  $\mathbf{e}_1, \mathbf{e}_2$  are the unit vectors along the  $x, y$  axes respectively,  $\theta$  is the angle formed by the position vector with the principal axis of the ellipse, and  $s$  is the arc length measured from an arbitrary origin on the perimeter of the ellipse. The tangent vector is given by:

$$\frac{d\mathbf{r}_c(\theta(s))}{ds} = \mathbf{u}_x + \mathbf{u}_y$$

where:

$$\mathbf{u}_x = [\dot{r}_c(s) \cos(\theta(s)) - r_c(\theta(s)) \dot{\theta}(s) \sin(\theta(s))] \mathbf{e}_1$$

$$\mathbf{u}_y = [\dot{r}_c(s) \sin(\theta(s)) + r_c(\theta(s)) \dot{\theta}(s) \cos(\theta(s))] \mathbf{e}_2$$

with:

$$\dot{r}_c(s) = \frac{dr_c(\theta(s))}{ds}, \quad \dot{\theta}(s) = \frac{d\theta(s)}{ds}$$

When the ellipse tends to become linear, i.e.,  $\theta(s) \rightarrow 0^+$ , then:

$$\mathbf{r}_x = r_c(0^+) \mathbf{e}_1$$

$$\mathbf{u}_x = \dot{r}_c(s) \mathbf{e}_1$$

$$\mathbf{r}_y = 0^+ \mathbf{e}_2$$

$$\mathbf{u}_y = 0^+ \mathbf{e}_2$$

As seen from the preceding relations, when  $\theta(s) \rightarrow 0^+$ , the position vector  $\mathbf{r}_c \rightarrow \mathbf{r}_x$  and the tangent vector  $\mathbf{u}_c \rightarrow \mathbf{u}_x$  at any point of the linear ellipse become collinear vectors in the same direction, parallel to the principal axis of the linear ellipse. If we make the mapping  $s \rightarrow t$ , where  $t$  is the time parameter, then  $\mathbf{u}_x, \mathbf{u}_y$  are no longer dimensionless quantities. From the closed conic sections, as illustrated in the following Figure 17.5, we select the simplest conic section, namely the unit circle with the equation:

$$f(x, y) = x^2 + y^2 = 1$$

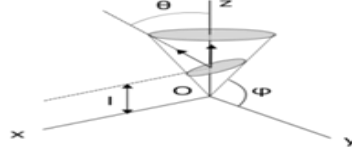


Figure 17.5: Conic Sections

With a proper choice of parameters  $(\phi, \theta)$ , all conic sections are produced. Thus, we obtain:

$$\nabla f(x, y) = \frac{\partial f(x, y)}{\partial x} \mathbf{e}_1 + \frac{\partial f(x, y)}{\partial y} \mathbf{e}_2 \propto x \mathbf{e}_1 + y \mathbf{e}_2 = \mathbf{r}_c \cdot 1 = \mathbf{r}_c \cdot f(x, y)$$

From this relation, and uniquely for conic sections, the position vector  $\mathbf{r}_c$  varies, thereby affecting other quantities of the conic sections just as the operator  $\nabla$  does. From this relation, the correspondence emerges:

$$\nabla \leftrightarrow \mathbf{r}_c$$

From the open conic sections, we select the simplest one, which is the unit parabola with the equation:

$$f(x, s) = \frac{x^2}{2s} = 1$$

Acting on  $f(x, s)$  with the 1D operator  $\nabla$ , where  $\nabla = \frac{\partial}{\partial x}$  (since in the linear limit the  $y$ -component tends to  $0^+$ ), we obtain:

$$\nabla f(x, s) = \left[ \frac{\partial}{\partial x} \left( \frac{x^2}{2s} \right) \right] \mathbf{e}_1 = \frac{x}{s} \mathbf{e}_1$$

For  $s \rightarrow t$ ,  $f(x, t) = 1$ , and  $\mathbf{u}_c = \frac{x}{t} \mathbf{e}_1$ , we obtain:

$$\nabla f(x, t) = \mathbf{u}_c f(x, t)$$

Thus, as in the case of the unit circle, we establish the correspondence:

$$\nabla \leftrightarrow \mathbf{u}_c$$

The equations defining the conic sections are:

$$\begin{aligned}\mathbf{L} &= \mathbf{r}_c \times \mathbf{u}_c \\ \mathbf{R} &= \mathbf{u}_c \times \mathbf{L} - \mathbf{r}_c \left( \frac{k}{r_c} \right)\end{aligned}$$

According to the previous correspondences, these equations are written as:

$$\begin{aligned}\mathbf{L} &= \nabla \times \mathbf{u}_c \\ \mathbf{R} &= \nabla \times \mathbf{L} - \nabla \left( \frac{k}{r_c} \right)\end{aligned}$$

The correspondences  $\nabla \leftrightarrow \mathbf{u}_c$  and  $\nabla \leftrightarrow \mathbf{r}_c$  hold when  $\mathbf{r}_c, \mathbf{u}_c$  are positioned on the left side of each term in the above equations. From these relations, Maxwell's equations for Electromagnetism and Einstein's equations of General Relativity are derived.

## 17.4 DEFINITION OF TEMPERATURE

### 17.4.1 Definitions of Magnitudes

Before examining what occurs during the expansion of the Universe, we must address the problem that arises for bound observers due to the random number of circles appearing in a random Planck patch, characterized on one hand by two different radii, and on the other hand by a random positive or negative area resulting from the position and tangent vectors that correspond to their orientation.

Free observers assume that the elementary particles of "matter," which compose the complex structures of the Universe, map to free quadrants of circles with radii  $1l_p$  and  $2l_p$  in  $\mathbf{P}^2$ , as shown in Figures 3.4, 3.5, and 3.6.

Given this fact, we examine a random quantity of two types of particles that are characterized by a vector magnitude, such as their spin. These two types of particles correspond to the same number of circles with radii  $1l_p$  and  $2l_p$  having a random orientation (up or down), as illustrated in the following figure.

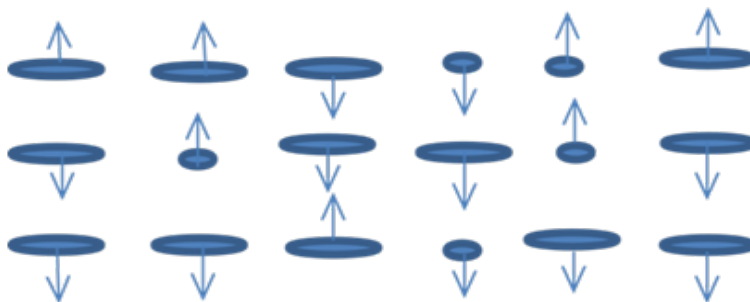


Figure 17.6: Random orientations of circles

Using methods of Statistical Physics, we find that:

1. The curvature of a region with this distribution of geometric objects is proportional to a macroscopic quantity that we call **temperature**.
2. Temperature is the common magnitude that the components of a system acquire when they are in equilibrium with each other.

As an example, we mention the temperature measurement of a patient, where the human body and the thermometer constitute a single unified system, and temperature is the common magnitude we measure once equilibrium is established between them after a certain period of time. Thus, in the initial Planck patch, the temperature is enormous because the curvature of the inscribed circle within it is immense due to its infinitesimal radius, as we saw in the preface.

As the Planck patch continuously expands, its temperature and curvature decrease because the radius of the circle inscribed in it continuously increases until it reaches an infinite value, at which point the temperature and curvature tend toward zero. The next question to be answered is how the expansion of the initial Planck patch will take place, such that, on one hand, we can enumerate the infinite Planck patches that will cover the infinite space  $\mathbf{P}^2$ , and on the other hand, their structuring occurs via a self-similar process.

The first Planck patch is a square with a side length equal to the Planck length  $l_p$ .

The second Planck patch is also a square with a side length of  $2l_p$ .

The third Planck patch is likewise a square with a side length of  $3l_p$ . If we were to continue with this process of generating Planck patches, the next Planck patch would have a side length exactly one Planck length  $l_p$  larger than the previous one.

However, this does not happen because, as we will see in the geometric model developed below, the expansion of the Planck patches initially followed an exponential expansion (inflation) rather than a linear expansion during the early stages of its creation. This process is illustrated in the first 4 Planck patches below.

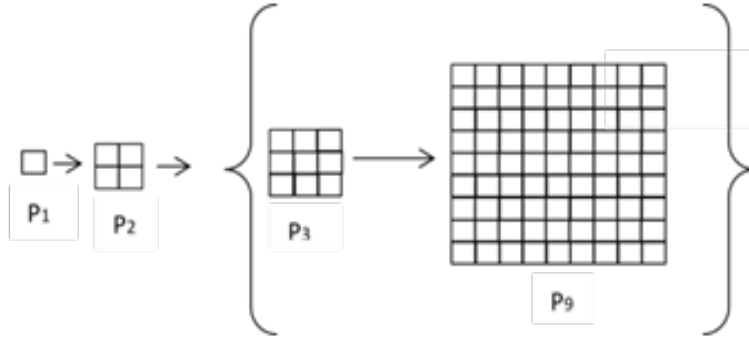


Figure 17.7: Self-similar expansion of the Planck area

In the early stages of the expansion of the Planck patches, the creation of complex geometric structures continues up to a certain critical Planck patch ( $P_c$ ) and then stops because the expansion conditions do not favor the creation of larger complex bodies or their aggregates.

Let there be a square Planck patch with a side length of  $L_p = N \cdot l_p$ , where  $N$  is the number of Planck lengths  $l_p$ . Within the Planck patch, there are circles of radii 1 and 2 from which, via correspondences, the fundamental constituents of matter—namely electrons  $e$  and quarks  $u, d$ —emerge. In this Planck patch, the above circles overlap because certain geometric points of the Planck patch act as centers of circles.

The sets of circles with radii 1 and 2 are denoted as  $A_1$  and  $A_2$ , respectively. The characteristics of a circle are its radius magnitude  $|r_i|$  and its orientation. Combining these two characteristics, we obtain the relation:

$$Q_i = \mu_0 \cdot |r_i| \frac{(\mathbf{I} \times \mathbf{b}) \cdot \mathbf{e}_3}{|\mathbf{I} \times \mathbf{b}|}$$

The vector  $\mathbf{e}_3$  is the positive unit vector perpendicular to the plane of the circle. The coordinate vector  $b = (\pm r_i, \pm r_i)$  specifies each quadrant of the circle,  $I = (\frac{1}{3}, \frac{2}{3})$  is a constant vector defined in the conic sections, and  $\mu_0$  is a proportionality constant. From this definition,  $Q_i$  takes the values  $\pm \mu_0 |r_i|$ .

What we want to achieve is to find the number of circles with radius  $r_1$  that have  $Q_1 > 0$  and  $Q_1 < 0$ , and the number of circles with radius  $r_2$  that have  $Q_2 > 0$  and  $Q_2 < 0$ , such that the sum  $Q_1 + Q_2$  yields the quantity:  $Q_p = \mu_0 \frac{N_\delta - 1}{2} l_p > 0$  which corresponds to the inscribed circle of a Planck patch.

The Planck patch contains  $N^2$  Planck areas, so the total number of circles with radii  $r_n = n \cdot l_p$  is  $N_n = (N_\delta - 2 \cdot n)^2$ . Thus, we obtain the number of circles  $N_1 = (N_\delta - 2 \cdot 1)^2$  with radius  $r_1$  that have  $Q_1 > 0$  or  $Q_1 < 0$ , and the number of circles  $N_2 = (N_\delta - 2 \cdot 2)^2$  with radius  $r_2$  that have  $Q_2 > 0$  or  $Q_2 < 0$ .

If, out of the circles with radius  $r_1$ , a number  $c_1^+$  have  $Q_1 > 0$  and a number  $c_1^-$  have  $Q_1 < 0$ , then:

$$c_1^+ + c_1^- = (N_\delta - 2 \cdot 1)^2 \quad (17.40)$$

For circles of radius  $r_2$ , we obtain:

$$c_2^+ + c_2^- = (N_\delta - 2 \cdot 2)^2 \quad (17.41)$$

To determine the equivalent radius  $R_p$  from the circles with radii  $r_1, r_2$ , we utilize the following relationship:

$$(c_1^+ + |c_1^-|)Q_1 + (c_2^+ + |c_2^-|)Q_2 = (N_\delta - 2 \cdot 2)^2 \quad (17.42)$$

Substituting  $c_1^-, c_2^-$  from the preceding equations, we get:

$$4c_1^+ + 8c_2^- = 2(N_\delta - 2)^2 + 4(N_\delta - 4)^2 + N_\delta - 1 \quad (17.43)$$

Taking  $N_\delta$  as an odd number, namely  $N_\delta = 2k + 1$ , the previous relationship is written as:

$$2c_1^+ + 4c_2^- = (2k - 1)^2 + 2(2k - 3)^2 + k \quad (17.44)$$

Equation 17.44 is a **Diophantine equation** for each  $k$ , with the integers  $c_1^+, c_2^-$  as unknowns. The left-hand side of Equation 17.44 is an even number, which requires the right-hand side to be an even number as well, thus yielding:

$$(2k - 1)^2 + k = 2\nu$$

where  $\nu$  is a natural number. Since the square of an odd number is an odd number,  $k$  can only take values from the subset of odd natural numbers, namely  $k = 1, 3, 5, \dots$ . The possible states of the composite system are obtained from the solutions of Equation 17.44. These concepts will be clarified through the example in the next section.

We consider that the composite system  $A_t = A_1 + A_2$  does not change or is in equilibrium. The quantity  $Q_1(c_1^+, c_2^+) = (\lambda_1, \lambda_2)$  can take many possible values, where  $\lambda_1, \lambda_2$  are the values of a magnitude in the subsystems  $A_1, A_2$ . Thus, the following question arises:

**“What is the probability  $P(Q_1(\dots))$  that the quantity  $Q_1(\dots)$  has the specified value  $Q_1 = (\lambda_1, \lambda_2)$ ?”**

To answer this question, we assume that the isolated system  $A_t$  possesses a number of accessible states  $\Omega_t$ , all of which, by definition, have the same probability of occurring. The question becomes:

**“How many of these states share the same  $\Omega_1 = (\lambda_1, \lambda_2)$  with those of  $A_1$ ?”** If  $\Omega_t(Q_1 = (\lambda_1, \lambda_2))$  represents the states of  $A_t$ , then:

$$P(Q_1(\lambda_1, \lambda_2)) = \frac{\Omega_t(Q_1(\lambda_1, \lambda_2))}{\Omega_t} = C_t \Omega_t(Q_1(\lambda_1, \lambda_2)) \quad (17.45)$$

where:

$$C_t = \frac{1}{\Omega_t}$$

When  $A_1$  has  $Q_1(\lambda_1, \lambda_2)$ , it is in one of the  $\Omega_1(Q_1(\lambda_1, \lambda_2))$  possible states, and  $A_2$  has  $Q_p - Q_1$  and is in one of the  $\Omega_2(Q_p - Q_1)$  possible states. The total number  $\Omega_t(Q_1)$  is:

$$\Omega_t(Q_1) = \Omega_1(Q_1) \cdot \Omega_2(Q_p - Q_1)$$

The probability  $P(Q_1)$  in this case is given by the relation:

$$P(Q_1) = C_t \Omega_t(Q_1) = \Omega_1(Q_1) \cdot \Omega_2(Q_p - Q_1) \quad (17.46)$$

The product  $\Omega_1(Q_1) \cdot \Omega_2(Q_p - Q_1)$  produces a very sharp maximum for the specific value  $Q_1(\lambda_1, \lambda_2)$  of  $Q_1$ .

Relation 17.46 expresses the equivalence of two infinite sets of points, where the infinite points obtained from the left side belong to the finite closed interval  $[0, 1]$ , while the infinite points of the right side belong to the semi-open infinite interval  $[0, +\infty)$ .

To establish a 1-1 correspondence between the points of the two intervals, we consider two point objects moving in these two intervals. The velocity of the object moving along the points of the finite interval decreases continuously at a rate proportional to the remaining distance to the other end of the interval. The velocity of the other moving object remains constant at every point of the infinite interval.

The velocity of both objects at the beginning of the two intervals has the same value, which we set equal to zero. Thus, both objects pass through all points of both intervals in the same time period. This was **Napier's** original idea to define the logarithmic function and its properties. Taking the logarithm of both sides of the last relation, we have:

$$\ln P(Q_1) = \ln C_t + \ln \Omega_1(Q_1) + \ln \Omega_2(Q_p - Q_1) \quad (17.47)$$

From relation 17.47, we can determine what the maximum value of  $\ln P(Q_1)$  will be and what the value of  $Q_1$  will be in this case. The maximum value of  $\ln P(Q_1)$  is given by the condition:

$$d \ln P(Q_1) = 0$$

Thus, we obtain:

$$\frac{\partial \ln \Omega_1(Q_1)}{\partial Q_1} + \left[ \frac{\partial(Q_p - Q_1)}{\partial Q_1} \right] \left[ \frac{\partial \ln \Omega_2(Q_p - Q_1)}{\partial(Q_p - Q_1)} \right] = 0 \quad (17.48)$$

or:

$$\frac{\partial \ln \Omega_1(Q_1)}{\partial Q_1} - \frac{\partial \ln \Omega_2(Q_p - Q_1)}{\partial(Q_p - Q_1)} = 0 \quad (17.49)$$

We define the quantities:

$$\theta_1 = \frac{\partial \ln \Omega_1(Q_1)}{\partial Q_1}, \quad \theta_2 = \frac{\partial \ln \Omega_2(Q_P - Q_1)}{\partial (Q_P - Q_1)} \quad (17.50)$$

Due to the definition of  $Q_1$ , the quantities  $\theta_1, \theta_2$  are written as:

$$\theta_1 = \frac{1}{\mu_0} \frac{\partial \ln \Omega_1(Q_1)}{\partial r_1}, \quad \theta_2 = \frac{1}{\mu_0} \frac{\partial \ln \Omega_2(Q_P - Q_1)}{\partial (R_P - r_1)} \quad (17.51)$$

From relations 17.49 and 17.51, we get:

$$\theta_1 = \theta_2 \quad (17.52)$$

The quantity  $\theta$  has dimensions of inverse length or curvature  $\kappa$ . In the equilibrium state, this is a constant quantity, so it is proportional to the curvature of the inscribed circle of the corresponding Planck patch, yielding:

$$\theta = b \cdot \kappa = \frac{b}{r} \quad \text{or} \quad T = b \cdot \kappa \quad (17.53)$$

where  $b$  is a proportionality constant. From this result, we conclude that the quantity  $\theta$  is the geometric analogue of temperature  $T$  in  $3D$  space.

### 17.4.2 Wien's Law

From Equation 17.53, with a constant value of  $b = 2.897 \dots \times 10^{-3}, \text{m} \cdot ^0\text{K}$  and  $\kappa = \frac{1}{\lambda_{\max}}, \text{m}^{-1}$  or  $\kappa = \frac{\nu_{\max}}{c}, \text{m}^{-1}$ , where  $\lambda_{\max}$  is the peak wavelength of radiation emitted by a black body, we arrive at Wien's law:

$$T = \frac{2.897 \dots \times 10^{-3}}{\lambda_{\max}} [^0\text{K}] = b \nu_{\max} \quad (17.54)$$

Through this relationship, the surface temperature of any star can be estimated by measuring  $\lambda_{\max}$  or  $\nu_{\max}$  in its radiation spectrum.

### 17.4.3 Derivation of Planck's Law

In a Planck patch, there are circles of various radii surrounded by ellipses. The radii of these circles have lengths  $1l_p, 2l_p, 3l_p, \dots$ , where  $l_p$  is the Planck length ( $l_p = 1.62 \cdot 10^{-35}, \text{m}$ ). The semi-minor axes of the surrounding ellipses have the same length as the radii of these circles. These ellipses appear linear because the length of their semi-minor axes is of the same order of magnitude as the Planck length. The semi-major axes of these ellipses are considered to tend to coincide with the corresponding branch of a linear hyperbola, as derived from the collapse of a cone where all types of conic sections coexist.

The equation of the hyperbola is formally identical to the wave equation, so the linear ellipse, which we consider a segment of the linear hyperbola,



will satisfy the same equation. Thus, the composite structure consisting of a circle of radius  $nl_p$  and its surrounding linear ellipse will correspond in 3D space, according to free observers, to a three-dimensional entity that simultaneously exhibits particle nature (circle) and wave nature (linear ellipse - a segment of a linear hyperbola). Free observers refer to these types of geometric objects as wave-particles, as previously mentioned.

We consider all circles in the Planck patch with the same radius  $r_i$  and hence the same curvature ( $\kappa_i$ ), whose number is  $n_i$ . As we have mentioned, the curvature of each circle defined by bound observers is proportional to the energy ( $E$ ) defined by free observers. From the relationships defining the concept of energy  $E$  in circles, we obtain:

$$E = \left| -\frac{k}{2r} \right| = \mu\kappa$$

with  $\mu = \frac{k}{2} = \text{const.}$  and  $\kappa = \frac{1}{r}$ . From these relations, it follows that  $E$  is proportional to the curvature  $\kappa$ . Thus, for the group of circles with radius  $r_i$ , we obtain the relation:

$$E_i(\nu) = \mu\kappa_i = \frac{\mu}{r_i}$$

In the event that this circle is surrounded by a linear ellipse corresponding to a wave, the relationship will hold:

$$u = \lambda \cdot \nu$$

where  $u$  is the wave velocity,  $\lambda$  is its wavelength, and  $\nu$  is its frequency. For light waves, the relationship holds:

$$c = \lambda \cdot \nu$$

where  $c$  is the speed of light. In the case of linear ellipses that are segments of linear hyperbolas, they are characterized by the constant radius  $r_i$  of the circle they enclose. Thus, we have:

$$c = \lambda_i \cdot \nu_i$$

For  $r_i = \lambda_i$  and  $\frac{\mu}{c} = h$ , the energy of these circles is written as:

$$E_i(\nu) = \mu\kappa_i = \frac{\mu}{\lambda_i} = \frac{\mu}{c}\nu_i = h\nu_i$$

If the number of circles with the same radius is  $n$ , then their total energy (or total curvature) is:

$$E_n(\nu) = nh\nu$$

Bound and free observers call this specific set of circles a mode  $\nu$ . The probability that a single mode has an energy  $E_n(\nu) = nh\nu$  is given by the Boltzmann factor:

$$p(n) = p'(n) \left( \sum_{n=0}^{\infty} p'(n) \right)^{-1}$$

where:

$$p'(n) = e^{\frac{-E_n}{kT}}$$

From this relationship, we find the average energy:

$$\bar{E} = \sum_{n=0}^{\infty} E_n p(n)$$

of the set of modes of circles with frequencies  $\nu$ , which is given by the following relation (Planck's formula):

$$\bar{E} = \frac{\hbar\nu}{e^{\frac{\hbar\nu}{kT}} - 1}$$

Due to the correspondence  $E_i \rightarrow \kappa_i$ , this relationship expresses the average curvature  $\bar{\kappa}$  of the circles found in a Planck patch. Since curvature is proportional to temperature, from  $\bar{\kappa}$  we find the average temperature  $\bar{T}$  of each Planck patch.

#### **Why do we use the exponential or logarithmic function?**

When dealing with large systems, we use the logarithm of the number of microstates  $N$  instead of the number  $N$  itself, namely:

$$N \rightarrow \ln(N)$$

Using the logarithm of the number  $N$  instead of the number  $N$  of microstates offers the advantage that when two systems are combined, the logarithms of the systems add up:

$$\ln(N_1 \cdot N_2) = \ln N_1 + \ln N_2$$

Logarithmic functions are highly effective at reducing huge numbers into much more manageable ones, for instance,  $\log(10^{23}) = 23$ . By convention, we will use the natural base  $e$ . Let us also multiply  $\ln$  by a constant that has units of Joules/<sup>0</sup>K. We will refer to this multiplicative constant simply as the "constant" for now. From the previous relationship, we define the quantity called "entropy"  $S$ :

$$S = \text{constant} \times \ln(N), \text{J}/^0\text{K}$$

Since  $S$  increases and decreases as the  $N$  microstates increase and decrease, we can reformulate the Second Law of Thermodynamics as follows:

**Large isolated physical systems will remain extremely close to the state that has the largest  $S$ .**

#### 17.4.4 Derivation of the Mass-Temperature Relationship (Hawking Effect)

As we have seen, bound observers define the curvature of a circle of radius  $r_c$  in the space  $\mathbf{P}^2$  as the ratio:

$$\kappa_c = \lambda_c \frac{r_c}{l_p^2}$$

where  $l_p^2 = \frac{hG}{c^3}$  is the area of the Planck point (a square of side  $1l_p$ ). As defined previously, the quantity that bound observers call the "mass of conic sections"  $\mu_c$  for a circle is the curvature of the circle. Thus, we obtain the relation:

$$\mu_c = \kappa_c = \lambda_c \frac{r_c}{l_p^2}$$

The quantity  $\mu_c$  is mapped onto the 3D space into the holographic mass  $M_H$  via the analogy:

$$M_H = \nu \mu_c = \bar{\nu} \frac{r_c}{l_p^2}$$

where  $\bar{\nu} = \nu \cdot \lambda_c$ . Free observers define the temperature  $\Theta$  of a body of mass  $M_H$  using the relation:

$$\Theta = \lambda_H \kappa_H = \lambda_H \frac{1}{r_c} = \lambda_H \cdot \tilde{\nu} \cdot \frac{1}{M_H l_p^2} = \tilde{\lambda}_H \cdot \frac{1}{M_H l_p^2}$$

where  $\kappa_H = \frac{1}{r_c}$  is the curvature of the circle in space  $\mathbf{P}^2$  as viewed by free observers, and  $\tilde{\lambda}_H = \lambda_H \cdot \tilde{\nu}$ .

The constant  $\tilde{\lambda}_H$  must be expressed in 3D space via physical constants. These constants will reflect, on one hand, the number of holographic particles that make up the macroscopic body, and on the other hand, their state of thermal equilibrium.

Thus, both the Boltzmann constant  $k_B$  and the Planck constant  $\hbar$  must be used. To find how these constants combine, we will utilize dimensional analysis. The units of the right-hand side of the previous relationship must be in units of temperature, just like the left-hand side. The constants  $\hbar, k_B$  have dimensions given by:

$$[\hbar] = \frac{[M][L^2]}{[T^2]}[T]$$

$$[k_B] = \frac{[M][L^2]}{[T^2]}[\Theta]$$

From the preceding relationships, we obtain:

$$[\Theta] = [\tilde{\lambda}_H] \cdot \frac{\frac{[M][L^2][T]}{[T^2]}}{\frac{[M][L^2]}{[T^2]}[\Theta]} \cdot \frac{1}{[M][L^2]} = [\tilde{\lambda}_H] \frac{[T][\Theta]}{[M][L^2]}$$

The dimensions of  $\tilde{\lambda}H$  must be  $[\tilde{\lambda}H] = \frac{[M][L^2]}{[T^2]} \cdot [T] = [\hbar]$ , because substituting  $[\tilde{\lambda}H]$  into the previous relationship yields matching units for both sides:

$$[\Theta] = \frac{[M][L^2]}{[T^2]} \cdot [T] \cdot [T] \cdot [\Theta] \cdot \frac{1}{[M][L^2]} = [\Theta]$$

Due to the relationship for temperature  $\Theta$ , both sides are written as:

$$\Theta = \frac{\hbar^2}{kB} \cdot \frac{1}{MH} \cdot lp^2 = \frac{\hbar^2 c^3}{kB \cdot MH \cdot \hbar \cdot G} = \frac{\hbar c^3}{kBGMH}$$

This procedure describes a stage of expansion-contraction of a random Planck patch. The next stage of expansion involves a fourfold increase of the space occupied by this Planck patch. Since the curvature of each Planck patch is characterized by its enclosing circle, then with each expansion, this circle will have a larger radius. Therefore, a constant must be defined that yields the same value at each stage of expansion for any Planck patch. This constant is defined as the ratio of the circumference of the circle to its radius, namely  $\nu = 2\frac{\pi r}{r} = 2\pi$ . Since at each stage of expansion, 4 new identical circles enclose each of the 4 new Planck patches—which are identical due to self-similarity to the previous Planck patch—the constant  $\nu$  quadruples, leading to the relationship:

$$4\tilde{\nu} = 8\pi$$

Since in each new Planck patch at the end of the expansion process, the radius of its circumscribed circle increases, its curvature decreases, and consequently its temperature decreases. This implies that the constant  $\tilde{\nu}$  must divide the expression that yields the temperature. The relationship for  $\Theta$  becomes:

$$\Theta = \frac{\hbar c^3}{4\tilde{\nu}kBGM_H}$$

or:

$$\Theta = \frac{\hbar c^3}{8\pi kBGM_H}$$

Free observers situated far from any celestial body, which may also be a black hole, detect a flux of radiation from the celestial body. This relationship was presented for the first time by Hawking in 1974 and was derived using General Relativity. However, the equations of General Relativity originate from the equations of conic sections, as we saw in a previous Chapter. This phenomenon is known as the **Hawking effect**.

### 17.4.5 Finding the Accessible States via an Example

Let us examine the case for  $k = 3$ . For this value of  $k$ , Equation 17.44 is written as:

$$2c_1^+ + 4c_2^+ = (2k-1)^2 + 2(2k-3)^2 + k$$

$$c_1^+ + 2c_2^+ = 23$$

Since the right-hand side of this equation is an odd number,  $c_1^+$  must also be an odd number. Thus, the values of  $c_1^+$  are drawn from the subset  $(1, 3, 5, \dots)$  containing the odd numbers of the set of natural numbers. In order to satisfy the conditions:

$$\text{a) } N_\delta = 2k + 1$$

$$\text{b) } (c_1^+ - c_1^-)Q_1 + (c_2^+ - c_2^-)Q_2 = Q_p$$

$$\text{c) } c_1^+ + c_1^- = (N_\delta - 2 \cdot 1)^2 = 25$$

$$\text{d) } c_2^+ + c_2^- = (N_\delta - 2 \cdot 2)^2 = 9$$

then the pairs  $(c_1^+, c_1^-)$  and  $(c_2^+, c_2^-)$  must be:  $(5, 20)$ ,  $(7, 18)$ ,  $(9, 16)$ ,  $(11, 14)$ ,  $(13, 12)$ ,  $(15, 10)$ ,  $(17, 8)$ ,  $(19, 6)$ ,  $(21, 4)$ ,  $(23, 2)$  and  $(9, 0)$ ,  $(8, 1)$ ,  $(7, 2)$ ,  $(6, 3)$ ,  $(5, 4)$ ,  $(4, 5)$ ,  $(3, 6)$ ,  $(2, 7)$ ,  $(1, 8)$ ,  $(0, 9)$

The Diophantine equation has as solutions the pairs:

$$(c_1^+, c_2^-) = (5, 9) + (7, 8) + (9, 7) + (11, 6) + (13, 5) + (15, 4) + (17, 3) + (19, 2) + (21, 1) + (23, 0)$$

From these solutions, we will examine the solution  $(5, 9)$ . For this solution, there are  $5c_1^+$  circles and  $20c_1^-$  circles with radius  $r_1$ . The quantity  $Q_1$  for this solution is:

$$Q_1(5, 20) = (+5\mu_0 - 20\mu_0) \cdot r_1 = -15\mu_0 l_p$$

The microstates for the circles with radius  $r_1$  that have  $Q_1(5, 20) = -15\mu_0 l_p$  are given by  $(25 \ 5)$

The microstates for all combinations are:

$$\Omega_1(Q_1(5, 20)) = 53, 130$$

$$\Omega_1(Q_1(7, 18)) = 480, 700$$

$$\Omega_1(Q_1(9, 16)) = 2, 042, 075$$

$$\Omega_1(Q_1(11, 14)) = 4, 457, 400$$

$$\Omega_1(Q_1(13, 12)) = 5, 200, 300$$

$$\Omega_1(Q_1(15, 10)) = 3, 268, 760$$

$$\Omega_1(Q_1(17, 8)) = 1, 081, 575$$

$$\Omega_1(Q_1(19, 6)) = 177, 100$$

$$\Omega_1(Q_1(21, 4)) = 12, 650$$

$$\Omega_1(Q_1(23, 2)) = 300$$

The microstates for the circles with radius  $r_2$  that have  $Q_2(9, 0) = +9\mu_0 r_2 - 0\mu_0 l_p = 18\mu_0 l_p$  are given by (9 0). The microstates for all combinations are:

$$\Omega_2(Q_2(9, 0)) = 1$$

$$\Omega_2(Q_2(8, 1)) = 9$$

$$\Omega_2(Q_2(7, 2)) = 36$$

$$\Omega_2(Q_2(6, 3)) = 63$$

$$\Omega_2(Q_2(5, 4)) = 189$$

$$\Omega_2(Q_2(4, 5)) = 189$$

$$\Omega_2(Q_2(3, 6)) = 63$$

$$\Omega_2(Q_2(2, 7)) = 36$$

$$\Omega_2(Q_2(1, 8)) = 9$$

$$\Omega_2(Q_2(0, 9)) = 1$$

The sum of  $Q_1(5, 20)$  and  $Q_2(9, 0)$  is:

$$Q_p = Q_1(5, 20) + Q_2(9, 0) = 3\mu_0 l_p$$

All solutions of the Diophantine equation for the same value of  $k$  yield the same value for the quantity  $Q_p$ . These solutions constitute the accessible states of the composite system. The accessible states of the composite system are the product of:

$$(\lambda_i + \lambda_j \ \lambda_i) \quad \text{and} \quad (\mu_i + \mu_j \ \mu_i)$$

Thus, we have:

$$\Omega_p(Q_1(\lambda_i, \lambda_j)) = \Omega_1(Q_1(\lambda_i, \lambda_j)) \cdot \Omega_2(Q_2(\mu_i, \mu_j))$$

The probability of obtaining the value  $Q_1(\lambda_i, \lambda_j)$  is  $\Omega_p(Q_1(\lambda_i, \lambda_j))$  divided by the sum  $\Omega_t$  of all  $\Omega_p(Q_1(\lambda_i, \lambda_j))$ , namely:

$$P(\Omega_p(Q_1(\lambda_i, \lambda_j))) = (\Omega_p(Q_1(\lambda_i, \lambda_j))) / \Omega_t$$

As the value of  $N_\delta$  increases, the quantity  $\Omega_p(Q_1)$  increases, as does  $P(\Omega_p(Q_1))$ , and the graph of  $\Omega_p(Q_1)$  with respect to  $Q_1$  tends toward a very sharp maximum.

## 17.5 THE GEOMETRIC MODEL

In this Appendix, we will develop a geometric model to interpret the creation and evolution of the Universe. To examine how the Universe evolved from its beginning until today, we will reduce the three-dimensional problem to a two-dimensional problem as a whole; that is, we will formulate the problem in  $\mathbf{P}^2$ .

Thus, we will see how the Universe expanded from a Planck point via a self-similar process to its present state, and how it will evolve in the future. The sole constraint we impose is the application of the definition criteria of  $\mathbf{P}^2$ . To determine the mathematical model from which answers to the preceding questions will emerge, we pose the following inquiry:

**“What is the length of the radius of the circle enclosing each Planck patch, its curvature, its temperature, its expansion velocity and acceleration, as well as the time interval within which all this takes place?”**

We will begin by finding the radius of the circle enclosing each Planck patch, its curvature, and its temperature. The  $n$ -th Planck patch contains  $N_n = 9^{2n}$  Planck squares  $P_9$ . The patch  $P_9$  is the first Planck patch that emerged after inflation from the third Planck patch  $P_3$ , as shown in Figure 17.7.

Thus, the total area of each Planck patch after inflation is given by the relationship  $A_n = 9^{2n} \cdot l_p^2$ , and the radius of the circle enclosing this Planck patch is:

$$R_n = \sqrt{A_n} = 9^n l_p$$

The exponents in this relationship will be determined by the phenomenon of inflation, as we will see below. The curvature of this circle is:

$$\kappa_n = \frac{1}{9^n} \cdot l_p^{-1} \quad (17.55)$$

The temperature is proportional to the curvature of the circle, so its numerical value is:

$$T_n = \nu \cdot \kappa_n = \nu \frac{1}{9^n} \cdot l_p^{-1} \sim 9^{-n} 6 \cdot 10^{32} [^0K] \quad (17.56)$$

where the units of  $\nu$  are taken as  $\text{cm} \cdot ^0\text{K}$ .

For  $n = 0$ , namely at the initial Planck point, we obtain a temperature of  $T_n \sim 10^{32} \text{ } ^0\text{K}$  and the quantities  $l_0 = l_p, t_0 = t_p, v_0 = v_p = c$ . The side length of the first Planck patch after inflation is  $l_1 = 9^1 l_p$ . To determine the velocity and acceleration of expansion of a Planck patch, as well as the time interval within which these occur, we construct the following simple model: We consider a hypothetical particle moving along the perimeter of the initial Planck square from one vertex to another with an initial velocity  $v_0 = c$ .

The beginning of the distance it covers is the point  $l_0 = 0$  and the other end of the distance is the point  $l_1$ . The interval has a length of  $\Delta l_1 = l_1 - l_0$ . The velocity of this hypothetical particle represents the expansion velocity of the Planck patch. The value of the time instance at the end of  $\Delta l_1$  is  $t_1 = \frac{\Delta l_1}{v_0}$ . The time elapsed during the motion of this point is  $\Delta t_1 = t_1 - t_0$ . We define the velocity of the hypothetical particle at the end of  $l_1$  as  $v_1 = \frac{\Delta l_1}{\Delta t_1}$ . This particle will move at this constant velocity over a distance  $l_2$ . This process repeats via the following recurrence relations:

$$l_n = 9^n l_p \quad (17.57)$$

$$\Delta l_n = l_n - l_{n-1} = 8 \cdot 9^{n-1} l_p \quad (17.58)$$

$$t_n = \frac{\Delta l_n}{v_{n-1}} \quad (17.59)$$

$$\Delta t_n = t_n - t_{n-1} \quad (17.60)$$

$$v_n = \frac{\Delta l_n}{\Delta t_n} \quad (17.61)$$

$$\Delta v_n = v_n - v_{n-1} \quad (17.62)$$

$$a_n = \frac{\Delta v_n}{\Delta t_n} \quad (17.63)$$

where  $a_n$  is the expansion acceleration of the respective Planck patch. The recurrence relations are applied for  $n = 1, 2, 3, \dots$ , yielding the following tables.

| Exponents | L {cm}   | T {K}    | Remarks                                           |
|-----------|----------|----------|---------------------------------------------------|
| 0         | 8.10E-34 | 2.47E+33 | Photon domination                                 |
| 1         | 1.62E-33 | 6.17E+32 |                                                   |
| 2         | 3.24E-33 | 3.09E+32 |                                                   |
| 5         | 9.57E-29 | 1.05E+28 |                                                   |
| 8         | 6.97E-26 | 1.43E+25 | First Inflation (sharp) or<br>Adiabatic Expansion |
| 10        | 5.65E-24 | 1.77E+23 |                                                   |
| 11        | 5.08E-23 | 1.97E+22 |                                                   |
| 12        | 4.58E-22 | 1.00E+00 |                                                   |
| 13        | 4.12E-21 | 2.43E+20 |                                                   |
| 14        | 3.71E-20 | 2.70E+19 |                                                   |
| 15        | 3.34E-19 | 3.00E+18 |                                                   |
| 16        | 3.00E-18 | 3.33E+17 |                                                   |
| 17        | 2.70E-17 | 3.70E+16 |                                                   |
| 18        | 2.43E-16 | 4.11E+15 |                                                   |
| 19        | 2.19E-15 | 4.57E+14 |                                                   |
| 20        | 1.97E-14 | 5.08E+13 |                                                   |
| 21        | 1.77E-13 | 5.64E+12 |                                                   |
| 22        | 1.60E-12 | 6.27E+11 |                                                   |
| 23        | 1.44E-11 | 6.96E+10 |                                                   |
| 24        | 1.29E-10 | 7.74E+09 | Matter domination                                 |
| 25        | 1.16E-09 | 8.60E+08 |                                                   |
| 26        | 1.05E-08 | 9.55E+07 |                                                   |
| 27        | 9.42E-08 | 1.06E+07 |                                                   |
| 28        | 8.48E-07 | 1.18E+06 | Second Inflation(mild)                            |
| 29        | 7.63E-06 | 1.31E+05 |                                                   |
| 30        | 6.87E-05 | 1.46E+04 | Erevoon domination                                |
| 31        | 6.18E-04 | 1.62E+03 |                                                   |
| 32        | 5.56E-03 | 1.80E+02 |                                                   |
| 33        | 5.01E-02 | 2.00E+01 |                                                   |
| 33.90659  | 3.67E-01 | 2.725056 | TODAY                                             |

Figure 17.8: Temperature of the Universe



| Remarks                    | Exponents | L (cm)   | dl(cm)   | t (sec)  | dt (sec) | v (cm/sec) | a(cm/sec <sup>2</sup> ) | age(years) |
|----------------------------|-----------|----------|----------|----------|----------|------------|-------------------------|------------|
| Photon domination          | 0         | 1.62E-33 |          | 5.40E-44 | 0.00E+00 | 3.00E+10   | 0                       | 1.71E-51   |
|                            | 1         | 1.46E-32 | 1.30E-32 | 4.32E-43 | 3.78E-43 | 3.43E+10   | 1.13E+52                | 1.37E-50   |
|                            | 2         | 1.31E-31 | 1.17E-31 | 3.40E-42 | 2.97E-42 | 3.93E+10   | 1.68E+51                | 1.08E-49   |
| First Inflation (sharp) or | 10        | 5.65E-24 | 5.65E-24 | 1.44E-34 | 1.44E-34 | 3.93E+10   | 6.46E+36                | 4.56E-42   |
| Adiabatic Expansion        | 16        | 3.00E-18 | 3.00E-18 | 7.64E-29 | 7.64E-29 | 3.93E+10   | 9.67E+32                | 2.42E-36   |
|                            | 20        | 1.97E-14 | 1.97E-14 | 5.01E-25 | 5.01E-25 | 3.93E+10   | 1.19E+31                | 1.59E-32   |
|                            | 22        | 1.60E-12 | 1.58E-12 | 4.01E-23 | 3.96E-23 | 3.98E+10   | 1.26E+31                | 1.27E-30   |
|                            | 24        | 1.29E-10 | 1.28E-10 | 3.21E-21 | 3.17E-21 | 4.03E+10   | 1.59E+29                | 1.02E-28   |
|                            | 26        | 1.05E-08 | 1.03E-08 | 2.57E-19 | 2.53E-19 | 4.08E+10   | 2.01E+27                | 8.14E-27   |
|                            | 28        | 8.48E-07 | 8.37E-07 | 2.05E-17 | 2.03E-17 | 4.13E+10   | 2.55E+25                | 6.51E-25   |
|                            | 30        | 6.87E-05 | 6.78E-05 | 1.64E-15 | 1.62E-15 | 4.18E+10   | 3.23E+23                | 5.21E-23   |
|                            | 32        | 5.56E-03 | 5.49E-03 | 1.38E-13 | 1.36E-13 | 4.03E+10   | 1.15E+22                | 4.38E-21   |
|                            | 34        | 4.51E-01 | 4.45E-01 | 1.11E-11 | 1.09E-11 | 4.08E+10   | 4.67E+19                | 3.51E-19   |
|                            | 36        | 3.65E+01 | 3.60E+01 | 8.84E-10 | 8.73E-10 | 4.13E+10   | 5.91E+17                | 2.80E-17   |
|                            | 38        | 2.96E+03 | 2.92E+03 | 7.07E-08 | 6.98E-08 | 4.18E+10   | 7.48E+15                | 2.24E-15   |
|                            | 40        | 2.39E+05 | 2.36E+05 | 5.66E-06 | 5.59E-06 | 4.23E+10   | 9.47E+13                | 1.79E-13   |
|                            | 42        | 1.94E+07 | 1.92E+07 | 4.53E-04 | 4.47E-04 | 4.29E+10   | 1.20E+12                | 1.43E-11   |
| Matter domination          | 44        | 1.57E+09 | 1.55E+09 | 3.62E-02 | 3.57E-02 | 4.34E+10   | 1.52E+10                | 1.15E-09   |
|                            | 46        | 1.27E+11 | 1.26E+11 | 2.90E+00 | 2.86E+00 | 4.40E+10   | 1.92E+08                | 9.18E-08   |
|                            | 48        | 1.03E+13 | 1.02E+13 | 2.32E+02 | 2.29E+02 | 4.45E+10   | 2.43E+06                | 7.34E-06   |
|                            | 50        | 8.35E+14 | 8.25E+14 | 1.85E+04 | 1.83E+04 | 4.51E+10   | 3.08E+04                | 5.87E-04   |
|                            | 52        | 6.76E+16 | 6.68E+16 | 1.48E+06 | 1.46E+06 | 4.57E+10   | 3.90E+02                | 4.70E-02   |
|                            | 54        | 5.48E+18 | 5.41E+18 | 1.19E+08 | 1.17E+08 | 4.62E+10   | 4.94E+00                | 3.76E+00   |
| Second Inflation(mild)     | 56        | 4.44E+20 | 4.38E+20 | 9.48E+09 | 9.36E+09 | 4.68E+10   | 6.25E-02                | 3.01E+02   |
| Erevo domination           | 58        | 3.59E+22 | 3.55E+22 | 7.58E+11 | 7.49E+11 | 4.74E+10   | 7.92E-04                | 2.40E+04   |
|                            | 60        | 2.91E+24 | 2.88E+24 | 6.06E+13 | 5.99E+13 | 4.80E+10   | 1.00E-05                | 1.92E+06   |
|                            | 61        | 2.62E+25 | 2.33E+25 | 4.85E+14 | 4.24E+14 | 5.49E+10   | 1.62E-05                | 1.54E+07   |
|                            | 62        | 2.36E+26 | 2.10E+26 | 3.82E+15 | 3.34E+15 | 6.28E+10   | 2.39E-06                | 1.21E+08   |
|                            | 63        | 2.12E+27 | 1.89E+27 | 3.00E+16 | 2.62E+16 | 7.20E+10   | 3.50E-07                | 9.52E+08   |
| TODAY                      | 64.251    | 3.32E+28 | 3.10E+28 | 4.31E+17 | 4.01E+17 | 7.74E+10   | 1.3448E-08              | 1.37E+10   |

Figure 17.9: Other magnitudes of the Universe

The units used in the columns of both Tables belong to the c.g.s. system of units. The first column of the table in Figure 17.8 gives the exponents for the increase in length in cm. This length is provided in the second column. The inverse of the radius of the circle, which is circumscribed around each Planck patch—that is, the curvature of the circle—is proportional to its temperature. Thus, the third column of this table lists the temperature in degrees Kelvin (<sup>0</sup>K) of the respective Planck patch.

The table indicates that the temperatures of the second and third rows in the third column are of the same order of magnitude, and the Planck patch tends to expand due to the tendency of its curvature to vanish. Furthermore, this Planck patch tends to contract due to self-similarity with its predecessor to acquire roughly the same curvature as it. Thus, two opposing tendencies coexist, with the tendency toward expansion ultimately prevailing in order to satisfy the *z.c.P*<sup>2</sup> criterion.

For free observers, this means that the criteria for an "isothermal and adiabatic expansion of an ideal fluid" within a limited region of three-dimensional space are met. The corresponding Planck patch of this region consists of mutually independent light geometric objects capable of moving isotropically, which map onto an ideal fluid. For bound observers, the result of the expansion of the space of the Planck patch is to increase its area in a nearly instantaneous time interval while maintaining the order of magnitude of its temperature.

Free observers have named this isothermal and adiabatic expansion of this Planck patch **inflation** in three-dimensional space, where the relation

holds:

$$TV^{\gamma-1} = \text{constant}$$

Bound observers for the two-dimensional Planck patch consider the correspondence:

$$V \rightarrow L^2$$

and the corresponding relationship in  $\mathbf{P}^2$  is:

$$T(L^2)^{\gamma-1} = \text{constant}$$

Since the temperature  $T$  is proportional to the curvature, we obtain:

$$T \propto L^{-1}$$

Thus, the preceding relationship for the two-dimensional Planck patch is written as:

$$L^{2\mu-1} = \text{constant}, \text{ where } \mu = \gamma - 1$$

According to the exponent  $2\mu - 1$ , the fourth exponent will take the value  $2 \cdot 3 = 5$  instead of 3 due to inflation. This value is larger by 3 than the previous value of 2. This difference is identical between this exponent and the next one; that is, the next exponent has a value of 8. Given that the phenomenon of inflation begins to fade, the next value of the exponent is 10, meaning it increases by two units, and thereafter the exponents resume their initial rate of increase, incrementing by one. The step increase taken by the exponents is: 3-3-2-1.

The gradual dampening of this expansion and the evolution that appears in subsequent Planck patches are shown in Table 17.8. The fourth column of Table 17.8 writes down certain observations relative to the evolution of the Universe. The fifth column of Table 17.9 provides the two initial exponents and the modified exponents after inflation. The sixth column yields the corresponding total length  $L$  in cm of the side of each Planck patch traversed by the moving hypothetical particle.

In the fourth row of the fifth column of Table 17.9, where the isothermal and adiabatic expansion begins, we double the exponent of  $L$  to obtain the length of one side of the Planck patch. The subsequent values of the exponents in the 4-th column of Table 17.9 will be double the exponents in the first column of Table 17.8. This doubling of the exponents tracks the doubling of the exponents that took place between the beginning and the end of inflation, namely, between the fourth and sixth rows of Table 17.8.

The exponent is doubled, except for the last rows which increase by half a unit. This is due to the emergence of the identical system behavior after the second inflation, which is much milder than the first and occurs between the 27-th and 28-th rows.

Just as in the first inflation the exponents increased after its conclusion, the same occurs during the second inflation. The last row in Table 4 has this value because the construction of the Planck patch has not been completed in the present era.

The orders of magnitude of the values of the various quantities we obtain from the Table of Figure 17.9 in the current era, such as the size of the Universe, its expansion acceleration, and its age, are nearly identical to those recorded by satellites.

Table 17.9 shows that as the age of the Universe increased from 1 to 100 seconds, the temperature of the Universe dropped from 10 billion to 1 billion degrees Kelvin. According to Table 17.9, the Universe was so hot at roughly 10 seconds from its beginning that protons and neutrons did not yet have a sufficiently low temperature to form nuclei. This occurred later in subsequent Planck patches during the radiation-dominated era, and following the formation of nucleons from quarks and gluons.

In this stage of the evolution of the Universe, the Universe consisted of a "gas of relatively hot particles" which were photons, gluons, neutrinos, electrons, protons, and neutrons. These particles constituted visible matter. Particles with circular radii of  $1^+l_p$  made up dark matter. All these particles were the holograms of geometric objects that corresponded to observable or unobservable objects.

Bound observers have a clearer picture of this state by following the geometric model. The first Planck patch at the beginning of the matter-dominated era contained circles of radii  $0, 0^+, 1, 2, 3, \dots$  in units of  $l_p$ . Circles of zero radius correspond to photons and gluons (when the temperature or curvature decreases sufficiently); circles of radius  $0^+l_p$  correspond to neutrinos; circles of radius  $1l_p$  correspond to electrons and  $d$ -quarks; circles of radius  $2l_p$  correspond to  $u$ -quarks; circles with a radius of  $3l_p$  contain the first complex geometric objects; and circles with a radius larger than  $3l_p$  correspond to entities that tend to decrease the curvature of the Planck patch in which they exist.

In three-dimensional space, the holograms of geometric objects, namely the particles of visible matter, were in thermal equilibrium, just as their geometric objects were. According to the *expansion-contraction* process, geometric objects were dispersed into each new Planck patch (expansion) and subsequently gathered to form increasingly complex geometric objects (contraction), the holograms of which were nuclei or atoms. In the next Planck patch, these structures were partially dissolved, only to form more complex geometric objects subsequently, whose parts are more loosely bound together. Free observers interpreted the dissolution of old structures and the creation of new ones through collisions with high-energy particles.

These nuclear reactions, which were driven by scattering processes, were expressed via the following 4 reactions:

$$p + n \leftrightarrow D + \gamma$$

$$n + n \leftrightarrow D + e + \bar{\nu}$$

$$n + \nu \leftrightarrow p + e$$

$$n \leftrightarrow p + e + \bar{\nu}$$

where  $D$  is deuterium,  $\gamma$  is the photon,  $e$  is the electron, and  $\bar{\nu}, \nu$  are the antineutrino and neutrino, respectively.

During this process, the following questions arise:

1. When was helium produced?
2. Why did the production of heavier chemical elements not continue after the helium production period?

#### ANSWERS

The production of geometric objects corresponding to hydrogen  $H$  and helium  $He$  occurred in the first Planck patch of the matter-dominated era.

This implies that the geometric objects corresponding to  $He$  must have been distributed isotropically in every section of the Planck patch that corresponds to a large "celestial body" such as a Galaxy. This has indeed been confirmed by observations in our Galaxy.

Deuterium  $D$  and tritium  $T$ , which are required for the creation of heavier nuclei, are largely consumed during the process of helium production.

## 17.6 VARIOUS TOPICS OF ASTRONOMY AND PHYSICS

### 17.6.1 The Chandrasekhar Limit

#### White Dwarfs

According to the geometric model, the electron and the  $d$ -quark correspond to a circle of radius  $1l_p$ , and the  $u$ -quark corresponds to a circle of radius  $2l_p$ . Consequently, the geometric object in  $\mathbf{P}^2$  that corresponds to the proton consists of a circle of radius  $5l_p$  containing  $2u$ -quarks and  $1d$ -quark. Similarly, the geometric object in  $\mathbf{P}^2$  corresponding to the neutron consists of a circle of radius  $4l_p$  containing  $1u$ -quark and  $2d$ -quarks. The geometric object representing a hydrogen atom consists of a circle of radius  $5l_p$  (proton) and a circle of radius  $1l_p$  (electron). Thus, its radius is greater than  $7l_p$  because the electron is located at least  $1l_p$  away from the proton. Therefore, the radius of the circle representing a hydrogen atom is  $r_H > 7l_p$ , as depicted in the following figures.

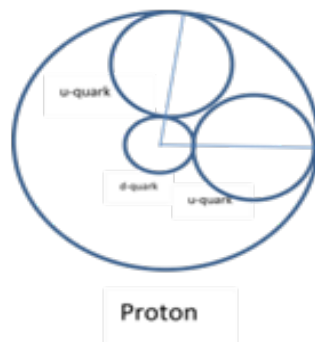


Figure 17.10: Proton

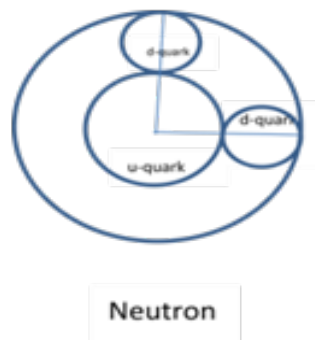


Figure 17.11: Neutron

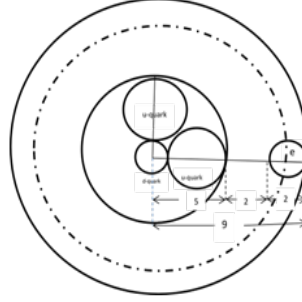


Figure 17.12: Hydrogen atom

Let us consider a main sequence star similar to our Sun in its initial stage, which contains only hydrogen atoms. The geometric object in  $\mathbf{P}^2$  corresponding to the Sun is a circle of radius  $R_{\text{Sun}}$  containing  $N_{\text{Sun}}$  hydrogen atoms. Each of these occupies an area of  $\pi(9l_p)^2$ , assuming that the electron is located at a distance of  $2l_p$  from the proton, as shown in Figure 17.12. Since the geometric objects corresponding to the hydrogen atoms cover the entire circle in  $\mathbf{P}^2$  that corresponds to the Sun, i.e., the area  $\pi R_{\text{Sun}}^2$ , we obtain:

$$\pi R_{\text{Sun}}^2 = 9^2 \pi N_{\text{Sun}} \cdot l_p^2 = 81 \pi \cdot N_{\text{Sun}} \cdot l_p^2$$

If the white dwarf (*wd*) is at a stage where it consists only of protons, neutrons, and electrons, then its geometric analogue is a circle whose area is  $\pi R_{\text{wd}}^2$ . Within this circle, the geometric objects corresponding to the protons  $p$  have a total count of  $N_{\text{wd}}$ .

The groups of neutrons and electrons have the same count, meaning  $N_e = N_p = N_n$ . Each of these counts equals the maximum count  $N_{\text{wd}}$  that can exist inside the circle corresponding to the white dwarf, i.e.,  $N_e = N_p = N_n = N_{\text{wd}}$ . This is due to the fact that the circle corresponding to the white dwarf consists of geometric objects that correspond to He, C, O atoms, which possess equal numbers of  $p, n$ , and  $e$ . In the initial phase of the Sun, there were only hydrogen atoms.

We assume that the maximum number  $N_{\text{wd}}$  of each type of geometric object corresponding to  $p, n, e$  equals the number  $N_{\text{Sun}}$  of geometric objects corresponding to the hydrogen atoms that composed the Sun, i.e.,  $N_{\text{Sun}} = N_{\text{wd}}$ . Because the area of the circle occupied by the protons  $p$  is  $\pi(5l_p)^2$ , the area of the circle occupied by the neutrons  $n$  is  $\pi(4l_p)^2$ , and the area of the circle occupied by the electrons  $e$  is  $\pi(1l_p)^2$ , we obtain:

$$\pi R_{\text{wd}}^2 = N_{\text{wd}} \cdot (1^2 + 5^2 + 4^2) \cdot l_p^2 = 42 N_{\text{wd}} \cdot l_p^2$$

Thus, from the relations that yield  $(R_{\text{Sun}})^2$  and  $(R_{\text{wd}})^2$ , we find:

$$\frac{R_{\text{Sun}}}{R_{\text{wd}}} = \sqrt{\frac{81}{42}} = 1.388$$

Curvature is proportional to mass in three-dimensional space. This curvature corresponds to the curvature of a circle in  $\mathbf{P}^2$ , which equals the inverse of its radius. Thus, we have the relation:

$$M_{wd} = \frac{R_{\text{Sun}}}{R_{wd}} M_{\text{Sun}}$$

or:

$$M_{wd} \sim 1.4 M_{\text{Sun}}$$

This is the maximum mass value that a white dwarf can have, which is known as the **Chandrasekhar limit**.

### 17.6.2 The Tolman-Oppenheimer-Volkoff Limit

#### Neutron Stars

Since the analysis for the formation of neutron stars ( $ns$ ) shows that in each group of identical geometric objects, the number of neutrons equals the number of electrons, this number is written as  $N_{ns}$ . The members of these two groups begin to expand, covering almost all the available space allocated for the geometric object that corresponds to the neutron star. Following the same arguments used in the case of white dwarfs, we write the relations:

$$\begin{aligned} \pi R_{\text{Sun}}^2 &= N_{\text{Sun}} \cdot \pi (9 \cdot l_p)^2 \\ \pi R_{ns}^2 &= N_{ns} \cdot \pi \cdot (1^2 + 4^2) l_p^2 = 17 N_{ns} \cdot \pi l_p^2 \end{aligned}$$

For equal counts  $N_{ns}$  and  $N_{\text{Sun}}$ , i.e.,  $N_{ns} = N_{\text{Sun}}$ , we obtain the relation:

$$\frac{R_{\text{Sun}}}{R_{ns}} = \sqrt{\frac{81}{17}} = 2.18$$

By the same reasoning as in the case of white dwarfs, we obtain the relationship for the curvatures of the circles corresponding to the Sun and the neutron star:

$$\kappa_{ns} = 2.18 \kappa_{\text{Sun}}$$

or:

$$M_{ns} = 2.18 M_{\text{Sun}}$$

This result is approximately twice the solar mass, which is in agreement with the results of the Tolman-Oppenheimer-Volkoff study. If we assume that the neutron star contains only neutrons, then we obtain:

$$\pi \cdot R_{ns}^2 = N_{ns} \cdot (4^2) l_p^2 = 16 \cdot N_{ns} \pi \cdot l_p^2$$

or by the same arguments as above, we have:

$$\frac{R_{\text{Sun}}}{R_{ns}} = \sqrt{\frac{81}{16}} = 2.25$$

and:

$$M_{ns} = 2.25 M_{\text{Sun}}$$

### 17.6.3 Solar Neutrinos

The neutrinos emitted from the core of the Sun are formed during the creation of  ${}^4_2\text{He}$  from 4 protons according to the reaction:

$$4p \rightarrow {}^4_2\text{He} + 2e^+ + 2\nu_e \quad (17.64)$$

The neutrinos in Equation 17.64 and the figure below are the focus of the mystery we want to explore here.

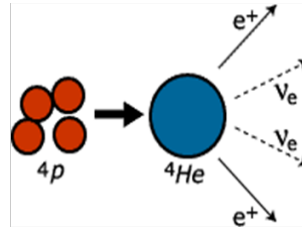


Figure 17.13: Nuclear fusion

According to the geometric model, neutrinos have an infinitesimal non-zero electric charge and an infinitesimal non-zero mass. Approximately 100 billion neutrinos from the Sun pass through us every second, but we do not feel them because they interact so rarely and so weakly with matter. Nuclear fusion in the Sun produces only electron neutrinos  $\nu_e$ . The two other types of neutrinos, muon neutrinos ( $\nu_\mu$ ) and tau neutrinos ( $\nu_\tau$ ), are produced, for example, in laboratory accelerators or in exploding stars, along with the more massive versions of the electron, which are the muon ( $\mu$ ) and tau ( $\tau$ ) particles of the other two generations.

#### The Missing Neutrinos

In 1964, an experiment was proposed to test whether the conversion of hydrogen nuclei into helium nuclei in the Sun is indeed the source of sunlight, as indicated by Equation 17.64. Thus, the number of neutrinos of different energies produced by the Sun was calculated using a detailed stellar model for the Sun. Furthermore, the number of radioactive argon atoms  ${}^{37}_{18}\text{Ar}$  that these solar neutrinos would produce in a large cleaning tank filled with chlorine-based liquid  $\text{C}_2\text{Cl}_4$  was computed. The first results of the experiment were announced in 1968. From these results, they detected only about one-third of the radioactive  ${}^{37}_{18}\text{Ar}$  atoms that had been predicted. This discrepancy between the number of detected  ${}^{37}_{18}\text{Ar}$  atoms and the predicted neutrinos soon became known as “**The solar neutrino problem**” or “**The mystery of the missing neutrinos**”.

#### Possible Explanations

Three categories of explanations were proposed to solve the mystery. **First:** perhaps the theoretical calculations are wrong. This could happen in two ways: either the predicted number of neutrinos is incorrect, or the



calculated production rate of Argon atoms is wrong. **Second:** perhaps the experiment itself is flawed. **Third:** perhaps physicists do not understand how neutrinos behave when traveling over astronomical distances. The theoretical calculations were refined and verified numerous times over the next two decades by different researchers. The data used in the calculations were improved, and the predictions became more precise. No significant error was found in the solar model or in the calculation of the cross-section for neutrino capture in the tank. But what about the third possible explanation, namely that neutrinos behave differently from what physicists had assumed?

Consequently, two questions arise:

1. What causes this behavior in neutrinos, and why?
2. Does the structure of space play a role in the trajectory of neutrinos from the Sun to the Earth?

Very few physicists took this idea seriously when it was first proposed, but the evidence favoring this possibility increased over time. For every one hundred billion solar neutrinos that pass through the Earth, only about one interacts with the material composing the Earth.

Because neutrinos interact so rarely, they can easily escape from the interior of the Sun where they are created, bringing direct information to us on Earth about nuclear fusion reactions. Nuclear fusion in the Sun produces only electron neutrinos. Electron neutrinos are generated in the core of the Sun. When they escape from the Sun, and during their journey toward the Earth, each group of  $9 \times 9\nu_e = 81\nu_e$  forms a Planck patch  $P_9$ , which is the structural element of  $\mathbf{P}^2$  after inflation.

In other words, if this hypothesis is verified, it confirms, along with the case of partial reflection of photons from glass, that the space  $\mathbf{P}^2$  is structured via a self-similar process with the Planck patch  $P_9$  as its building block. Thus, we will examine what happens in an initial Planck patch  $P_9$  that consists purely of electron neutrinos  $\nu_e$ . The nature of the inner neutrinos changes from  $\nu_e$  to  $\nu_\mu, \nu_\tau$ . If the percentages of the 3 neutrino generations found experimentally emerge from the process we apply, then our hypothesis verifies the creation of  $\mathbf{P}^2$  from the Planck patches  $P_9$ .

This situation is analogous to the partial reflection of light (photons) by glass. The total number of electron neutrinos present in  $P_9$  is equal to the number of Planck points within  $P_9$ , because at each Planck point, the inscribed circle corresponds to an electron neutrino, as shown in figure (a) of the Planck patch  $P_9$ .

|         |         |         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ | $\nu_e$ |

Figure 17.14: (a) Neutrinos from the Sun

|         |           |            |            |            |            |            |            |           |         |
|---------|-----------|------------|------------|------------|------------|------------|------------|-----------|---------|
| $\nu_e$ | $\nu_e$   | $\nu_e$    | $\nu_e$    | $\nu_e$    | $\nu_e$    | $\nu_e$    | $\nu_e$    | $\nu_e$   | $\nu_e$ |
| $\nu_e$ | $\nu_\mu$ | $\nu_\mu$  | $\nu_\mu$  | $\nu_\mu$  | $\nu_\mu$  | $\nu_\mu$  | $\nu_\mu$  | $\nu_\mu$ | $\nu_e$ |
| $\nu_e$ | $\nu_\mu$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\mu$ | $\nu_e$ |
| $\nu_e$ | $\nu_\mu$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\mu$ | $\nu_e$ |
| $\nu_e$ | $\nu_\mu$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\mu$ | $\nu_e$ |
| $\nu_e$ | $\nu_\mu$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\mu$ | $\nu_e$ |
| $\nu_e$ | $\nu_\mu$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\mu$ | $\nu_e$ |
| $\nu_e$ | $\nu_\mu$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\mu$ | $\nu_e$ |
| $\nu_e$ | $\nu_\mu$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\tau$ | $\nu_\mu$ | $\nu_e$ |
| $\nu_e$ | $\nu_e$   | $\nu_e$    | $\nu_e$    | $\nu_e$    | $\nu_e$    | $\nu_e$    | $\nu_e$    | $\nu_e$   | $\nu_e$ |

Figure 17.15: (b) Three generations of neutrinos at the Earth

The neutrinos located on the 1st perimeter of  $P_9$  in figure (b) are electron neutrinos ( $\nu_e$ ). These have greater freedom of movement due to the fact that they are free to move in one direction or in all directions between the angles in the range from  $\theta = 0^0$  to  $\theta = 90^0$ , such as the neutrinos at the corners of  $P_9$ . If the  $\nu_e$  of the four corners of  $P_9$  move away from  $P_9$ , then the remaining set of neutrinos equals 77.

The neutrinos of the 2nd perimeter of  $P_9$  are muon neutrinos ( $\nu_\mu$ ); they have less freedom of movement than  $\nu_e$  because they are blocked by them.

The neutrinos in the interior of  $P_9$  are tau neutrinos ( $\nu_\tau$ ), which have even less freedom of movement because they are blocked by both  $\nu_\mu$  and  $\nu_e$ , and are thus considered almost stationary.

These observations lead free observers to attribute the difficulty of movement of elementary particles to the property of having greater curvature, and therefore mass, meaning they belong to the other two neutrino generations. According to table (b), the percentages of  $\nu_e$ ,

$\nu_\mu$ , and  $\nu_\tau$  are:

$$\nu_e: \frac{28}{77} = 0.36$$

$$\nu_\mu: \frac{24}{77} = 0.31$$

$$\nu_\tau: \frac{25}{77} = 0.32$$

These percentages have also been verified experimentally. From this table, we can explain the “mystery of the missing solar neutrinos” during their measurement by various apparatuses, such as the Super-Kamiokande detector, GALLEX (Gallium Experiment), SAGE (Soviet-American Gallium Experiment), and SNO (Solar Neutrino Observatory).

#### **Finding Neutrino Masses via the Boundary of the Universe $P^2$**

From the Table of Figure 17.15(b), we consider the central neutrino, located at the intersection of the fifth row and the fifth column, as the common center of circles with radii  $2.5l_p$ ,  $3.5l_p$ , and  $4.5l_p$ .

Since mass is proportional to the curvature of a circle, from the relations  $\kappa = \lambda_\kappa \cdot \frac{r}{l_p^2}$  and  $m = \lambda_m \cdot \kappa$ , we obtain:

$$m = \left( \frac{\lambda_m \cdot \lambda_\kappa}{l_p^2} \right) \cdot r$$

where  $[\lambda m] = [M]$ ,  $[\lambda_\kappa] = [L]$ ,  $[r] = [L]$ , and  $[l_p^2] = [L^2]$ .

From this relationship, it follows that the quantity that varies is the radius of the circle, because  $l_p^2$  is a constant whose numerical value equals the numerical value of the product  $\lambda_m \cdot \lambda_\kappa = 2.6 \cdot 10^{-70}$ . Thus, it “appears” that mass is proportional to the radius of the circle.

According to this relationship, the circle with the largest radius also has the largest mass, according to bound observers. From this procedure, we theoretically obtain the masses of the 3 neutrino generations, which are:

$$\begin{aligned} m_{\nu_e} &= 2.5l_p = 2.5 \cdot 1.6 \cdot 10^{-35}[M] = 4 \cdot 10^{-35}[M] \\ m_{\nu_\mu} &= 3.5l_p = 3.5 \cdot 1.6 \cdot 10^{-35}[M] = 5.6 \cdot 10^{-35}[M] \\ m_{\nu_\tau} &= 4.5l_p = 4.5 \cdot 1.6 \cdot 10^{-35}[M] = 7.2 \cdot 10^{-35}[M] \end{aligned}$$

These results are in full agreement with the results of the KATRIN experiment, where the electron mass is of the order of magnitude:

$$m_e \sim 10^{-30}$$

hence it is  $10^5$  times larger than the neutrino mass.

#### **17.6.4 Interpretation of the Tunneling Phenomenon in Uranium**

The creation of nuclei and atoms in space  $\mathbf{P}^2$  is due, on one hand, to the satisfaction of the zero criterion of  $\mathbf{P}^2$  and, on the other hand, to the Expansion-Contraction process. The creation of the uranium nucleus  $92^{238}\text{U}$  is driven by the Expansion-Contraction process of Planck patches, as is every element of the Periodic Table.

Within the uranium nucleus, the expansion-contraction process continues for smaller Planck patches. Thus, in some of them, during the Contraction process,  ${}^4_2\text{He}$  nuclei are created. The  ${}^4_2\text{He}$  nuclei are the most stable, meaning they have the highest abundance. They consist of 2 protons ( $p$ ) and 2 neutrons ( $n$ ), so the circle enclosing them in  $\mathbf{P}^2$  has a radius of  $r_{pn} = 18l_p$ , because the radius of the circle corresponding to  $p$  is  $r_p = 5l_p$  ( $r_{2p} = 10l_p$ ) and the radius corresponding to  $n$  is  $r_n = 4l_p$  ( $r_{2n} = 8l_p$ ). These radii are derived from the structure of  $p$  and  $n$ , which consist of  $2u$ -quarks,  $1d$ -quark and  $2d$ -quarks,  $1u$ -quark respectively.

Each  $u$ -quark corresponds to a circle of radius  $r_u = 2l_p$  and each  $d$ -quark corresponds to a circle of radius  $r_d = 1l_p$ . According to the above, the nucleus of  ${}^4_2\text{He}$  corresponds to a circle with a diameter of  $r_{\text{He}} = 36l_p \sim 10^{-31}$ , cm. For a helium nucleus  ${}^4_2\text{He}$  to escape from the interior of the uranium nucleus  ${}^{238}_{92}\text{U}$ , a tunnel must open near them during an expansion process, with a minimum opening length of  $36l_p$ . The opening of a “tunnel” is possible for the following reason:

The components of the uranium nucleus that correspond to the geometric objects of a Planck patch are relatively close to each other during the contraction process. When the expansion process begins, the geometric objects formed in smaller Planck patches (such as the  ${}^4_2\text{He}$  nuclei) do not move away from each other in the short term (in order to maintain the curvature of the Planck patch they formed) during each expansion process. Thus, the  ${}^4_2\text{He}$  nuclei preserve their composition and tend to move away from the uranium nucleus during an expansion process.

Their escape pathways are created by the tendency of all geometric objects corresponding to the uranium nucleus to disperse throughout the available space of the new Planck patch. In the next stage, which is the contraction process, they will gather again to form the new compact geometric object corresponding to the thorium nucleus  ${}^{234}_{90}\text{Th}$ . During the expansion process, the  ${}^4_2\text{He}$  nuclei leave the uranium nucleus not because of electrical repulsion with the rest of the uranium nucleus, as interpreted by free observers, but due to the expansion process itself.

## 17.7 PERIHELION PRECESSION

Planets move around the Sun in elliptical orbits. The perihelion is located at one end of the major axis of these ellipses, and the aphelion of each planet is at the other end. At the midpoint of the major axis of each planet lies the center  $c$  of a circle with radius  $R_c = \frac{1}{2} \cdot (r_P + r_A)$ , where  $r_P$  and  $r_A$  are the Sun–perihelion and Sun–aphelion distances, respectively. Let us assume that a planet is at its perihelion, i.e., at position  $P_1$  in Figure 17.17. To satisfy the isotropy criterion of  $\mathbf{P}^2$  with respect to the Sun, the planet shifts briefly to position  $P_2$  on circle  $c$ . Each planet covers the distance  $d_{P_1P_2}$ , which is

nearly tangent to circle  $c$  at point  $P_1$ . Point  $P_2$  is considered to belong to circle  $c$  as well. The aphelion of the planet moves with velocity  $V_A$  from position  $A_1$  to position  $A_2$ , because these are the two ends of the major axis of any elliptical planetary orbit, as shown in Figure 17.16.

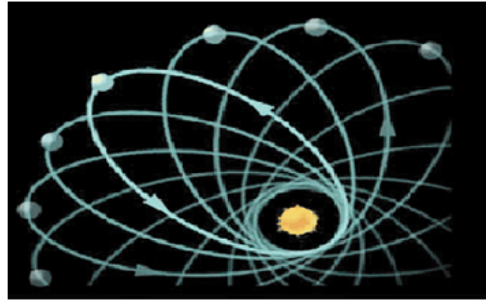


Figure 17.16: Precession of a planet's perihelion

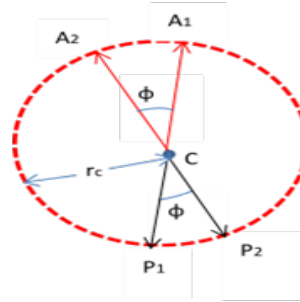


Figure 17.17: Geometric calculations

A planet completes its elliptical orbit in  $D$  Earth days. In one Earth year, a planet completes  $N_1 = \frac{365}{D}$  orbits, and in a century it completes  $N_{100} = 100 \cdot N_1$  orbits. A planet at its perihelion moves along an arc that has a length of  $S_P = V_P \cdot \Delta t_P$ , and at its aphelion, it moves along an arc that has a length of  $S_A = V_A \cdot \Delta t_A$ . The variables  $V_P$  and  $V_A$  represent the velocities of the planet at its perihelion and aphelion, respectively. Since these two arcs belong to the same circle  $c$  and are subtended by the same angle  $\phi$ , the two arcs are equal. From this condition, we obtain the relationship:

$$V_A \cdot \Delta t_A = V_P \cdot \Delta t_P$$

or:

$$\Delta t_P = \frac{V_A}{V_P} \cdot \Delta t_A$$

We group the planets of our Solar System into three categories based on the radius of the circle  $c$  of each planet. The first group has the planet Mercury as its first member and the planet Mars as its last member. The second group has the planet Jupiter as its first member—which has nearly twice the radius

of circle  $c$  compared to Mars—and the planet Saturn as its last member. The third group has the planet Uranus as its first member—which has more than twice the radius of circle  $c$  compared to Saturn—and the planet Neptune as its last member. We form the ratios of the radius of each planet's circle to the radius of Mercury's circle, denoted as  $R_1^1$ . The superscript indicates the group, and the subscript indicates the planet according to its distance from the Sun, namely:

$$\text{Mercury} \rightarrow 1, \quad \text{Venus} \rightarrow 2, \quad \text{Earth} \rightarrow 3, \quad \text{Mars} \rightarrow 4$$

$$\text{Jupiter} \rightarrow 5, \quad \text{Saturn} \rightarrow 6, \quad \text{Uranus} \rightarrow 7, \quad \text{Neptune} \rightarrow 8$$

Thus, we obtain for the three groups:

$$\frac{R_1^j}{R_1^1} = 2^0, \quad j = 1, 2, 3, 4$$

$$\frac{R_2^j}{R_1^1} = 2^1, \quad j = 5, 6$$

$$\frac{R_3^j}{R_1^1} = 2^2, \quad j = 7, 8$$

Let us examine the continuity equation in the form:

$$V_i \cdot A_u = V_j \cdot A_u$$

where  $A_u$  is the unit area and  $V_i$  is the constant rate of increase of the radius of the circles of these groups, meaning:

$$V_i = V_u$$

We apply this continuity equation for:

$$V_i = \frac{R_1^1}{\Delta t_{\text{unit}}} = \frac{R_j^i}{\Delta t_{A,i}}$$

from which we obtain:

$$\Delta t_{A,i} = \frac{R_j^i}{R_1^1} \cdot \Delta t_{\text{unit}}$$

where we define  $\Delta t_{\text{unit}} = 1 \text{ sec}$ . Thus, we have:

$$\Delta t_{A,i} = \frac{R_1^i}{R_1^1} \cdot \Delta t_{\text{unit}} = 1 \text{ sec}$$

$$\Delta t_{A,i} = \frac{R_2^i}{R_1^1} \cdot \Delta t_{\text{unit}} = 2 \text{ sec}$$

$$\Delta t_{A,i} = \frac{R_3^i}{R_1^1} \cdot \Delta t_{\text{unit}} = 4 \text{ sec}$$

The distance  $d_{P_1P_2}$  is given by  $d_{P_1P_2} = V_m \cdot \Delta t_P$ . As mentioned previously, the distance  $d_{P_1P_2}$  is nearly tangent to circle  $c$  at  $P_1$  and is therefore perpendicular to its radius  $R_c$  at  $P_1$ . The right triangle formed has the very small angle  $\phi$  opposite to the side  $d_{P_1P_2}$ . The tangent of angle  $\phi$  is approximately equal to the angle  $\phi$  itself, so we get:

$$\tan(\phi) \simeq \phi = \frac{d_{P_1P_2}}{R_c} = \frac{V_m \cdot \Delta t_P}{R_c} \cdot \frac{\text{radians}}{\text{orbit}}$$

where  $1 \text{ radian} = 57.3 \text{ degrees} = 57.3 \cdot 3600 \text{ arcsec} = 206,280 \text{ arcsec}$ . Thus, we obtain the following relationship for the precession of a planet's perihelion over a century:

$$\phi \cdot N_{100} \quad \frac{\text{arcsec}}{\text{century}}$$

Following this procedure, we will obtain the precession angle for all the planets in our solar system and then compare them with their observed values.

### 17.7.1 Mercury

1. Characteristic quantities of Mercury's orbit:

$$D = 87.969 \text{ days}$$

$$R_c = 5.79 \cdot 10^7 \text{ km}$$

$$V_m = 47.4 \frac{\text{km}}{\text{sec}}$$

$$V_P = 59.0 \frac{\text{km}}{\text{sec}}$$

$$V_A = 39.0 \frac{\text{km}}{\text{sec}}$$

2. Calculation of perihelion precession:

$$\Delta t_A = 1 \text{ sec}$$

$$\Delta t_P = \frac{V_A}{V_P} \cdot \Delta t_A = 0.66 \text{ sec}$$

$$d_{P_1P_2} = V_m \cdot \Delta t_P = 31.3 \text{ km}$$

$$\tan(\phi) \sim \phi = \frac{d_{P_1P_2}}{R_c} \sim 5.4 \cdot 10^{-7} \frac{\text{rad}}{\text{orb}} = 3.1 \cdot 10^{-5} \frac{\text{deg}}{\text{orb}} = 0.1034 \frac{\text{arcsec}}{\text{orb}}$$

$$N_1 = 365/D = 4.15 \frac{\text{orb}}{\text{year}}$$

$$N_{100} = 415 \frac{\text{orb}}{\text{century}}$$

Therefore:

$$N_{100} \cdot \phi = 415 \cdot 0.1034 \frac{\text{arcsec}}{\text{century}} = 42.9 \frac{\text{arcsec}}{\text{century}} \sim 43 \frac{\text{arcsec}}{\text{century}}$$

$$\text{OBSERVED VALUE} = 43 \frac{\text{arcsec}}{\text{century}}$$

### 17.7.2 Venus

Characteristic quantities of Venus's orbit:

$$D = 224.595 \text{ days}$$

$$R_c = 108.210 \cdot 10^6 \text{ km}$$

$$V_m = 35.02 \frac{\text{km}}{\text{sec}}$$

$$V_P = 35.26 \frac{\text{km}}{\text{sec}}$$

$$V_A = 34.78 \frac{\text{km}}{\text{sec}}$$

Calculation of perihelion precession:

$$\Delta t_A = 1 \text{ sec}$$

$$\Delta t_P = \frac{V_A}{V_P} \cdot \Delta t_A = 0.986 \text{ sec}$$

$$d_{P_1 P_2} = V_m \cdot \Delta t_P = 34.5 \text{ km}$$

$$\tan(\phi) \sim \phi = \frac{d_{P_1 P_2}}{R_c} \sim 3.19 \cdot 10^{-7} \frac{\text{rad}}{\text{orb}} = 1.83 \cdot 10^{-5} \frac{\text{deg}}{\text{orb}} = 0.0658 \frac{\text{arcsec}}{\text{orb}}$$

$$N_1 = 365/D = 1.625 \frac{\text{orb}}{\text{year}}$$

$$N_{100} \cdot \phi = 162.5 \left( \frac{\text{orb}}{\text{century}} \right) \cdot 0.0658 \frac{\text{arcsec}}{\text{orb}} \sim 10.7 \frac{\text{arcsec}}{\text{century}}$$

$$\text{OBSERVED VALUE} = 8.6 \frac{\text{arcsec}}{\text{century}}$$



### 17.7.3 Earth

Characteristic quantities of the Earth's orbit:

$$D = 365.242 \text{ days}$$

$$R_c = 149.598 \cdot 10^6 \text{ km}$$

$$V_m = 29.78 \frac{\text{km}}{\text{sec}}$$

$$V_P = 30.29 \frac{\text{km}}{\text{sec}}$$

$$V_A = 29.29 \frac{\text{km}}{\text{sec}}$$

Calculation of perihelion precession:

$$\Delta t_A = 1 \text{ sec}$$

$$\Delta t_P = \frac{V_A}{V_P} \cdot \Delta t_A = 0.967 \text{ sec}$$

$$d_{P_1 P_2} = V_m \cdot \Delta t_P = 28.8 \text{ km}$$

$$\tan(\phi) \sim \phi = \frac{d_{P_1 P_2}}{R_c} \sim 0.192 \cdot 10^{-6} \frac{\text{rad}}{\text{orb}} = 1.10 \cdot 10^{-5} \frac{\text{deg}}{\text{orb}} = 0.0397 \frac{\text{arcsec}}{\text{orb}}$$

$$N_1 = 365/D = 1 \frac{\text{orb}}{\text{year}}$$

$$N_{100} \cdot \phi = 100 \left( \frac{\text{orb}}{\text{century}} \right) \cdot 0.0397 \frac{\text{arcsec}}{\text{orb}} \sim 4 \frac{\text{arcsec}}{\text{century}}$$

$$\text{OBSERVED VALUE} = 4 \frac{\text{arcsec}}{\text{century}}$$

### 17.7.4 Mars

Characteristic quantities of Mars's orbit:

$$D = 686.973 \text{ days}$$

$$R_c = 227.956 \cdot 10^6 \text{ km}$$

$$V_m = 24.08 \frac{\text{km}}{\text{sec}}$$

$$V_P = 26.50 \frac{\text{km}}{\text{sec}}$$

$$V_A = 21.97 \frac{\text{km}}{\text{sec}}$$

Calculation of perihelion precession:

$$\Delta t_A = 1 \text{ sec}$$

$$\Delta t_P = \frac{V_A}{V_P} \cdot \Delta t_A = 0.83 \text{ sec}$$

$$d_{P_1 P_2} = V_m \cdot \Delta t_P = 19.98 \text{ km}$$

$$\tan(\phi) \sim \phi = \frac{d_{P_1 P_2}}{R_c} \sim 0.087 \cdot 10^{-6} \frac{\text{rad}}{\text{orb}} = 0.5 \cdot 10^{-5} \frac{\text{deg}}{\text{orb}} = 0.018 \frac{\text{arcsec}}{\text{orb}}$$

$$N_1 = 365/D = 0.53 \frac{\text{orb}}{\text{year}}$$

$$N_{100} \cdot \phi = 53 \left( \frac{\text{orb}}{\text{century}} \right) \cdot 0.018 \frac{\text{arcsec}}{\text{orb}} \sim 1 \frac{\text{arcsec}}{\text{century}}$$

$$\text{OBSERVED VALUE} = 1.35 \frac{\text{arcsec}}{\text{century}}$$

### 17.7.5 Jupiter

Characteristic quantities of Jupiter's orbit:

$$D = 4,330.595 \text{ days}$$

$$R_c = 778.479 \cdot 10^6 \text{ km}$$

$$V_m = 13.06 \frac{\text{km}}{\text{sec}}$$

$$V_P = 13.72 \frac{\text{km}}{\text{sec}}$$

$$V_A = 12.44 \frac{\text{km}}{\text{sec}}$$

Calculation of perihelion precession:

$$\Delta t_A = 2 \text{ sec}$$

$$\Delta t_P = \frac{V_A}{V_P} \cdot \Delta t_A = 1.81 \text{ sec}$$

$$d_{P_1 P_2} = V_m \cdot \Delta t_P = 23.6 \text{ km}$$

$$\tan(\phi) \sim \phi = \frac{d_{P_1 P_2}}{R_c} \sim 0.03 \cdot 10^{-6} \frac{\text{rad}}{\text{orb}} = 0.17 \cdot 10^{-5} \frac{\text{deg}}{\text{orb}} = 0.0062 \frac{\text{arcsec}}{\text{orb}}$$

$$N_1 = 365/D = 0.0843 \frac{\text{orb}}{\text{year}}$$

$$N_{100} \cdot \phi = 8.43 \left( \frac{\text{orb}}{\text{century}} \right) \cdot 0.0062 \frac{\text{arcsec}}{\text{orb}} = 0.0523 \frac{\text{arcsec}}{\text{century}}$$

$$\text{OBSERVED VALUE} = 0.06 \left( \frac{\text{arcsec}}{\text{century}} \right)$$

**17.7.6 Saturn**

Use code with caution. Characteristic quantities of Saturn's orbit:

$$D = 10,746.94, \text{ days}$$

$$R_c = 1,432.041 \cdot 10^6, \text{ km}$$

$$V_m = 9.67, \frac{\text{km}}{\text{sec}}$$

$$V_P = 10.14, \frac{\text{km}}{\text{sec}}$$

$$V_A = 9.14, \frac{\text{km}}{\text{sec}}$$

Calculation of perihelion precession:

$$\Delta t_A = 2, \text{ sec}$$

$$\Delta t_P = \frac{V_A}{V_P} \cdot \Delta t_A = 1.80, \text{ sec}$$

$$d_{P_1 P_2} = V_m \cdot \Delta t_P = 17.4, \text{ km}$$

$$\tan(\phi) \sim \phi = \frac{d_{P_1 P_2}}{R_c} \sim 0.01215 \cdot 10^{-6}, \frac{\text{rad}}{\text{orb}} = 2.5 \cdot 10^{-3}, \frac{\text{arcsec}}{\text{orb}}$$

$$N_1 = 365/D = 0.034, \left( \frac{\text{orb}}{\text{year}} \right)$$

$$N_{100} \cdot \phi = 3.4, \left( \frac{\text{orb}}{\text{century}} \right) \cdot 2.5 \cdot 10^{-3}, \frac{\text{arcsec}}{\text{orb}} \sim 0.0085, \frac{\text{arcsec}}{\text{century}}$$

$$\text{OBSERVED VALUE} = 0.01, \left( \frac{\text{arcsec}}{\text{century}} \right)$$

**17.7.7 Uranus**

Characteristic quantities of Uranus's orbit:

$$D = 30,588.740, \text{ days}$$

$$R_c = 2,867.043 \cdot 10^6, \text{ km}$$

$$V_m = 6.79, \left( \frac{\text{km}}{\text{sec}} \right)$$

$$V_P = 7.13, \left( \frac{\text{km}}{\text{sec}} \right)$$

$$V_A = 6.49, \left( \frac{\text{km}}{\text{sec}} \right)$$

Calculation of perihelion precession:

$$\begin{aligned}
 \Delta t_A &= 4, \text{ sec} \\
 \Delta t_P &= \frac{V_A}{V_P} \cdot \Delta t_A = 3.64, \text{ sec} \\
 d_{P_1 P_2} &= V_m \cdot \Delta t_P = 24.7, \text{ km} \\
 \tan(\phi) \sim \phi &= \frac{d_{P_1 P_2}}{R_c} \sim 0.0086 \cdot 10^{-6}, \frac{\text{rad}}{\text{orb}} = 1.78 \cdot 10^{-3}, \frac{\text{arcsec}}{\text{orb}} \\
 N_1 &= \frac{365}{D} = 0.0119, \frac{\text{orb}}{\text{year}} \\
 N_{100} \cdot \phi &= 1.19, \left( \frac{\text{orb}}{\text{century}} \right) \cdot 1.78 \cdot 10^{-3}, \left( \frac{\text{arcsec}}{\text{orb}} \right) \sim 0.0021, \left( \frac{\text{arcsec}}{\text{century}} \right) \\
 \text{OBSERVED VALUE} &= 0.0024, \left( \frac{\text{arcsec}}{\text{century}} \right)
 \end{aligned}$$

### 17.7.8 Neptune

Characteristic quantities of Neptune's orbit:

$$\begin{aligned}
 D &= 59,799.9, \text{ days} \\
 R_c &= 4,514.953 \cdot 10^6, \text{ km} \\
 V_m &= 5.45, \left( \frac{\text{km}}{\text{sec}} \right) \\
 V_P &= 5.47, \left( \frac{\text{km}}{\text{sec}} \right) \\
 V_A &= 5.37, \left( \frac{\text{km}}{\text{sec}} \right)
 \end{aligned}$$

Calculation of perihelion precession:

$$\begin{aligned}
 \Delta t_A &= 4, \text{ sec} \\
 \Delta t_P &= \frac{V_A}{V_P} \cdot \Delta t_A = 3.926, \text{ sec} \\
 d_{P_1 P_2} &= V_m \cdot \Delta t_P = 21.4, \text{ km} \\
 \tan(\phi) \sim \phi &= \frac{d_{P_1 P_2}}{R_c} \sim 4.74 \cdot 10^{-9}, \left( \frac{\text{rad}}{\text{orb}} \right) \sim 0.978 \cdot 10^{-3}, \left( \frac{\text{arcsec}}{\text{orb}} \right) \\
 N_1 &= \frac{365}{D} = 0.61 \cdot 10^{-2}, \left( \frac{\text{orb}}{\text{year}} \right) \\
 N_{100} \cdot \phi &= 0.61, \left( \frac{\text{orb}}{\text{century}} \right) \cdot 0.978 \cdot 10^{-3}, \left( \frac{\text{arcsec}}{\text{orb}} \right) \sim 6 \cdot 10^{-4}, \left( \frac{\text{arcsec}}{\text{century}} \right) \\
 \text{OBSERVED VALUE} &= 7 \cdot 10^{-4}, \left( \frac{\text{arcsec}}{\text{century}} \right)
 \end{aligned}$$

## 17.8 ASTRONOMICAL ISSUES

### 17.8.1 Binary Pulsars

Following the same procedure as before, we will determine the periastron precession of the Hulse–Taylor pulsar as it orbits around another pulsar, as illustrated in Figure 17.18.

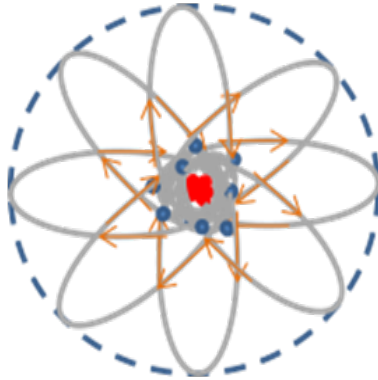


Figure 17.18: Hulse–Taylor Pulsar

The observational data are as follows:

$$r_P = 1.1 \cdot R_{\text{Sun}}$$

$$r_A = 4.8 \cdot R_{\text{Sun}}$$

$$R_{\text{Sun}} = 6.957 \cdot 10^5 \text{ km}$$

$$R_c = 2.045 \cdot 10^6 \text{ km}$$

$$v_P = 450 \left( \frac{\text{km}}{\text{sec}} \right)$$

$$v_A = 110 \left( \frac{\text{km}}{\text{sec}} \right)$$

$$M_{\text{pulsar}} \sim 1.441 \cdot M_{\text{Sun}}$$

$$N = 10^3 \cdot \left( \frac{\text{orbits}}{\text{year}} \right)$$

From the equality of the arcs forming the periastron and the apastron of the orbiting pulsar on the circle of radius  $R_c$ , we obtain the same relationship as in the case of Mercury, namely:

$$\Delta t_P = \frac{v_A}{v_P} \cdot \Delta t_A$$

Furthermore, from the continuity equation, we get:

$$M_{\text{Sun}} \cdot \Delta t_A = M_{\text{pulsar}} \cdot 1 \text{ sec}$$

From this relationship, it follows that:

$$\Delta t_A = \frac{M_{\text{pulsar}}}{M_{\text{Sun}}} \cdot 1 \text{ sec} = 1.441 \text{ sec}$$

consequently, we obtain:

$$\Delta t_P = \frac{v_A}{v_P} \cdot \Delta t_A = \frac{110}{450} \cdot 1.441 \text{ sec} = 0.352 \text{ sec}$$

The distance  $d_{P_1 P_2}$  is:

$$d_{P_1 P_2} = v_P \cdot \Delta t_P = 450 \cdot 0.352 \text{ km} = 158.4 \text{ km}$$

By the same reasoning, we find the angle  $\phi$  from the relationship:

$$\tan \phi \sim \phi = \frac{d_{P_1 P_2}}{R_c} = 77.45 \cdot 10^{-6} \cdot 57.3 \frac{\text{degrees}}{\text{orbits}} \sim 4.2 \cdot 10^{-3} \frac{\text{degrees}}{\text{orbits}}$$

For  $N = 10^3 \frac{\text{orbits}}{\text{year}}$ , we find:

$$N \cdot \phi = 4.2 \frac{\text{degrees}}{\text{year}}$$

This value is in exceptionally close agreement with the observed value. Based on the above methodology and for different values of various parameters, such as the aphelion velocity, the perihelion velocity, and the length of the semi-major axis, we can calculate the perihelion or periastron precession of any celestial body.

### 17.8.2 Bending of Light Rays Near Celestial Bodies

We will describe the motion of a photon in space  $\mathbf{P}^2$  according to the contraction-expansion process defined in this space. The rectilinear trajectory of a photon when passing very close to the Sun is transformed into a circular arc to satisfy the isotropy criterion of  $\mathbf{P}^2$ . Thus, it traverses the arc  $\Delta s_1$  of the circle with radius  $R_{\text{Sun}} = 6.95 \cdot 10^{10} \text{ cm}$ , as illustrated in Figure 17.19.

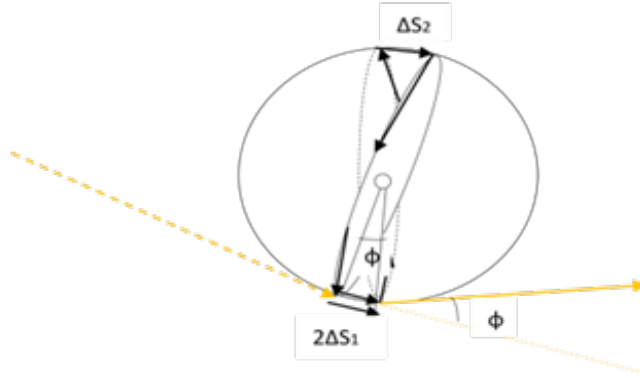


Figure 17.19: Deflection of the rectilinear path of light near the Sun

The motion of the photon takes place during the expansion process. The subsequent motion of the photon is upward along an ellipse into which the previous circle has contracted. This motion of the photon takes place during the contraction process. The next motion of the photon is on the same circle as before, during the expansion process. Thus, it traverses a circular arc  $\Delta s_2$  that is equal to the arc  $\Delta s_1$ , since their velocities are identical and equal to the linear rotational velocity of the Sun, which is  $V_{\text{Sun}} \sim 2 \cdot 10^5 \frac{\text{cm}}{\text{sec}}$ .

Following the expansion process, the contraction process takes place, during which the photon moves along an ellipse until it reaches its entry point on the circle. It follows the circular motion (expansion process) until it covers the arc  $\Delta s_1$ , at which point it moves away from the Sun. To calculate the deflection angle of the velocity vector of the incoming photon from the velocity vector of the outgoing photon, we consider a right-angled triangle. One vertical side of this triangle has a length equal to the total length of the arcs that the photon traveled along the circle, i.e.,

$$\Delta s_{\text{total}} = \Delta s_1 + \Delta s_2 + \Delta s_1 = 3 \cdot \Delta s_1$$

The other perpendicular side of this triangle has a length equal to the radius  $R_{\text{Sun}}$  of the circle. The angle  $\hat{\phi}$  opposite to the side  $\Delta s_{\text{total}}$  is equal to the angle formed by the velocity vectors of the photon upon entering and exiting the circle of radius  $R_{\text{Sun}}$ , because they have perpendicular sides. Thus, for the tangent of angle  $\hat{\phi}$ , we obtain the relationship:

$$\tan(\phi) \approx \phi = \frac{\Delta s_{\text{total}}}{R_{\text{Sun}}} = \frac{6 \cdot 10^5}{6.95 \cdot 10^{10}} = 0.85 \cdot 10^{-5} \text{ radians}$$

or:

$$\phi = 0.85 \cdot 10^{-5} \cdot 57.3 \cdot 3.6 \cdot 10^3 \text{ arcsec} = 1.75 \text{ arcsec}$$

This angle is equal to the angle measured by free observers. This methodology is similar to the one used for finding the precession of the perihelion or periastron.

In this methodology, neither the concept of force nor the mass of the Sun is introduced, as required on one hand by Newtonian theory, nor is the concept of a metric change due to the mass of the Sun introduced, as required on the other hand by Einstein's theory to account for cause and effect. In our case, the proof of the bending of the light ray was achieved via geometric objects in the flat space  $\mathbf{P}^2$  and was based on the properties of this space combined with its repetitive construction process, namely the contraction-expansion process (C-E-C...).

### 17.8.3 Time Delay of a Radio Signal in the Shapiro Experiment

Figure 17.20 shows how the emission, delays, and reflection of a radio beam transmitted from the Earth to Venus and passing near the Sun take place. Venus has a radius of  $R_A = 6052 \cdot 10^5$  cm. We assume that the reflection

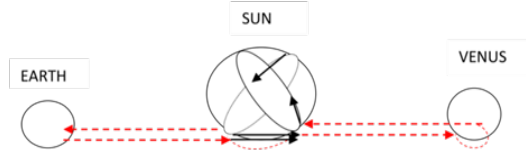


Figure 17.20: Shapiro Experiment

of the radio waves takes place at a reasonable distance, for instance, at a distance equal to one-hundredth of its radius, i.e., at a distance  $\bar{R}_A = \frac{R_A}{100}$  from the South (or North) pole of Venus. As the radio signal enters it, the radio beam will move briefly around the surface and then onto the surface of Venus along  $\bar{R}_A$  within the time interval:

$$\Delta t_V = \frac{R_V}{100 \cdot c} = \frac{6.052 \cdot 10^8 \text{ cm}}{100 \cdot 3 \cdot 10^{10} \frac{\text{cm}}{\text{sec}}} = 202 \mu\text{s} \quad (1 \mu\text{s} = 10^{-6} \text{ s})$$

In the Sun, we also have a double delay because the radio beam passes near it twice on its way to and from Venus. Applying the same procedure to the Sun as we did for calculating the bending of the light ray, we find:

$$S_{\text{total}} = 3 \cdot S_1 = 3 \cdot v_S \cdot t_S = 6 \cdot 10^5 \cdot t_S$$

Considering  $S_{\text{total}}$  as a unit interval, we obtain:

$$t_S = \frac{10^{-5}}{6} \text{ sec} = 1.7 \mu\text{s}$$

Since this happens twice, the total delay from the Sun will be  $3.4 \mu\text{s}$ . Adding the two delays together, we obtain:

$$\Delta t_{\text{delay}} \approx 206 \mu\text{s}$$



### 17.8.4 Saha Ionization Equation

Let the energies of two states  $A$  and  $B$  be  $E_A$  and  $E_B$ , and let their statistical weights be  $g_A$  and  $g_B$ , respectively. In local thermodynamic equilibrium, the number of particles in the two states,  $N_A$  and  $N_B$ , satisfies the Boltzmann equation:

$$\frac{N_A}{N_B} = \frac{g_A}{g_B} \cdot \exp \left[ -\frac{E_A - E_B}{kT} \right]$$

In physics, the Saha ionization equation is an expression that relates the ionization state of a gas in thermal equilibrium to its temperature and pressure. The equation is used to explain the spectral classification of stars. The expression was developed by the physicist Meghnad Saha in 1920.

#### Statistical Weights

In quantum mechanics, an energy level is degenerate if it corresponds to two or more different measurable states of a quantum system. Conversely, two or more different states of a quantum mechanical system are said to be degenerate if they yield the same energy value during measurement. The number of different states corresponding to a specific energy level is known as the **degree of degeneracy** (or simply degeneracy) of the level. Mathematically, it is expressed by the Hamiltonian of the system having more than one linearly independent eigenstate with the same energy eigenvalue. Quantum numbers are required to characterize the exact state when a distinction is desired. In classical mechanics, this can be understood in terms of different possible orbits corresponding to the same energy. Degeneracy plays a fundamental role in quantum statistical mechanics. For an  $N$ -particle system in three dimensions, a single energy level can correspond to many different wave functions or energy states. These degenerate states at the same energy level all have an equal probability of occurring. The number of such states determines the degeneracy of a specific energy level.

#### Description

According to free observers, in a gas with sufficiently high density and temperature (measured in keV or J), thermal collisions of atoms with one another will ionize some of the atoms, creating an ionized gas. Use code with caution. According to bound observers, during the expansion process of the Planck patch containing the geometric objects that correspond to the electrons of the gas atoms, these electrons will begin to move away from their nuclei to reduce the curvature of the enclosing circle. When many or most of the electrons normally bound to each atom in orbits around its atomic nucleus are freed, they form an independent electron gas cloud that coexists with the aggregate of atomic ions and neutral atoms. With sufficient ionization, the gas can become a **plasma**, which is the state of matter consisting of free ions and electrons. The Saha equation describes the degree of ionization for any gas in thermal equilibrium as a function of temperature, density, and the ionization energies of the atoms.

The Saha equation holds only for a weakly ionized plasma for which the Debye length is small. This implies that the screening of the Coulomb interaction of ions and electrons by other ions and electrons is negligible. The subsequent reduction of ionization potentials and the “cut-off” of the partition function are therefore also helical/negligible. For a gas consisting of a single atomic species, the Saha equation is written as:

$$\frac{n_{i+1}}{n_i} = \frac{g_{i+1}}{g_i} \frac{2}{n_e \lambda_{\text{th}}^3} \exp \left[ -\frac{\epsilon_{i+1} - \epsilon_i}{k_B T} \right]$$

where:  $n_i$  : is the density of atoms in the  $i$ -th ionization state, i.e., after the removal of  $i$  electrons.

$g_i$  : is the degeneracy of states for the ions.

$\epsilon_i$  : is the energy required to remove the  $i$  electrons from a neutral atom (ionization energy), creating an  $i$ -level ion.

$n_e$  : is the electron density.

$k_B$  : is the Boltzmann constant.

$\lambda_{\text{th}} = \left( \frac{h^2}{2\pi m_e k_B T} \right)^{\frac{1}{2}}$  : is the thermal de Broglie wavelength of an electron.

$m_e$  : is the electron mass.

$h$  : is the Planck constant.

The Saha equation is the mathematical relationship between the observed spectra of stars and their temperatures. The Saha equation expresses how the ionization state of any given element in a star changes with varying temperatures and pressures. The spectrum of a star is directly related to the relative numbers of atoms and ions it contains, because each atom or ion can absorb or emit radiation of a specific set of wavelengths.

The expression  $(\epsilon_{i+1} - \epsilon_i)$  is the energy required to remove the  $(i + 1)$ -th electron. In the case where only one ionization level is significant, we have  $n_1 = n_e$ , and defining the total density as  $n = n_0 + n_1$ , the Saha equation simplifies as follows:

$$\frac{n_e^2}{n - n_e} = \left( \frac{2}{\lambda_{\text{th}}^3} \right) \left( \frac{g_1}{g_0} \right) \exp \left[ -\frac{\epsilon}{k_B T} \right]$$

where  $\epsilon$  is the ionization energy. We can define the degree of ionization  $x = \frac{n_1}{n}$ , and find:

$$\frac{x^2}{1 - x} = A = \left( \frac{2}{\lambda_{\text{th}}^3 \cdot n} \right) \left( \frac{g_1}{g_0} \right) \exp \left[ -\frac{\epsilon}{k_B T} \right]$$

This is a quadratic equation that can be solved in closed form: for small  $A$ ,  $x \sim A^{\frac{1}{2}} \propto n^{-\frac{1}{2}}$ , meaning that ionization decreases with density.

As a simple example, consider a gas of monatomic hydrogen atoms, setting  $g_1 = g_0$ , and let  $\epsilon = 13.6 \text{ eV} = 158,000^\circ \text{K}$  be the ionization energy of hydrogen from its ground state.

Let  $n = 2.69 \times 10^{25} \text{ m}^{-3}$ , which is the Loschmidt constant or the particle density of the Earth's atmosphere at standard pressure and temperature. For  $T = 300^0 \text{ K}$ , the ionization is essentially zero, i.e., when  $x$  has the value  $x = 5 \times 10^{-115}$ , there are certainly no ionized atoms in the volume of the Earth's atmosphere. The value  $x$  increases rapidly with  $T$ , reaching a value of 0.35 for  $T = 20,000^0 \text{ K}$ . Substantial ionization occurs for this  $T$ , which is much smaller than the ionization energy (though this depends somewhat on the density). This is a common phenomenon.

Naturally, it stems from the fact that at a given temperature, the particles have an energy distribution, including some with multiple times the average energy. These high-energy particles are much more efficient at ionizing atoms.

In the Earth's atmosphere, ionization is actually governed not by the Saha equation but by cosmic rays with very high energy. Cosmic rays consist largely of muons and potentially dark matter particles according to their energies, which are in the range of  $10^{19} - 10^{21} \text{ eV}$ , as previously mentioned in section (4.14) of Chapter 4. These particles are not in thermal equilibrium with the atmosphere, meaning they do not share its temperature, so Saha's relation does not apply.

## 17.9 COSMOLOGICAL ISSUES

### 17.9.1 The Friedmann-Robertson-Walker Metric for an Isotropic Universe

A photon moves in a straight line (which is part of a geodesic line in an extended Planck patch). Assuming that the photon can move freely with a constant velocity equal to  $c$  and the space in which it moves is stationary, we obtain the relationship:

$$dt = \frac{dr}{c} \quad (17.65)$$

If the space in which the photon moves expands due to the expansion process, we obtain the relationship:

$$dt = -a(t)dr \quad (17.66)$$

The factor  $a(t)$  is a scale factor determining that the Universe or the Planck patch continuously grows over time. The negative sign in this equation is defined because when  $r$  decreases, the time  $t$  increases, meaning their differences have opposite signs. The distance  $r$  is the distance from a luminous object to the observer, which a photon must traverse. Given that the proper time interval  $d\tau$  is zero—since we are considering a moving photon—we obtain:

$$d\tau^2 = dt^2 - a^2(t)dr^2 \quad (17.67)$$

Today, it has been verified that the tendency to reduce the curvature of the expanding Planck patch, in contrast to the tendency to increase the curvature of other geometric objects present within it, results in the growth of the Planck patch. A free observer today observes the expansion of the Universe due to the observational fact that the frequency of light reaching them from a luminous celestial body is redshifted. The metric of the observed space incorporates all the characteristics of this three-dimensional space, namely isotropy, homogeneity, and expansion. Thus, we obtain the relationship:

$$d\tau^2 = dt^2 - a^2(t)[dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)] \quad (17.68)$$

where  $r, \theta, \phi$  are the polar coordinates of a point with respect to a coordinate system in three-dimensional space. This metric for a continuously expanding three-dimensional space, as considered by current free observers, is called the Friedmann-Robertson-Walker metric (*FRW* metric). Equation 17.68 defines the geometry of the extended Planck patch via the general relation:

$$d\tau^2 = dt^2 - g_{\mu\nu}dx^\mu dx^\nu \quad (17.69)$$

Generally, in any differentiable manifold, the equations of motion for freely moving entities are:

$$\frac{d^2x^\mu}{du^2} + \Gamma_{\nu\kappa}^\mu \frac{dx^\nu}{du} \frac{dx^\kappa}{du} = 0 \quad (17.70)$$

where  $u$  is an appropriate parameter for positions along a spacetime curve, analogous to  $\tau$  for entities with mass.  $\Gamma_{\nu\kappa}^\mu$  are the symmetric intrinsic connections since  $\Gamma_{\nu\kappa}^\mu = \Gamma_{\kappa\nu}^\mu$ . These connections determine the correlation between two adjacent spacetime points of a differentiable manifold and are given by the relationship:

$$\Gamma_{\nu\kappa}^\mu = \frac{1}{2}g^{\mu\lambda} \left[ \frac{\partial g_{\lambda\nu}}{\partial x^\kappa} + \frac{\partial g_{\lambda\kappa}}{\partial x^\nu} - \frac{\partial g_{\nu\kappa}}{\partial x^\lambda} \right] \quad (17.71)$$

A body at rest in these coordinates will continue to be at rest; therefore, these coordinates are called **co-moving coordinates**, which follow the motion of typical observers. For co-moving coordinates, the relation  $dl = 0$  holds, so:

$$(d\tau)^2 = -g_{\mu\nu}dx^\mu dx^\nu = -g_{00}(dx^0)^2 - (dl)^2 = (dt)^2$$

with:

$$g_{00} = -1$$

For a co-moving clock, the proper time is  $d\tau = dt$ , so  $t$  is the time measured by this clock. The factor  $a(t)$  in the FRW metric becomes clear when we calculate the **proper distance** at time  $t$  from the origin to a co-moving observer at radial distance  $r$  (with  $d\theta = d\phi = 0$ ):

$$d(r, t) = \int_0^r a(t) dr = a(t)r \quad (17.72)$$

If the position of a co-moving observer is arbitrary, then the distance between any two co-moving observers, wherever they are located, will be proportional to  $a(t)$ . The rate of change of any such distance is  $\dot{d}(r, t)$ :

$$\dot{d} = \frac{dd(r, t)}{dt} = d \left( \frac{\dot{a}(t)}{a(t)} \right)$$

or:

$$v = H(t)d \quad (17.73)$$

This relationship represents **Hubble's law**, where the Hubble constant is:

$$H(t) = \frac{\dot{a}(t)}{a(t)}$$

### 17.9.2 The Cosmological Redshift

Present observers can experimentally verify the hypothesis of the continuous growth of the observed Universe and interpret it accordingly in the expansion of the Planck patch as follows:

By observing the spectral lines of distant galaxies in terrestrial laboratories, it was found that their frequencies are shifted toward the red. To calculate these frequency changes, we write Equation 17.72 as follows:

$$\int_{t_1}^{t_0} \frac{dt}{a(t)} = - \int_r^0 dr = r \quad (17.74)$$

Taking the differential  $\delta t$  of this relationship with respect to time, we have:

$$\delta t \int_{t_1}^{t_0} \frac{dt}{a(t)} = \delta t \cdot r = 0$$

or:

$$\frac{\delta t_0}{a(t_0)} = \frac{\delta t_1}{a(t_1)} \quad (17.75)$$

If the signals emitted by the light source are considered as successive wave crests, the emitted frequency is  $\nu_1 = \frac{1}{\delta t_1}$ . The observed frequency is  $\nu_0 = \frac{1}{\delta t_0}$ . Thus, we obtain:

$$\frac{\nu_0}{\nu_1} = \frac{\delta t_1}{\delta t_0} = \frac{a(t_1)}{a(t_0)} \quad (17.76)$$

What an observer detects is the relation  $\nu_0 < \nu_1$ , i.e., a decrease in the observed frequency value, or in other words, a shift toward the red. Due to Equation 17.76, we have  $a(t_1) < a(t_0)$ .

This relationship implies that for  $t_1 < t_0$ , the factor  $a(t_0)$  grows, meaning the Universe expands or, correspondingly, the Planck patch expands. Thus, due to self-similarity, the expansion of the Planck patch will occur at any moment of its evolution after the inflationary period. Regarding the wavelength  $\lambda$ , Equation 17.75 is written as:

$$\frac{\lambda_1}{\lambda_0} = \frac{a(t_1)}{a(t_0)} < 1 \quad (17.77)$$

or:

$$\lambda_0 > \lambda_1$$

From this relationship, it follows that the wavelength of light reaching the observer increases. We define the quantity:

$$z = \frac{\lambda_0 - \lambda_1}{\lambda_1}$$

Thus, we obtain:

$$1 + z = \frac{a(t_0)}{a(t_1)} \quad (17.78)$$

In an expanding Universe like ours, the scale factor  $a(t)$  increases monotonically over time, where  $z$  is a positive number and visible distant celestial bodies appear redshifted.

### 17.9.3 The Friedmann Equation and the Flatness of the Universe

From Einstein's equations and the Friedmann-Robertson-Walker metric, we take:

$$d\tau^2 = dt^2 - a^2(t) \left[ \frac{dr^2}{1 - \kappa r^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right]$$

This relationship expresses a spatially isotropic and homogeneous space that evolves over time (via the parameter  $a(t)$ ), yielding the **Friedmann equation**:

$$\left( \frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{\kappa}{a^2}$$

The dot denotes differentiation with respect to  $t$ . The sign of the parameter  $\kappa$  expresses the type of spatial curvature. The rate of expansion of space—that is, the left side of the Friedmann equation—is the **Hubble parameter**, namely  $H(t)$ . From the fundamental Friedmann equation and for each value of the Hubble parameter, we define a critical density for each epoch:

$$\rho_{\text{crit}} = \frac{3H^2}{8\pi G}$$

Another useful quantity is the  $\Omega(t)$ -density parameter:

$$\Omega(t) = \frac{8\pi G}{3H^2} \rho = \frac{\rho}{\rho_{\text{crit}}}$$

From the fundamental Friedmann equation and the definition of  $\Omega(t)$ , we obtain the relationship:

$$\Omega(t) - 1 = \frac{\kappa}{H^2 a^2}$$

Thus, if we have:

$$(1) \quad \rho < \rho_{\text{crit}} \leftrightarrow \Omega(t) < 1 \leftrightarrow \kappa < 0$$

then the Universe is open;

$$(2) \quad \rho = \rho_{\text{crit}} \leftrightarrow \Omega(t) = 1 \leftrightarrow \kappa = 0$$

then the Universe is flat;

$$(3) \quad \rho > \rho_{\text{crit}} \leftrightarrow \Omega(t) > 1 \leftrightarrow \kappa > 0$$

then the Universe is closed.

From current measurements of the anisotropies in the cosmic microwave background radiation, we obtain a value for  $\Omega(t)$  close to unity, namely:

$$\Omega(t) \simeq 1$$

This result agrees with our assumption that the Universe is a two-dimensional flat space, i.e.,  $\mathbf{P}^2$ , and its interior is a three-dimensional space that we perceive as beings within this space. Furthermore, the curvature that the three-dimensional space appears to have as a whole is due to the different curvatures of the flat Planck patches in which various complex geometric structures have formed. This view is identical to the definition of a differentiable manifold. In the future, the value of  $\Omega(t)$  will be exactly unity because no form of matter will appear to exist anywhere, due to the immense distance that will separate the parts composing it, as observed by us today.

#### 17.9.4 Why is the Sky Dark at Night?

##### Olbers' Paradox

This question might seem amusing to many, as they would give the obvious answer that due to the rotation of the Earth around its axis, for a certain period of the twenty-four hours, our position finds itself on the dark side of the Earth away from where the Sun's rays illuminate. In this case, i.e., during the night, there is an insufficient number of stars to illuminate the night sky.

However, this answer did not satisfy the German astronomer Heinrich Olbers about 200 years ago, because he reasoned that the infinite stars of the Universe would be capable of sufficiently illuminating the night sky. Thus, in 1826, after making some assumptions that were obvious for his time, he arrived at a striking result that contradicted daily observation. For the first time, an answer to this and other cosmological problems was provided in the famous work *Eureka* by the American author and poet Edgar Allan Poe. On February 3, 1848, Poe gave a lecture in New York regarding the *Cosmogony of the Universe*. Poe conceived, for the first time, the idea of a dynamic, evolving Universe—something that Einstein found difficult to accept 70 years later! Poe's Universe evolves under the influence of two competing forces: a universal repulsion, which breaks the unity of the primordial atom," and the attraction of gravity," which tends to restore this unity.

Astronomers tried for 150 years to discover what the mistake was in the assumptions set by Olbers, which is why it was named "Olbers' paradox." Let us look at the assumptions set by Olbers:

1. **The Universe is a Euclidean space.**
2. **The Universe is static.**
3. **The average value of emitted energy per unit time and surface area is uniform.**
4. **The average radius of stars is identical everywhere and at all times.**

Free observers assume that if the density  $\rho$  of stars is constant, then the stars located in a spherical shell of radius  $r$  and thickness  $dr$  centered on the Earth emit luminous energy per unit time equal to:

$$E_0 = 16\pi^2 \epsilon \rho R^2 r^2 dr \quad (17.79)$$

The Earth receives an energy of  $4\pi^2 R^2 r R^2 \epsilon \rho$  per unit time, so its unit surface area receives an illumination of  $\pi R^2 \rho \epsilon dr$ . Integrating this quantity from 0 to  $\infty$ , we get infinity, i.e.,

$$\int_0^\infty 0\pi R^2 \epsilon \rho dr \rightarrow \infty$$

If at a distance  $r$  from a source of power  $E_0$ , the total energy received by the surface of a sphere of radius  $r$  per unit time equals  $E$ , then the percentage of energy intercepted by stars in the spherical shell of thickness  $dr$  is:

$$\frac{dE}{E} = -4\pi r^2 \rho R^2 \frac{1}{4\pi r^2} dr \quad (17.80)$$

or:

$$\ln \frac{E}{E_0} = -\pi \rho R^2 r \quad (17.81)$$



The unit surface area of the Earth receives on average an illumination of:

$$\frac{E}{4\pi r^2} \cdot \frac{\pi R^2 r}{4\pi R^2 r} = \frac{E_0 \cdot e^{-(\pi \rho R^2 r)}}{16\pi r^2} \quad (17.82)$$

We substitute  $E_0$  from Equation 17.79 into Equation 17.82, so as to obtain the total illumination from the relation:

$$\int_0^\infty \pi \rho \epsilon R^2 e^{-(\pi \rho R^2 r)} dr = \epsilon \quad (17.83)$$

Performing some calculations, we find that the intensity of radiation on the surface of the Earth should equal the intensity of radiation on the surface of the stars!! This could not occur simultaneously with our existence on Earth due to its immense temperature. Therefore, we must examine whether one or more of the assumptions set by Olbers do not hold.

Let us examine the first one:

We consider that the interior of the Universe is not a Euclidean space but a differentiable manifold, in contrast to its entirety. Thus, the first assumption does not hold.

Let us examine the second one:

As we know today, stars do not always maintain the same luminosity during their lifetime because they evolve into various celestial bodies that emit changing radiation in both intensity and frequency. Thus, the second assumption also does not hold.

Let us examine the third one:

Today, we know that the Universe is expanding such that its expansion is accelerated. This means that galaxies, and consequently the stars within them, are continuously moving away, meaning the Universe is not static.

The expansion of the Universe is demonstrated by the following diagrams (Figure 17.21) for a static Universe where the black lines are distributed everywhere in the spectrum, and for an expanding Universe where the black lines accumulate toward the red.

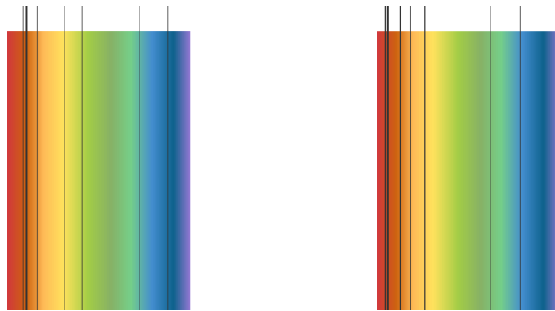


Figure 17.21: (a) Stationary light source (b) Moving light source

The expansion of the Universe is also visible in the subsequent photographs (Figure 17.22). In Figure 17.22, the first column on the left shows

the clusters of nebulae in the constellations of Virgo, Ursa Major, Corona Borealis, Boötes, and Hydra;

the second column in the middle shows their distance in light-years;

and the third column on the right shows their velocities derived from the shift of their spectrum toward the red.

The real spectrum is located in the center, while above and below lies the spectrum of a laboratory source in which dark lines (absorption lines) appear.

The spectrum of the laboratory source is called the “comparison spectrum.” In the real spectrum, there are one or two dark lines that have shifted toward the red end compared to the lines of the comparison spectrum, as shown in the subsequent figures. As clear from the photographs, the clusters of nebulae closest to us are more distinct, while the more distant ones are nearly invisible, thus refuting Olbers’ assumptions and consequently reaching the solution to the paradox.

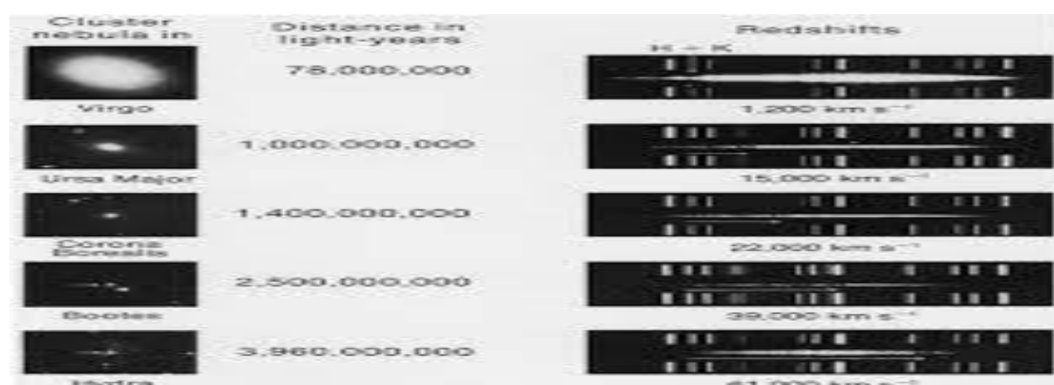


Figure 17.22: MOTION OF GALAXIES - EXPANSION OF THE UNIVERSE

(Photographs from the Palomar Observatory (California Institute of Technology))

## 17.10 INTRODUCTION TO DIFFERENTIAL FORMS

In reference [20], there is a comprehensive presentation of differential forms and their applications in physics. Here, we select certain useful results that are utilized in this text.

### 1. Notes and Definitions

If  $K$  denotes an  $n$ -dimensional space, then its boundary surface is denoted by  $\partial K$ . If  $K$  is *closed*, it includes its boundary surface ( $S$ ), whereas if it is *open*, it does not include its boundary surface.

Let  $M$  be an  $n$ -dimensional unit ball. The boundary surface of this ball is an  $(n - 1)$ -dimensional sphere  $\partial M$  that encloses it. Thus, we write:

$$M = \{x \in 3D: |x| \leq 1\} \quad \text{closed 3-ball}$$

$$M = \{x \in 3D: |x| < 1\} \quad \text{open 3-ball}$$

$$\partial M = \{x \in 3D: |x| = 1\} \quad \text{2-sphere}$$

For functions in  $3D$ , we write  $f : S \rightarrow 3D$ , which means that the argument of  $f$  is a point  $p \in S$  and  $f(p)$  yields a real number. According to this definition, the coordinates of a point of  $f$  are functions of that point.

## 2. Algebra of Differential Forms

We begin with a function  $f(x, y, z) \in 3D$ . From the formula  $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$  and the definition of  $x, y, z$  as functions of a point in  $3D$ , it follows that  $df$  and  $dx, dy, dz$  share the same form. Thus, all objects of this type (differentials of functions) are collected in the set  $\Lambda^1(3D)$ , which are called *1-forms*. It should also be noted that functions  $f$  commute with the differentials of functions, meaning for example  $f dx = dx f$ . If we multiply a pair of 1-forms, we obtain the objects:

$$f dx \wedge g dy = f g dx \wedge dy \quad (17.84)$$

The multiplication symbol  $\wedge$  (wedge) reminds us that the interchange rule applies, i.e.,

$$dx \wedge dy = -dy \wedge dx \quad (17.85)$$

Given that  $dx \wedge dy$  denotes a surface element, the orientation of the surface on which the element lies is also specified. Therefore, the interchange rule determines the orientation of  $dx \wedge dy$ . The algebra that includes this rule is called *exterior algebra* or *Grassmann algebra*, in honor of Hermann Grassmann (Grassmann, H. G.), who wanted to find an algebra that reflects elementary geometry. Exterior algebra is non-commutative due to the interchange rule. Thus, we have generally:

$$df \wedge dg = -dg \wedge df \quad (17.86)$$

If  $f = g$ , then from the preceding relation we obtain  $df \wedge df = 0$ , meaning that the product of a 1-form with itself is zero. All pairs of this type belong to the set  $\Lambda^2(3D)$ . Linear combinations of these objects are called *2-forms*. Finally, in  $3D$  there are objects  $f dx \wedge dy \wedge dz$  that belong to the set  $\Lambda^3(3D)$  and are called *3-forms*. These forms of  $3D$ , which are the top-order forms of  $3D$  (*top forms*), appear only as  $dx \wedge dy \wedge dz$ , due to the identity  $df \wedge df = 0$ . To complete the sets of forms in  $3D$ , we consider that all its functions belong to the set  $\Lambda^0(3D)$  and are called *0-forms*.

### 3. The Operator $d$

The operator  $d$  possesses the following properties:

1. It is linear.
2. For a function  $f \in \Lambda^0(3D)$ ,  $df$  is given by the formula:

$$df = \frac{\partial f}{\partial u^1} du^1 + \cdots + \frac{\partial f}{\partial u^n} du^n$$

3. If  $\omega \in \Lambda^j(3D)$  and  $\eta \in \Lambda^k(3D)$ , then  $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^j \omega \wedge d\eta$  (Leibniz rule).
4.  $dd = 0$  (Poincaré lemma: if  $\omega$  is a differential form, then we obtain  $dd\omega = 0$ ).

Let  $u^i, i = 1, 2, \dots, n$  be variables in  $nD$ . We select an ordering of  $du^i, i = 1, 2, \dots, n$ , which form the element  $du^1 \wedge du^2 \wedge \cdots \wedge du^n$  of a hypersurface in  $nD$ . We define that this element has the “correct” or positive orientation. Any other arrangement of  $du^i$  expressing an element of this hypersurface different from the correct element is transformed into the correct element via an interchange between the components  $du^i$ . If we obtain the correct ordering after interchanges among  $du^i$  but with a negative sign, then this hypersurface of  $R^n$  has a negative orientation. If the correct ordering is obtained after interchanges among  $du^i$  with a positive sign, then this hypersurface of  $R^n$  has a positive orientation.

### 4. Orientation

Let  $u^i, i = 1, 2, \dots, n$  be variables in  $D^n$ . We select an ordering of  $du^i, i = 1, 2, \dots, n$ , which form the element  $du^1 \wedge du^2 \wedge \cdots \wedge du^n$  of a hypersurface in  $D^n$ . We define that this element has the “correct” or positive orientation. Any other arrangement of  $du^i$  expressing an element of this hypersurface different from the correct element is transformed into the correct element via an interchange among the components  $du^i$ . If we obtain the correct ordering after interchanges among  $du^i$  but with a negative sign, then this hypersurface of  $D^n$  has a negative orientation. If we obtain the correct ordering after interchanges among  $du^i$  with a positive sign, then this hypersurface of  $D^n$  has a positive orientation.

### 5. Differential Forms and Vectors

In  $3D$ , there are typical analogies between forms and vectors; that is, differential forms function as vectors in many instances. In practice, things work smoothly if you make the following correspondences:

1. For 1-forms:  $dx \leftrightarrow \mathbf{e}_1$ ,  $dy \leftrightarrow \mathbf{e}_2$ ,  $dz \leftrightarrow \mathbf{e}_3$
2. For 2-forms:  $dy \wedge dz \leftrightarrow \mathbf{e}_1$ ,  $dz \wedge dx \leftrightarrow \mathbf{e}_2$ ,  $dx \wedge dy \leftrightarrow \mathbf{e}_3$

The ordering of 1-forms must be correct so that we can obtain the proper results due to the interchange rule. Through these correspondences, we can express the grad of a scalar function and the curl and div of a vector  $\mathbf{V}$  of vector analysis using differential forms. Thus, for:

1. 0-forms  $f$ , we obtain  $df \leftrightarrow \text{grad } f$
2. 1-forms  $\phi \leftrightarrow \mathbf{V}$ , we obtain  $d\phi \leftrightarrow \text{curl } \mathbf{V}$
3. 2-forms  $\eta \leftrightarrow \mathbf{V}$ , we obtain  $d\eta \leftrightarrow (\text{div } \mathbf{V})dx dy dz$
6. Integrals of Forms

Depending on the type of form we wish to integrate, we will find geometric loci of the same dimension as the corresponding forms, namely:

1. 1-forms must be integrated over a curve  $C$ .
2. 2-forms must be integrated over a surface  $S$ .
3. 3-forms must be integrated over a region  $M$ .

The calculation of an integral follows three steps:

1. The surface and volume elements must have the correct ordering.
2. We omit the wedges ( $\wedge$ ) to obtain an ordinary multiple integral.
3. We evaluate the integral according to known methods.

All integrals referred to in fundamental integral calculus as *Green's theorem*, the divergence theorem (*Gauss's theorem*), and *Stokes's theorem* are unified into the *generalized Stokes's theorem*, which is expressed as follows: Let  $K$  be an  $n$ -dimensional subset of  $dD$  and let  $\partial K$  be the appropriately oriented boundary surface of  $K$ . If  $\omega$  is an  $(n-1)$ -form, then:

$$\oint_K d\omega = \oint_{\partial K} \omega \quad (17.87)$$

From Equation 17.87, all the aforementioned theorems of integral calculus are derived. In an integral, it is more convenient to change variables in order to facilitate the calculation. For example, we want to calculate the integral:

$$\int f(x, y) dx \wedge dy$$

In region  $A$ , the old variables  $x, y$  are connected to the new variables  $u, v$  via the following relations:

$$x = x(u, v)$$

$$y = y(u, v)$$

Thus, we obtain:

$$f(x, y) = \tilde{f}(u, v)$$

and:

$$dx \wedge dy = \left( \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right) du \wedge dv$$

Thus, the integral is written as: Use code with caution.

$$\begin{aligned} \int f(x, y) dx \wedge dy &= \int \tilde{f}(u, v) \left( \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right) du \wedge dv = \\ &= \int \tilde{f}(u, v) \det \left( \frac{\partial(x, y)}{\partial(u, v)} \right) du \wedge dv \end{aligned} \quad (17.88)$$

### 7. The Inner Product of a Vector Space, the Function $\Phi$ , and the Operator $\star$

Let  $u^1, u^2, \dots, u^n$  be curvilinear coordinates in a neighborhood of a point on a manifold. From these, we obtain the position vector  $\mathbf{r}$ , its derivatives that span a basis in this neighborhood, and the inner product of two arbitrary vectors of this basis, namely:

$$\mathbf{r} = \mathbf{r}(u^1, u^2, \dots, u^n) \quad (17.89)$$

$$\mathbf{e}_i = \frac{\partial \mathbf{r}}{\partial u^i} \quad (17.90)$$

$$g_{ij} = (\mathbf{e}_i, \mathbf{e}_j) = \mathbf{e}_i \cdot \mathbf{e}_j \quad (17.91)$$

$$g^{ij} = (g_{ij})^{-1} \quad (17.92)$$

The components  $g_{ij}$  are the coefficients of the metric that allow us to work with distances and angles in the coordinate system  $u^1, u^2, \dots, u^n$ . The vectors  $\mathbf{e}_i$  are basis vectors of a point with position vector  $\mathbf{r}$ . Any vector  $\mathbf{v}$  that originates from this point is expanded in the basis  $(\mathbf{e}_i)$  as follows:

$$\mathbf{v} = v^i \mathbf{e}_i$$

In order to interpret  $v^i$ , we seek a function  $b^i$  whose argument is the vector  $\mathbf{v}$ , and the result of the action of  $b^i$  on  $\mathbf{v}$  is the component  $v^i$ , namely:

$$b^i(\mathbf{v}) = b^i(v^j \mathbf{e}_j) = v^j b^i(\mathbf{e}_j) = v^i \quad (17.93)$$

For the preceding relation to be valid, the condition  $b^i(\mathbf{e}j) = \delta j^i$  must be applied. This means that  $b^i$  is the basis element of a basis that is *dual to the basis that has the vector  $\mathbf{e}j$  as a basis element*. This basis has the 1-forms  $du^i$  as elements. Given that  $du^i : V \rightarrow R$ , all  $du^i$  belong to the *dual space*  $V^*$ , where  $V^*$  is the vector space of linear functions on  $V$ . We define the function  $\Phi : V \rightarrow V^*$  and its inverse  $\Phi^{-1} : V^* \rightarrow V$  as follows:

$$\Phi(\mathbf{v}) = g_{ij}v^i du^j \quad (17.94)$$

$$\Phi^{-1}(\omega) = g^{ij}\omega_i \mathbf{e}j \quad (17.95)$$

The function  $\Phi$  takes a vector as an argument and outputs a 1-form, and the function  $\Phi^{-1}$  takes a 1-form as an argument and outputs a vector. The symbol  $\star$  (*Hodge star operator*) is an operator defined on an  $n$ -dimensional space  $V$  as follows:

$$\star : \Lambda^r(V) \rightarrow \Lambda^{n-r}(V)$$

The operator  $\star$  possesses the following properties:

$$\star du^i = \sum_k (-1)^{k-1} g^{ik} \sqrt{g} du^1 \wedge \cdots \wedge du^{k-1} \wedge du^{k+1} \wedge \cdots \wedge du^n$$

$$\star \star \omega = (-1)^{r(n-r)} \omega \quad \text{with } \omega \in \Lambda^r(V^*)$$

$$\Omega_0 = \sqrt{g} du^1 \wedge \cdots \wedge du^n$$

$$\star \Omega_0 = 1$$

$$\star 1 = \Omega_0$$

#### 8. Formulas for grad, curl, and div in Curvilinear Coordinates

We will find the curl in 3D. As is well known, the argument of curl is a vector, and the result of this operation is also a vector. If we define  $\omega = v^1 dx + v^2 dy + v^3 dz$ , then:

$$d\omega = \left( \frac{\partial v^3}{\partial y} - \frac{\partial v^2}{\partial z} \right) dydz + \left( \frac{\partial v^1}{\partial z} - \frac{\partial v^3}{\partial x} \right) dzdx + \left( \frac{\partial v^2}{\partial x} - \frac{\partial v^1}{\partial y} \right) dxdy$$

Starting with the vector  $\mathbf{v} = v^1 \mathbf{e}1 + v^2 \mathbf{e}2 + v^3 \mathbf{e}3$ , we find  $\omega$  using the function  $\Phi$  and the formulas from the preceding paragraph. Thus, for orthogonal basis vectors, we obtain:

$$\begin{aligned} \Phi(\mathbf{v}) &= g_{ij}v^i du^j = (\mathbf{e}i, \mathbf{e}j)v^i du^j \\ &= \delta_{ij}v^i du^j = v^1 dx + v^2 dy + v^3 dz = \omega \end{aligned}$$

Given that  $\star d\omega = \left( \frac{\partial v^3}{\partial y} - \frac{\partial v^2}{\partial z} \right) dx + \left( \frac{\partial v^1}{\partial z} - \frac{\partial v^3}{\partial x} \right) dy + \left( \frac{\partial v^2}{\partial x} - \frac{\partial v^1}{\partial y} \right) dz$ , meaning that  $\star d\omega$  is a 1-form, we can then find  $\Phi^{-1}(\star d\omega)$ .

Thus, we obtain:

$$\Phi^{-1}(\star d\omega) = \left(\frac{\partial v^3}{\partial y} - \frac{\partial v^2}{\partial z}\right) \mathbf{e}_1 + \left(\frac{\partial v^1}{\partial z} - \frac{\partial v^3}{\partial x}\right) \mathbf{e}_2 + \left(\frac{\partial v^2}{\partial x} - \frac{\partial v^1}{\partial y}\right) \mathbf{e}_3$$

Combining these relations, we obtain:

$$\Phi^{-1}(\star d\omega) = \Phi^{-1}(\star d\Phi(\mathbf{v})) = \text{curl}(\mathbf{v}) \quad (17.96)$$

If we use the symbol  $\circ$  for the composition of functions, then the preceding relation is written as:

$$\Phi^{-1} \circ \star \circ d \circ \Phi = \text{curl} \quad (17.97)$$

Given that  $\star, d, \Phi$  are independent of coordinate systems,  $\text{curl}$  is also independent of coordinate systems. For  $\text{grad}$ , the composition of functions is directly apparent because when the operator  $d$  acts on a function  $f$ , it outputs a 1-form which becomes vector-valued through the action of  $\Phi^{-1}$  upon it; therefore, we have:

$$\Phi^{-1} \circ d = \text{grad} \quad (17.98)$$

In general, for curvilinear coordinates (not necessarily orthogonal), we have:

$$\begin{aligned} \text{grad}, f &= \Phi^{-1} df = \Phi^{-1} \left( \frac{\partial f}{\partial u^i} du^i \right) \\ &= \frac{\partial f}{\partial u^i} \Phi^{-1}(du^i) = g^{ij} \frac{\partial f}{\partial u^i} \mathbf{e}_j \end{aligned}$$

To find the expression for the  $\text{div}$  of a vector, which is a 0-form, we proceed as follows: we must transform a vector into an  $(n-1)$ -form. This vector is transformed by  $\Phi$  into a 1-form, and this 1-form is transformed into an  $(n-1)$ -form by the operator  $\star$ . The  $(n-1)$ -form is transformed into an  $n$ -form by the operator  $d$ , and finally it is transformed by the operator  $\star$  into a 0-form. This implies that the sequence of operators  $\star \circ d \circ \star \circ \Phi$  is a suitable choice to express  $\text{div}$ . We will apply all of this in orthogonal coordinates. Thus, we assume:

$$\mathbf{v} = v^i \mathbf{e}_i$$

$$\Phi(\mathbf{v}) = v^i du^i$$

$$\star \Phi(\mathbf{v}) = \sum_{k=1}^n v^k (-1)^{k-1} du^1 \wedge \cdots \wedge du^{k-1} \wedge \check{du}^k \wedge du^{k+1} \wedge \cdots \wedge du^n$$

The symbol  $\check{du}^k$  means that this factor is missing.



If the operator  $d$  acts on  $\star\Phi(\mathbf{v})$ , then the missing factor will appear in each term of the sum rather than in one of the already existing factors, due to the identity  $du^k \wedge du^k = 0$ . Thus, we finally obtain:

$$\begin{aligned} d \star \Phi(\mathbf{v}) &= \sum_{k=1}^n \frac{\partial v^k}{\partial u^k} du^1 \wedge du^2 \wedge \cdots \wedge du^n \\ \star d \star \Phi(\mathbf{v}) &= \sum_{k=1}^n \frac{\partial v^k}{\partial u^k} = \operatorname{div}(\mathbf{v}) \end{aligned} \quad (17.99)$$

If we take into account the generalized coordinates for  $\Phi$ , then  $\Phi(\mathbf{v}) = g_{ij}v^i du^j$  and the formula for  $\operatorname{div}(\mathbf{v})$  is expressed as:

$$\operatorname{div}(\mathbf{v}) = \frac{1}{\sqrt{g}} \sum_{k=1}^n \frac{\partial(\sqrt{g}v^k)}{\partial u^k} \quad (17.100)$$

Finally, we will find the expression for the Laplacian ( $\Delta$ ) of functions in any coordinate system. From the definition of the Laplacian equation, we obtain:

$$\Delta = \operatorname{div}(\operatorname{grad}, f) = \operatorname{div} \left( g^{ik} \frac{\partial f}{\partial u^i} \right) \quad (17.101)$$

$$\Delta L - B = \frac{1}{\sqrt{g}} \sum_{i,k=1}^n \frac{\partial}{\partial u^k} \left( \sqrt{g} g^{ik} \frac{\partial f}{\partial u^i} \right) \quad (17.102)$$

The operator  $\Delta_{L-B}$  in Equation 17.102 is called the *Laplace-Beltrami operator*.

### 9. Surfaces and Manifolds

*Surface* is the term commonly used for a two-dimensional object. The corresponding term for an  $m$ -dimensional subspace of  $nD$  is an *immersed manifold*, where this geometric object is enclosed by an  $n$ -dimensional space. If  $m = n - 1$ , then the immersed manifold is usually called a *hypersurface*. A ball in  $nD$  is the set of points located within a specified distance from a central point, i.e.,  $B_r(p) = \{x \in nD : |x - p| \leq r\}$ .

A homeomorphism is a *1-1* and *onto* mapping that is continuous and has a continuous inverse, from one set to another. Let  $p$  be a point of a manifold. We assume that there exists a radius  $r$  that is small enough, such that there is a homeomorphism between the intersection of  $B_r(p)$  with the manifold and a disk (which is an open set) in  $mD$ . In other words, the neighborhood of each point of the manifold resembles a disk in  $mD$ . Through this mapping, we say equivalently that the manifold has  $m$ -dimensions.

Now we must define a coordinate system in this region of the manifold.

The disk of  $mD$  has coordinates  $u^1, \dots, u^m$ , and if a point  $q$  of the manifold corresponds to the point  $y$  of the disk with coordinates  $u^1, \dots, u^m$ , then these

are used for the point  $p$ . Each point  $p'$  in the neighborhood of  $p$  has a region defined as  $B_s(p')$  and there is a homeomorphism with its own disk in  $mD$ , which yields the coordinates  $u'^1, \dots, u'^m$ . Every point of intersection of  $B_r(p)$  and  $B_s(p')$  will have two sets of coordinates  $u^1, \dots, u^m$  and  $u'^1, \dots, u'^m$ , where one set is a function of the other, namely:

$$u'^i = u'^i(u^1, \dots, u^m)$$

$$u^i = u^i(u'^1, \dots, u'^m)$$

The issue is that these functions must have multiple derivatives. For physics, we require at least three continuous derivatives. The manifold in this case is denoted by  $C^3$ . We will not mention the condition that must be met for embedding a manifold into a manifold with a larger dimension. Details can be found in the primary reference [20]. As we saw in paragraph 7 of this Appendix, the basis vectors at a point of a manifold are easily found from Equation 17.90. The vector space defined by the basis vectors at a point of the manifold  $M$  is called the *tangent space* and is denoted by  $T_p(M)$ . The embedding condition of a manifold into a manifold with more dimensions certifies that the basis vectors of  $T_p(M)$  are linearly independent. The dual space of  $T_p(M)$  is the space of 1-forms  $du^i$ , which is denoted by  $T_p^*(M)$ .

## 17.11 AFFINE CONNECTIONS - GEODESIC EQUATION

### 17.11.1 Affine Connections

As mentioned previously, for each point  $P$  of an  $n$ -dimensional differentiable manifold  $M$ , a tangent space  $T_P(M)$  of dimension  $n$  is defined. At points  $P, Q$  of  $M$ , the tangent spaces  $T_P(M)$  and  $T_Q(M)$  are isomorphic because both share the same dimensions. To establish a definite isomorphism that correlates  $T_P(M)$  with  $T_Q(M)$ , it is necessary to introduce an additional structure on the manifold called a **connection**.

This structure "connects" the tangent spaces at different points of a manifold. This connection can be achieved, for example, via a non-singular linear mapping of a basis ( $\mathbf{e}_P$ ) of  $T_P(M)$  to a basis ( $\mathbf{e}_Q$ ) of  $T_Q(M)$ . Let there be a differentiable curve in  $M$  connecting  $P$  and  $Q$ . A specific isomorphism  $\phi_{PQ}$  is required to map  $T_P(M)$  to  $T_Q(M)$ . The isomorphism  $\phi_{PQ}$  depends solely on the points  $P, Q$  and the curve joining them. Furthermore,  $\phi_{PQ}$  must satisfy two conditions:

1. If  $R$  is a point on the curve between  $P$  and  $Q$ , then it is required that:

$$\phi_{QP} = \phi_{QR} \circ \phi_{RP} \quad (17.103)$$

2. If  $\phi_{PQ}$  denotes the isomorphism  $T_Q(M) \rightarrow T_P(M)$ , then  $\phi_{PQ}$  must be the inverse of the isomorphism  $\phi_{QP}$ .

We assume that the points  $P, Q$  lie within a coordinate neighborhood that also contains the curve connecting these points. The curve is expressed by the relations introduced at the beginning of Section 17.10:

$$x^i = f^i(t) \quad , \quad 0 \leq t \leq 1 \quad (17.104)$$

where  $f(0) = P$  and  $f(1) = Q$ . We seek a function  $\phi_t$  at each point of the curve parameterized by  $t$ , such that the mapping:

$$\phi_t: T_0 \rightarrow T_t \quad (17.105)$$

is an isomorphism from the tangent space  $T_P$  to the tangent space  $T_t$ . In other words, we require that the relation:

$$\phi_t(\alpha \mathbf{a} + \beta \mathbf{b}) = \alpha \phi_t(\mathbf{a}) + \beta \phi_t(\mathbf{b}) \quad (17.106)$$

holds for any  $\mathbf{a}, \mathbf{b} \in T_0$ , where  $\alpha, \beta$  are real numbers. The function  $\phi_t$  can be expressed, with respect to a basis  $(\mathbf{e}_i)$  of  $T_P$ , by a non-singular matrix, and  $\mathbf{a}$  by a column vector. Thus, the mapping 17.105 is written in matrix form as:

$$\mathbf{a}_t = \phi_t \mathbf{a}_0 \quad (17.107)$$

Let us assume that the elements of the matrix  $\phi_t$  are functions of  $t$  possessing as many continuous derivatives as required. Differentiating Equation 17.107 with respect to  $t$ , we obtain:

$$\dot{\mathbf{a}}_t = \dot{\phi} \mathbf{a}_0 = (\dot{\phi} \phi^{-1}) \mathbf{a}_t$$

or:

$$\dot{\mathbf{a}}_t + \omega \mathbf{a}_t = 0 \quad (17.108)$$

where  $\dot{\phi} = \frac{d\phi}{dt}$  and  $\dot{\phi} \phi^{-1} = -\omega$ . The quantity  $\omega$  depends on the curve and the parameter  $t$ . Suppose now that  $\omega$  is restricted such that it depends only on  $x(t)$  and  $\dot{x}(t)$ , namely:

$$\omega = \omega(x, \dot{x}) \quad (17.109)$$

In this case, Equation 17.108 is expressed as:

$$\dot{\mathbf{a}}_t + \omega(x, \dot{x}) \mathbf{a}_t = 0 \quad (17.110)$$

From the theory of first-order linear differential equations, we know that a unique solution to Equation 17.110 exists for a given initial value  $\mathbf{a}_0$  at  $t = 0$  (*existence theorem*). This means that for a given initial vector  $\mathbf{a}_0$  at  $P$ , we obtain a vector field along the curve via Equation 17.110.

It is also necessary that the isomorphism  $\phi_t$  remains independent of the specific parameterization of the curve.

This implies that Equation 17.110 should be invariant under coordinate transformations of the variable  $x$ , meaning it depends only on the infinitesimal differences  $dx$  of the variable  $x$ . From the structure of Equation 17.110, this is possible only if  $\omega(x, \dot{x})$  is homogeneous of first degree with respect to  $\dot{x}$ .

From this, we conclude that an isomorphic mapping between the tangent spaces at all points of a curve is achieved by associating a matrix  $\omega$  with each of these points, whose elements are homogeneous of first degree with respect to  $\dot{x}$ . In this context, the matrix  $\omega$  is called an **affine connection**. According to the definitions above, the elements of the matrix  $\omega$  are written as:

$$\omega_j^i = L_{jk}^i \dot{x}^k \quad (17.111)$$

The quantities  $L_{jk}^i$  are called the **coefficients of the affine connection**. From the transformation laws between the coefficients of affine connections, we obtain:

$$L_{j'k'}^{i'} p_j^{j'} p_k^{k'} = L_{jk}^i p_i^{i'} - p_{jk}^{i'} \quad (17.112)$$

where  $p_k^{k'} = \frac{\partial x^{k'}}{\partial x^k}$  and  $p_{jk}^{i'} = \frac{\partial^2 x^{i'}}{\partial x^j \partial x^k}$ . From Equation 17.112, it is clear that the coefficients of the affine connections are not components of a tensor. Let us consider the coefficients  $L_{jk}^i$  and  $\bar{L}_{jk}^i$  of two affine connections in two overlapping coordinate neighborhoods. From Equation 17.112, it follows that the functions:

$$\alpha_{jk}^i = L_{jk}^i - \bar{L}_{jk}^i \quad (17.113)$$

are components of a tensor field. Due to Equation 17.111, the coefficients  $L_{jk}^i$  are not necessarily symmetric with respect to their lower indices. From a given set of connection coefficients  $L_{jk}^i$ , two other connections can be obtained: one with coefficients  $L_{kj}^i$ , and another symmetric connection given by the coefficients  $\Gamma_{jk}^i$  (**Christoffel symbols**), where:

$$\Gamma_{jk}^i = \frac{1}{2}(L_{jk}^i + L_{kj}^i) \quad (17.114)$$

The tensor with components  $T_{jk}^i$  defined by the relation:

$$T_{jk}^i = \frac{1}{2}(L_{jk}^i - L_{kj}^i) \quad (17.115)$$

is called the **torsion tensor** of the affine connection. From Equations 17.114 and 17.115, it follows that the torsion tensor of a symmetric affine connection  $\Gamma_{jk}^i$  is zero.

### 17.11.2 Covariant Differentiation

Let  $L_{jk}^i$  be the components of an affine connection in a coordinate neighborhood  $U$  of the differentiable manifold  $M$ , and let  $\gamma$  be the arc of a curve

lying in  $U$  with parametric equations  $x^i = f^i(t)$ , where  $0 \leq t \leq 1$ . Let  $P$  be a point on  $\gamma$  corresponding to a parameter value of 0, and let  $a_0^i$  be the components of a tangent vector  $\mathbf{a}_0$  at  $P$ . From Equation 17.110 and the assumptions regarding the matrix  $\omega$ , we obtain the following system of linear differential equations:

$$\frac{da^i}{dt} + L_{jk}^i a^j \frac{dx^k}{dt} = 0 \quad (17.116)$$

where  $L_{jk}^i$  are expressed as functions of  $t$  via the equations of the curve  $\gamma$ .

By virtue of the existence theorem for Equation 17.116, the functions  $a^i(t)$  define a vector field  $\mathbf{a}_t$  along  $\gamma$ , which is generated from  $\mathbf{a}_0$  at  $P$  **via parallel transport along the arc**  $\gamma$ . The solutions to the subsequent equations 17.117 represent the components  $a^i$  of the vector field  $\mathbf{a}_t$  at any point on the curve with parametric value  $t$ . The coefficients of the affine connection at this point are  $L_{jk}^i$ . Equations 17.110 take the identical form if we use a symmetric connection at this point, where the affine connection coefficients are the Christoffel symbols, namely:

$$\frac{da^i}{dt} + \Gamma_{jk}^i a^j \frac{dx^k}{dt} = 0 \quad (17.117)$$

Since Einstein's equations of General Relativity are derived from symmetric affine connections, Equations 17.117 will be used in physical applications. The components of the spacetime metric tensor, which constitute a solution to Einstein's equations of General Relativity, are correlated with  $\Gamma_{jk}^i$ ; therefore, by knowing the expressions for  $\Gamma_{jk}^i$ , we can theoretically solve Equations 17.117.

Thus, we can determine, within the tangent space of a point on a differentiable manifold, the components of an invariant vector that form its basis. These basis vectors are considered to arise from the basis vectors of a tangent space through their parallel transport along a curve. Let us assume that  $\mu_i(t)$  are the components of a given covariant vector field  $\mu_t$  that is constant along the arc  $\gamma$ . By differentiating the scalar function  $a^i \mu_i$  with respect to  $t$ , it is easily shown that the relation for  $\mu_i$  is given by:

$$\frac{D\mu_j}{dt} = \frac{d\mu_j}{dt} - \mu_i L_{jk}^i \frac{dx^k}{dt} \quad (17.118)$$

where  $\frac{D\mu_j}{dt}$  are the components of a covariant vector field along the arc  $\gamma$ . The covariant vector specified by these elements is called the absolute derivative (intrinsic derivative) of  $\mu_t$  and is denoted by  $\frac{D\mu_t}{dt}$ . If we set:

$$\frac{D\mu_j}{dt} = \frac{d\mu_j}{dt} - \mu_i L_{jk}^i \frac{dx^k}{dt} = 0 \quad (17.119)$$

we define the equation that governs the parallel transport of the covariant vector field  $\mu_t$  along the arc  $\gamma$ .

### 17.11.3 Geodesic Equation and Killing Vectors

Use code with caution. Equation 17.117, when evaluated for  $a^j = U^j = \frac{dx^j}{d\tau}$  (i.e., with respect to the four-velocity components), is written as:

$$\frac{d^2 x^i}{d\tau^2} + \Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dx^k}{d\tau} = 0 \quad (17.120)$$

Equation 17.120 is the **geodesic equation**. This can be expressed in terms of the four-momentum  $p^j = mU^j$  as [8]:

$$p^j \partial_j p^k - \Gamma_{jk}^i p^j p^k = 0$$

or:

$$m \frac{dp^i}{d\tau} = \frac{1}{2} (\partial_i g_{jk}) p^j p^k \quad (17.121)$$

If all components of the metric tensor are independent of the coordinate  $x^i$ , then  $p_i$  is a conserved momentum quantity because:

$$(\partial_i g_{jk}) = 0 \implies \frac{dp_i}{d\tau} = 0 \quad (17.122)$$

From Equation 17.122, it follows that this type of basis vectors, denoted by  $K = \partial_i$ , implies symmetries known as isometries, which are highly useful in studying particle motion in a curved spacetime. The components of  $K$  in the basis  $\mathbf{e}_j$  and its dual basis  $\mathbf{e}^k$  of a tangent space in curved spacetime are given by the following relations:

$$K_j = (\partial_i)j = \mathbf{e}_j \cdot \mathbf{e}_i = g_{ji} \quad (17.123)$$

$$K^j = (\partial_i)^j = \mathbf{e}^j \cdot \mathbf{e}_i = \delta_i^j \quad (17.124)$$

The quantities  $K_j$  or  $K^j$  are called **Killing vectors**. From Equation 17.123, we obtain:

$$K^j p_j = K_j p^j = p_i = \text{constant} \quad (17.125)$$

Equation 17.125 yields the corresponding *conserved quantity* for every Killing vector.

## 17.12 HOLOGRAPHIC METHOD

First, we will describe the procedure through which a hologram is recorded and reconstructed in 3D, using the following diagrams.

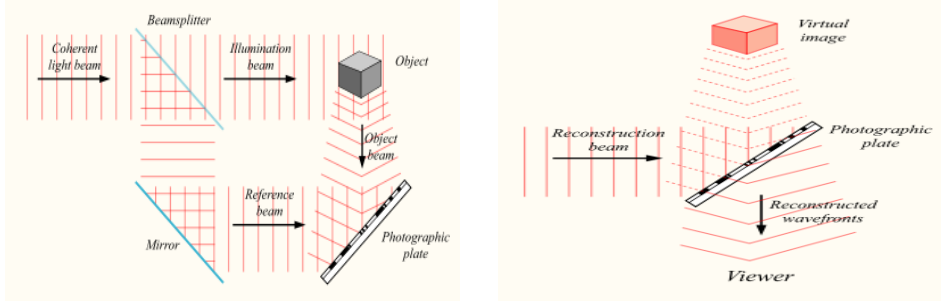


Figure 17.23: (a) Recording a hologram (b) Reconstructing a hologram (Hariharan, Basics of holography, Amazon, 2002)

A LASER beam is split into two beams by a beam splitter. One beam illuminates an object, and the other, with the aid of a mirror, is directed straight to the photographic plate. The image of the object is stored on the film by the interference of these two light beams. When the film is illuminated, the image of the object is reproduced in space. From this setup, we make two important observations:

1. Each piece of the film contains the image of the entire object.
2. If the LASER beams hit a section of the film at different angles, they will record multiple different images of the object from that single surface.

If these two observations are modified in their formulation, they can become the assumptions of the *holographic principle* as proposed by Gerard 't Hooft [12]. Thus, we obtain the assumptions of the holographic principle:

1. All the information contained in a region of space is mapped onto a hologram of the information present on the boundary surface enclosing this region.
2. The information on this surface should contain at most one degree of freedom per Planck area  $L_p^2$ , where the Planck length is  $L_p = 1.6 \cdot 10^{-33}$  cm.

Every object appearing in our observational space can be considered a holographic representation of information present on the surface bounding this space.

*The first question* that arises concerns the first assumption of the holographic principle:

If we assume that "objective reality" is determined by the information stored on the surface of the horizon of the observable universe, then everything inside it will be a hologram.

Thus, the human brain, according to the above assumption, is a hologram that additionally has the capacity to record information in its cerebral cortex.

The question that arises is what the human brain perceives and accepts as reality. Let us return to the recording process of holograms. It is clear that we obtain different holograms if the photographic plate is illuminated by light beams when they hit it at different angles.

From this observation, it appears that an infinite number of holograms can exist, which can be perceived by the human brain depending on how "a tangent plane of the cosmic horizon is illuminated," where all the information regarding the internal structures is stored.

This process also determines the consciousness of each human brain (**individual consciousness**), that is, the belief that what someone "sees" is what they call **reality**. Thus, consciousness determines reality!

From this individual consciousness, a collective consciousness is created, where a group of human brains (bound or free observers) accepts a specific reality, as described in Plato's allegory of the cave.

The second question that arises concerns the second assumption of the holographic principle:

If we assume that a Planck region  $L_p^2$  contains the minimum amount of information, then how many such regions must be collected to construct a new unit area? The information for creating all the holograms existing within the Observable Universe must reside on this new unit area.

The question that arises is:

**How can this vast amount of information exist within such a finite region?**

This impasse can be overcome by quantizing each new unit area through the sequential application of the natural equation of conic sections to a single initial circle, which possesses one degree of freedom in each  $L_p^2$ . The circle is chosen because it exhibits all the characteristics of  $2D$  space (isotropy, homogeneity, and homotopy). Through these properties of  $2D$  space, a large number of conic sections of all types (closed, closed-open, and open) are created within  $L_p^2$  of the  $2D$  space, interacting with each other. These conic sections are projected as holograms into  $3D$  space.



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